

Almost-Sure Radial Laws for Nested Random Polynomials and Erdős Problem #522

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Abstract

Let $\epsilon_0, \epsilon_1, \dots$ be independent random signs, each equal to ± 1 with probability $1/2$, and let $f_n(z) = \sum_{k=0}^n \epsilon_k z^k$. Erdős asked in 1961 whether, almost surely, these random Littlewood polynomials have $n/2 + o(n)$ zeros in the closed unit disk, counted with multiplicity. We answer this question, Erdős Problem #522, in the affirmative. In fact, we prove that, almost surely, $\frac{1}{n} \#\{z : f_n(z) = 0, |z| \leq 1 + x/n\}$ converges uniformly on compact subsets of $\mathbb{R}$ to $\frac{1}{2}(1 + \coth x - 1/x)$, interpreted as $1/2$ when $x = 0$. The same unit-disk and radial laws hold when the coefficients are independent and identically distributed with the Steinhaus distribution, the standard real Gaussian, the standard complex Gaussian, or any nondegenerate bounded centrally symmetric distribution. For random signs we also prove the almost-sure bound $\#\{z : f_n(z) = 0, |z| \leq 1\} = n/2 + O(n/\log n)$. Since different degrees share their coefficients, we first prove the result along a subsequence of degrees, and then fill in the other degrees with Rouché's theorem, once we show that few zeros near the unit circle have a small derivative. All main results have been formally verified in Lean.

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1 Introduction

We consider random polynomials whose coefficients are taken from the same sequence for every degree. Throughout, zeros are counted with multiplicity. Our main result is the following radial law.

Theorem 1.1 (Radial law). *Let $(\xi_k)_{k \geq 0}$ be independent and identically distributed, with the Rademacher, Steinhaus, standard real Gaussian or standard complex Gaussian distribution, or with a nondegenerate bounded centrally symmetric distribution on $\mathbb{R}$ or $\mathbb{C}$. Put $f_n(z) = \sum_{k=0}^n \xi_k z^k$, let $\nu_n(r)$ be the number of zeros of f_n in $\{|z| \leq r\}$, counted with multiplicity, with $\nu_n = 0$ when $f_n \equiv 0$, and put*

$$\Phi(x) = \frac{\int_0^1 t e^{2xt} dt}{\int_0^1 e^{2xt} dt}.$$

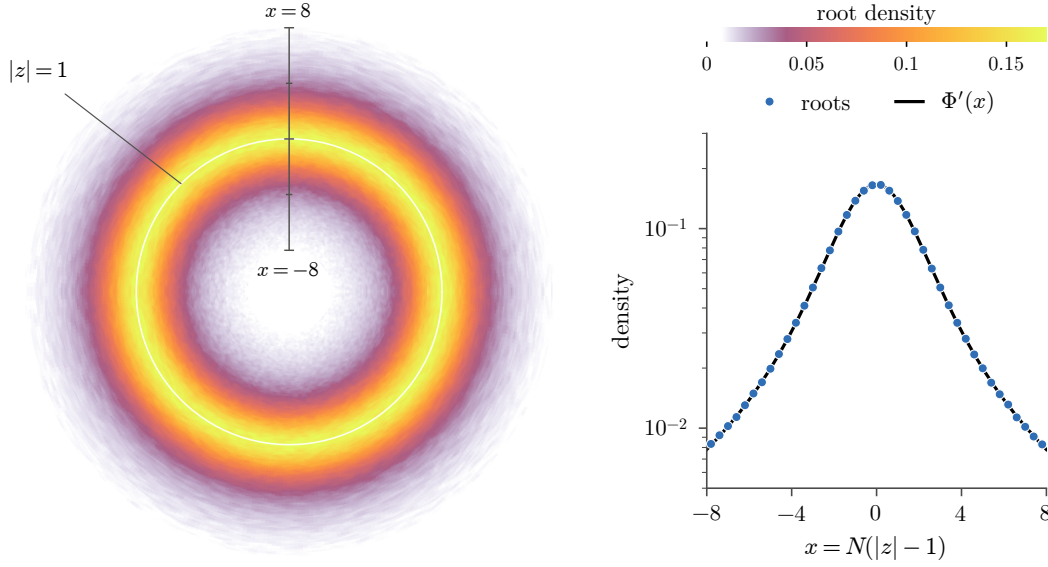
On one event of probability one, for every $A > 0$,

$$\lim_{n \rightarrow \infty} \sup_{|x| \leq A} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| = 0.$$

Explicitly,

$$\Phi(0) = \frac{1}{2}, \quad \Phi(x) = \frac{e^{2x}}{e^{2x} - 1} - \frac{1}{2x} \quad (x \neq 0).$$

We prove Theorem 1.1 for Rademacher coefficients in Section 6, and for the other distributions in Section 7. Here the Rademacher distribution gives ± 1 with probability $\frac{1}{2}$ each, the Steinhaus distribution is uniform on the unit circle, and the standard complex Gaussian distribution has density $\pi^{-1} e^{-|z|^2}$ on $\mathbb{C}$.



7.7×10^6 roots of 77 polynomials with random signs, $N=10^5$

Figure 1: Density of the zeros of 77 random Littlewood polynomials of degree 10^5 near the unit circle (left), and its angular average with the limiting density Φ' (right).

In particular, almost surely we have $\nu_n(1 + x/n)/n \rightarrow \Phi(x)$ simultaneously for every $x \in \mathbb{R}$, and the convergence is uniform on each compact interval of x . To obtain the explicit formula, note that the quotient defining Φ is the derivative of $\frac{1}{2} \log \int_0^1 e^{2xt} dt$, and that $\int_0^1 e^{2xt} dt = (e^{2x} - 1)/(2x)$ for $x \neq 0$, so that

$$\Phi(x) = \frac{d}{dx} \frac{1}{2} \log \frac{e^{2x} - 1}{2x} = \frac{e^{2x}}{e^{2x} - 1} - \frac{1}{2x} = \frac{1}{2} \left(1 + \coth x - \frac{1}{x} \right).$$

In particular $\Phi(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $\Phi(x) \rightarrow 1$ as $x \rightarrow \infty$. In Figure 1 we compare the limiting density Φ' with the roots of 77 polynomials with random signs. Taking $x = 0$ in Theorem 1.1, we obtain the following strong law.

Corollary 1.2 (The unweighted strong law). *In the setting of Theorem 1.1, almost surely*

$$\frac{\nu_n(1)}{n} \longrightarrow \frac{1}{2}.$$

For Rademacher coefficients, in the notation of Erdős problem #522 [4], where $R_n = \nu_n(1)$ is the number of roots in the closed unit disk, Corollary 1.2 states that $R_n/(n/2) \rightarrow 1$ almost surely. For these coefficients we prove Theorem 1.1 in Section 6, and Corollary 1.2 is its case $x = 0$.

For coefficients in $\{0, 1\}$, the polynomial f_n has the nonzero mean $D_n/2$, where $D_n(z) = \sum_{k=0}^n z^k$; by Remark B.8, its normalized size $|D_n(z)|/\sigma_n(|z|)$, where

$\sigma_n(r)^2 = \sum_{k=0}^n r^{2k}$, is $O(n^{-1/2})$ for $||z| - 1| \leq K/n$, with K fixed, at angles bounded away from 0.

For Rademacher coefficients we also obtain a rate of convergence in the unit disk.

Theorem 1.3 (Logarithmic rate). *In the setting of Theorem 1.1, for Rademacher coefficients,*

$$\limsup_{n \rightarrow \infty} (\log n) \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| \leq 128 \quad \text{almost surely.}$$

In Section 8 we also determine sufficient exponents for a sparse subsequence of degrees in the construction (Proposition 8.1), bound the number of roots in sublevel components that meet circles near the unit circle (Corollary 8.2), and compute the logarithmic variance profile of polynomially weighted polynomials with random signs (Proposition 8.3).

The question of convergence in probability appears in Hayman's problem collection [8, 9, Problem 4.15]. Yakir [25] proved that the proportion of roots in the unit disk converges to $1/2$ in probability, by combining concentration of an angular logarithmic integral with Jensen's formula. In his convention of n coefficients, with probability tending to one, the number of roots in the unit disk differs from $n/2$ by less than $n^{9/10}$. Do [7] proved that, for nested Kac polynomials with iid real coefficients of mean zero, variance one, and finite $(2 + \varepsilon)$ th moment, the number of real roots in $[-1, 1]$, divided by $\log n$, tends to $1/\pi$ almost surely. That proof uses maximal inequalities to pass from a sparse subsequence of degrees to all degrees. For the complex roots considered here, we compare consecutive degrees by root matching under the appended coefficient tail.

Conventions. Sections 2–6 treat Rademacher coefficients, written ϵ_k , and Section 7 the other distributions, written ξ_k . In $\nu_n(r)$ the disk is closed, the denominator in every limit is the nominal degree n , and $\nu_n = 0$ whenever f_n is identically zero. The radii $1 + x/n$ with $|x| \leq A$ are positive once $n > A$, and only sufficiently large n are used when $x < 0$. Whenever j indexes blocks of degrees, $N = j^8$ is the first degree of the block $j^8 \leq n < (j+1)^8$ (in the proof of Proposition 8.1, $N = \lfloor j^q \rfloor$), and these first degrees form the sparse subsequence of degrees. We write μ for the normalized arc measure $d\theta/(2\pi)$ on the circle $\mathbb{T}$ of angles θ , and $\log$ for the natural logarithm. For constants:

- a named constant, such as H_K , $\mathcal{B}_K$ or S_K , is fixed once defined, and Appendix A lists all of them;

- C_* and c_{HW} are the absolute constants of Nazarov–Nishry–Sodin [17] and of Rudelson–Vershynin [20];
- $C(K)$ is a positive constant that depends only on K (and, in Section 7, on the coefficient distribution), not on N , the radius, the angle or the coefficients; each $C(K)$ is used once and may change between estimates, and $C(K, B)$ may also depend on B ;
- $O_K(f(N))$ means at most $C(K)f(N)$ for all large N ;
- Section 8.1 uses only named constants.

The proof

The main difficulty is to pass from the sparse subsequence of degrees to every degree, which requires control of root movement under every partial appended tail. We show that most roots lie in a narrow annulus, and then control the roots at which the derivative is small. Figure 2 shows how the steps depend on one another.

We first estimate the angular average of the logarithm of the normalized polynomial (Theorem 3.1), using one- and two-point Gaussian approximations to the occupation of small disks, and logarithmic integrability near zero. The probability bounds are summable along the sparse subsequence of degrees. Jensen’s formula at four radii (Lemmas 2.1 and 2.4, and Corollary 3.3) then gives

$$\frac{1}{N} \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\} \leq \frac{2 \log 2}{K} + o(1),$$

so that K controls the fraction of roots outside the annulus.

Near each annular root with small derivative there is a mesh point at which the polynomial and its derivative are both small. We bound the first two moments of the number of such mesh points through covariance matrices of four real coordinates for one point and eight for a pair, and the number of roots per mesh point by Hanson–Wright concentration and Jensen’s formula. By Lemma 4.5, outside a summable exceptional event at most $C(K)N^{15/16} \log N$ annular roots have $|f'_N(\alpha)| < N^{3/2-1/64}$.

A Taylor expansion (Lemma 5.4) localizes each remaining root: disks of a common small radius about these roots are pairwise disjoint, each with exactly one root. We compare every degree n of a block $j^8 \leq n < (j+1)^8$ with its first degree $N = j^8$. A maximal bound on every appended tail (Lemma 4.6) gives the hypothesis of Rouché’s theorem on the boundary of each selected disk, and the tail amplitude sets the localization radius. The exponent eight makes the probability

bounds summable and the localization radius smaller than the half-width of the thin band.

By root matching (Theorem 5.1) we have

$$|\nu_n(1) - \nu_N(1)| \leq c_N + U_N + (\deg f_n - \deg f_N),$$

where c_N is the number of selected disks whose closures meet the unit circle and U_N is the number of unselected roots. The crossing disks have their roots in a thinner annulus, so the local radial secants of Lemma B.3 and the same concentration estimate show that $c_N/N \rightarrow 0$. At the radius $1+x/n$ the argument is the same, with the probes about $1+x/N$ and a band that also contains the circles $|z| = 1+x/n$ of the block. The unselected roots contribute at most $2 \log 2/K + o_K(1)$. For fixed K the exceptional probabilities are summable over the blocks, so by the Borel–Cantelli lemma these bounds hold for all large blocks on an event Ω_K of probability one, and we let $K \rightarrow \infty$ on $\bigcap_{K \geq 2} \Omega_K$, on the original probability space.

The probabilistic estimates use Raič’s convex-set normal approximation [18], the logarithmic integrability of Nazarov–Nishry–Sodin [17], and the Hanson–Wright inequality [20]. Appendix A computes the constants and their dependence on the annular width.

The Lean formalization [14] proves the unit-disk and radial laws on the shared coefficient space. Appendix C describes the theorem correspondence.

Reader’s guide

Sections 2–6 prove Theorem 1.1 and Corollary 1.2 for signs. Section 2 turns root counts into the angular logarithm by Jensen’s formula, Section 3 proves its concentration, Section 4 bounds local counts, small derivatives and appended tails, Section 5 matches roots by Rouché’s theorem, and Section 6 assembles the proof, with its parameters in Table 2. Section 7 states five hypotheses (H1)–(H5) under which that proof applies and verifies them for the other distributions. Section 8 proves Theorem 1.3 and further refinements, Section 9 poses open problems, and Section 10 describes the computations behind the figures. Appendices A, B and C give the explicit constants, the general forms of the probabilistic estimates, and the Lean correspondence. For Erdős #522 alone, Sections 2–6 at $x = 0$ suffice, with the general estimates of Appendix B.

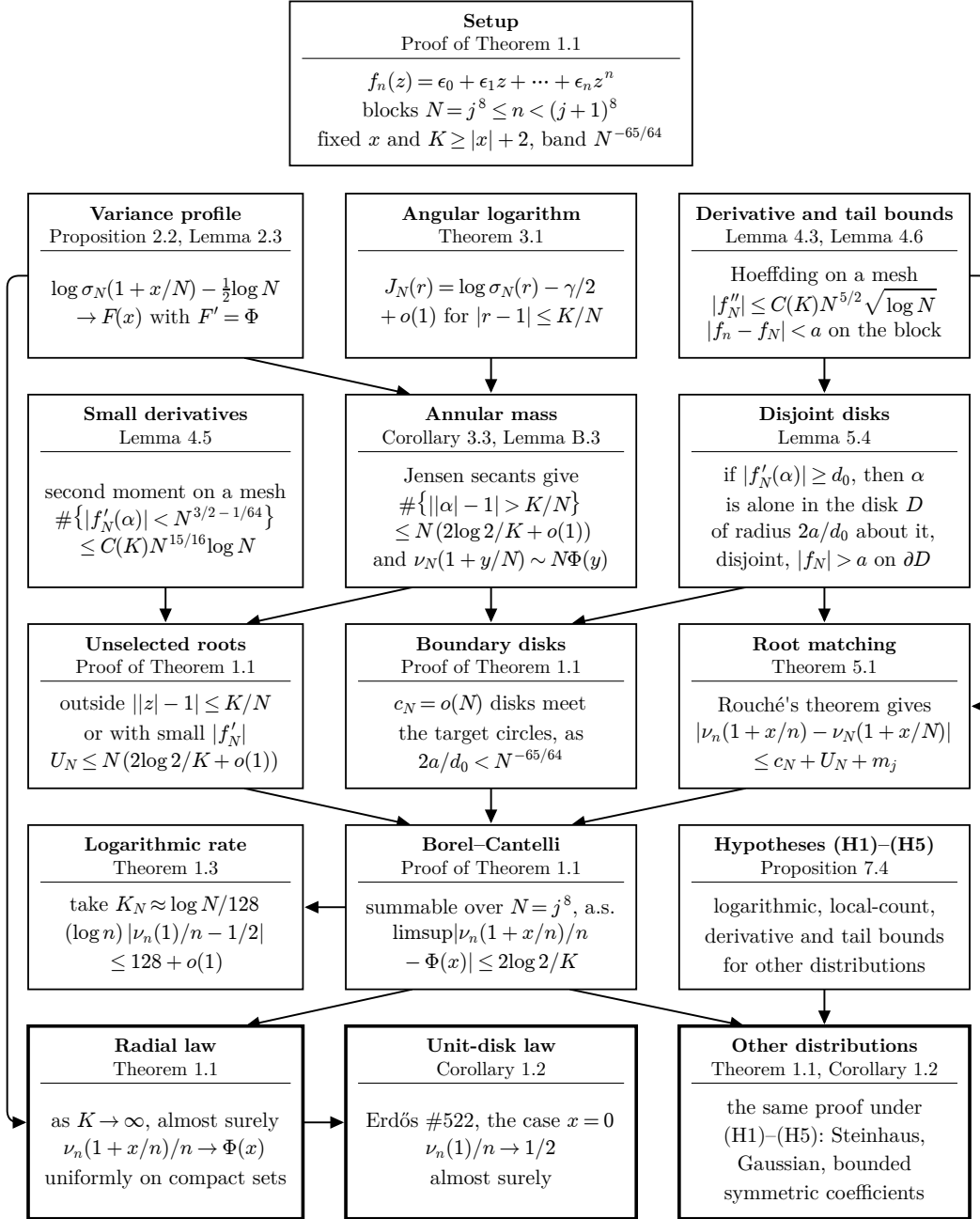


Figure 2: Structure of the proof. Arrows point from inputs to the steps that use them.

2 Jensen's formula and the variance profile

By Jensen's formula, the number of roots in a disk is the slope of the angular logarithm $J_N(e^t)$, a convex function of t , so bounds on J_N at four radii bound the roots outside an annulus. Replacing J_N by its Gaussian value, which Proposition 2.2

computes, gives the limiting fraction. Let $f_N(z) = \sum_{k=0}^N \epsilon_k z^k$, and set

$$\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}, \quad J_N(r) = \int_{\mathbb{T}} \log |f_N(re^{i\theta})| d\mu(\theta).$$

The results of this section hold for every nonzero polynomial f_N of degree at most N ; randomness enters only through Section 3. We write γ for Euler's constant.

Lemma 2.1 (Four-radius bound for radial mass). *Let f_N be a nonzero polynomial of degree at most N , and fix $K > 0$, $0 < \eta < 1$ and $N > K$. We set*

$$(x_1, x_2, x_3, x_4) = (-K, -(1-\eta)K, (1-\eta)K, K), \quad r_i = 1 + x_i/N.$$

Let $\sigma_N(r) > 0$ be a normalization, and suppose that at each of these four radii

$$(1) \quad |J_N(r_i) - \log \sigma_N(r_i) + \gamma/2| \leq \varepsilon_i, \quad i = 1, 2, 3, 4.$$

Then

$$(2) \quad \frac{\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\}}{N} \leq \frac{\deg f_N}{N} + \frac{\log \sigma_N(r_2) - \log \sigma_N(r_1) + \varepsilon_1 + \varepsilon_2}{N \log(r_2/r_1)} - \frac{\log \sigma_N(r_4) - \log \sigma_N(r_3) - \varepsilon_3 - \varepsilon_4}{N \log(r_4/r_3)}.$$

In particular, suppose that along a sequence of degrees $\deg f_N/N \rightarrow 1$, that the errors ε_i tend to zero, and that $\log \sigma_N(1 + x/N)$, minus a deterministic centering that may depend on N but not on x , converges to $F(x)$ uniformly on $[-K, K]$. Then the limiting upper bound for $\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\}/N$ is

$$(3) \quad 1 + \frac{1}{\eta K} \{F(-(1-\eta)K) - F(-K) - F(K) + F((1-\eta)K)\}.$$

We call $r_1, \dots, r_4$ the secant radii. Here the centering $-\gamma/2$ is $\mathbb{E} \log |G|$, where G has density $\pi^{-1} e^{-|z|^2}$ on $\mathbb{C}$, and it is the same at all four radii. If the four bounds (1) hold outside four exceptional events, then (2) holds outside their union, whose probability is at most the sum of their probabilities. The limiting bound (3) holds for every sample sequence on which the four bounds hold for all sufficiently large degrees.

Proof. We bound the roots in $\{|z| < r_1\}$ and in $\{|z| > r_4\}$ by the slopes of two secants of the convex function $t \mapsto J_N(e^t)$, and then replace J_N by $\log \sigma_N$ using (1). Write $f_N(z) = a \prod_{j=1}^{\deg f_N} (z - \alpha_j)$. By Jensen's formula [23, §3.61] for one factor,

$\int_{\mathbb{T}} \log |e^t e^{i\theta} - \alpha| d\mu(\theta) = \max(t, \log |\alpha|)$ for $\alpha \neq 0$, and the integral is t for $\alpha = 0$, so that

$$J_N(e^t) = \log |a| + \sum_{j=1}^{\deg f_N} \max(t, \log |\alpha_j|), \quad t \in \mathbb{R},$$

where a term with $\alpha_j = 0$ is read as t . The term of α_j has right derivative 1 if $|\alpha_j| \leq e^t$ and 0 otherwise, so $t \mapsto J_N(e^t)$ is convex and piecewise linear with right derivative $\nu_N(e^t)$. Integrating it and substituting $r = e^s$, we get

$$(4) \quad J_N(e^{t'}) - J_N(e^t) = \int_t^{t'} \nu_N(e^s) ds = \int_{e^t}^{e^{t'}} \frac{\nu_N(r)}{r} dr, \quad t < t'.$$

Since $N > K$, we have $0 < r_1 < r_2 < r_3 < r_4$, and since ν_N is nondecreasing, (4) gives

$$\begin{aligned} \nu_N(r_1) &\leq \frac{1}{\log(r_2/r_1)} \int_{r_1}^{r_2} \frac{\nu_N(r)}{r} dr = \frac{J_N(r_2) - J_N(r_1)}{\log(r_2/r_1)}, \\ \frac{J_N(r_4) - J_N(r_3)}{\log(r_4/r_3)} &= \frac{1}{\log(r_4/r_3)} \int_{r_3}^{r_4} \frac{\nu_N(r)}{r} dr \leq \nu_N(r_4). \end{aligned}$$

By (1), in which $\gamma/2$ cancels in differences, we have

$$\begin{aligned} J_N(r_2) - J_N(r_1) &\leq \log \sigma_N(r_2) - \log \sigma_N(r_1) + \varepsilon_1 + \varepsilon_2, \\ J_N(r_4) - J_N(r_3) &\geq \log \sigma_N(r_4) - \log \sigma_N(r_3) - \varepsilon_3 - \varepsilon_4. \end{aligned}$$

Let m be the number of roots with $||\alpha| - 1| > K/N$. At most $\nu_N(r_1)$ of them lie in $\{|z| < r_1\}$ and exactly $\deg f_N - \nu_N(r_4)$ in $\{|z| > r_4\}$, so by the last two displays

$$\begin{aligned} \frac{m}{N} &\leq \frac{\nu_N(r_1)}{N} + \frac{\deg f_N - \nu_N(r_4)}{N} \leq \frac{\deg f_N}{N} + \frac{J_N(r_2) - J_N(r_1)}{N \log(r_2/r_1)} - \frac{J_N(r_4) - J_N(r_3)}{N \log(r_4/r_3)} \\ &\leq \frac{\deg f_N}{N} + \frac{\log \sigma_N(r_2) - \log \sigma_N(r_1) + \varepsilon_1 + \varepsilon_2}{N \log(r_2/r_1)} \\ &\quad - \frac{\log \sigma_N(r_4) - \log \sigma_N(r_3) - \varepsilon_3 - \varepsilon_4}{N \log(r_4/r_3)}. \end{aligned}$$

This proves (2).

For the limit, let c be the centering of the hypothesis and $F_N(x) = \log \sigma_N(1 + x/N) - c$, so that $\log \sigma_N(r_i) = F_N(x_i) + c$. Then

$$\begin{aligned} N \log(r_2/r_1) &= \int_{-K}^{-(1-\eta)K} \frac{dx}{1 + x/N} \longrightarrow \eta K, \\ N \log(r_4/r_3) &= \int_{(1-\eta)K}^K \frac{dx}{1 + x/N} \longrightarrow \eta K. \end{aligned}$$

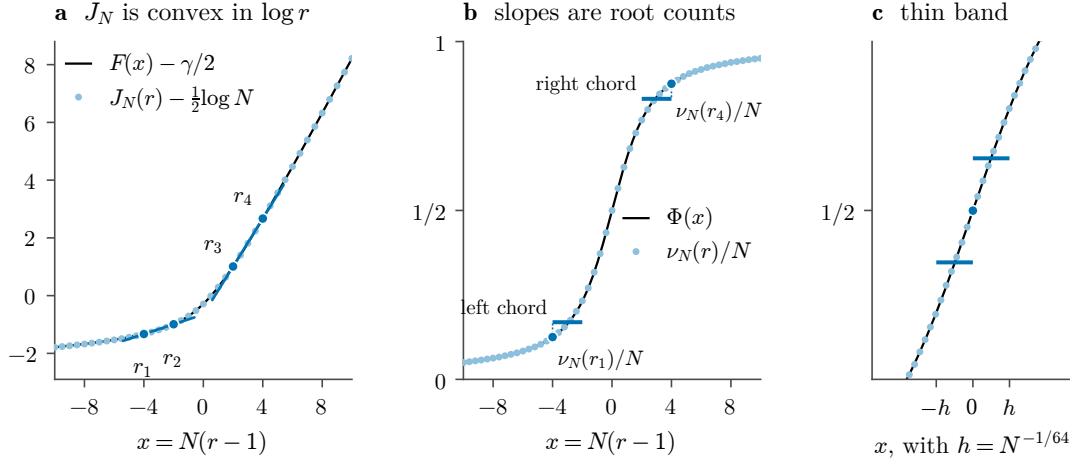


Figure 3: Jensen secants for one sequence at $N = 10^5$. The chord slopes bound the normalized zero counts, which approach Φ .

Since c cancels in differences, the right-hand side of (2) is the first line below, and since $\deg f_N/N \rightarrow 1$, $\varepsilon_i \rightarrow 0$ and $F_N(x_i) \rightarrow F(x_i)$,

$$\begin{aligned} & \frac{\deg f_N}{N} + \frac{F_N(x_2) - F_N(x_1) + \varepsilon_1 + \varepsilon_2}{N \log(r_2/r_1)} - \frac{F_N(x_4) - F_N(x_3) - \varepsilon_3 - \varepsilon_4}{N \log(r_4/r_3)} \\ & \rightarrow 1 + \frac{F(x_2) - F(x_1)}{\eta K} - \frac{F(x_4) - F(x_3)}{\eta K}, \end{aligned}$$

so that the limit superior of m/N is at most (3). $\square$

Figure 3 shows these secants for one sequence at $N = 10^5$. Section 8.1 lets η tend to zero with N , and all other uses take $\eta = 1/2$.

Proposition 2.2 (The unweighted logarithmic variance profile). *For the normalization $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$ of this section,*

$$\log \sigma_N(1 + x/N) - \frac{1}{2} \log N \rightarrow F(x) := \frac{1}{2} \log \int_0^1 e^{2xt} dt$$

uniformly on compact intervals of x . The profile satisfies

$$F'(x) = \Phi(x), \quad F(x) = x + F(-x), \quad F'(0) = \frac{1}{2}.$$

For $0 < \eta < 1$, the bound in (3) is

$$(5) \quad \frac{2}{\eta K} \{F(-(1-\eta)K) - F(-K)\} = \frac{1}{\eta K} \log \frac{1 - e^{-2(1-\eta)K}}{(1-\eta)(1 - e^{-2K})} \leq \frac{1}{\eta K} \log \frac{1}{1-\eta},$$

which at $\eta = 1/2$ is

$$\frac{2}{K} \log \frac{2}{1 + e^{-K}} \leq \frac{2 \log 2}{K}.$$

Proof. We first compare each term $(1 + x/N)^{2k}$ of $\sigma_N(1 + x/N)^2$ with $e^{2xk/N}$, which turns $\sigma_N(1 + x/N)^2/N$ into a Riemann sum for $\int_0^1 e^{2xt} dt$, and then derive the identities for F and the value of (3) from this integral.

For $|x| \leq K$, $N \geq 2K$ and $0 \leq k \leq N$, expanding the logarithm in its power series and using $|x/N| \leq K/N \leq 1/2$, we obtain

$$\begin{aligned} |2k \log(1 + x/N) - 2xk/N| &= 2k \left| -\frac{1}{2} \left(\frac{x}{N} \right)^2 + \frac{1}{3} \left(\frac{x}{N} \right)^3 - \cdots \right| \\ &\leq k \left(\frac{x}{N} \right)^2 \left(1 + \left| \frac{x}{N} \right| + \left(\frac{x}{N} \right)^2 + \cdots \right) = \frac{k(x/N)^2}{1 - |x/N|} \leq 2k \left(\frac{x}{N} \right)^2 \leq 2K^2/N. \end{aligned}$$

Exponentiating, we get

$$(6) \quad e^{-2K^2/N} \leq \frac{(1 + x/N)^{2k}}{e^{2xk/N}} \leq e^{2K^2/N}.$$

Since $(x, t) \mapsto e^{2xt}$ is uniformly continuous on $[-K, K] \times [0, 1]$ and the term $k = 0$ is $1/N$, the Riemann sums $\frac{1}{N} \sum_{k=0}^N e^{2xk/N}$ converge to $\int_0^1 e^{2xt} dt$ uniformly for $|x| \leq K$. Multiplying (6) by $e^{2xk/N}/N$ and summing over $0 \leq k \leq N$, we see that $\sigma_N(1 + x/N)^2/N = \frac{1}{N} \sum_{k=0}^N (1 + x/N)^{2k}$ lies between $e^{-2K^2/N}$ and $e^{2K^2/N}$ times the Riemann sum, so that it has the same limit. Since this limit is at least e^{-2K} and the logarithm is uniformly continuous on $[e^{-2K}/2, \infty)$, we obtain, uniformly for $|x| \leq K$,

$$\log \sigma_N(1 + x/N) - \frac{1}{2} \log N = \frac{1}{2} \log \frac{\sigma_N(1 + x/N)^2}{N} \longrightarrow \frac{1}{2} \log \int_0^1 e^{2xt} dt = F(x).$$

Differentiating the positive integral under the integral sign, we have

$$F'(x) = \frac{1}{2} \frac{\int_0^1 2te^{2xt} dt}{\int_0^1 e^{2xt} dt} = \frac{\int_0^1 te^{2xt} dt}{\int_0^1 e^{2xt} dt} = \Phi(x),$$

and in particular $F'(0) = \int_0^1 t dt = \frac{1}{2}$. Thus $\Phi(x)$ is the mean of t under the probability density proportional to e^{2xt} on $[0, 1]$, so that $0 \leq \Phi \leq 1$, and differentiating this mean under the integral sign we get

$$\begin{aligned} \Phi'(x) &= \frac{d}{dx} \frac{\int_0^1 te^{2xt} dt}{\int_0^1 e^{2xt} dt} = \frac{\int_0^1 2t^2 e^{2xt} dt}{\int_0^1 e^{2xt} dt} - \frac{\int_0^1 te^{2xt} dt \int_0^1 2te^{2xt} dt}{\left(\int_0^1 e^{2xt} dt \right)^2} \\ (7) \quad &= 2 \left(\frac{\int_0^1 t^2 e^{2xt} dt}{\int_0^1 e^{2xt} dt} - \left(\frac{\int_0^1 te^{2xt} dt}{\int_0^1 e^{2xt} dt} \right)^2 \right) = 2 \text{Var}_x(t) \geq 0, \end{aligned}$$

where $\text{Var}_x(t)$ is the variance of t under the same density. The change of variables $t \mapsto 1 - t$ gives the symmetry

$$F(x) = \frac{1}{2} \log \int_0^1 e^{2x(1-t)} dt = \frac{1}{2} \log \left(e^{2x} \int_0^1 e^{-2xt} dt \right) = x + F(-x).$$

By integration we also obtain, for $y > 0$, $\int_0^1 e^{-2yt} dt = (1 - e^{-2y})/(2y)$, and hence

$$\begin{aligned} F(-(1-\eta)K) - F(-K) &= \frac{1}{2} \log \frac{(1 - e^{-2(1-\eta)K})/(2(1-\eta)K)}{(1 - e^{-2K})/(2K)} \\ &= \frac{1}{2} \log \frac{1 - e^{-2(1-\eta)K}}{(1-\eta)(1 - e^{-2K})} \leq \frac{1}{2} \log \frac{1}{1-\eta}, \end{aligned}$$

since $1 - e^{-2(1-\eta)K} \leq 1 - e^{-2K}$. At $\eta = 1/2$ the expression before the inequality is

$$F(-K/2) - F(-K) = \frac{1}{2} \log \frac{2(1 - e^{-K})}{1 - e^{-2K}} = \frac{1}{2} \log \frac{2}{1 + e^{-K}}.$$

By $F(x) = x + F(-x)$ at $x = K$ and at $x = (1-\eta)K$,

$$\begin{aligned} F(K) - F((1-\eta)K) &= K + F(-K) - (1-\eta)K - F(-(1-\eta)K) \\ &= \eta K + F(-K) - F(-(1-\eta)K), \end{aligned}$$

and substituting this into (3) we get

$$\begin{aligned} 1 + \frac{1}{\eta K} \{F(-(1-\eta)K) - F(-K) - F(K) + F((1-\eta)K)\} \\ = 1 + \frac{1}{\eta K} \{2F(-(1-\eta)K) - 2F(-K) - \eta K\} \\ = \frac{2}{\eta K} \{F(-(1-\eta)K) - F(-K)\}, \end{aligned}$$

which is (5) by the evaluation of $F(-(1-\eta)K) - F(-K)$ above, and at $\eta = 1/2$ is

$$\frac{2}{K} \log \frac{2}{1 + e^{-K}} \leq \frac{2 \log 2}{K}.$$

□

This proof also yields a rate of convergence in the profile limit.

Lemma 2.3 (Rate in the variance profile). *For $K > 0$, $N \geq 2K$ and $|x| \leq K$,*

$$\left| \log \sigma_N(1 + x/N) - \frac{1}{2} \log N - F(x) \right| \leq \frac{K^2 + e^{4K}/2}{N}.$$

Proof. We use (6) to compare $\sigma_N(1 + x/N)^2/N$ with a Riemann sum of e^{2xt} , and the monotonicity of $t \mapsto e^{2xt}$ to compare the Riemann sum with the integral. Let

$$S_N(x) = \frac{1}{N} \sum_{k=0}^N e^{2xk/N}, \quad I(x) = \int_0^1 e^{2xt} dt,$$

so that $F = \frac{1}{2} \log I$. Multiplying (6) by $e^{2xk/N}/N$ and summing over $0 \leq k \leq N$, we get $e^{-2K^2/N} S_N(x) \leq \sigma_N(1 + x/N)^2/N \leq e^{2K^2/N} S_N(x)$, and hence

$$\left| \log \frac{\sigma_N(1 + x/N)^2}{N} - \log S_N(x) \right| \leq \frac{2K^2}{N}.$$

The function $g(t) = e^{2xt}$ is nondecreasing on $[0, 1]$ for $x \geq 0$ and nonincreasing for $x \leq 0$, so that $I(x)$ lies between the Riemann sums $\frac{1}{N} \sum_{k=0}^{N-1} g(k/N)$ and $\frac{1}{N} \sum_{k=1}^N g(k/N)$, in the order that gives

$$\begin{aligned} I(x) &\leq \frac{1}{N} \sum_{k=1}^N g(k/N) \leq S_N(x) = \frac{1}{N} \sum_{k=0}^{N-1} g(k/N) + \frac{g(1)}{N} \leq I(x) + \frac{g(1)}{N}, & x \geq 0, \\ I(x) &\leq \frac{1}{N} \sum_{k=0}^{N-1} g(k/N) \leq S_N(x) = \frac{1}{N} \sum_{k=1}^N g(k/N) + \frac{g(0)}{N} \leq I(x) + \frac{g(0)}{N}, & x \leq 0. \end{aligned}$$

In either case, since $g(0) = 1$ and $g(1) = e^{2x} \leq e^{2K}$,

$$I(x) \leq S_N(x) \leq I(x) + \frac{\max(g(0), g(1))}{N} \leq I(x) + \frac{e^{2K}}{N}.$$

For $|x| \leq K$ we also have $I(x) \geq e^{-2K}$, since $e^{2xt} \geq e^{-2K}$ for $0 \leq t \leq 1$. By the mean value theorem, $|\log a - \log b| \leq |a - b|/\min(a, b)$ for $a, b > 0$, and hence

$$0 \leq \log S_N(x) - \log I(x) \leq \frac{e^{2K}/N}{e^{-2K}} = \frac{e^{4K}}{N}.$$

Combining the two estimates, we obtain

$$\begin{aligned} \left| \log \sigma_N(1 + x/N) - \frac{1}{2} \log N - F(x) \right| &= \frac{1}{2} \left| \log \frac{\sigma_N(1 + x/N)^2}{N} - \log I(x) \right| \\ &\leq \frac{1}{2} \left| \log \frac{\sigma_N(1 + x/N)^2}{N} - \log S_N(x) \right| + \frac{1}{2} \left| \log S_N(x) - \log I(x) \right| \\ &\leq \frac{1}{2} \left(\frac{2K^2}{N} + \frac{e^{4K}}{N} \right) = \frac{K^2 + e^{4K}/2}{N}. \end{aligned}$$

□

Figure 4 shows $\Phi(x)$ as the mean of t under the limit, in $t = k/N$, of the tilted coefficient mass $Nr^{2k}/\sigma_N(r)^2$ at $r = 1 + x/N$.

2.1 Radial mass and thin bands

In the next two lemmas we turn bounds on J_N at a few radii into bounds on root counts, and in Section 3 these bounds are supplied by Theorem 3.1 outside exceptional events with summable probabilities.

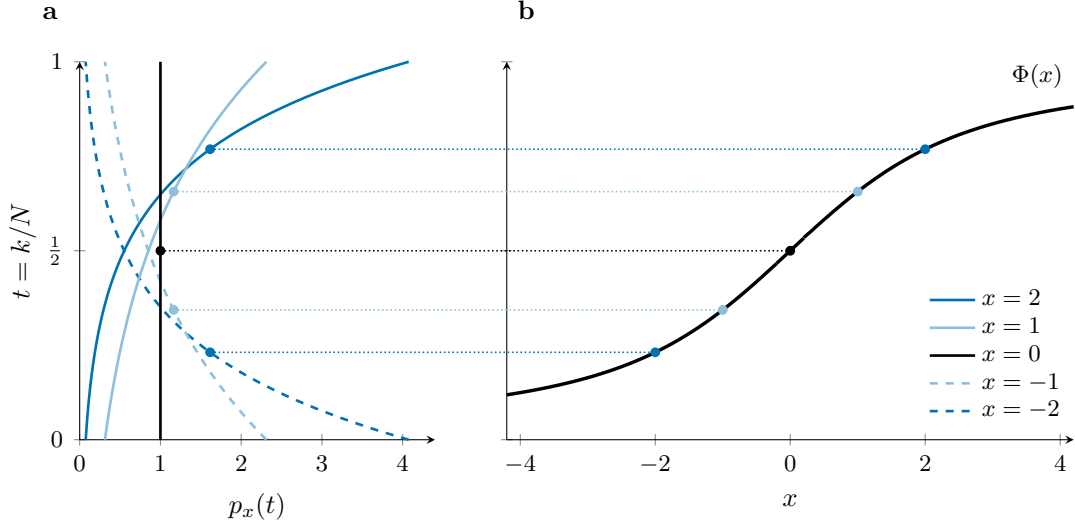


Figure 4: The densities $p_x(t) = 2xe^{2xt}/(e^{2x} - 1)$ (a) and their means $\Phi(x)$ (b).

Lemma 2.4 (Annular mass). *Let $K > 0$, $0 < \eta < 1$ and $N \geq 2K$, let f_N be a nonzero polynomial of degree at most N , and let $r_i = 1 + x_i/N$ with $(x_1, x_2, x_3, x_4) = (-K, -(1 - \eta)K, (1 - \eta)K, K)$. Let $\varepsilon, D \geq 0$ be such that*

$$\begin{aligned} |J_N(r_i) - \log \sigma_N(r_i) + \gamma/2| &\leq \varepsilon, \\ \left| \log \sigma_N(r_i) - \frac{1}{2} \log N - F(x_i) \right| &\leq D/N, \quad i = 1, 2, 3, 4. \end{aligned}$$

Then

$$\begin{aligned} &\frac{\#\{\alpha : f_N(\alpha) = 0, ||\alpha| - 1| > K/N\}}{N} \\ &\leq \frac{2}{\eta K} \{F(-(1 - \eta)K) - F(-K)\} + \frac{4}{\eta K} \left(\varepsilon + \frac{D}{N} \right) \\ &\leq \frac{1}{\eta K} \log \frac{1}{1 - \eta} + \frac{4}{\eta K} \left(\varepsilon + \frac{D}{N} \right). \end{aligned}$$

In particular, for $\eta = 1/2$,

$$\begin{aligned} \frac{\#\{\alpha : f_N(\alpha) = 0, ||\alpha| - 1| > K/N\}}{N} &\leq \frac{2}{K} \log \frac{2}{1 + e^{-K}} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right) \\ &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right). \end{aligned}$$

By Lemma 2.3, the second condition holds with $D = K^2 + e^{4K}/2$.

Proof. We divide the two secants of Lemma 2.1 by the linear spacing ηK in place of the logarithmic one, replace J_N by $\frac{1}{2} \log N + F$, and evaluate the leading term

by Proposition 2.2. Let m be the number of roots with $||\alpha| - 1| > K/N$. Since $0 < 1 + x/N \leq 1$ for $-K \leq x \leq 0$, because $N > K$, and $1 + x/N \geq 1$ for $x \geq 0$, the denominators satisfy

$$\begin{aligned} N \log(r_2/r_1) &= \int_{-K}^{-(1-\eta)K} \frac{dx}{1+x/N} \geq \eta K, \\ N \log(r_4/r_3) &= \int_{(1-\eta)K}^K \frac{dx}{1+x/N} \leq \eta K. \end{aligned}$$

By (4), $J_N(r)$ is nondecreasing in r , so that the numerators $J_N(r_2) - J_N(r_1)$ and $J_N(r_4) - J_N(r_3)$ are nonnegative. By the two hypotheses, in which $\gamma/2$ and $\frac{1}{2} \log N$ cancel in differences,

$$\begin{aligned} J_N(r_2) - J_N(r_1) &\leq F(x_2) - F(x_1) + 2\left(\varepsilon + \frac{D}{N}\right), \\ J_N(r_4) - J_N(r_3) &\geq F(x_4) - F(x_3) - 2\left(\varepsilon + \frac{D}{N}\right). \end{aligned}$$

By the second inequality of the bound on m/N in the proof of Lemma 2.1, $\deg f_N \leq N$, the bounds on the denominators and the two bounds above, we have

$$\begin{aligned} \frac{m}{N} &\leq 1 + \frac{J_N(r_2) - J_N(r_1)}{\eta K} - \frac{J_N(r_4) - J_N(r_3)}{\eta K} \\ &\leq 1 + \frac{1}{\eta K} \{F(x_2) - F(x_1) - F(x_4) + F(x_3)\} + \frac{4}{\eta K} \left(\varepsilon + \frac{D}{N}\right) \\ &= \frac{2}{\eta K} \{F(-(1-\eta)K) - F(-K)\} + \frac{4}{\eta K} \left(\varepsilon + \frac{D}{N}\right), \end{aligned}$$

where the last line is the evaluation of (3) in the proof of Proposition 2.2, and (5) bounds its first term. $\square$

The two secants contribute equally to the leading term $\frac{2}{\eta K} \{F(-(1-\eta)K) - F(-K)\}$ of Lemma 2.4: the inner secant of Lemma 2.1 controls the roots in $|z| < 1 - K/N$, and the outer secant controls those in $|z| > 1 + K/N$. By Lemma 2.3 and $N \log(r_2/r_1) \rightarrow \eta K$, the inner secant $(\log \sigma_N(r_2) - \log \sigma_N(r_1))/(N \log(r_2/r_1))$ tends to

$$\frac{F(-(1-\eta)K) - F(-K)}{\eta K}.$$

There are at most $N - \nu_N(r_4)$ roots in $|z| > 1 + K/N$, and $\nu_N(r_4)/N$ is at least the outer secant, as in the proof of Lemma 2.1. By the same limits and the symmetry $F(x) = x + F(-x)$, 1 minus the outer secant tends to

$$1 - \frac{F(K) - F((1-\eta)K)}{\eta K} = 1 - \frac{\eta K + F(-K) - F(-(1-\eta)K)}{\eta K}$$

$$= \frac{F(-(1-\eta)K) - F(-K)}{\eta K}.$$

As $\eta \rightarrow 0$ the leading term tends to the fraction of roots outside the annulus given by Theorem 1.1, since $F' = \Phi$ and $\Phi(x) + \Phi(-x) = 1$:

$$\lim_{\eta \rightarrow 0} \frac{2}{\eta K} \{F(-(1-\eta)K) - F(-K)\} = 2\Phi(-K) = 1 - \{\Phi(K) - \Phi(-K)\}.$$

Its upper bound $\log(1/(1-\eta))/(\eta K)$ tends to $1/K$, and the value $\eta = 1/2$ gives $2 \log 2/K$, which suffices when only $K \rightarrow \infty$ matters.

Lemma 2.5 (Thin bands). *Fix $0 < \kappa \leq 1/64$, let $N \geq 4$, and let f_N be a nonzero polynomial of degree at most N . Suppose that for some $\varepsilon \geq 0$,*

$$\left| J_N(1 + qN^{-1-\kappa}) - \log \sigma_N(1 + qN^{-1-\kappa}) + \gamma/2 \right| \leq \varepsilon, \quad q = -2, -1, 0, 1, 2.$$

Then, for $q = -1, 0, 1$,

$$\left| \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa} + 4\varepsilon N^\kappa.$$

Proof. We follow the proof of Lemma 2.1, now at five radii spaced by $N^{-1-\kappa}$. At each of the three middle radii, the count ν_N lies between the slopes of two secants of $t \mapsto J_N(e^t)$, and a bound on the second derivative of $\log \sigma_N(e^t)$ shows that the slopes of the corresponding secants of $\log \sigma_N(e^t)$, divided by N , are within $2N^{-\kappa}$ of $1/2$. Let $g(t) = \log \sigma_N(e^t)$. Differentiating the finite sum, we see that

$$g'(t) = \frac{\sum_{k=0}^N k e^{2kt}}{\sum_{k=0}^N e^{2kt}},$$

$$g''(t) = 2 \left\{ \frac{\sum_{k=0}^N k^2 e^{2kt}}{\sum_{k=0}^N e^{2kt}} - \left(\frac{\sum_{k=0}^N k e^{2kt}}{\sum_{k=0}^N e^{2kt}} \right)^2 \right\}.$$

That is, under the probability weights proportional to e^{2kt} on $\{0, 1, \dots, N\}$, $g'(t)$ is the mean of k and $g''(t)$ is twice its variance $\text{Var}_t(k)$. At $t = 0$ the weights are uniform, so that $g'(0) = \frac{1}{N+1} \sum_{k=0}^N k = \frac{N}{2}$. For a random variable Y with values in $[m, M]$, since $\mathbb{E}(Y - m)(M - Y) \geq 0$, we have

$$\begin{aligned} \text{Var}(Y) &= (\mathbb{E}Y - m)(M - \mathbb{E}Y) - \mathbb{E}(Y - m)(M - Y) \\ &\leq (\mathbb{E}Y - m)(M - \mathbb{E}Y) \leq \frac{(M - m)^2}{4}. \end{aligned}$$

Applying this bound with $Y = k$, $m = 0$ and $M = N$, we obtain

$$(8) \quad g''(t) = 2 \text{Var}_t(k) \leq 2 \cdot \frac{N^2}{4} = \frac{N^2}{2}.$$

Let $t_q = \log(1 + qN^{-1-\kappa})$ for $q = -2, -1, 0, 1, 2$. Since $N \geq 4$, we have $N^{-1-\kappa} \leq 1/4$, and since $|\log(1 + y)| \leq |y|/(1 - |y|) \leq 2|y|$ for $|y| \leq 1/2$, the logarithmic radii and their gaps satisfy

$$|t_q| \leq 2|q|N^{-1-\kappa} \leq 4N^{-1-\kappa},$$

$$t_{q+1} - t_q = \int_{1+qN^{-1-\kappa}}^{1+(q+1)N^{-1-\kappa}} \frac{du}{u} \geq \frac{N^{-1-\kappa}}{1 + (q+1)N^{-1-\kappa}} \geq \frac{N^{-1-\kappa}}{2}, \quad q = -2, -1, 0, 1.$$

By the hypothesis at $t_{-2}, \dots, t_2$, in which $\gamma/2$ cancels in differences, by (4) and since $\nu_N(e^t)$ is nondecreasing in t , for $q = -1, 0, 1$ we have

$$\begin{aligned} \frac{g(t_q) - g(t_{q-1}) - 2\varepsilon}{N(t_q - t_{q-1})} &\leq \frac{J_N(e^{t_q}) - J_N(e^{t_{q-1}})}{N(t_q - t_{q-1})} = \frac{1}{N(t_q - t_{q-1})} \int_{t_{q-1}}^{t_q} \nu_N(e^t) dt \\ &\leq \frac{\nu_N(1 + qN^{-1-\kappa})}{N} \leq \frac{1}{N(t_{q+1} - t_q)} \int_{t_q}^{t_{q+1}} \nu_N(e^t) dt \\ &= \frac{J_N(e^{t_{q+1}}) - J_N(e^{t_q})}{N(t_{q+1} - t_q)} \leq \frac{g(t_{q+1}) - g(t_q) + 2\varepsilon}{N(t_{q+1} - t_q)}. \end{aligned}$$

Let $[a, b]$ be $[t_{q-1}, t_q]$ or $[t_q, t_{q+1}]$. Since $g'(0) = N/2$, and $|g'(t) - g'(0)| \leq \frac{N^2}{2}|t|$ by (8) and the mean value theorem, with $|t| \leq 4N^{-1-\kappa}$ on $[a, b]$, we have

$$\begin{aligned} \left| \frac{g(b) - g(a)}{N(b - a)} - \frac{1}{2} \right| &= \frac{1}{N(b - a)} \left| \int_a^b (g'(t) - g'(0)) dt \right| \\ &\leq \frac{1}{N(b - a)} \int_a^b |g'(t) - g'(0)| dt \leq \frac{1}{N(b - a)} \int_a^b \frac{N^2}{2} |t| dt \\ &\leq \frac{N}{2} \cdot 4N^{-1-\kappa} = 2N^{-\kappa}. \end{aligned}$$

Moreover, since $b - a \geq N^{-1-\kappa}/2$, we have $2\varepsilon/(N(b - a)) \leq 4\varepsilon N^\kappa$. By the secant bounds and the last two estimates, we have

$$\begin{aligned} \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} &\leq \left(\frac{g(t_{q+1}) - g(t_q)}{N(t_{q+1} - t_q)} - \frac{1}{2} \right) + \frac{2\varepsilon}{N(t_{q+1} - t_q)} \leq 2N^{-\kappa} + 4\varepsilon N^\kappa, \\ \frac{1}{2} - \frac{\nu_N(1 + qN^{-1-\kappa})}{N} &\leq \left(\frac{1}{2} - \frac{g(t_q) - g(t_{q-1})}{N(t_q - t_{q-1})} \right) + \frac{2\varepsilon}{N(t_q - t_{q-1})} \leq 2N^{-\kappa} + 4\varepsilon N^\kappa. \end{aligned}$$

□

3 Logarithmic concentration for polynomials with random signs

Section 2 reduces root counts to the angular logarithm $J_N(r)$, so we now show that $J_N(r)$ is close to its Gaussian value $\log \sigma_N(r) - \gamma/2$. At one angle, and at

two separated angles, the normalized polynomial is close in law to one and to two independent copies of G , which controls how often it is small; logarithmic integrability controls the rest. Let $X_r(\theta) = f_N(re^{i\theta})/\sigma_N(r)$, so that $J_N(r) - \log \sigma_N(r) = \int \log |X_r| d\mu$, where G has density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$. We fix $K \geq 2$ before letting N tend to infinity; exact bounds and degree thresholds are in Appendix A.

Theorem 3.1 (Concentration of the angular logarithm). *Fix $K \geq 2$. For all sufficiently large N , uniformly for $r \in [1 - K/N, 1 + K/N]$,*

$$(9) \quad \begin{aligned} & \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > C(K)N^{-1/16}(\log N)^7\} \\ & \leq C(K)N^{-3/8} + N^{-12} + C(K)N^{-3/8}(\log N)^{-12}. \end{aligned}$$

Here the constants and the degree threshold depend only on K . In particular, these probabilities are summable along $N = j^8$, since there $N^{-3/8} = j^{-3}$.

Theorem 3.1 gives, at every radius of the annulus $[1 - K/N, 1 + K/N]$, the logarithmic bounds that Lemmas 2.4 and 2.5 require (Corollary 3.3). Section 3.1 controls the occupation of small disks, and Section 3.2 adds the logarithmic integrability of Nazarov–Nishry–Sodin and proves the theorem.

Figure 5 illustrates the theorem for one nested sequence.

3.1 One- and two-point bounds

We show that, for radii $r, s \in [1 - K/N, 1 + K/N]$ and for angles θ, ϕ outside small exceptional sets, the probabilities that $X_r(\theta)$ lies in a disk centered at 0 and that $(X_r(\theta), X_s(\phi))$ lies in a product of two such disks differ by $O_K(N^{-1/2})$ from those for one and for two independent copies of G . Summation by parts shows that the covariance matrices are close to $\frac{1}{2}I$, Raič's theorem and an interpolation of densities compare the laws with that of G , and we then deduce bounds for the mean and the variance of the angular occupation of a disk.

By the definition of X_r ,

$$\begin{pmatrix} \Re X_r(\theta) \\ \Im X_r(\theta) \end{pmatrix} = \sum_{k=0}^N \epsilon_k \frac{r^k}{\sigma_N(r)} \begin{pmatrix} \cos(k\theta) \\ \sin(k\theta) \end{pmatrix}.$$

Since the signs are independent with mean zero and variance one, the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta))$ is given by

$$V(r, \theta) = \frac{1}{\sigma_N(r)^2} \sum_{k=0}^N r^{2k} \begin{pmatrix} \cos^2(k\theta) & \cos(k\theta) \sin(k\theta) \\ \cos(k\theta) \sin(k\theta) & \sin^2(k\theta) \end{pmatrix}.$$

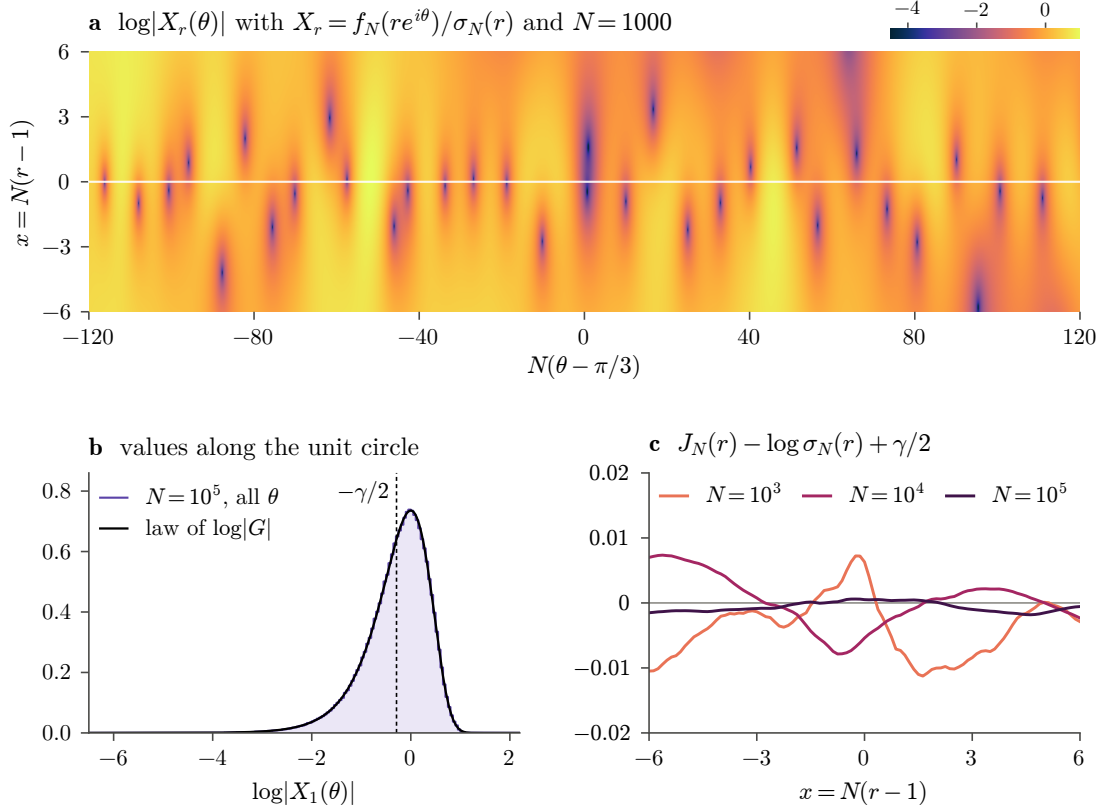


Figure 5: The angular logarithm near $e^{i\pi/3}$ for the sequence of Figure 9, with the zeros as dark spots (a), its distribution (b), and its centered mean at three degrees (c).

For two points, the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta), \Re X_s(\phi), \Im X_s(\phi))$ is the 4×4 matrix

$$\begin{pmatrix} V(r, \theta) & V(r, s, \theta, \phi) \\ V(r, s, \theta, \phi)^T & V(s, \phi) \end{pmatrix},$$

where

$$V(r, s, \theta, \phi) = \frac{1}{\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k \begin{pmatrix} \cos(k\theta) \cos(k\phi) & \cos(k\theta) \sin(k\phi) \\ \sin(k\theta) \cos(k\phi) & \sin(k\theta) \sin(k\phi) \end{pmatrix},$$

so that $V(r, \theta) = V(r, r, \theta, \theta)$. By the product-to-sum identities with $x, y \in \{k\theta, k\phi\}$, and since $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$, we get

$$V(r, \theta) = \frac{1}{2} I_2 + \frac{1}{2\sigma_N(r)^2} \sum_{k=0}^N r^{2k} \begin{pmatrix} \cos(2k\theta) & \sin(2k\theta) \\ \sin(2k\theta) & -\cos(2k\theta) \end{pmatrix},$$

$$V(r, s, \theta, \phi) = \frac{1}{2\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k$$

$$\times \begin{pmatrix} \cos(k(\theta - \phi)) + \cos(k(\theta + \phi)) & \sin(k(\theta + \phi)) - \sin(k(\theta - \phi)) \\ \sin(k(\theta + \phi)) + \sin(k(\theta - \phi)) & \cos(k(\theta - \phi)) - \cos(k(\theta + \phi)) \end{pmatrix}.$$

In terms of the rotation and the reflection

$$(10) \quad R(\psi) = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}, \quad F(\psi) = \begin{pmatrix} \cos \psi & \sin \psi \\ \sin \psi & -\cos \psi \end{pmatrix},$$

the matrix summands are $F(2k\theta)$ and $\Theta_k = R(k(\theta - \phi)) + F(k(\theta + \phi))$, so that

$$V(r, \theta) = \frac{1}{2}I_2 + \frac{1}{2\sigma_N(r)^2} \sum_{k=0}^N r^{2k} F(2k\theta),$$

$$V(r, s, \theta, \phi) = \frac{1}{2\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k \Theta_k.$$

The oscillatory sums are small unless 2θ , 2ϕ , $\theta - \phi$ or $\theta + \phi$ is close to $2\pi\mathbb{Z}$, and we remove these angles.

For one point, we remove the set $\text{dist}(\theta, \pi\mathbb{Z}) < N^{-1/2}$: two arcs of length $2N^{-1/2}$, of normalized measure $4N^{-1/2}/(2\pi) = 2N^{-1/2}/\pi \leq 2N^{-1/2}$. Every retained angle satisfies $\text{dist}(2\theta, 2\pi\mathbb{Z}) = 2 \text{dist}(\theta, \pi\mathbb{Z}) \geq 2N^{-1/2}$. For two points, we remove this set in each coordinate, and the sets $\text{dist}(\theta \mp \phi, 2\pi\mathbb{Z}) < N^{-1/2}$, each an arc of θ of normalized length $N^{-1/2}/\pi$ for fixed ϕ . The removed part of the angular square has measure at most $2N^{-1/2} + 2N^{-1/2} + 2N^{-1/2}/\pi \leq 10N^{-1/2}$.

The entries of these matrices are real and imaginary parts of weighted exponential sums, which we bound by summation by parts. Let $r, s \in [1 - K/N, 1 + K/N]$, $N \geq 2K$ and $\beta \in \{0, 1, 2\}$, and for $\text{dist}(\psi, 2\pi\mathbb{Z}) > 0$ set

$$b_k = (rs)^k (k/N)^\beta, \quad S_\ell(\psi) = \sum_{k=0}^{\ell} e^{ik\psi},$$

where $(k/N)^0 = 1$, including $k = 0$, and $S_{-1}(\psi) = 0$. Summing the geometric series, and since $|1 - e^{i\psi}| = 2|\sin(\psi/2)| \geq 2 \text{dist}(\psi, 2\pi\mathbb{Z})/\pi$, we get

$$S_\ell(\psi) = \frac{1 - e^{i(\ell+1)\psi}}{1 - e^{i\psi}}, \quad |S_\ell(\psi)| \leq \frac{2}{|1 - e^{i\psi}|} \leq \frac{\pi}{\text{dist}(\psi, 2\pi\mathbb{Z})}.$$

Summation by parts yields

$$\sum_{k=0}^N b_k e^{ik\psi} = \sum_{k=0}^N b_k (S_k(\psi) - S_{k-1}(\psi)) = b_N S_N(\psi) + \sum_{k=0}^{N-1} (b_k - b_{k+1}) S_k(\psi).$$

For $\beta = 0$ the weights are geometric. For $\beta > 0$ we have $b_0 = 0$ and the ratios $b_{k+1}/b_k = (rs)(1 + 1/k)^\beta$, $k \geq 1$, are nonincreasing, so the weights first

increase and then decrease (either part may be empty). Let $b_* = \max_{k \leq N} b_k$. The variation of such weights is the rise from b_0 to b_* plus the fall from b_* to b_N , and $b_k \leq (rs)^k \leq \max(1, rs)^N$ with $rs \leq (1 + K/N)^2$, so that

$$\begin{aligned} b_* &\leq (1 + K/N)^{2N} \leq e^{2K}, \\ \sum_{k=0}^{N-1} |b_{k+1} - b_k| &= 2b_* - b_0 - b_N, \\ b_N + \sum_{k=0}^{N-1} |b_{k+1} - b_k| &\leq b_0 + b_N + 2b_* \leq 4e^{2K}. \end{aligned}$$

Consequently, since $4\pi \leq 10e^{2K}$ for $K \geq 2$,

$$\begin{aligned} (11) \quad \left| N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik\psi} \right| &\leq \frac{\pi}{N \operatorname{dist}(\psi, 2\pi\mathbb{Z})} \left(b_N + \sum_{k=0}^{N-1} |b_{k+1} - b_k| \right) \\ &\leq \frac{4\pi e^{2K}}{N \operatorname{dist}(\psi, 2\pi\mathbb{Z})} \leq \frac{10e^{4K}}{N \operatorname{dist}(\psi, 2\pi\mathbb{Z})}. \end{aligned}$$

We keep the larger constant $10e^{4K}$, from which the later bounds are computed.

If $N \geq 2K$ and $r \in [1 - K/N, 1 + K/N]$, then each term of $\sigma_N(r)^2$ satisfies $e^{-4K} \leq (1 - K/N)^{2N} \leq r^{2k} \leq (1 + K/N)^{2N} \leq e^{2K}$, since $\log(1 - x) \geq -2x$ for $0 \leq x \leq K/N \leq 1/2$. Summing the $N + 1$ terms, we get the radial bounds

$$(12) \quad Ne^{-4K} \leq (N + 1)e^{-4K} \leq \sigma_N(r)^2 \leq (N + 1)e^{2K} \leq 2Ne^{2K}.$$

The entries of $R(\psi)$ and $F(\psi)$ are $\pm \cos \psi$ and $\pm \sin \psi$. Hence every entry of $V(r, \theta) - \frac{1}{2}I_2$ is $N/(2\sigma_N(r)^2)$ times $\pm$ the real or imaginary part of the sum in (11) with $s = r$, $\beta = 0$ and $\psi = 2\theta$, and every entry of $V(r, s, \theta, \phi)$ is $N/(2\sigma_N(r)\sigma_N(s))$ times the sum of two such terms, with $\beta = 0$ and $\psi = \theta \mp \phi$. At the retained angles, by (11) and the lower bound in (12), these entries are at most, in absolute value,

$$\begin{aligned} (13) \quad \frac{N}{2\sigma_N(r)^2} \left| \frac{1}{N} \sum_{k=0}^N r^{2k} e^{2ik\theta} \right| &\leq \frac{e^{4K}}{2} \cdot \frac{10e^{4K}}{N \cdot 2N^{-1/2}} \leq 10e^{8K} N^{-1/2}, \\ \frac{N}{2\sigma_N(r)\sigma_N(s)} \sum_{\pm} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k e^{ik(\theta \pm \phi)} \right| &\leq \frac{e^{4K}}{2} \cdot 2 \cdot \frac{10e^{4K}}{N \cdot N^{-1/2}} = 10e^{8K} N^{-1/2}. \end{aligned}$$

The first bound also holds for $V(s, \phi) - \frac{1}{2}I_2$. The operator norm of a $d \times d$ matrix is at most d times its largest entry, so at the retained angles $V(r, \theta) - \frac{1}{2}I_2$ and the 4×4 covariance matrix minus $\frac{1}{2}I_4$ have norm at most $40e^{8K} N^{-1/2}$. From now on $N \geq (160e^{8K})^2$, so that this norm is at most $1/4$ and every eigenvalue lies in $[1/4, 3/4]$.

Raič's theorem [18, Theorem 1.1] states that, for independent centered real vectors Y_k with total covariance I_d and a measurable convex set D ,

$$\left| \mathbb{P}\left(\sum_k Y_k \in D\right) - \mathbb{P}(G_d \in D) \right| \leq (42d^{1/4} + 16) \sum_k \mathbb{E}\|Y_k\|^3,$$

where G_d is standard Gaussian in $\mathbb{R}^d$. For a covariance matrix Σ we write G_Σ for a centered Gaussian vector with covariance Σ , so that $G_d = G_{I_d}$; distances and entropies between such vectors are those of their laws. For two points, the summands and their covariance are

$$u_k = \begin{pmatrix} r^k \cos(k\theta)/\sigma_N(r) \\ r^k \sin(k\theta)/\sigma_N(r) \\ s^k \cos(k\phi)/\sigma_N(s) \\ s^k \sin(k\phi)/\sigma_N(s) \end{pmatrix}, \quad \Sigma = \sum_{k=0}^N u_k u_k^T,$$

so that $\sum_{k=0}^N \epsilon_k u_k$ is the vector $(\Re X_r(\theta), \Im X_r(\theta), \Re X_s(\phi), \Im X_s(\phi))^T$, whose covariance Σ is the 4×4 matrix above. Let $Y_k = \epsilon_k \Sigma^{-1/2} u_k$. These are independent and centered, since $\mathbb{E}\epsilon_k = 0$ and $\mathbb{E}\epsilon_k^2 = 1$, and $\|Y_k\| \leq \|\Sigma^{-1/2}\| \|u_k\| \leq 2\|u_k\|$, since every eigenvalue of Σ is at least $1/4$. By $r^{2k}, s^{2k} \leq e^{2K}$ and the lower bound in (12),

$$\begin{aligned} \sum_{k=0}^N \mathbb{E}[Y_k Y_k^T] &= \Sigma^{-1/2} \left(\sum_{k=0}^N u_k u_k^T \right) \Sigma^{-1/2} = I_4, \\ \|u_k\|^2 &= \frac{r^{2k}}{\sigma_N(r)^2} + \frac{s^{2k}}{\sigma_N(s)^2} \leq \frac{2e^{6K}}{N}, \\ \sum_{k=0}^N \mathbb{E}\|Y_k\|^3 &\leq (N+1) \left(\frac{2\sqrt{2}e^{3K}}{\sqrt{N}} \right)^3 \leq 32\sqrt{2}e^{9K}N^{-1/2} < 46e^{9K}N^{-1/2}. \end{aligned}$$

For one point we use the first two coordinates, with $\Sigma = V(r, \theta)$, and get the smaller bound

$$\|u_k\|^2 = \frac{r^{2k}}{\sigma_N(r)^2} \leq \frac{e^{6K}}{N}, \quad \sum_{k=0}^N \mathbb{E}\|Y_k\|^3 \leq (N+1) \left(\frac{2e^{3K}}{\sqrt{N}} \right)^3 \leq 16e^{9K}N^{-1/2}.$$

Disks, products of disks and their images under whitening are measurable and convex.

To compare G_Σ with $G_{I_d/2}$, where $d = 2$ or 4 , let $\Delta = \Sigma - \frac{1}{2}I_d$, so that $\|\Delta\| \leq 40e^{8K}N^{-1/2} \leq 1/4$, and let p_λ be the centered Gaussian density with covariance $\Sigma_\lambda = \frac{1}{2}I_d + \lambda\Delta$, $0 \leq \lambda \leq 1$, whose eigenvalues lie in $[1/4, 3/4]$. Then

$$\begin{aligned} \partial_\lambda p_\lambda(x) &= \frac{p_\lambda(x)}{2} \left\{ x^T \Sigma_\lambda^{-1} \Delta \Sigma_\lambda^{-1} x - \text{tr}(\Sigma_\lambda^{-1} \Delta) \right\}, \\ \int_{\mathbb{R}^d} |\partial_\lambda p_\lambda(x)| dx &= \frac{1}{2} \mathbb{E} \left| G_d^T \Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2} G_d - \text{tr}(\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}) \right| \end{aligned}$$

$$\begin{aligned}
&\leq \frac{\sqrt{2d}}{2} \|\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}\| \leq 2\sqrt{2d} \|\Delta\|, \\
(14) \quad \text{TV}(G_\Sigma, G_{I_d/2}) &\leq \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\partial_\lambda p_\lambda(x)| \, dx \, d\lambda \leq \sqrt{2d} \|\Delta\| \leq 320e^{8K} N^{-1/2}.
\end{aligned}$$

The four lines of (14) hold for these reasons.

1. The derivative uses $\partial_\lambda \log \det \Sigma_\lambda = \text{tr}(\Sigma_\lambda^{-1} \Delta)$ and $\partial_\lambda \Sigma_\lambda^{-1} = -\Sigma_\lambda^{-1} \Delta \Sigma_\lambda^{-1}$.
2. The change of variables $x \mapsto \Sigma_\lambda^{-1/2} x$ carries p_λ to the law of G_d .
3. In an eigenbasis of $\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}$, with eigenvalues $a_1, \dots, a_d$, the quadratic form minus the trace is $\sum_i a_i (g_i^2 - 1)$ with independent standard real Gaussian g_i . It has mean zero and variance $2 \sum_i a_i^2 \leq 2d \|\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}\|^2$, so by Cauchy–Schwarz its mean absolute value is at most $\sqrt{2d}$ times this norm, which is at most $4\|\Delta\|$ since $\|\Sigma_\lambda^{-1}\| \leq 4$.
4. Total variation is half the L^1 distance, $p_1 - p_0 = \int_0^1 \partial_\lambda p_\lambda \, d\lambda$, and $\sqrt{2d} \leq 2\sqrt{2} \leq 8$ for $d \leq 4$.

The same computation gives, for positive-definite Σ_0, Σ_1 in dimension d such that every $\Sigma_0 + \lambda(\Sigma_1 - \Sigma_0)$, $0 \leq \lambda \leq 1$, has least eigenvalue at least $\mu > 0$, $\text{TV}(G_{\Sigma_1}, G_{\Sigma_0}) \leq \sqrt{2d} \|\Sigma_1 - \Sigma_0\| / (4\mu)$, since the norm of $\Sigma_\lambda^{-1/2} (\Sigma_1 - \Sigma_0) \Sigma_\lambda^{-1/2}$ is at most $\|\Sigma_1 - \Sigma_0\| / \mu$.

Let $F_G(t) = \mathbb{P}(|G| \leq t) = 1 - e^{-t^2}$, and for $t, t' \geq 0$ let D be the disk $\{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq t^2\}$ when $d = 2$ and the product of disks $\{x \in \mathbb{R}^4 : x_1^2 + x_2^2 \leq t^2, x_3^2 + x_4^2 \leq t'^2\}$ when $d = 4$. Since $\sum_k \epsilon_k u_k = \Sigma^{1/2} \sum_k Y_k$ and $\Sigma^{1/2} G_d$ has the law of G_Σ , we have $\mathbb{P}(\sum_k \epsilon_k u_k \in D) = \mathbb{P}(\sum_k Y_k \in \Sigma^{-1/2} D)$ and $\mathbb{P}(G_\Sigma \in D) = \mathbb{P}(G_d \in \Sigma^{-1/2} D)$. Hence, by Raič's theorem with the third-moment bounds above and by (14), at retained angles and pairs,

$$\begin{aligned}
&\left| \mathbb{P}\left(\sum_k \epsilon_k u_k \in D\right) - \mathbb{P}(G_{I_d/2} \in D) \right| \\
&\leq \left| \mathbb{P}\left(\sum_k Y_k \in \Sigma^{-1/2} D\right) - \mathbb{P}(G_d \in \Sigma^{-1/2} D) \right| + \left| \mathbb{P}(G_\Sigma \in D) - \mathbb{P}(G_{I_d/2} \in D) \right| \\
(15) \quad &\leq (42d^{1/4} + 16) \cdot 46e^{9K} N^{-1/2} + 320e^{8K} N^{-1/2} \leq C(K) N^{-1/2}.
\end{aligned}$$

For $d = 4$ the left-hand side is $|\mathbb{P}(|X_r(\theta)| \leq t, |X_s(\phi)| \leq t') - F_G(t)F_G(t')|$, since the two coordinate pairs of $G_{I_4/2}$ are independent copies of G , and for $d = 2$ it is $|\mathbb{P}(|X_r(\theta)| \leq t) - F_G(t)|$.

Let $A_t = \mu\{\theta : |X_r(\theta)| \leq t\}$ be the angular occupation. By Tonelli's theorem,

$$\mathbb{E}A_t = \int \mathbb{P}(|X_r(\theta)| \leq t) \, d\mu(\theta),$$

$$\mathbb{E}A_t^2 = \iint \mathbb{P}(|X_r(\theta)| \leq t, |X_r(\phi)| \leq t) d\mu(\theta) d\mu(\phi).$$

On the removed sets, of measures at most $2N^{-1/2}$ and $10N^{-1/2}$, the integrands differ from $F_G(t)$ and $F_G(t)^2$ by at most one; elsewhere we use (15), with $s = r$ and $t' = t$ for the pairs. Hence, uniformly for $t \geq 0$, since $0 \leq \mathbb{E}A_t + F_G(t) \leq 2$,

$$\begin{aligned} |\mathbb{E}A_t - F_G(t)| &\leq 2N^{-1/2} + C(K)N^{-1/2} \leq C(K)N^{-1/2}, \\ |\mathbb{E}A_t^2 - F_G(t)^2| &\leq 10N^{-1/2} + C(K)N^{-1/2} \leq C(K)N^{-1/2}, \\ \text{Var}(A_t) &= \mathbb{E}A_t^2 - F_G(t)^2 - (\mathbb{E}A_t - F_G(t))(\mathbb{E}A_t + F_G(t)) \\ (16) \quad &\leq |\mathbb{E}A_t^2 - F_G(t)^2| + |\mathbb{E}A_t - F_G(t)| |\mathbb{E}A_t + F_G(t)| \leq C(K)N^{-1/2}. \end{aligned}$$

The two-point argument also permits different radii and different disk thresholds.

3.2 Logarithmic integrability and angular energy

The normalized Fourier coefficients satisfy $\sum_{k=0}^N |r^k/\sigma_N(r)|^2 = 1$, which is the normalization in $L^2(\Omega \times \mathbb{T})$ of Corollary 1.2 of Nazarov–Nishry–Sodin [17]. By that corollary,

$$(17) \quad \mathbb{E} \int |\log |X_r||^p d\mu \leq (C_* p)^{6p}, \quad p \geq 1,$$

where $C_* \geq 1$ is an absolute constant; zeros on the circle give only integrable logarithmic singularities. By Parseval's identity, for every choice of signs, the angular energy is

$$(18) \quad \int |X_r|^2 d\mu = \frac{1}{\sigma_N(r)^2} \sum_{k=0}^N \epsilon_k^2 r^{2k} = 1.$$

We clip the logarithm at $\pm T$, control the clipped integral by the occupation bounds, and control the two tails by the logarithmic moments and the angular energy. Table 1 lists what depends on T . Any c in $(0, 3/32)$ would do. By Hölder's

Quantity	Bound	At $T = c \log N$	Requirement
Gaussian mass of $\{ z \leq e^{-T}\}$	$F_G(e^{-T}) \leq e^{-2T}$	N^{-2c}	tends to zero
Chebyshev bound for $A_{e^{-T}}$	$C(K)N^{-1/2}e^{4T}$	$C(K)N^{4c-1/2}$	summable in j
Lower tail ($e^{-2T} \geq N^{-1/2}$)	order $(\log N)^7 e^{-2T}$	$(\log N)^7 N^{-2c}$	tends to zero

Table 1: The quantities in Lemma 3.2 that depend on the clipping level $T = c \log N$, for signs; the last row needs $c \leq 1/4$. At $N = j^8$ the second is $C(K)j^{32c-4}$, summable when $c < 3/32$.

inequality in Theorem B.1 the lower tail is of order e^{-2T} , up to the logarithmic factor, so a larger c gives a smaller error and a larger failure probability. We take $c = 1/32$: then $T = (1/32) \log N$, $e^{-2T} = N^{-1/16} \geq N^{-1/2}$, the lower tail is $(\log N)^7 N^{-1/16}$, the error term of Theorem 3.1, and $N^{-1/2} e^{4T} = N^{-3/8}$, which is j^{-3} at $N = j^8$. This failure probability stays summable when $C(K)$ grows like a small power of N , as in Section 8.1. We state the argument for coefficients that need not be signs, for use in Section 7.

Theorem B.1, which we apply below, uses two facts about the standard complex Gaussian variable G . For every $T \geq 0$, since $F_G(e^{-\tau}) \leq e^{-2\tau}$ and $\mathbb{P}(|G| > e^\tau) \leq e^{-2\tau} \mathbb{E}|G|^2 = e^{-2\tau}$,

$$(19) \quad \int_T^\infty \mathbb{P}(|G| < e^{-\tau}) d\tau \leq \frac{1}{2} e^{-2T}, \quad \int_T^\infty \mathbb{P}(|G| > e^\tau) d\tau \leq \frac{1}{2} e^{-2T}.$$

Since $|G|^2$ has the exponential distribution with mean one, the classical integral for Euler's constant gives

$$(20) \quad \mathbb{E} \log |G| = \frac{1}{2} \mathbb{E} \log(|G|^2) = \frac{1}{2} \int_0^\infty e^{-y} \log y dy = -\frac{\gamma}{2}.$$

Lemma 3.2 (Clipping). *Let $N \geq e$ and $r > 0$, and let X_r and A_t be defined as above for a random polynomial f_N whose coefficients need not be signs. Let $H \geq 0$, $C > 0$ and $B_E > 0$ be constants and E an event such that, for all $t \geq 0$ and $p \geq 1$,*

$$|\mathbb{E} A_t - F_G(t)| \leq H N^{-1/2}, \quad \text{Var}(A_t) \leq H N^{-1/2},$$

$$\int_E \int |\log |X_r||^p d\mu d\mathbb{P} \leq (Cp)^{6p},$$

and $\int |X_r|^2 d\mu \leq B_E$ on E . Let $T = (1/32) \log N$ and

$$\begin{aligned} \varepsilon &= N^{-1/16} (\log N)^7 + 2HTN^{-1/2} \\ &\quad + e^{1/2} (4eC)^6 (2N^{-1/16} + H N^{-1/2}) (\log N)^7 + (B_E/2 + 1) N^{-1/16} \\ &\leq \left(2 + 2e^{1/2} (4eC)^6 + B_E/2\right) N^{-1/16} (\log N)^7 \\ &\quad + \left(\frac{1}{16} + e^{1/2} (4eC)^6\right) H N^{-1/2} (\log N)^7. \end{aligned}$$

Then

$$\begin{aligned} &\mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > \varepsilon\} \\ &\leq H N^{-3/8} + N^{-12} + \frac{H}{256} N^{-3/8} (\log N)^{-12} + \mathbb{P}(E^c). \end{aligned}$$

Proof. This is Theorem B.1 in the variant form given at the end of its statement, with $X = X_r$, $d_1 = v = H N^{-1/2}$, $\beta = 6$, $E_H = E$, $T = (1/32) \log N$, $u = N^{-1/16} (\log N)^7$,

$b = N^{-1/16}$, $q = \lceil \log N \rceil$ and $\lambda = (2e \log N/q)^6 \log N$. Its hypotheses hold, since the occupation bounds of the lemma are the bounds on $\mathbb{E}A_t$ and $\text{Var}(A_t)$ of that form, the moment bound of the lemma is (72) with $E_H = E$ and $\beta = 6$, and $\int |X_r|^2 d\mu \leq B_E$ on E . For $N \geq e$ we have $\log N \leq q \leq \log N + 1 \leq 2 \log N$, so that $\lambda \geq e^6 \log N > 1$,

$$\lambda(2Cq)^6 = \left(\frac{2e \log N}{q} \right)^6 (\log N)(2Cq)^6 = (4eC)^6 (\log N)^7$$

and

$$\lambda^{-2q} = \left[\left(\frac{q}{2e \log N} \right)^{12} (\log N)^{-2} \right]^q \leq [e^{-12} (\log N)^{-2}]^q \leq N^{-12},$$

where the last two steps use $q \leq 2 \log N$, $\log N \geq 1$ and $q \geq \log N$. By the first identity, the Markov event for W in the proof of Theorem B.1 is $E \cap \{W > ((4eC)^6 (\log N)^7)^{2q}\}$, where $W = \int |\log |X_r||^{2q} d\mu$. Since $e^{-2T} = b = N^{-1/16}$, the base $x = e^{-2T} + d_1 + b = 2N^{-1/16} + HN^{-1/2}$ satisfies $x \geq N^{-1}$, and hence $x^{-1/(2q)} \leq N^{1/(2q)} \leq e^{1/2}$, since $q \geq \log N$, so that

$$x^{1-1/(2q)} \leq e^{1/2} x = e^{1/2} (2N^{-1/16} + HN^{-1/2}),$$

and the error $\eta_{\log}$ of (75) is at most ε . The first three terms of (76) are $v/b^2 = HN^{-1/2}N^{1/8} = HN^{-3/8}$, $\lambda^{-2q} \leq N^{-12}$ and $4T^2 v/u^2 = 4HT^2 N^{-3/8} (\log N)^{-14} = (H/256)N^{-3/8} (\log N)^{-12}$, since $4T^2 = (\log N)^2/256$, and $p_E + p_H$ is replaced by $\mathbb{P}(E^c)$. Since $J_N(r) - \log \sigma_N(r) = \int \log |X_r| d\mu$ and $\mathbb{E} \log |G| = -\gamma/2$ by (20), this is the bound of the lemma. The bound on ε in the statement uses $2TN^{-1/2} = \frac{1}{16}N^{-1/2} \log N \leq \frac{1}{16}N^{-1/2} (\log N)^7$ and $N^{-1/16} \leq N^{-1/16} (\log N)^7$ for $N \geq e$, and keeps the terms with H , which carry the factor $N^{-1/2}$, apart from the others. $\square$

Proof of Theorem 3.1. By (16), (17) and (18), Lemma 3.2 applies with H the constant $C(K)$ of (16), $C = C_*$, $E = \Omega$ and $B_E = 1$. Since C_* is an absolute constant and $N^{-1/2} \leq N^{-1/16}$, its error ε is at most $C(K)N^{-1/16} (\log N)^7$, and, since $\mathbb{P}(E^c) = 0$, its probability bound is at most the right-hand side of (9). All constants above depend only on K , and all bounds hold uniformly for $r \in [1 - K/N, 1 + K/N]$. $\square$

3.3 Radial mass and thin bands with high probability

Corollary 3.3 (Annular mass and thin bands). *Fix $K \geq 2$ and $0 < \kappa \leq 1/64$, and take N sufficiently large. Outside four events, each of probability at most the right-hand side of (9),*

$$(21) \quad \frac{\#\{\alpha : ||\alpha| - 1| > K/N\}}{N} \leq \frac{2 \log 2}{K} + C(K)N^{-1/16} (\log N)^7.$$

Outside five further events of the same kind, for $q = -1, 0, 1$,

$$(22) \quad \left| \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa} + C(K)N^{\kappa-1/16}(\log N)^7.$$

By Theorem 3.1, the sum of the probabilities of these nine exceptional events is summable at $N = j^8$.

Proof. We apply the deterministic Lemmas 2.4 and 2.5 outside the exceptional events of Theorem 3.1 at their radii; $f_N \neq 0$ since $f_N(0) = \epsilon_0 \neq 0$.

The four secant radii $r_i = 1 + x_i/N$ of Lemma 2.4 with $\eta = 1/2$ lie in $[1 - K/N, 1 + K/N]$. Outside the four logarithmic exceptional events at these radii, the first hypothesis of that lemma holds with $\varepsilon = C(K)N^{-1/16}(\log N)^7$, by Theorem 3.1, and the second with $D = K^2 + e^{4K}/2$, by Lemma 2.3, since $|x_i| \leq K$ and $N \geq 2K$. Therefore Lemma 2.4 at $\eta = 1/2$ and $1/N \leq N^{-1/16}(\log N)^7$, for $N \geq 3$, yield

$$\begin{aligned} \frac{\#\{\alpha : ||\alpha| - 1| > K/N\}}{N} &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(C(K)N^{-1/16}(\log N)^7 + \frac{K^2 + e^{4K}/2}{N} \right) \\ &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(C(K) + K^2 + \frac{e^{4K}}{2} \right) N^{-1/16}(\log N)^7 \\ &\leq \frac{2 \log 2}{K} + C(K)N^{-1/16}(\log N)^7, \end{aligned}$$

which is (21).

The five radii $1 + qN^{-1-\kappa}$, $q = -2, \dots, 2$, lie in the same annulus, since $|qN^{-1-\kappa}| \leq 2N^{-1-\kappa} \leq K/N$. Outside the five logarithmic exceptional events at these radii, Lemma 2.5 applies with $\varepsilon = C(K)N^{-1/16}(\log N)^7$ (and $N \geq 4$), and bounds the left-hand side of (22) by $2N^{-\kappa} + 4C(K)N^{-1/16}(\log N)^7N^\kappa = 2N^{-\kappa} + 4C(K)N^{\kappa-1/16}(\log N)^7$. Both terms tend to zero, since $\kappa \leq 1/64 < 1/16$. $\square$

Figure 6 compares the fraction of roots outside the annulus with $2 \log 2/K$, the part of (21) that does not tend to zero. Panel (a) also shows the limit $1 - \{\Phi(K) - \Phi(-K)\}$ of the fraction, and panel (b) shows K times its difference from this limit, with pointwise 95% bootstrap bands, both over 77 sequences.

4 Local zeros and small derivatives

Besides the logarithmic concentration, the proof of the radial law uses four estimates near the unit circle: local zero counts, a bound on f_N'' , few annular zeros with a small derivative, and a uniform bound on appended tails. The zero counts come from Jensen's formula at a point where $|f_N|$ is large, which the Hanson–Wright

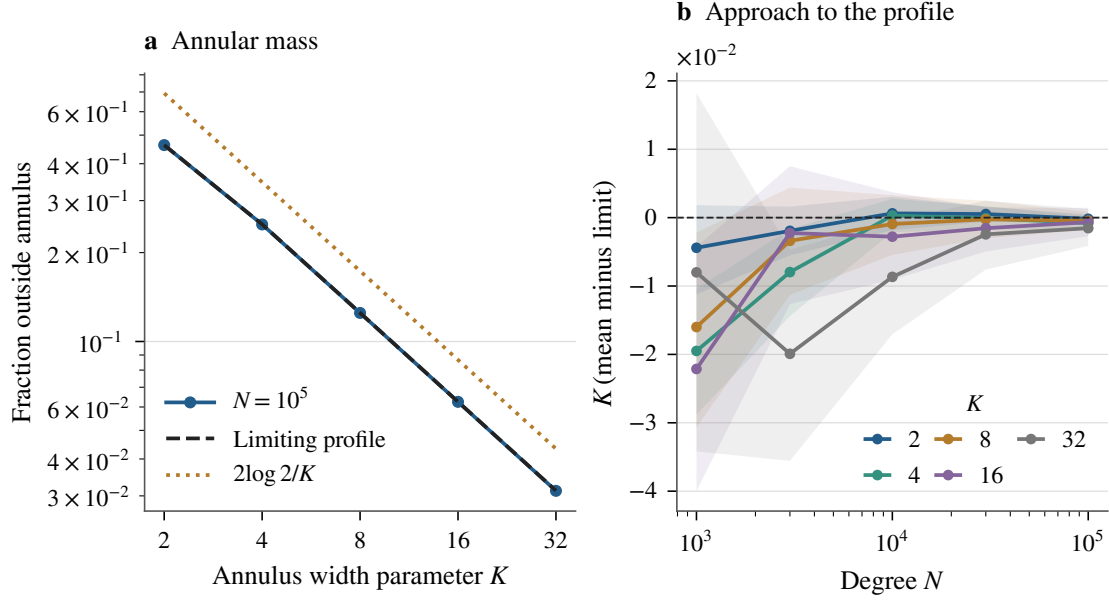


Figure 6: The fraction of zeros outside the annulus of width K/N .

inequality provides on each arc of a fine grid. The zeros with a small derivative are detected on a mesh by Theorem B.4, using Gaussian approximations of the jet (f_N, zf'_N) .

4.1 Local zero counts

We will use the Hanson–Wright inequality of Rudelson–Vershynin [20, Theorem 1.1]: if ξ is a random vector with independent centered real coordinates whose ψ_2 norm is at most Λ , M is a real matrix and $t > 0$, then

$$\mathbb{P}\{|\xi^T M \xi - \mathbb{E} \xi^T M \xi| > t\} \leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{t^2}{\Lambda^4 \|M\|_{\text{HS}}^2}, \frac{t}{\Lambda^2 \|M\|} \right\} \right],$$

where $c_{\text{HW}} > 0$ is an absolute constant. Rudelson and Vershynin use the moment convention for ψ_2 , and signs satisfy both it and the equivalent exponential-moment convention with $\Lambda = 2$, since $p^{-1/2}(\mathbb{E}|\epsilon_k|^p)^{1/p} = p^{-1/2} \leq 2$ for $p \geq 1$ and $\mathbb{E} \exp(\epsilon_k^2/2^2) = e^{1/4} \leq 2$. Hence, for independent uniform signs $\epsilon = (\epsilon_0, \dots, \epsilon_N)$, a real $(N+1) \times (N+1)$ matrix M and $t > 0$,

$$(23) \quad \mathbb{P}\{|\epsilon^T M \epsilon - \mathbb{E} \epsilon^T M \epsilon| > t\} \leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{t^2}{16 \|M\|_{\text{HS}}^2}, \frac{t}{4 \|M\|} \right\} \right].$$

We first treat independent signs times bounded amplitudes, and then specialize to signs.

Lemma 4.1 (Local zero counts for bounded amplitudes). *Fix $K \geq 2$, $B \geq 1$ and $\frac{1}{2} \leq \eta \leq 1$, and put*

$$A(B, \eta) = \max \left(1, \frac{1024\pi B^2}{c_{\text{HW}}\eta} \right), \quad t_N(B, \eta) = \frac{A(B, \eta) \log N}{N},$$

$$C_1(K, B, \eta) = \frac{4A(B, \eta) + 4K + 4 + \log B}{\log 2}.$$

Let $N \geq e$ with $t_N(B, \eta) \leq 1$. Let $\epsilon_0, \dots, \epsilon_N$ be independent uniform signs, and let $a_0, \dots, a_N$ be real or complex random variables, independent of the signs, with $|a_k| \leq B$ for every k . Let $\mathcal{E}$ be an event determined by $a_0, \dots, a_N$ on which

$$\sum_{k=0}^N |a_k|^2 \geq \eta(N+1),$$

and put $P_N(z) = \sum_{k=0}^N \epsilon_k a_k z^k$. Then the probability that $\mathcal{E}$ occurs and some disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains more than $C_1(K, B, \eta) \log N$ zeros of P_N is at most N^{-3} .

Proof. The idea is to show that the mean of $|P_N|^2$ over each arc of a fine angular grid is at least half its conditional expectation, and then to apply Jensen's formula at a point of the arc at which $|P_N|$ is large.

We condition on $a_0, \dots, a_N$ and assume that $\mathcal{E}$ occurs, so that the signs remain independent and uniform. We take $J = \lceil 2\pi/t_N(B, \eta) \rceil$ equally spaced angular centers, with the closed arc of radian length $2t_N(B, \eta)$ around each center. Since $t_N(B, \eta) \leq 1 < \pi$, every such arc I has $\mu(I) = t_N(B, \eta)/\pi$, and since $A(B, \eta) \geq 1$ and $\log N \geq 1$,

$$J \leq \frac{2\pi}{t_N(B, \eta)} + 1 = \frac{2\pi N}{A(B, \eta) \log N} + 1 \leq 2\pi N + 1 \leq 8N.$$

For each such arc I , the conditional real Gram matrix is defined by

$$(A_I)_{jk} = \int_I \Re(a_j \overline{a_k} e^{i(j-k)\theta}) d\mu(\theta), \quad 0 \leq j, k \leq N.$$

For every real vector x , since the x_k are real and by Parseval's identity on the whole circle, we have

$$\begin{aligned} 0 \leq x^T A_I x &= \int_I \left| \sum_{k=0}^N x_k a_k e^{ik\theta} \right|^2 d\mu(\theta) \leq \int \left| \sum_{k=0}^N x_k a_k e^{ik\theta} \right|^2 d\mu(\theta) \\ &= \sum_{k=0}^N x_k^2 |a_k|^2 \leq B^2 \|x\|^2. \end{aligned}$$

Hence the eigenvalues of the real symmetric matrix A_I lie in $[0, B^2]$, and since they are nonnegative,

$$\begin{aligned} \|A_I\| &\leq B^2, & \|A_I\|_{\text{HS}}^2 &\leq \|A_I\| \operatorname{tr} A_I \leq B^2 \operatorname{tr} A_I, \\ \operatorname{tr} A_I &= \mu(I) \sum_{k=0}^N |a_k|^2 \geq \eta(N+1)\mu(I) = \frac{\eta(N+1)t_N(B, \eta)}{\pi}, \end{aligned}$$

where the trace identity follows from $(A_I)_{kk} = |a_k|^2 \mu(I)$, and the inequality after it from $\mathcal{E}$. Taking $x = \epsilon$ in the identity above and using $\mathbb{E}\epsilon_j \epsilon_k = \mathbf{1}\{j = k\}$, we obtain

$$Q_I := \int_I |P_N(e^{i\theta})|^2 d\mu(\theta) = \epsilon^T A_I \epsilon, \quad \mathbb{E}_\epsilon Q_I = \operatorname{tr} A_I.$$

If $Q_I < \frac{1}{2} \operatorname{tr} A_I$, then $|Q_I - \mathbb{E}_\epsilon Q_I| > \frac{1}{2} \operatorname{tr} A_I$. By (23) with $M = A_I$ and $t = \frac{1}{2} \operatorname{tr} A_I$, and the bounds above, we have

$$\begin{aligned} \mathbb{P}_\epsilon \{Q_I < \tfrac{1}{2} \operatorname{tr} A_I\} &\leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{(\operatorname{tr} A_I)^2}{64 \|A_I\|_{\text{HS}}^2}, \frac{\operatorname{tr} A_I}{8 \|A_I\|} \right\} \right] \\ &\leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{\operatorname{tr} A_I}{64 B^2}, \frac{\operatorname{tr} A_I}{8 B^2} \right\} \right] = 2 \exp \left[-\frac{c_{\text{HW}} \operatorname{tr} A_I}{64 B^2} \right] \\ &\leq 2 \exp \left[-\frac{c_{\text{HW}} \eta(N+1) A(B, \eta) \log N}{64 \pi B^2 N} \right] \leq 2e^{-16 \log N} = 2N^{-16}, \end{aligned}$$

where the last inequality holds since $A(B, \eta) \geq 1024\pi B^2/(c_{\text{HW}}\eta)$. By the union bound over the at most $8N$ arcs, we get

$$\mathbb{P}_\epsilon \{Q_I < \tfrac{1}{2} \operatorname{tr} A_I \text{ for some arc } I\} \leq (8N)(2N^{-16}) = 16N^{-15} \leq N^{-3}.$$

After the expectation over $a_0, \dots, a_N$ on $\mathcal{E}$, the probability that $\mathcal{E}$ occurs and some arc has $Q_I < \frac{1}{2} \operatorname{tr} A_I$ is at most N^{-3} .

It remains to bound the number of zeros when $\mathcal{E}$ occurs and every arc has $Q_I \geq \frac{1}{2} \operatorname{tr} A_I$. In this case, averaging over each closed arc and using the lower bound on $\operatorname{tr} A_I$, we get

$$\max_{\theta \in I} |P_N(e^{i\theta})|^2 \geq \frac{Q_I}{\mu(I)} \geq \frac{\operatorname{tr} A_I}{2\mu(I)} \geq \frac{\eta(N+1)}{2}.$$

By continuity, there exists $z_1 = e^{i\theta_1}$ on the arc such that $|P_N(z_1)|^2 \geq \eta(N+1)/2$. Now let $||z_0| - 1| \leq 1/N$. Then the angular center nearest to $z_0/|z_0|$ is at angular distance at most $\pi/J \leq t_N(B, \eta)/2$ from it, and such a point z_1 on its arc is at angular distance at most $t_N(B, \eta)$ from the center, so that $|z_0/|z_0| - z_1| \leq 3t_N(B, \eta)/2$, since a chord is at most its arc. Let $R = (2K+1)/N + 2t_N(B, \eta)$. For $|z - z_0| \leq 2K/N$, since $|z_0 - z_0/|z_0|| = ||z_0| - 1| \leq 1/N$, we have

$$|z - z_1| \leq |z - z_0| + |z_0 - z_0/|z_0|| + |z_0/|z_0| - z_1| \leq \frac{2K}{N} + \frac{1}{N} + \frac{3t_N(B, \eta)}{2} < R,$$

so that the target disk lies strictly within $\{|z - z_1| < R\}$. For $|z - z_1| \leq 2R$ we have $|z| \leq 1 + 2R$, and hence

$$|P_N(z)| \leq \sum_{k=0}^N |a_k| |z|^k \leq B(N+1)(1+2R)^N \leq B(N+1)e^{2NR}.$$

Let $n(s)$ denote the number of zeros of P_N in $|z - z_1| \leq s$. Since $P_N(z_1) \neq 0$, Jensen's formula applies on the disk $|z - z_1| \leq 2R$, and using $2NR = 4K + 2 + 4A(B, \eta) \log N$ we get

$$\begin{aligned} \#\{\alpha : |\alpha - z_1| < R\} &\leq n(R) \leq \frac{1}{\log 2} \int_R^{2R} \frac{n(s)}{s} ds \\ &\leq \frac{\max_{|z-z_1|=2R} \log |P_N(z)| - \log |P_N(z_1)|}{\log 2} \\ &\leq \frac{\log(B(N+1)e^{2NR}) - \frac{1}{2} \log(\eta(N+1)/2)}{\log 2} \\ &= \frac{\frac{1}{2} \log(N+1) + \log B + \frac{1}{2} \log(2/\eta) + 2NR}{\log 2} \\ &= \frac{\frac{1}{2} \log(N+1) + \log B + \frac{1}{2} \log(2/\eta) + 4K + 2 + 4A(B, \eta) \log N}{\log 2} \\ &\leq \frac{\log N + (4K + 3 + \log B) \log N + 4A(B, \eta) \log N}{\log 2} = C_1(K, B, \eta) \log N. \end{aligned}$$

The second inequality is the zero-count step of [23, §8.21], the third is Jensen's formula [23, §3.61] for $P_N(z_1 + z)$, and the last uses $\frac{1}{2} \log(N+1) \leq \log N$ for $N \geq 2$, $\frac{1}{2} \log(2/\eta) \leq \frac{1}{2} \log 4 = \log 2$ for $\eta \geq \frac{1}{2}$, and $\log B + \log 2 + 4K + 2 \leq (4K + 3 + \log B) \log N$, which holds for $\log N \geq 1$ since $\log B \geq 0$ and $\log 2 < 1$. Since the containment is strict, this count includes the zeros on the boundary of the target disk. The angular cover is finite and does not depend on z_0 , so the bound holds for every admissible center at once. $\square$

Lemma 4.2 (Local zero counts). *Fix $K \geq 2$ and take N sufficiently large. Except on an event of probability at most N^{-3} , every disk*

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C(K) \log N$ zeros of f_N .

Appendix A lists the local-count constant.

Proof. We apply Lemma 4.1 with $a_k = 1$ for every k , $B = 1$, $\eta = 1$ and $\mathcal{E} = \Omega$, which we may do since $\sum_{k=0}^N |a_k|^2 = N + 1$. Then $P_N = f_N$, and since $\log 1 = 0$, the constants of that lemma are

$$A(1, 1) = \max(1, 1024\pi/c_{\text{HW}}) = A, \quad t_N(1, 1) = \frac{A \log N}{N} = t_N,$$

$$C_1(K, 1, 1) = \frac{4A + 4K + 4}{\log 2} = C_1(K),$$

which are the constants of Appendix A. The conditions $N \geq e$ and $t_N \leq 1$ hold for all sufficiently large N , since $t_N \rightarrow 0$. This proves the lemma with $C(K) = C_1(K)$. $\square$

4.2 Derivative detection and small-ball estimates

In this subsection we fix $K \geq 2$, let $L = N^{1/64}$, and consider the annulus $||z| - 1| \leq K/N$.

Lemma 4.3 (Second derivative). *For every $N \geq \max\{2, (K+1)/3\}$, outside an event of probability at most N^{-10} ,*

$$(24) \quad \sup_{|z| \leq 1 + (K+1)/N} |f_N''(z)| \leq C(K) N^{5/2} \sqrt{\log N},$$

and one can take $C(K) = 100e^{K+4}$.

Proof. We first bound f_N'' at finitely many points of the circle $|z| = 1 + (K+1)/N$ by Hoeffding's inequality, then pass to the whole circle using a deterministic bound on f_N''' , and finally to the disk by the maximum principle. Bounds of this order for the supremum of a random trigonometric polynomial go back to Salem and Zygmund [13, Chapter 6].

We choose $\lceil N^3 \rceil \leq 2N^3$ equally spaced points on the boundary circle, and at each sample z we take $b = 100e^{K+4}N^{5/2}\sqrt{\log N}$. Since $|z| = 1 + (K+1)/N$ and $0 \leq 2k-4 \leq 2N$ for $2 \leq k \leq N$, we have $|z|^{2k-4} \leq (1 + (K+1)/N)^{2N} \leq e^{2K+2}$, and hence

$$\sum_{k=2}^N k^2(k-1)^2 |z|^{2k-4} \leq e^{2K+2} \sum_{k=2}^N k^4 \leq 2e^{2K+2} N^5.$$

By Hoeffding's inequality [10, Theorem 2], applied to both tails, $\mathbb{P}\{|\sum_k \epsilon_k c_k| \geq t\} \leq 2 \exp[-t^2/(2 \sum_k c_k^2)]$ for real numbers c_k and $t > 0$. Applying this real bound to the real and imaginary parts $\Re f_N''(z) = \sum_{k=2}^N \epsilon_k k(k-1) \Re(z^{k-2})$ and $\Im f_N''(z) = \sum_{k=2}^N \epsilon_k k(k-1) \Im(z^{k-2})$, whose squared coefficients are at most $k^2(k-1)^2 |z|^{2k-4}$, we get

$$\begin{aligned} \mathbb{P}\{|f_N''(z)| \geq b/2\} &\leq \mathbb{P}\left\{|\Re f_N''(z)| \geq \frac{b}{2\sqrt{2}}\right\} + \mathbb{P}\left\{|\Im f_N''(z)| \geq \frac{b}{2\sqrt{2}}\right\} \\ &\leq 4 \exp\left[-\frac{(b/2)^2}{4 \sum_{k=2}^N k^2(k-1)^2 |z|^{2k-4}}\right] \leq 4 \exp\left[-\frac{b^2}{32e^{2K+2}N^5}\right] \\ &= 4 \exp\left[-\frac{10000e^6}{32} \log N\right] \leq 4 \exp[-100 \log N] = 4N^{-100}, \end{aligned}$$

where the first step holds since $|f_N''(z)| \geq b/2$ implies that one of the two parts is at least $b/(2\sqrt{2})$ in absolute value.

Since $N \geq (K+1)/3$, the boundary circle has radius $1 + (K+1)/N \leq 4$ and length at most 8π , so that each point of it lies within arc length $4\pi/N^3$ of a sample. Moreover, we have the deterministic bound

$$\begin{aligned} \sup_{|z| \leq 1+(K+1)/N} |f_N'''(z)| &\leq \sup_{|z| \leq 1+(K+1)/N} \sum_{k=3}^N k(k-1)(k-2)|z|^{k-3} \\ &\leq N^3 \cdot N \cdot \left(1 + \frac{K+1}{N}\right)^N \leq e^{K+1} N^4. \end{aligned}$$

Integrated along the arc from a point of the circle to its nearest sample, f_N''' gives an interpolation error of at most $4\pi e^{K+1} N$, which is less than $b/2$ for every $N \geq 2$:

$$\begin{aligned} \frac{4\pi}{N^3} \sup_{|z| \leq 1+(K+1)/N} |f_N'''(z)| &\leq \frac{4\pi}{N^3} e^{K+1} N^4 = 4\pi e^{K+1} N \\ &< 50e^{K+4} N^{5/2} \sqrt{\log N} = \frac{b}{2}, \end{aligned}$$

since $4\pi < 50e^3 \sqrt{\log 2}$ and $N^{3/2} \sqrt{\log N} \geq \sqrt{\log 2}$. By the union bound over the samples, the probability that $|f_N''(z)| \geq b/2$ for some sample z is at most $(2N^3)(4N^{-100}) = 8N^{-97} \leq N^{-10}$. Outside this event, every point ζ of the circle of radius $1 + (K+1)/N$ and its nearest sample z satisfy $|f_N''(\zeta)| \leq |f_N''(z)| + 4\pi e^{K+1} N$, so that $|f_N''| < b/2 + b/2 = b$ on the circle $|z| = 1 + (K+1)/N$, and hence, by the maximum principle, on the disk $|z| \leq 1 + (K+1)/N$. This proves (24) with the constant $100e^{K+4}$. $\square$

The following lemma compares two centered Gaussian laws through relative entropy. Devroye, Mehrabian and Reddad [6] give upper and lower bounds on the total variation distance between Gaussians with equal means that match up to constant factors; we need only an upper bound. For the block diagonal comparison of the paired jet covariance in the proof of Lemma 4.5, the interpolation bound (14) gives one more directly.

Lemma 4.4 (An entropy comparison). *Let p, q be probability densities, with p absolutely continuous with respect to q , and set*

$$D(p||q) = \int p \log(p/q), \quad \text{TV}(p, q) = \frac{1}{2} \int |p - q|.$$

Then

$$\text{TV}(p, q) \leq \sqrt{D(p||q)/2}.$$

For positive-definite covariance matrices Σ, Σ_0 in dimension d , let G_Σ and G_{Σ_0} be centered Gaussian vectors with these covariances, and set

$$R = \Sigma_0^{-1/2} \Sigma \Sigma_0^{-1/2}.$$

Then

$$D(G_\Sigma \| G_{\Sigma_0}) = \frac{1}{2} \{ \text{tr } R - d - \log \det R \}.$$

If the eigenvalues of R are $1 + \eta_1, \dots, 1 + \eta_d$ and $|\eta_j| \leq \delta \leq 1/2$, then

$$D(G_\Sigma \| G_{\Sigma_0}) \leq \frac{1}{2} \sum_{j=1}^d \eta_j^2 \leq \frac{d}{2} \delta^2,$$

$$\text{TV}(G_\Sigma, G_{\Sigma_0}) \leq \frac{\sqrt{d}}{2} \delta.$$

Proof. The first inequality is Pinsker's inequality [24, Lemma 2.5(i)]. The formula for $D(G_\Sigma \| G_{\Sigma_0})$ is [19, eq. (A.23)] with zero means and with Σ, Σ_0 in place of the covariances Σ_0, Σ_1 there, as $\text{tr}(\Sigma_0^{-1} \Sigma) = \text{tr } R$ and $\log \det(\Sigma_0 \Sigma^{-1}) = -\log \det R$. Since the eigenvalues of R are $1 + \eta_1, \dots, 1 + \eta_d$, we have $\text{tr } R = \sum_j (1 + \eta_j)$ and $\log \det R = \sum_j \log(1 + \eta_j)$, so that

$$\text{tr } R - d - \log \det R = \sum_{j=1}^d \{ \eta_j - \log(1 + \eta_j) \}.$$

If $|\eta_j| \leq \delta \leq 1/2$, then $0 \leq \eta_j - \log(1 + \eta_j) \leq \eta_j^2$, since $y - \log(1 + y) = \int_0^y t/(1+t) dt$, $t/(1+t)$ has the sign of t for $t > -1$, and $|t/(1+t)| \leq 2|t|$ for $|t| \leq 1/2$. It follows that

$$D(G_\Sigma \| G_{\Sigma_0}) = \frac{1}{2} \sum_{j=1}^d \{ \eta_j - \log(1 + \eta_j) \} \leq \frac{1}{2} \sum_{j=1}^d \eta_j^2 \leq \frac{d}{2} \delta^2,$$

and hence, by the first inequality,

$$\text{TV}(G_\Sigma, G_{\Sigma_0}) \leq \sqrt{D(G_\Sigma \| G_{\Sigma_0})/2} \leq \sqrt{\frac{d}{4} \delta^2} = \frac{\sqrt{d}}{2} \delta.$$

□

Lemma 4.5 (Few small derivatives). *Let $B_{K,N}$ count the roots α of f_N with $|\alpha| - 1| \leq K/N$ and $|f'_N(\alpha)| < N^{3/2}/L$. For all sufficiently large N ,*

$$\mathbb{P}\{B_{K,N} > C(K)NL^{-4} \log N\} \leq N^{-3} + N^{-10} + C(K)N^{-5/16}(\log N)^2.$$

Since $L = N^{1/64}$, the bound on $B_{K,N}$ is $C(K)N^{15/16} \log N$.

Proof. The idea is to apply Theorem B.4. Outside two thin sectors around the real axis, it detects each annular root with a small derivative at a nearby point of a fixed mesh at which both f_N and zf'_N are small, and it bounds the number of such mesh points by their mean and variance. We construct the mesh, verify the one- and two-point Gaussian approximations that the theorem requires, and bound the roots in the two sectors directly by the local zero counts.

By the Taylor step in the proof of Theorem B.4, at a mesh point z within h of a root, $|f_N(z)|/\sqrt{N}$ is at most a constant times $Nh/L + N^2h^2\sqrt{\log N}$. For the mean count below, both terms must be at most a constant times $L^{-2}(\log N)^{-1/2}$, which holds when h is at most a constant times $1/(NL\sqrt{\log N})$. A smaller h would only add mesh points, so we choose a deterministic mesh with spacing

$$h = \frac{c}{NL\sqrt{\log N}},$$

where $c > 0$ is sufficiently small and depends only on K . The bounds (25) below require $100e^{K+4}c \leq 1/2$, which implies $c \leq 1$. The construction in Section A.4 takes $\lceil 2K/(Nh) \rceil + 1$ rows, at equally spaced radii from $1 - K/N$ to $1 + K/N$, and on each row the same $\lceil 8\pi/h \rceil$ equally spaced angles, so that adjacent rows are at most h apart, and adjacent angles at most $h/4$ apart. Since $L\sqrt{\log N}/c \geq 1$ for $c \leq 1$ and $N \geq e$, the number of mesh points satisfies

$$\begin{aligned} \left(\left\lceil \frac{2K}{Nh} \right\rceil + 1 \right) \left\lceil \frac{8\pi}{h} \right\rceil &\leq \left(\frac{2K}{Nh} + 2 \right) \left(\frac{8\pi}{h} + 1 \right) \\ &= \left(\frac{2KL\sqrt{\log N}}{c} + 2 \right) \left(\frac{8\pi NL\sqrt{\log N}}{c} + 1 \right) \\ &\leq \frac{(2K+2)L\sqrt{\log N}}{c} \cdot \frac{(8\pi+1)NL\sqrt{\log N}}{c} \leq \frac{53(K+1)}{c^2} NL^2 \log N, \end{aligned}$$

and since c depends only on K , the mesh has at most $C(K)NL^2 \log N$ points.

We retain the mesh points $re^{i\theta}$ such that $\text{dist}(\theta, \pi\mathbb{Z}) \geq N^{-1/2}$, or equivalently $\text{dist}(2\theta, 2\pi\mathbb{Z}) \geq 2N^{-1/2}$. These are the angles retained in Section 3.1, and we use them in the covariance bounds below. Let α be a root with $||\alpha| - 1| \leq K/N$ and $\text{dist}(\arg \alpha, \pi\mathbb{Z}) \geq 2N^{-1/2}$. We assign to it the mesh point z on the row nearest to $|\alpha|$ whose angle is nearest to $\arg \alpha$. Then $||\alpha| - |z|| \leq h/2$, the angles of α and z differ by at most $h/8$ modulo 2π , and $|z| \leq 1 + K/N \leq 2$ for $N \geq K$, so that, since a chord is at most its arc,

$$|\alpha - z| \leq ||\alpha| - |z|| + |z| |e^{i\arg \alpha} - e^{i\arg z}| \leq \frac{h}{2} + 2 \cdot \frac{h}{8} < h.$$

Moreover, since $h \leq 1/N \leq N^{-1/2}$ for $c \leq 1$ and $N \geq e$, we have

$$\text{dist}(\arg z, \pi\mathbb{Z}) \geq \text{dist}(\arg \alpha, \pi\mathbb{Z}) - \frac{h}{8} \geq 2N^{-1/2} - N^{-1/2} = N^{-1/2}.$$

Thus every annular root α with $\text{dist}(\arg \alpha, \pi\mathbb{Z}) \geq 2N^{-1/2}$ has a retained mesh point within h . We write $z_1, \dots, z_m$ for the retained mesh points, so that $m \leq 53(K+1)c^{-2}NL^2 \log N$.

Now let α be such a root with $|f'_N(\alpha)| < N^{3/2}/L$, and let z be its assigned mesh point. The segment from α to z lies in the disk $D = \{|\zeta| \leq 1 + (K+1)/N\}$, since α and z lie in the convex disk $|\zeta| \leq 1 + K/N$. By Lemma 4.3, outside an event of probability at most N^{-10} we have $\sup_D |f''_N| \leq M_2 = C(K)N^{5/2}\sqrt{\log N}$ with $C(K) = 100e^{K+4}$. On this event, since $|z - \alpha| \leq h$ and $|z| \leq 2$, the Taylor step in the proof of Theorem B.4, with $\tau = h$ and radius 2, shows that $|f_N(z)|/\sqrt{N} \leq u$ and $|zf'_N(z)|/N^{3/2} \leq v_0$. Since $Nh = c/(L\sqrt{\log N})$ and $M_2h/N^{3/2} = C(K)c/L$, the thresholds of that theorem satisfy

$$(25) \quad u = \frac{Nh}{L} + \frac{C(K)}{2}N^2h^2\sqrt{\log N} \leq \frac{C(K)}{L^2\sqrt{\log N}}, \quad v_0 = \frac{2(1 + C(K)c)}{L} \leq \frac{3}{L}.$$

The first bound holds with the same $C(K)$, since $Nh/L + (C(K)/2)N^2h^2\sqrt{\log N} = (c + C(K)c^2/2)/(L^2\sqrt{\log N})$ and $c + C(K)c^2/2 = c(1 + C(K)c/2) \leq 5c/4 \leq C(K)$ by $C(K)c \leq 1/2$ and $c \leq 1$, and the second holds since $C(K)c \leq 1/2$. If a root α is assigned to a mesh point z , then, since $|z - \alpha|/|z| = ||z| - 1| \leq K/N$ and $h \leq 1/N$,

$$|\alpha - z/|z|| \leq |\alpha - z| + |z - z/|z|| \leq h + K/N < 2(K+2)/N.$$

Hence, by Lemma 4.2 with $K+2$ in place of K , applied to the disk $\{\zeta : |\zeta - z/|z|| \leq 2(K+2)/N\}$, whose center lies on the unit circle, at most $C_1(K+2) \log N$ roots are assigned to each cell.

We consider

$$W(z) = (W_0(z), W_1(z)) = \left(\frac{f_N(z)}{\sqrt{N}}, \frac{zf'_N(z)}{N^{3/2}} \right),$$

as a real vector with coordinates $\Re W_0(z), \Im W_0(z), \Re W_1(z), \Im W_1(z)$. For $z = re^{i\theta}$, we have $W_0(z) = \sum_{k=0}^N \epsilon_k r^k e^{ik\theta} / \sqrt{N}$ and $W_1(z) = \sum_{k=0}^N \epsilon_k (k/N) r^k e^{ik\theta} / \sqrt{N}$, and since the signs are independent with mean zero and variance one, the real covariance is

$$A_z = \frac{1}{N} \sum_{k=0}^N r^{2k} \times \begin{pmatrix} \cos^2(k\theta) & \cos(k\theta) \sin(k\theta) & \frac{k}{N} \cos^2(k\theta) & \frac{k}{N} \cos(k\theta) \sin(k\theta) \\ \cos(k\theta) \sin(k\theta) & \sin^2(k\theta) & \frac{k}{N} \cos(k\theta) \sin(k\theta) & \frac{k}{N} \sin^2(k\theta) \\ \frac{k}{N} \cos^2(k\theta) & \frac{k}{N} \cos(k\theta) \sin(k\theta) & \frac{k^2}{N^2} \cos^2(k\theta) & \frac{k^2}{N^2} \cos(k\theta) \sin(k\theta) \\ \frac{k}{N} \cos(k\theta) \sin(k\theta) & \frac{k}{N} \sin^2(k\theta) & \frac{k^2}{N^2} \cos(k\theta) \sin(k\theta) & \frac{k^2}{N^2} \sin^2(k\theta) \end{pmatrix}.$$

Using

$$\cos^2(k\theta) = \frac{1 + \cos(2k\theta)}{2}, \quad \sin^2(k\theta) = \frac{1 - \cos(2k\theta)}{2}, \quad \cos(k\theta) \sin(k\theta) = \frac{\sin(2k\theta)}{2},$$

we write $A_z = A_z^{(0)} + E_z$, where

$$A_z^{(0)} = \frac{1}{2N} \sum_{k=0}^N r^{2k} \begin{pmatrix} 1 & 0 & k/N & 0 \\ 0 & 1 & 0 & k/N \\ k/N & 0 & (k/N)^2 & 0 \\ 0 & k/N & 0 & (k/N)^2 \end{pmatrix}$$

and

$$E_z = \frac{1}{2N} \sum_{k=0}^N r^{2k} \begin{pmatrix} \cos(2k\theta) & \sin(2k\theta) & \frac{k}{N} \cos(2k\theta) & \frac{k}{N} \sin(2k\theta) \\ \sin(2k\theta) & -\cos(2k\theta) & \frac{k}{N} \sin(2k\theta) & -\frac{k}{N} \cos(2k\theta) \\ \frac{k}{N} \cos(2k\theta) & \frac{k}{N} \sin(2k\theta) & \frac{k^2}{N^2} \cos(2k\theta) & \frac{k^2}{N^2} \sin(2k\theta) \\ \frac{k}{N} \sin(2k\theta) & -\frac{k}{N} \cos(2k\theta) & \frac{k^2}{N^2} \sin(2k\theta) & -\frac{k^2}{N^2} \cos(2k\theta) \end{pmatrix}.$$

Every entry of E_z is half the real or imaginary part of an oscillatory sum in (11), with $r = s$ and $\psi = 2\theta$, namely, up to sign, of $N^{-1} \sum_{k=0}^N (r \cdot r)^k (k/N)^\beta e^{ik \cdot 2\theta}$ with $\beta \in \{0, 1, 2\}$. For retained angles we have $\text{dist}(2\theta, 2\pi\mathbb{Z}) \geq 2N^{-1/2}$, and the norm of a 4×4 matrix is at most four times its largest entry in absolute value. Thus, by (11), which applies since $r \in [1 - K/N, 1 + K/N]$ and $N \geq 2K$, we have

$$\begin{aligned} \|E_z\| &\leq 4 \cdot \frac{1}{2} \cdot \frac{10e^{4K}}{N \text{dist}(2\theta, 2\pi\mathbb{Z})} \leq \frac{20e^{4K}}{N \cdot 2N^{-1/2}} \\ &= 10e^{4K} N^{-1/2} \leq 20e^{4K} N^{-1/2} \leq e^{-4K}/320, \end{aligned}$$

where we keep the cruder constant $20e^{4K}$, from which the threshold is computed. The last step holds if and only if $N^{1/2} \geq 20 \cdot 320e^{8K} = 6400e^{8K}$, that is, at the appendix threshold $N \geq (6400e^{8K})^2$.

To bound $A_z^{(0)}$ from below, we list the coordinates in the order $\Re W_0(z)$, $\Re W_1(z)$, $\Im W_0(z)$, $\Im W_1(z)$. In this order $A_z^{(0)}$ is block diagonal with two equal blocks

$$\frac{1}{2N} \sum_{k=0}^N r^{2k} \begin{pmatrix} 1 & k/N \\ k/N & (k/N)^2 \end{pmatrix},$$

and reordering the coordinates does not change the eigenvalues. Each matrix in this sum is positive semidefinite, since it is the product of the column $(1, k/N)^T$ with its transpose. The radial weights are at least e^{-4K} , since $r^{2k} \geq \min(1, r)^{2N} \geq (1 - K/N)^{2N} \geq e^{-4K}$ for $0 \leq k \leq N$, by $r \geq 1 - K/N$, $K/N \leq 1/2$ and $\log(1-x) \geq -2x$

for $0 \leq x \leq 1/2$. It follows that each block is at least $e^{-4K}/2$ times the unweighted matrix

$$\begin{aligned} M_N &= \frac{1}{N} \sum_{k=0}^N \begin{pmatrix} 1 & k/N \\ k/N & (k/N)^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{N} \sum_{k=0}^N 1 & \frac{1}{N^2} \sum_{k=0}^N k \\ \frac{1}{N^2} \sum_{k=0}^N k & \frac{1}{N^3} \sum_{k=0}^N k^2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{N+1}{N} & \frac{N+1}{2N} \\ \frac{N+1}{2N} & \frac{(N+1)(2N+1)}{6N^2} \end{pmatrix} \end{aligned}$$

in the positive semidefinite order, and hence $\lambda_{\min}(A_z^{(0)}) \geq (e^{-4K}/2)\lambda_{\min}(M_N)$. By a direct calculation, we have

$$\begin{aligned} \det M_N &= \frac{N+1}{N} \cdot \frac{(N+1)(2N+1)}{6N^2} - \left(\frac{N+1}{2N}\right)^2 = \frac{(N+1)^2\{2(2N+1) - 3N\}}{12N^3} \\ &= \frac{(N+1)^2(N+2)}{12N^3}, \\ \operatorname{tr} M_N &= \frac{N+1}{N} + \frac{(N+1)(2N+1)}{6N^2} = \frac{(N+1)(8N+1)}{6N^2}. \end{aligned}$$

Since the smaller eigenvalue of the positive semidefinite matrix M_N is $\det M_N$ divided by the larger one, and the larger one is at most $\operatorname{tr} M_N$, we get

$$\lambda_{\min}(M_N) \geq \frac{\det M_N}{\operatorname{tr} M_N} = \frac{(N+1)(N+2)}{2N(8N+1)} \geq \frac{1}{16}.$$

By Weyl's inequality, at the appendix threshold $N \geq (6400e^{8K})^2$ we have

$$\begin{aligned} \lambda_{\min}(A_z) &\geq \lambda_{\min}(A_z^{(0)}) - \|E_z\| \geq \frac{e^{-4K}}{2} \lambda_{\min}(M_N) - \|E_z\| \\ (26) \quad &\geq \frac{e^{-4K}}{32} - \frac{e^{-4K}}{320} \geq \frac{e^{-4K}}{80} = \lambda, \end{aligned}$$

where the last line defines λ .

For two points $z = re^{i\theta}$ and $w = se^{i\phi}$, we consider the real vector $(W(z), W(w))$ in $\mathbb{R}^8$, with the four coordinates of $W(z)$ followed by those of $W(w)$, and denote its covariance matrix by

$$\Sigma = \begin{pmatrix} A_z & C_{z,w} \\ C_{z,w}^T & A_w \end{pmatrix},$$

where A_z, A_w are the four-dimensional covariance matrices above. By the independence of the signs, the cross block is

$$C_{z,w} = \frac{1}{N} \sum_{k=0}^N (rs)^k$$

$$\times \begin{pmatrix} \cos(k\theta) \cos(k\phi) & \cos(k\theta) \sin(k\phi) & \frac{k}{N} \cos(k\theta) \cos(k\phi) & \frac{k}{N} \cos(k\theta) \sin(k\phi) \\ \sin(k\theta) \cos(k\phi) & \sin(k\theta) \sin(k\phi) & \frac{k}{N} \sin(k\theta) \cos(k\phi) & \frac{k}{N} \sin(k\theta) \sin(k\phi) \\ \frac{k}{N} \cos(k\theta) \cos(k\phi) & \frac{k}{N} \cos(k\theta) \sin(k\phi) & \frac{k^2}{N^2} \cos(k\theta) \cos(k\phi) & \frac{k^2}{N^2} \cos(k\theta) \sin(k\phi) \\ \frac{k}{N} \sin(k\theta) \cos(k\phi) & \frac{k}{N} \sin(k\theta) \sin(k\phi) & \frac{k^2}{N^2} \sin(k\theta) \cos(k\phi) & \frac{k^2}{N^2} \sin(k\theta) \sin(k\phi) \end{pmatrix}.$$

Each of its four 2×2 blocks is $(k/N)^\beta$, $\beta \in \{0, 1, 2\}$, times the matrix of the products $\cos(k\theta) \cos(k\phi)$, $\cos(k\theta) \sin(k\phi)$, $\sin(k\theta) \cos(k\phi)$ and $\sin(k\theta) \sin(k\phi)$. Using the trigonometric identities

$$\begin{aligned} 2 \cos(k\theta) \cos(k\phi) &= \cos(k(\theta - \phi)) + \cos(k(\theta + \phi)), \\ 2 \cos(k\theta) \sin(k\phi) &= \sin(k(\theta + \phi)) - \sin(k(\theta - \phi)), \\ 2 \sin(k\theta) \cos(k\phi) &= \sin(k(\theta + \phi)) + \sin(k(\theta - \phi)), \\ 2 \sin(k\theta) \sin(k\phi) &= \cos(k(\theta - \phi)) - \cos(k(\theta + \phi)), \end{aligned}$$

we see that this matrix of products is $\frac{1}{2} \Theta_k$, where

$$\Theta_k = \begin{pmatrix} \cos(k(\theta - \phi)) + \cos(k(\theta + \phi)) & \sin(k(\theta + \phi)) - \sin(k(\theta - \phi)) \\ \sin(k(\theta + \phi)) + \sin(k(\theta - \phi)) & \cos(k(\theta - \phi)) - \cos(k(\theta + \phi)) \end{pmatrix},$$

and hence

$$C_{z,w} = \frac{1}{2N} \sum_{k=0}^N (rs)^k \begin{pmatrix} \Theta_k & \frac{k}{N} \Theta_k \\ \frac{k}{N} \Theta_k & \frac{k^2}{N^2} \Theta_k \end{pmatrix}.$$

For retained pairs, both $\text{dist}(\theta - \phi, 2\pi\mathbb{Z})$ and $\text{dist}(\theta + \phi, 2\pi\mathbb{Z})$ are at least $N^{-1/2}$. Every entry of $C_{z,w}$ is half a sum or a difference of the real or imaginary parts of $N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta-\phi)}$ and $N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta+\phi)}$, with $\beta \in \{0, 1, 2\}$. By (11) with $\psi = \theta \mp \phi$, each of these two sums is at most $10e^{4K}/(N \cdot N^{-1/2}) = 10e^{4K}N^{-1/2}$ in absolute value, and hence so is the largest entry of $C_{z,w}$. In particular, for

$$\Sigma_0 = \begin{pmatrix} A_z & 0 \\ 0 & A_w \end{pmatrix}, \quad \Sigma - \Sigma_0 = \begin{pmatrix} 0 & C_{z,w} \\ C_{z,w}^T & 0 \end{pmatrix},$$

we have

$$\|\Sigma - \Sigma_0\| = \|C_{z,w}\| \leq 4 \cdot 10e^{4K}N^{-1/2} = 40e^{4K}N^{-1/2},$$

where the first step holds since the eigenvalues of the symmetric matrix $\Sigma - \Sigma_0$, which maps (x, y) to $(C_{z,w}y, C_{z,w}^Tx)$ for $x, y \in \mathbb{R}^4$, are plus and minus the singular values of $C_{z,w}$. Since Σ_0 is block diagonal and z and w are retained, $\lambda_{\min}(\Sigma_0) = \min\{\lambda_{\min}(A_z), \lambda_{\min}(A_w)\} \geq \lambda$. By Weyl's inequality, we have

$$\lambda_{\min}(\Sigma) \geq \lambda_{\min}(\Sigma_0) - \|\Sigma - \Sigma_0\| \geq \lambda - 40e^{4K}N^{-1/2} \geq \lambda/2$$

for $N \geq (6400e^{8K})^2$, since the last step holds if and only if $40e^{4K}N^{-1/2} \leq \lambda/2 = e^{-4K}/160$, that is, $N^{1/2} \geq 40 \cdot 160e^{8K} = 6400e^{8K}$.

To apply Raič's theorem, we write the paired jet as a sum of independent vectors and whiten it by its covariance. We set

$$b_k(z) = \frac{r^k}{\sqrt{N}} \begin{pmatrix} \cos(k\theta) \\ \sin(k\theta) \\ \frac{k}{N} \cos(k\theta) \\ \frac{k}{N} \sin(k\theta) \end{pmatrix}, \quad b_k = \begin{pmatrix} b_k(z) \\ b_k(w) \end{pmatrix}, \quad Y_k = \epsilon_k \Sigma^{-1/2} b_k.$$

Then $(W(z), W(w)) = \sum_{k=0}^N \epsilon_k b_k$ and $\mathbb{E}Y_k = 0$, and the covariances $\text{Cov}(Y_k)$, $0 \leq k \leq N$, sum to the identity, since $\text{Cov}(Y_k) = \Sigma^{-1/2} b_k b_k^T \Sigma^{-1/2}$ and $\sum_{k=0}^N b_k b_k^T = \Sigma$. Since $r^{2k}, s^{2k} \leq (1 + K/N)^{2N} \leq e^{2K}$ for $0 \leq k \leq N$ and $\|\Sigma^{-1/2}\| = \lambda_{\min}(\Sigma)^{-1/2} \leq (2/\lambda)^{1/2}$, we have

$$\begin{aligned} \|b_k\|^2 &= \frac{(r^{2k} + s^{2k})(1 + (k/N)^2)}{N} \leq \frac{4e^{2K}}{N}, \\ \sum_{k=0}^N \mathbb{E}\|Y_k\|^3 &\leq \sum_{k=0}^N \|\Sigma^{-1/2}\|^3 \|b_k\|^3 \leq (N+1)(2/\lambda)^{3/2} \left(\frac{2e^K}{\sqrt{N}} \right)^3 \\ &\leq 16e^{3K} (2/\lambda)^{3/2} N^{-1/2}. \end{aligned}$$

Let $U \subset \mathbb{R}^4$ be the product of the disks $x_1^2 + x_2^2 \leq u^2$ and $x_3^2 + x_4^2 \leq v_0^2$, so that the event E_i of Theorem B.4 at $z = z_i$ is $\{W(z) \in U\}$, and $E_i \cap E_j$ is $\{(W(z), W(w)) \in U \times U\}$ for $w = z_j$. The products of disks that define these events are convex, and so are their images under whitening. Since $\Sigma^{-1/2}(W(z), W(w)) = \sum_k Y_k$ and $\Sigma^{-1/2}G_\Sigma$ has the law of G_8 , it follows from Raič's theorem, as stated in Section 3.1, with $d = 8$ and the convex set $\Sigma^{-1/2}(U \times U)$, that

$$\begin{aligned} &|\mathbb{P}\{(W(z), W(w)) \in U \times U\} - \mathbb{P}\{G_\Sigma \in U \times U\}| \\ &= \left| \mathbb{P}\left\{ \sum_{k=0}^N Y_k \in \Sigma^{-1/2}(U \times U) \right\} - \mathbb{P}\{G_8 \in \Sigma^{-1/2}(U \times U)\} \right| \\ &\leq (42 \cdot 8^{1/4} + 16) \sum_{k=0}^N \mathbb{E}\|Y_k\|^3 \\ &\leq (87 \cdot 16)e^{3K} (2/\lambda)^{3/2} N^{-1/2} \leq 2000e^{3K} (2/\lambda)^{3/2} N^{-1/2}. \end{aligned}$$

The four-dimensional marginal, whose summands have smaller norms and whose covariance has least eigenvalue at least λ , satisfies the same bound. Indeed, $W(z) = \sum_k \epsilon_k b_k(z)$ has covariance A_z , $\|b_k(z)\|^2 = r^{2k}(1 + (k/N)^2)/N \leq 2e^{2K}/N$ and $\|A_z^{-1/2}\| \leq \lambda^{-1/2}$. Therefore, by Raič's theorem with $d = 4$, applied to the

vectors $\epsilon_k A_z^{-1/2} b_k(z)$ and the convex set $A_z^{-1/2} U$, we have

$$\begin{aligned} |\mathbb{P}\{W(z) \in U\} - \mathbb{P}\{G_{A_z} \in U\}| &\leq (42 \cdot 4^{1/4} + 16) \sum_{k=0}^N \|A_z^{-1/2}\|^3 \|b_k(z)\|^3 \\ &\leq 76(N+1)\lambda^{-3/2} \left(\frac{2e^{2K}}{N}\right)^{3/2} = 76(N+1)e^{3K}(2/\lambda)^{3/2} N^{-3/2} \\ &\leq 152e^{3K}(2/\lambda)^{3/2} N^{-1/2} \leq 2000e^{3K}(2/\lambda)^{3/2} N^{-1/2}. \end{aligned}$$

These Raič bounds, and the interpolation bound below, hold for every product of disks, so that they apply to the events E_i . With $g_i = \mathbb{P}\{G_{A_{z_i}} \in U\}$, the density and volume bound $g_i \leq u^2 v_0^2 / (4\lambda^2)$ in the proof of Theorem B.4 and (25) give the one-point bound

$$\begin{aligned} \mathbb{P}(E_i) &\leq g_i + 2000e^{3K}(2/\lambda)^{3/2} N^{-1/2} \\ &\leq \frac{u^2 v_0^2}{4\lambda^2} + 2000e^{3K}(2/\lambda)^{3/2} N^{-1/2} \\ &\leq \frac{9C(K)^2}{4\lambda^2} \cdot \frac{L^{-6}}{\log N} + 2000e^{3K}(2/\lambda)^{3/2} N^{-1/2} \\ (27) \quad &\leq C(K)L^{-6}/\log N + C(K)N^{-1/2}, \end{aligned}$$

where the first step is the four-dimensional Raič bound, and the last holds since $\lambda = e^{-4K}/80$ depends only on K .

For a separated pair, we compare its Gaussian law with the product of the actual marginal Gaussian laws by the interpolation bound (14), in the general form stated after it, with Σ_0 in place of $\frac{1}{2}I_d$ and $d = 8$. Since $\lambda_{\min}(\Sigma_0) \geq \lambda$ and $\|\Sigma - \Sigma_0\| \leq 40e^{4K}N^{-1/2} \leq \lambda/2$ for $N \geq (6400e^{8K})^2$, every matrix $\Sigma_0 + \tau(\Sigma - \Sigma_0)$, $0 \leq \tau \leq 1$, has least eigenvalue at least $\lambda/2$, and hence

$$(28) \quad \text{TV}(G_\Sigma, G_{\Sigma_0}) \leq \frac{\sqrt{16}}{4} \cdot \frac{\|\Sigma - \Sigma_0\|}{\lambda/2} \leq \frac{80e^{4K}}{\lambda\sqrt{N}} \leq C(K)N^{-1/2}.$$

The Gaussian vector G_{Σ_0} is a pair of independent vectors G_{A_z} and G_{A_w} , and the two marginal covariances can differ.

For each retained mesh point $re^{i\theta}$, at most $10mN^{-1/2}$ partners lie in the removed pair bands. Indeed, a mesh point $se^{i\phi}$ lies in a removed pair band of $re^{i\theta}$ when $\text{dist}(\theta - \phi, 2\pi\mathbb{Z}) < N^{-1/2}$ or $\text{dist}(\theta + \phi, 2\pi\mathbb{Z}) < N^{-1/2}$, which on each row confines ϕ to two arcs of length $2N^{-1/2}$, centered at θ and at $-\theta$, of total length at most $4N^{-1/2}$. The $\lceil 8\pi/h \rceil$ angles of a row are $2\pi/\lceil 8\pi/h \rceil$ apart, so that these arcs contain at most $2(N^{-1/2}\lceil 8\pi/h \rceil/\pi + 1) \leq N^{-1/2}\lceil 8\pi/h \rceil$ of them, since $\lceil 8\pi/h \rceil \geq 8\pi/h \geq 8\pi N$ and $(1 - 2/\pi)8\pi N^{1/2} > 2$. The removed real sectors $\text{dist}(\phi, \pi\mathbb{Z}) < N^{-1/2}$ are also two arcs of length $2N^{-1/2}$, so that every row keeps at least $(1 - N^{-1/2})\lceil 8\pi/h \rceil \geq \frac{1}{2}\lceil 8\pi/h \rceil$

retained angles for $N \geq 4$, and the whole mesh has at most $2m$ points. Summing over the rows, at most $N^{-1/2} \cdot 2m \leq 10mN^{-1/2}$ partners of $re^{i\theta}$ lie in the removed pair bands. Together with the diagonal, which lies in the band $\text{dist}(\theta - \phi, 2\pi\mathbb{Z}) < N^{-1/2}$, at most $10m^2N^{-1/2}$ ordered pairs of retained mesh points lie in the removed pair bands.

It remains to count the roots in the sectors $\text{dist}(\arg z, \pi\mathbb{Z}) < 2N^{-1/2}$, $||z| - 1| \leq K/N$. We place centers on the unit circle, with angular gaps at most $1/N$, on the two intervals of total length $8N^{-1/2}$. Then every point of the sectors is within $(K+1)/N$ of a center, since its modulus differs from one by at most K/N and its angle differs from that of the nearest center by at most $1/(2N)$. Including endpoints and roundings, this requires at most $40\sqrt{N}$ centers, since each interval, of length $4N^{-1/2}$, needs at most $\lceil 4N^{1/2} \rceil + 1 \leq 4\sqrt{N} + 2$ of them. Hence the disks of radius $2(K+2)/N > (K+1)/N$ around these centers cover the sectors, and on the common local-count event of Lemma 4.2, with $K+2$ in place of K , each of them contains at most $C_1(K+2)\log N$ roots, so that there are at most $40C_1(K+2)\sqrt{N}\log N$ roots in their union. The cover uses only the local zero count and no lower bound on the constant and leading coefficients, so it also applies when these are random and may be small, as for the laws of Section 7.

We now apply Theorem B.4, whose count B_N is $B_{K,N}$. Its mesh hypothesis holds with the annulus $||z| - 1| \leq K/N$, radius 2, $\tau = h$, the retained mesh points $z_1, \dots, z_m$, the disk D and M_2 above, $p_2 = N^{-10}$, the cell bound $q_0 = C_1(K+2)\log N$ and the sector bound $b_0 = 40C_1(K+2)\sqrt{N}\log N$, which hold on the common local-count event of Lemma 4.2, with $K+2$ in place of K , so that $p_0 = N^{-3}$. Its jet approximation holds with $H_i = G_{A_{z_i}}$, whose covariance has least eigenvalue at least λ by (26), with $\delta = \delta_* = 2000e^{3K}(2/\lambda)^{3/2}N^{-1/2}$ by the four- and eight-dimensional Raič bounds, with $\chi = 80e^{4K}/(\lambda\sqrt{N})$ by (28), since $g_i g_j = \mathbb{P}\{G_{\Sigma_0} \in U \times U\}$ by the independence of $G_{A_{z_i}}$ and $G_{A_{z_j}}$, and with $b_* = 10N^{-1/2}$ by the count of pairs in the removed bands. By (27) and this bound on m , the quantities of the theorem satisfy

$$\mu_* \leq C(K)\{NL^{-4} + N^{1/2}L^2\log N\}, \quad V_* = b_* + \delta_* + \chi + 2\delta \leq C(K)N^{-1/2}.$$

We take $t = NL^{-4}$, the leading term of the bound on μ_* . Since $L = N^{1/64}$, we have $\sqrt{N} \leq NL^{-4}$ and $N^{1/2}L^2\log N = N^{17/32}\log N \leq N^{15/16} = NL^{-4}$ for $N \geq 2$, the latter since $\log N \leq N^{13/32}$, so that

$$\begin{aligned} b_0 + q_0(\mu_* + t) &\leq C_1(K+2)(\log N) \left[40\sqrt{N} + NL^{-4} \right. \\ &\quad \left. + C(K)\{NL^{-4} + N^{1/2}L^2\log N\} \right] \end{aligned}$$

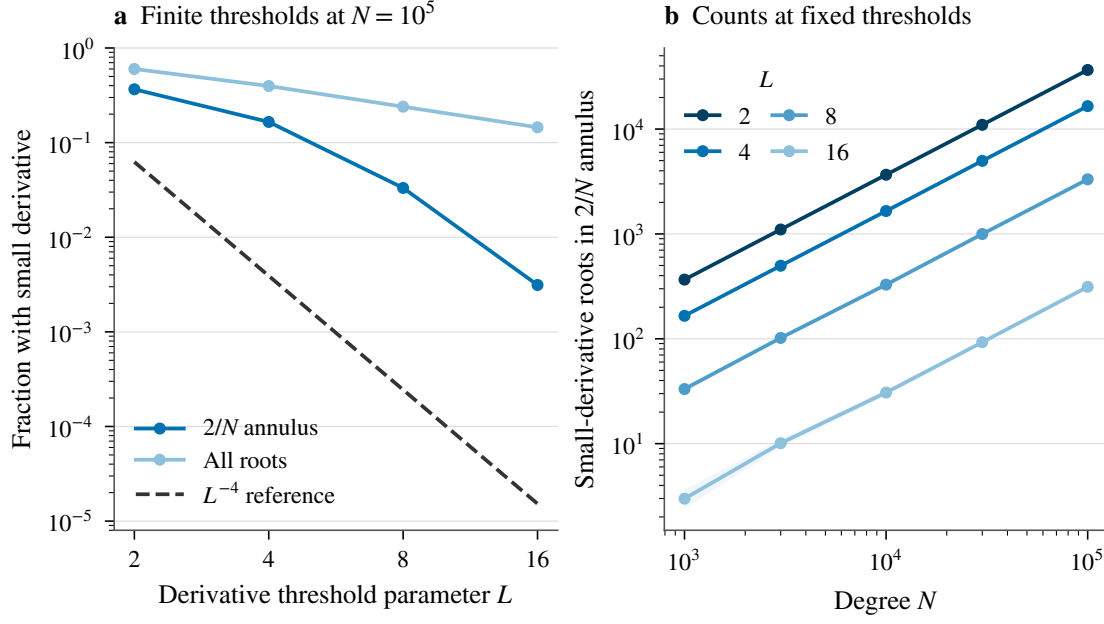


Figure 7: Zeros with small derivative, their fraction (a) and number (b) over 77 sequences.

$$\begin{aligned}
 &\leq C(K)NL^{-4}\log N, \\
 \frac{m^2V_*}{t^2} &\leq C(K)N^2L^4(\log N)^2 \cdot N^{-1/2} \cdot \frac{L^8}{N^2} = C(K)N^{-1/2}L^{12}(\log N)^2 \\
 &= C(K)N^{-5/16}(\log N)^2.
 \end{aligned}$$

Therefore (78) gives

$$(29) \quad \mathbb{P}\{B_{K,N} > C(K)NL^{-4}\log N\} \leq N^{-3} + N^{-10} + C(K)N^{-5/16}(\log N)^2,$$

whose three terms are the local-count exception of Lemma 4.2, the derivative exception of Lemma 4.3, and the mesh-deviation exception. The explicit constants are given in Section A.4. $\square$

Figure 7 shows the fraction of annular roots with a small derivative for fixed L , whereas the lemma takes $L = N^{1/64}$. Panel (a) compares the number of such zeros at $N = 10^5$, in the annulus and among all zeros, divided by N , with L^{-4} . This is the power of L in the bound $C(K)NL^{-4}\log N$ of the lemma and in the leading term of the bound on μ_* , the mean number of mesh points that detect such zeros.

4.3 Bounds for appended tails

Lemma 4.6 (Appended tails). *Let $K \geq 2$, $N \geq \max\{4, K\}$ and $1 \leq m_{\max} \leq N$. Set*

$$\rho = 1 + K/N, \quad g_m(z) = \sum_{k=N+1}^{N+m} \epsilon_k z^k, \quad 0 \leq m \leq m_{\max},$$

and put

$$a = 15e^{2K} \sqrt{m_{\max} \log N}.$$

Outside an event of probability at most N^{-10} ,

$$\max_{m \leq m_{\max}} \sup_{|z| \leq \rho} |g_m(z)| < a.$$

Proof. As in the proof of Lemma 4.3, we combine Hoeffding's inequality at the samples of the circle $|z| = \rho$, a deterministic bound on g'_m between the samples, and the maximum principle, now with the constants of the tails.

Note that the empty tail is identically zero. We choose $\lceil N^3 \rceil \leq 2N^3$ equally spaced samples on the boundary circle. For every prefix, since $\rho \geq 1$ and $k \leq N + m \leq 2N$, we have $\sum_{k=N+1}^{N+m} \rho^{2k} \leq m\rho^{4N} \leq m_{\max} e^{4K}$. For a sample z and $1 \leq m \leq m_{\max}$, we apply Hoeffding's inequality to $\Re g_m(z) = \sum_{k=N+1}^{N+m} \epsilon_k \Re(z^k)$ and $\Im g_m(z) = \sum_{k=N+1}^{N+m} \epsilon_k \Im(z^k)$, whose squared coefficients are at most $|z|^{2k} = \rho^{2k}$. As in the proof of Lemma 4.3, and since $(2a/3)^2 = 100e^{4K} m_{\max} \log N$, we get

$$\begin{aligned} \mathbb{P}\{|g_m(z)| \geq 2a/3\} &\leq \mathbb{P}\left\{|\Re g_m(z)| \geq \frac{2a}{3\sqrt{2}}\right\} + \mathbb{P}\left\{|\Im g_m(z)| \geq \frac{2a}{3\sqrt{2}}\right\} \\ &\leq 4 \exp\left[-\frac{(2a/3)^2}{4 \sum_{k=N+1}^{N+m} \rho^{2k}}\right] \leq 4 \exp\left[-\frac{(2a/3)^2}{4m_{\max} e^{4K}}\right] \\ &= 4 \exp[-25 \log N] = 4N^{-25}. \end{aligned}$$

Since there are at most $2N^3$ samples and at most $m_{\max} \leq N$ nonempty prefixes, there are at most $2N^4$ such pairs, and by the union bound the probability that $|g_m(z)| \geq 2a/3$ for one of them is at most $(2N^4)(4N^{-25}) = 8N^{-21} \leq N^{-10}$. For the derivative we have the deterministic bound

$$\sup_{|z| \leq \rho} |g'_m(z)| \leq \sum_{k=N+1}^{N+m} k \rho^{k-1} \leq m \cdot 2N \cdot \rho^{2N} \leq 2N m_{\max} e^{2K}.$$

Since $N \geq K$, the circle $|z| = \rho$ has radius at most 2 and length at most 4π , so that every point of it lies within arc length $2\pi/N^3$ of a sample. Therefore, since $m_{\max} \leq \sqrt{m_{\max}} \sqrt{N}$ and $4\pi < 40 \leq 5N^{3/2} \sqrt{\log N}$ for $N \geq 4$, the error of nearest-sample interpolation is at most

$$\frac{2\pi}{N^3} \cdot 2N m_{\max} e^{2K} = \frac{4\pi m_{\max} e^{2K}}{N^2} \leq 4\pi e^{2K} \sqrt{m_{\max}} N^{-3/2} < 5e^{2K} \sqrt{m_{\max} \log N} = a/3.$$

Outside this event, every point ζ of the circle of radius ρ and its nearest sample z therefore satisfy, for $1 \leq m \leq m_{\max}$,

$$|g_m(\zeta)| \leq |g_m(z)| + \frac{4\pi m_{\max} e^{2K}}{N^2} < \frac{2a}{3} + \frac{a}{3} = a.$$

By the maximum principle, the boundary bound extends to the disk. Hence $\max_{m \leq m_{\max}} \sup_{|z| \leq \rho} |g_m(z)| < a$ outside an event of probability at most N^{-10} , and this one event controls every appended tail g_m at once. $\square$

5 Root matching under perturbations

We compare the roots of a polynomial P with those of a perturbation $Q = P + G$. If $|G| < |P|$ on the boundaries of disjoint disks, each containing one root of P , then Rouché's theorem gives one root of Q in each disk, and only the disks that meet a circle and the other roots can change the count inside it. Lemma 5.4 provides such disks about the roots with a large derivative.

For a nonzero polynomial P , denote by $\nu_P(r)$ the number of roots of P in the closed disk $\{|z| \leq r\}$, counted with multiplicity.

Theorem 5.1 (Root matching). *Let $P, Q = P + G$ be nonzero complex polynomials. Let $D_1, \dots, D_s$ be pairwise disjoint open disks, each containing exactly one root of P counted with multiplicity. Suppose for a fixed $a > 0$ that*

$$|P(z)| \geq a > |G(z)| \quad (z \in \partial D_j, \ 1 \leq j \leq s).$$

Denote by $c(r)$ the number of these disks whose closures meet $\{|z| = r\}$. Then

$$(30) \quad |\nu_Q(r) - \nu_P(r)| \leq c(r) + (\deg P - s) + \max(0, \deg Q - \deg P).$$

The same assertion holds for pairwise disjoint bounded open domains whose boundaries are finite unions of smooth Jordan curves, with exactly one root in each domain and the same boundary domination.

Proof. The idea is to find exactly one root of Q in each selected domain by Rouché's theorem, and to count the other roots crudely, using the fact that for each polynomial the number of them in $\{|z| \leq r\}$ lies between zero and their total number.

We use Rouché's theorem in the form of [1, Chapter 4, Section 5.2, p. 153]. Let γ be a cycle that is homologous to zero in a region Ω and has winding number $n(\gamma, z) \in \{0, 1\}$ at every point z off γ , and let f, g be analytic in Ω with $|f - g| < |f|$ on γ . Then f and g have equally many zeros, counted with multiplicity,

in $\{z : n(\gamma, z) = 1\}$. We take $\Omega = \mathbb{C}$, in which every cycle is homologous to zero, $f = P$, $g = Q$, and $\gamma = \partial D_j$, with its outer curve oriented counterclockwise and its inner curves, if any, clockwise. Since D_j is the set of points inside its outer curve and outside its inner curves, and a smooth Jordan curve winds once about each point inside it and zero times about each point outside it, with the sign of its orientation, we have $n(\partial D_j, z) = 1$ for $z \in D_j$ and $n(\partial D_j, z) = 0$ for $z \notin \overline{D_j}$. On ∂D_j we have $|f - g| = |G| < a \leq |P| = |f|$ by hypothesis. Hence D_j contains as many roots of Q as of P , that is, exactly one, counted with multiplicity.

Since the selected domains are pairwise disjoint and each of them contains exactly one root of P and exactly one root of Q , exactly s roots of each polynomial lie in these domains. We call the other roots unmatched. Their numbers are $\deg P - s$ and $\deg Q - s = \deg P - s + \deg Q - \deg P$. If the closure of a selected domain misses $\{|z| = r\}$, then, since the domain is connected and the complement of the circle is the union of the disjoint open sets $\{|z| < r\}$ and $\{|z| > r\}$, it lies entirely inside or entirely outside that circle. Its root of P is therefore counted by $\nu_P(r)$ if and only if its root of Q is counted by $\nu_Q(r)$, so that this domain does not change the count. On the other hand, each crossing domain, that is, one whose closure meets the circle, changes the count by at most one, since each of its two roots adds 0 or 1 to its count. Hence the total change from selected roots is at most $c(r)$.

The numbers of unmatched roots inside the circle belong to $[0, \deg P - s]$ and $[0, \deg Q - s]$, and their difference therefore has absolute value at most

$$\max(\deg P - s, \deg Q - s) = \deg P - s + \max(0, \deg Q - \deg P),$$

by the above expression for $\deg Q - s$. Since $\nu_Q(r) - \nu_P(r)$ is the change from selected roots plus the difference of the unmatched counts, we obtain (30) by the triangle inequality. Roots on the target circle are included in the inside count, since the disk in $\nu_P(r)$ and $\nu_Q(r)$ is closed. Each selected root is simple, since its total multiplicity is one. $\square$

Remark 5.2. If $\deg Q \geq \deg P$, the last term in (30) is the degree difference. For arbitrary degrees the positive part is necessary. Indeed, take $P(z) = z^2$, $Q(z) = 1$, no disks, and $0 < r < 1$. With the signed degree difference we would have

$$2 \not\leq 2 + (0 - 2).$$

In this example we have $s = c(r) = 0$, $\nu_P(r) = 2$, since the double root 0 lies in $\{|z| \leq r\}$, and $\nu_Q(r) = 0$, so that $|\nu_Q(r) - \nu_P(r)| = 2$, $\deg P - s = 2$ and $\deg Q - \deg P = 0 - 2$.

In the same way we can compare the root counts of P and Q in two different disks.

Corollary 5.3 (Matching across two radii). *Let $P, Q = P + G$ and the domains $D_1, \dots, D_s$ satisfy the hypotheses of Theorem 5.1, in either of its forms. Let $r, r' > 0$, and denote by $c(r, r')$ the number of these domains whose closures meet the closed annulus $\{\min(r, r') \leq |z| \leq \max(r, r')\}$. Then*

$$|\nu_Q(r') - \nu_P(r)| \leq c(r, r') + (\deg P - s) + \max(0, \deg Q - \deg P).$$

Proof. As in the proof of Theorem 5.1, each domain D_j contains exactly one root of P , by hypothesis, and exactly one root of Q , by Rouché's theorem on ∂D_j , where $|G| < a \leq |P|$. That step does not involve the circle, so only the counting changes. We call the remaining $\deg P - s$ roots of P and $\deg Q - s$ roots of Q unmatched. Let A be the closed annulus between the two circles. A domain D_j whose closure misses A lies in $\{|z| < \min(r, r')\}$ or in $\{|z| > \max(r, r')\}$, since it is connected and the complement of A is the union of these two disjoint open sets. In the first case its root of P is counted by $\nu_P(r)$ and its root of Q by $\nu_Q(r')$, and in the second case neither is counted, so that the domain contributes the same amount to $\nu_Q(r')$ and to $\nu_P(r)$. On the other hand, each of the $c(r, r')$ remaining domains changes the difference by at most one, since each of its two roots adds 0 or 1 to its count. The numbers of unmatched roots counted by $\nu_P(r)$ and by $\nu_Q(r')$ belong to $[0, \deg P - s]$ and $[0, \deg Q - s]$, so that, as in the proof of Theorem 5.1, their difference has absolute value at most $\deg P - s + \max(0, \deg Q - \deg P)$. Since $\nu_Q(r') - \nu_P(r)$ is the sum of the two contributions, the corollary follows from the triangle inequality by adding their bounds. For $r = r'$ the annulus is the circle $\{|z| = r\}$, so that $c(r, r) = c(r)$, and the bound reduces to (30). $\square$

Lemma 5.4 (Disjoint disks near regular roots). *Let $a > 0$ and let α be a root of a nonconstant polynomial P , with $|P'(\alpha)| \geq d_0 > 0$. Suppose $|P''| \leq M_2$ throughout the closed disk of radius $4a/d_0$ around α , and $4M_2a \leq d_0^2$. Then*

$$|P| \geq 3a/2 \quad \text{on } |z - \alpha| = 2a/d_0,$$

the closed disk of radius $2a/d_0$ about α contains exactly one root, the component of $\{|P| < a\}$ containing α lies inside it, and every other root β of P satisfies $|\beta - \alpha| \geq 4a/d_0$. For any finite family of such roots with common a, d_0 and M_2 , the open disks of radius $2a/d_0$ about them are therefore pairwise disjoint, each with exactly one root, and they satisfy the hypotheses of Theorem 5.1 whenever $|G| < a$ on their boundary circles.

Proof. The idea is to compare P near α with its linear part $P'(\alpha)(z - \alpha)$ at α , using Taylor's theorem for the lower bound on the circle and for the separation of the roots, and Rouché's theorem for the single root in the disk.

Let $|z - \alpha| \leq 4a/d_0$. Then

$$\begin{aligned} |P(z) - P'(\alpha)(z - \alpha)| &= |P(z) - P(\alpha) - P'(\alpha)(z - \alpha)| \\ &= |z - \alpha|^2 \left| \int_0^1 (1 - u) P''(\alpha + u(z - \alpha)) du \right| \\ &\leq M_2 |z - \alpha|^2 \int_0^1 (1 - u) du = \frac{M_2}{2} |z - \alpha|^2, \end{aligned}$$

where the second equality is Taylor's theorem with integral remainder for the function $u \mapsto P(\alpha + u(z - \alpha))$ on $[0, 1]$, and the inequality holds since $|P''| \leq M_2$ on the segment from α to z , which lies in the closed disk of radius $4a/d_0$ around α . For $|z - \alpha| = 2a/d_0$, we therefore have

$$\begin{aligned} |P(z)| &\geq |P'(\alpha)(z - \alpha)| - |P(z) - P'(\alpha)(z - \alpha)| \geq |P'(\alpha)| |z - \alpha| - \frac{M_2}{2} |z - \alpha|^2 \\ &\geq d_0 \cdot \frac{2a}{d_0} - \frac{M_2(2a/d_0)^2}{2} = 2a - \frac{2M_2a^2}{d_0^2} \geq 2a - \frac{a}{2} = \frac{3a}{2}, \end{aligned}$$

where the third inequality uses $|P'(\alpha)| \geq d_0$, and the last is the hypothesis $4M_2a \leq d_0^2$ multiplied by $a/(2d_0^2)$.

On the same circle, as in the chain above, we also have

$$\begin{aligned} |P(z) - P'(\alpha)(z - \alpha)| &\leq \frac{M_2}{2} |z - \alpha|^2 = \frac{M_2(2a/d_0)^2}{2} = \frac{2M_2a^2}{d_0^2} \leq \frac{a}{2} \\ &< 2a = d_0 \cdot \frac{2a}{d_0} \leq |P'(\alpha)| |z - \alpha| = |P'(\alpha)(z - \alpha)|. \end{aligned}$$

Thus $f = P'(\alpha)(z - \alpha)$ and $g = P$ satisfy $|f - g| < |f|$ on the circle. We apply Rouché's theorem, in the form used in the proof of Theorem 5.1, with $\Omega = \mathbb{C}$ and γ the counterclockwise circle, which winds once about each point of the open disk and zero times about each point outside the closed disk. Hence P has as many roots in the disk $|z - \alpha| < 2a/d_0$ as the linear polynomial $P'(\alpha)(z - \alpha)$, counted with multiplicity. This linear polynomial is nonzero, since $|P'(\alpha)| \geq d_0 > 0$, and its only root α lies in the disk. Hence the open disk contains exactly one root, and so does the closed disk, since $|P| \geq 3a/2 > 0$ on the circle. Since $|P| \geq 3a/2 > a$ on the boundary circle, the component of $\{|P| < a\}$ containing α does not meet this circle. It therefore lies inside the disk, since any connected set meeting both the interior and the exterior of the disk meets its boundary.

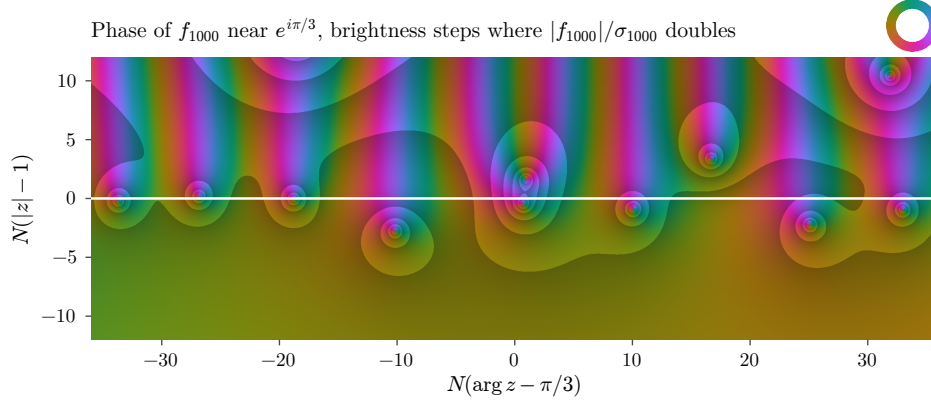


Figure 8: Phase portrait of f_{1000} near $e^{i\pi/3}$ for the sequence of Figure 9: the hue is $\arg f_{1000}$, all hues meet at each zero, the brightness steps where $|f_{1000}|/\sigma_{1000}(|z|)$ doubles, and the white line is the unit circle. Near a zero with a large derivative, f_{1000} is nearly linear, so the small level curves are nearly circles about it, like the boundaries of the disjoint disks about the selected roots in Lemma 5.4.

Let $\beta \neq \alpha$ be a root of P , and suppose that $|\beta - \alpha| < 4a/d_0$. By the first display at $z = \beta$, where $P(\beta) = 0$, and by $4M_2a \leq d_0^2$, we have

$$d_0|\beta - \alpha| \leq |P'(\alpha)(\beta - \alpha)| \leq \frac{M_2}{2}|\beta - \alpha|^2 \leq \frac{2M_2a}{d_0}|\beta - \alpha| \leq \frac{d_0}{2}|\beta - \alpha|,$$

which is impossible since $d_0 > 0$ and $\beta \neq \alpha$. Hence $|\beta - \alpha| \geq 4a/d_0$. For a finite family of such roots, with common a , d_0 and M_2 , two of the open disks of radius $2a/d_0$ about distinct roots α and β meet only if $|\beta - \alpha| < 4a/d_0$, by the triangle inequality, so that they are pairwise disjoint. Each contains exactly one root, and $|P| \geq 3a/2 > a > |G|$ on its boundary circle when $|G| < a$ there, which are the hypotheses of Theorem 5.1. $\square$

Figure 8 illustrates Lemma 5.4 for one polynomial with random signs.

6 Proof of the radial law

Sections 2–4 control the roots of f_N along the sparse subsequence of degrees, and root matching carries most of them to every degree of the block. The count at $1 + x/n$ then differs from the count at $1 + x/N$ only through roots near the target circle, unselected roots and the degree difference, and each of these is small. We prove Theorem 1.1 for one x at a time, and then for all x by monotonicity; Corollary 1.2 is the case $x = 0$, since $\Phi(0) = 1/2$.

Proof of Theorem 1.1. The idea is to compare every degree n in a block $j^8 \leq n \leq (j+1)^8$ with the first degree $N = j^8$ of the block, by root matching (Corollary 5.3) at the radii $1 + x/N$ and $1 + x/n$. A root of f_N is matched unless it lies outside the annulus $||z| - 1| \leq K/N$ or has a small derivative there, and a matched root can cross between the two target circles only if its disk meets the annulus between them.

Step 1: parameters. Fix x , an integer $R \geq |x| + 2$ and an integer $K \geq R$, and compare the degrees $N = j^8 \leq n \leq (j+1)^8 = N + m_j$. Table 2 lists the parameters, the conditions they must satisfy and where they are used. The annulus is $||z| - 1| \leq K/N$, and $L = N^{1/64}$ as in Section 4.2.

The conditions hold for all large j because the block exponent 8 makes m_j of order $N^{7/8}$, and hence a of order $N^{7/16}\sqrt{\log N}$, while $d_0 = N^{3/2-1/64}$. With blocks j^β , m_j would be of order $N^{1-1/\beta}$, and $2a/d_0$ and $4M_2a/d_0^2$ of orders $N^{(1-1/\beta)/2-95/64}$ and $N^{(1-1/\beta)/2-15/32}$, up to logarithmic factors, so Steps 2 and 4 would need $(1 - 1/\beta)/2 < 15/32$, that is, $\beta < 16$. The slowest exceptional probability in Step 6, of order $N^{-5/16}(\log N)^2$, is summable along $N = j^\beta$ only if $\beta > 16/5$. Proposition 8.1 treats other block exponents. For $j \geq 60$ we have

$$\begin{aligned} m_j &= (j+1)^8 - j^8 \leq 8(j+1)^7 = 8\left(1 + \frac{1}{j}\right)^7 N^{7/8} \\ (31) \quad &\leq 8\left(\frac{61}{60}\right)^7 N^{7/8} \leq 9N^{7/8}, \end{aligned}$$

$$(32) \quad a \leq 45e^{2(K+2)}N^{7/16}\sqrt{\log N},$$

$$\begin{aligned} \frac{2a}{d_0} &= \frac{2a}{N^{3/2-1/64}} \leq 90e^{2(K+2)}N^{7/16+1/64-3/2}\sqrt{\log N} \\ (33) \quad &= 90e^{2(K+2)}N^{-67/64}\sqrt{\log N}, \end{aligned}$$

$$(34) \quad \frac{4Na}{d_0} \leq 180e^{2(K+2)}N^{-3/64}\sqrt{\log N} \longrightarrow 0,$$

$$\begin{aligned} \frac{4M_2a}{d_0^2} &= \frac{400e^{K+4}N^{5/2}\sqrt{\log N}a}{N^{3-1/32}} \leq 180 \cdot 100e^{K+4}e^{2(K+2)}N^{7/16+1/32-1/2}\log N \\ (35) \quad &= 180 \cdot 100e^{K+4}e^{2(K+2)}N^{-1/32}\log N \longrightarrow 0. \end{aligned}$$

In (31) we use the mean value theorem for the eighth power on $[j, j+1]$, $j^7 = N^{7/8}$, $j \geq 60$ and $8(61/60)^7 < 9$, and (32) follows from it. In the last three bounds we substitute (32) and $d_0 = N^{3/2-1/64}$.

Step 2: selected disks. We call an annular root α , that is, a root of f_N with $||\alpha| - 1| \leq K/N$, good if $|f'_N(\alpha)| \geq N^{3/2}/L$. If $|z - \alpha| \leq 4a/d_0$, then $|z| \leq |\alpha| + 4a/d_0 \leq 1 + K/N + 1/N = 1 + (K+1)/N < \rho$, since $4a/d_0 \leq 1/N$ by Step 1. Hence, outside the exceptional event of Lemma 4.3, we have $|f''_N| \leq M_2$ throughout

Parameter	Value	Condition	Where used
N	j^8 , first degree of the block	$j \geq 60$	throughout
m_j	$(j+1)^8 - j^8 \leq 9N^{7/8}$, by (31)	$m_j \leq N$, since $N^{1/8} = j \geq 9$	Lemma 4.6; Steps 4 and 5
K	an integer, $K \geq R$; the annulus is $ z - 1 \leq K/N$	$N \geq K+2$ and $N \geq 2K$	Lemma 4.6; Steps 2, 3 and 6
R	an integer, $R \geq x + 2$	$N \geq 2R$	the probes of Step 4
x	fixed	$ x \leq R - 2$	target radius $r_N = 1 + x/N$
L	$N^{1/64}$	none	Lemma 4.5; d_0
d_0	$N^{3/2}/L = N^{3/2-1/64}$, the derivative level of Lemma 4.5	$4a/d_0 \leq 1/N$, by (34); $4M_2a \leq d_0^2$, by (35)	Step 2; Lemma 5.4
M_2	$100e^{K+4}N^{5/2}\sqrt{\log N}$, the bound (24) on $ f_N'' $ over $ z \leq 1 + (K+1)/N$	as for d_0	Step 2
a	$15e^{2(K+2)}\sqrt{m_j \log N}$, the tail bound of Lemma 4.6 on $ z \leq \rho$, at width $K+2$ with $m_{\max} = m_j$	as for d_0 and h	Steps 2 and 4
ρ	$1 + (K+2)/N$	none	Lemma 4.6; Step 2
h	$N^{-1/64}$, the offset of the probes	Table 3; $2a/d_0 + x m_j/N^2 < h/N = N^{-65/64}$, by (40)	Step 4
κ	$1/64$, so that $h = N^{-\kappa}$	$\kappa < 1/16$	Step 4; Proposition 8.1

Table 2: The parameters of the proof of Theorem 1.1 for fixed x , R and K . Every condition holds for all large j .

this disk, and Lemma 5.4, applied with $P = f_N$, $|f'_N(\alpha)| \geq d_0$ and $4M_2a \leq d_0^2$, shows that the open disk of radius $2a/d_0$ about α contains exactly one zero, that $|f_N| \geq 3a/2$ on its boundary, and that these disks, for the family of good annular roots with the common a , d_0 and M_2 , are pairwise disjoint. We call them the selected disks.

Step 3: unselected roots. Let U_N denote the number of unselected roots. Since every good annular root selects a disk that contains no other root, the unselected roots are the roots with $||\alpha| - 1| > K/N$ and the annular roots that are not good, which $B_{K,N}$ counts. Consequently,

$$\begin{aligned} U_N &\leq \#\{\alpha : f_N(\alpha) = 0, ||\alpha| - 1| > K/N\} + B_{K,N}, \\ \frac{U_N}{N} &\leq \frac{2 \log 2}{K} + C(K)N^{-1/16}(\log N)^7 + C(K)N^{-1/16} \log N \\ (36) \quad &= \frac{2 \log 2}{K} + O_K(N^{-1/16}(\log N)^7), \end{aligned}$$

where the first count is bounded by (21) of Corollary 3.3, and, by Lemma 4.5, $B_{K,N}$ is at most $C(K)NL^{-4} \log N$, which is $C(K)N^{15/16} \log N$, outside the exceptional events of these two results.

Step 4: the target circles. The profile in Proposition 2.2 satisfies

$$F' = \Phi, \quad 0 \leq \Phi \leq 1, \quad 0 \leq \Phi' = 2 \operatorname{Var}_x(t) \leq \frac{1}{2}.$$

Indeed, $\Phi(x)$ is the mean of $t \in [0, 1]$ under the density proportional to e^{2xt} , and $\Phi'(x) = 2 \operatorname{Var}_x(t)$ by (7), where the variance is at most the mean square deviation from $\frac{1}{2}$:

$$\operatorname{Var}_x(t) \leq \frac{\int_0^1 (t - \frac{1}{2})^2 e^{2xt} dt}{\int_0^1 e^{2xt} dt} \leq \frac{1}{4}.$$

We use the logarithmic bounds at the radii $1 + (x + qh)/N$, $|q| \leq 2$, about $r_N = 1 + x/N$. The offset h must meet the three requirements of Table 3, and $h = N^{-1/64}$ does, since the entries of its last column tend to zero.

We apply Theorem 3.1, with R in place of K , at the five probes $r_q = 1 + y_q/N$, where $y_q = x + qh$ and $q = -2, \dots, 2$. They lie in $[1 - R/N, 1 + R/N]$, since $|y_q| \leq |x| + 2h < R$. For all large N , outside five exceptional events, one at each probe, the theorem gives $|J_N(r_q) - \log \sigma_N(r_q) + \gamma/2| \leq C(R)N^{-1/16}(\log N)^7$, and, for $N \geq 2R$, Lemma 2.3 with R in place of K gives $|\log \sigma_N(r_q) - \frac{1}{2} \log N - F(y_q)| \leq (R^2 + e^{4R}/2)/N$. Adding the two bounds, and using $1/N \leq N^{-1/16}(\log N)^7$ for $N \geq 3$, we get, outside the five exceptional events,

$$\left| J_N(r_q) - \frac{1}{2} \log N + \frac{\gamma}{2} - F(y_q) \right| \leq C(R)N^{-1/16}(\log N)^7 + \frac{R^2 + e^{4R}/2}{N}$$

Requirement	Reason	At $h = N^{-1/64}$
$h \rightarrow 0$	the secants of F over steps of length h approximate $\Phi(x)$ with an error of order h	$N^{-1/64}$
$N^{-1/16}(\log N)^7/h \rightarrow 0$	the secants divide the logarithmic error $N^{-1/16}(\log N)^7$ of Theorem 3.1 by h	$N^{-3/64}(\log N)^7$
h/N exceeds a bound of order $N^{-67/64}\sqrt{\log N} + N^{-9/8}$	the band of half-width h/N about $ z = 1 + x/N$ contains the root of every selected disk that meets the annulus between $ z = 1 + x/N$ and $ z = 1 + x/n$, for every degree n of the block	$N^{-1/32}\sqrt{\log N}$ and $N^{-7/64}$

Table 3: The requirements on the offset h in Step 4, for fixed K and x . The last column evaluates each requirement at $h = N^{-1/64}$.

$$(37) \quad \leq \eta_N := (C(R) + R^2 + e^{4R}/2)N^{-1/16}(\log N)^7.$$

For $q = -1, 0, 1$ we apply Lemma B.3 at y_q , with R in place of K , the centering $\frac{1}{2} \log N$, the profile F , $M_F = 1/2$ and $B_F = 1$ (by the bounds on $\Phi = F'$ above), and the logarithmic and profile errors $\eta_{\log} = C(R)N^{-1/16}(\log N)^7$ and $\varepsilon = (R^2 + e^{4R}/2)/N$ of (37), so that $\eta_{\log} + \varepsilon \leq \eta_N$. Its hypotheses hold outside the five exceptional events, since:

- $J_N(r_q)$ is finite there, so that $f_N \neq 0$;
- $|y_q| + h \leq |x| + 2h < R < N$, the last since $N \geq 2R$;
- its three radii $1 + (y_q \mp h)/N$ and r_q are among the five probes.

It gives

$$\left| \frac{\nu_N(r_q)}{N} - \Phi(y_q) \right| \leq \frac{h}{4} + \frac{R}{N} + \frac{2\eta_N(1 + R/N)}{h}.$$

Since $r_N + qh/N = r_q$ and $|\Phi(y_q) - \Phi(x)| \leq |q|h/2$ by $0 \leq \Phi' \leq 1/2$, and since $|q|h/2 + h/4 \leq 2h$ for $|q| \leq 1$ and $1 + R/N \leq 2$, we get, for $q = -1, 0, 1$,

$$\begin{aligned} \left| \frac{\nu_N(r_N + qh/N)}{N} - \Phi(x) \right| &\leq \frac{|q|h}{2} + \frac{h}{4} + \frac{R}{N} + \frac{2\eta_N(1 + R/N)}{h} \\ &\leq \varepsilon_N := 2h + \frac{R}{N} + \frac{4\eta_N}{h} \end{aligned}$$

$$\begin{aligned}
&= 2N^{-1/64} + \frac{R}{N} + 4(C(R) + R^2 + e^{4R}/2)N^{-3/64}(\log N)^7 \\
(38) \quad &= O_R(N^{-1/64}).
\end{aligned}$$

Explicit constants for this estimate are given in (65). Subtracting the two endpoint counts and using (38) with $q = \pm 1$, we get

$$\begin{aligned}
&\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - r_N| < h/N\} \\
&= \#\{\alpha : f_N(\alpha) = 0, \ |\alpha| < r_N + h/N\} - \nu_N(r_N - h/N) \\
&\leq \nu_N(r_N + h/N) - \nu_N(r_N - h/N) \\
&= N \left(\frac{\nu_N(r_N + h/N)}{N} - \Phi(x) \right) + N \left(\Phi(x) - \frac{\nu_N(r_N - h/N)}{N} \right) \\
(39) \quad &\leq 2\varepsilon_N N,
\end{aligned}$$

where the first equality holds since the inner inequality is strict.

For $N \leq n \leq N + m_j$ the target radius moves by $|(1+x/n) - (1+x/N)| = |x|(n-N)/(nN) \leq |x|m_j/N^2$, and $m_j \leq 9N^{7/8}$ for $j \geq 60$ by (31). Since $N/h = N^{65/64}$, these bounds and (33) yield

$$\begin{aligned}
(40) \quad \frac{N}{h} \left(\frac{2a}{d_0} + \frac{|x|m_j}{N^2} \right) &\leq 90e^{2(K+2)} N^{-67/64+65/64} \sqrt{\log N} + 9|x|N^{7/8-2+65/64} \\
&= 90e^{2(K+2)} N^{-1/32} \sqrt{\log N} + 9|x|N^{-7/64} \longrightarrow 0.
\end{aligned}$$

Hence $2a/d_0 + |x|m_j/N^2 < h/N$ for all large N , depending on K and x . Let D be a selected disk whose closure meets the closed annulus between the circles of radii r_N and $1 + x/n$, let α be its center, the root of f_N in it, and let z be a point of this annulus in the closure of D . Then $|z - \alpha| \leq 2a/d_0$ and $|z|$ lies between r_N and $1 + x/n$, so that

$$\begin{aligned}
||\alpha| - r_N| &\leq ||\alpha| - |z|| + ||z| - r_N| \leq |\alpha - z| + ||z| - r_N| \\
&\leq \frac{2a}{d_0} + \left| \left(1 + \frac{x}{n} \right) - r_N \right| \leq \frac{2a}{d_0} + \frac{|x|m_j}{N^2} < \frac{h}{N}.
\end{aligned}$$

Hence the root in such a disk lies in the strict band of (39). Only such disks can carry matched roots to different sides of their target circles, since a disk whose closure misses the closed annulus lies inside both circles or outside both (proof of Corollary 5.3). Distinct disks have distinct roots, so by (39) the number $c(r_N, 1 + x/n)$ of these disks is at most $2\varepsilon_N N$. For $x = 0$ the two target circles are the unit circle, the annulus between them is that circle, and the band is $1 - N^{-65/64} < |\alpha| < 1 + N^{-65/64}$.

Step 5: comparison within a block. For $N < n \leq N + m_j$, we write $f_n = f_N + g_{n-N}$, with g_m as in Lemma 4.6 at width $K + 2$. That lemma bounds all the partial tails at once, so outside the maximal-tail event $|g_{n-N}| < a$ on $|z| \leq \rho$ for all these n . This disk contains the closed selected disks (Step 2), on whose boundaries $|f_N| \geq 3a/2 > a$. Hence the selected disks satisfy the hypotheses of Theorem 5.1 with $P = f_N$, $G = g_{n-N}$ and $Q = f_n$, for the entire block and all target radii at once. Moreover, $\deg f_n = n$ and $\deg f_N = N$, since $\epsilon_n, \epsilon_N \neq 0$, and each of the s selected disks contains exactly one root of f_N , the others being unselected, so that $\deg f_N - s = N - s = U_N$. By Corollary 5.3, with $P = f_N$, $Q = f_n$, $r = r_N$ and $r' = 1 + x/n$, and by Step 4, we have

$$\begin{aligned} & \max_{N \leq n \leq N+m_j} |\nu_n(1 + x/n) - \nu_N(r_N)| \\ & \leq \max_{N \leq n \leq N+m_j} \left\{ c(r_N, 1 + x/n) + (\deg f_N - s) + \max(0, \deg f_n - \deg f_N) \right\} \\ & = \max_{N \leq n \leq N+m_j} \left\{ c(r_N, 1 + x/n) + U_N + (n - N) \right\} \leq 2\varepsilon_N N + U_N + m_j, \end{aligned}$$

which holds trivially for $n = N$. Splitting $\nu_n(1 + x/n)/n - \Phi(x)$ into three terms and using (38) with $q = 0$, the matching bound above and $0 \leq \Phi \leq 1$, we get, for $N \leq n \leq N + m_j$,

$$\begin{aligned} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| &= \left| \frac{N}{n} \left(\frac{\nu_N(r_N)}{N} - \Phi(x) \right) \right. \\ & \quad \left. + \frac{\nu_n(1 + x/n) - \nu_N(r_N)}{n} - \frac{n - N}{n} \Phi(x) \right| \\ &\leq \frac{N}{n} \left| \frac{\nu_N(r_N)}{N} - \Phi(x) \right| + \frac{|\nu_n(1 + x/n) - \nu_N(r_N)|}{n} \\ & \quad + \frac{n - N}{n} \Phi(x) \\ (41) \quad &\leq \varepsilon_N + \frac{2\varepsilon_N N + U_N + m_j}{N} + \frac{m_j}{N} = 3\varepsilon_N + \frac{U_N}{N} + \frac{2m_j}{N}. \end{aligned}$$

Step 6: exceptional events, and $K \rightarrow \infty$. Let $E_{K,R,x,j}$ denote the common exceptional event in block j , the union of the events of Table 4, where $\Pi_K(N)$ is the right-hand side of (9) and $\Pi_R(N)$ is the same with R in place of K . For all large j ,

$$(42) \quad \mathbb{P}(E_{K,R,x,j}) \leq 4\Pi_K(N) + 5\Pi_R(N) + N^{-3} + 2N^{-10} + C(K)N^{-5/16}(\log N)^2.$$

Here the shared local-count and derivative events are counted once each, and the maximal-tail event once for the whole block, since outside it Step 5 holds for all degrees of the block and all target radii. With $N = j^8$ we have $\Pi_K(j^8) =$

$C(K)j^{-3} + j^{-96} + C(K)j^{-3}(8 \log j)^{-12}$, since $N^{-3/8} = j^{-3}$, $N^{-12} = j^{-96}$ and $\log N = 8 \log j$, and (42) reads

$$\mathbb{P}(E_{K,R,x,j}) \leq 4\Pi_K(j^8) + 5\Pi_R(j^8) + j^{-24} + 2j^{-80} + C(K)j^{-5/2}(8 \log j)^2.$$

For fixed K , R and x , every term is summable over j , since the exponents of j are -3 , -96 , -24 , -80 and $-5/2$, and powers of $\log j$ do not affect convergence. In finite form, (42) is (67), and the root-count bound of Step 3 is (64).

Event	Source	Used in	Probability	At $N = j^8$, ignoring constants
Four radial events	(21)	Step 3, U_N	$\Pi_K(N)$ each	$j^{-3}, j^{-96},$ $j^{-3}(\log j)^{-12}$
Five probe events	(9), width R	Step 4, $\nu_N(r_N)$, crossings	$\Pi_R(N)$ each	$j^{-3}, j^{-96},$ $j^{-3}(\log j)^{-12}$
Local-count event	Lemma 4.2	Step 3, $B_{K,N}$	N^{-3}	j^{-24}
Derivative event	Lemma 4.3	Steps 1–3	N^{-10}	j^{-80}
Mesh-deviation event	Lemma 4.5	Step 3, $B_{K,N}$	$C(K)N^{-5/16}(\log N)^2$	$j^{-5/2}(\log j)^2$
Maximal-tail event	Lemma 4.6	Step 5	N^{-10}	j^{-80}

Table 4: The exceptional events whose union is $E_{K,R,x,j}$ in Step 6. The three terms of $\Pi_K(N)$ and $\Pi_R(N)$ come from the occupation, the logarithmic norm and the clipped integral.

Let $\Omega_{K,R,x} = (\limsup_j E_{K,R,x,j})^c$. Since $\limsup_j E_{K,R,x,j} \subset \bigcup_{j \geq J} E_{K,R,x,j}$ for every J , the summability above gives $\mathbb{P}(\Omega_{K,R,x}^c) \leq \lim_{J \rightarrow \infty} \sum_{j \geq J} \mathbb{P}(E_{K,R,x,j}) = 0$. On $\Omega_{K,R,x}$ the preceding bounds hold for all large j , and hence, by (41), (36) and $m_j \leq 9N^{7/8}$, for all large j and $N \leq n \leq N + m_j$ we get

$$\left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq 3\varepsilon_N + \frac{2 \log 2}{K} + C(K)N^{-1/16}(\log N)^7 + 18N^{-1/8}.$$

Every large n lies in such a block, and each term on the right other than $2 \log 2/K$ tends to zero as $j \rightarrow \infty$. Hence, on the intersection Ω_x of the events $\Omega_{K,R,x}$ over integers $K \geq R$, which has probability one as a countable intersection of events of probability one,

$$0 \leq \limsup_{n \rightarrow \infty} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq \frac{2 \log 2}{K} \longrightarrow 0 \quad (K \rightarrow \infty).$$

Step 7: all x . As a countable intersection, the rational event $\bigcap_{x \in \mathbb{Q}} \Omega_x$ also has probability one.

For any real x , choose rationals $x_- < x < x_+$. By monotonicity we have $\nu_n(1 + x_-/n)/n \leq \nu_n(1 + x/n)/n \leq \nu_n(1 + x_+/n)/n$, and taking limits on the rational event we get

$$\Phi(x_-) \leq \liminf_{n \rightarrow \infty} \frac{\nu_n(1 + x/n)}{n} \leq \limsup_{n \rightarrow \infty} \frac{\nu_n(1 + x/n)}{n} \leq \Phi(x_+).$$

Letting x_- increase to x and x_+ decrease to x through rationals, the continuity of Φ shows that the limit inferior and the limit superior are both $\Phi(x)$. To prove the uniform convergence, fix a compact interval I and $\delta > 0$, write $\hat{\Phi}_n(x) = \nu_n(1 + x/n)/n$, and choose rational numbers $x_0 < x_1 < \dots < x_M$ with $I \subset (x_0, x_M)$ and $x_{i+1} - x_i < \delta$. For $x \in [x_i, x_{i+1}] \cap I$ we have

$$\hat{\Phi}_n(x_i) - \Phi(x_{i+1}) \leq \hat{\Phi}_n(x) - \Phi(x) \leq \hat{\Phi}_n(x_{i+1}) - \Phi(x_i),$$

since $\hat{\Phi}_n$ is nondecreasing, as ν_n is, and Φ is nondecreasing, as $\Phi' \geq 0$. Taking the supremum over $x \in I$, we get

$$\begin{aligned} \sup_{x \in I} |\hat{\Phi}_n(x) - \Phi(x)| &\leq \max_{0 \leq i \leq M} |\hat{\Phi}_n(x_i) - \Phi(x_i)| + \max_{0 \leq i < M} \{\Phi(x_{i+1}) - \Phi(x_i)\} \\ &\leq \max_{0 \leq i \leq M} |\hat{\Phi}_n(x_i) - \Phi(x_i)| + \frac{\delta}{2}, \end{aligned}$$

where the second inequality follows from $|\Phi'| \leq 1/2$ and $x_{i+1} - x_i < \delta$ by the mean value theorem. On the rational event the maximum tends to zero, and we then let δ tend to zero. Finally, we intersect over intervals with integer endpoints. All these intersections take place on the given probability space. Since the rational event does not depend on I , the uniform convergence holds on it for every compact interval I at once, and in particular for $I = [-A, A]$ for every $A > 0$, which is the assertion of Theorem 1.1. $\square$

Proof of Corollary 1.2. This is the case $x = 0$ of Theorem 1.1, since $1 + 0/n = 1$ and $\Phi(0) = 1/2$. In the proof above, at $x = 0$ we take $R = 2$, so that $r_N = 1$, both target circles are the unit circle, and the probes lie at the radii $1 + qN^{-65/64}$, $q = -2, \dots, 2$. For every integer $K \geq 2$ and all large j , outside $E_{K,2,0,j}$ and uniformly for $N \leq n \leq N + m_j$, (41) reads

$$(43) \quad \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| \leq 3\varepsilon_N + \frac{U_N}{N} + \frac{2m_j}{N},$$

where $\varepsilon_N = 2N^{-1/64} + 2/N + 4(C(2) + 4 + e^8/2)N^{-3/64}(\log N)^7$ does not depend on K . $\square$

Remark 6.1 (A deterministic block envelope). There are deterministic numbers $\varepsilon_j \rightarrow 0$ and events E_j , defined on the original probability space, such that $\sum_j \mathbb{P}(E_j) < \infty$ and, off E_j ,

$$\max_{j^8 \leq n < (j+1)^8} \frac{|\nu_n(1) - \nu_{j^8}(1)|}{j^8} \leq \varepsilon_j.$$

Proof. We apply Lemma B.7 with the exponent 8, $n_j = j^8$ and $X_n = \nu_n(1)$, for which its conclusion is the assertion of the remark. It requires, for each positive integer k , numbers a_k , $b_k(j)$ and $p_k(j)$ such that, for every j ,

$$\mathbb{P} \left\{ \max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} > a_k + b_k(j) \right\} \leq p_k(j),$$

where $a_k \rightarrow 0$ as $k \rightarrow \infty$, and where $b_k(j) \rightarrow 0$ as $j \rightarrow \infty$ and $\sum_j p_k(j) < \infty$ for each fixed k .

Let k be a positive integer, and put $K = k + 1$, so that $K \geq 2 = R$, as the proof of Corollary 1.2 above requires at $x = 0$. Since every threshold in that proof depends only on K , there is a deterministic $j_0(K) \geq 60$ such that for every $j \geq j_0(K)$ the bound (42) with $R = 2$ and $x = 0$ holds and, outside the event $E_{K,2,0,j}$, with $N = j^8$, the matching bound of Step 5, (36), the bound $\varepsilon_N \leq C_0 N^{-1/64}$ with an absolute constant C_0 , and (31) give

$$\begin{aligned} \max_{N < n \leq N+m_j} |\nu_n(1) - \nu_N(1)| &\leq 2\varepsilon_N N + U_N + m_j, \\ \frac{U_N}{N} &\leq \frac{2 \log 2}{K} + C(K) N^{-1/16} (\log N)^7, \\ 2\varepsilon_N &\leq 2C_0 N^{-1/64}, \\ \frac{m_j}{N} &\leq 9N^{-1/8}. \end{aligned}$$

We set $a_k = 2 \log 2 / K$,

$$b_k(j) = C(K) N^{-1/16} (\log N)^7 + 2C_0 N^{-1/64} + 9N^{-1/8},$$

and let $p_k(j)$ be the bound (42) for $\mathbb{P}(E_{K,2,0,j})$ when $j \geq j_0(K)$, and $p_k(j) = 1$ when $j < j_0(K)$. Since the block $n_j \leq n < n_{j+1}$ is contained in $N \leq n \leq N + m_j$, and the increment vanishes at $n = N$, the four bounds above give, for $j \geq j_0(K)$ and outside $E_{K,2,0,j}$,

$$\begin{aligned} \max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} &\leq \frac{1}{N} \max_{N < n \leq N+m_j} |\nu_n(1) - \nu_N(1)| \\ &\leq 2\varepsilon_N + \frac{U_N}{N} + \frac{m_j}{N} \leq a_k + b_k(j). \end{aligned}$$

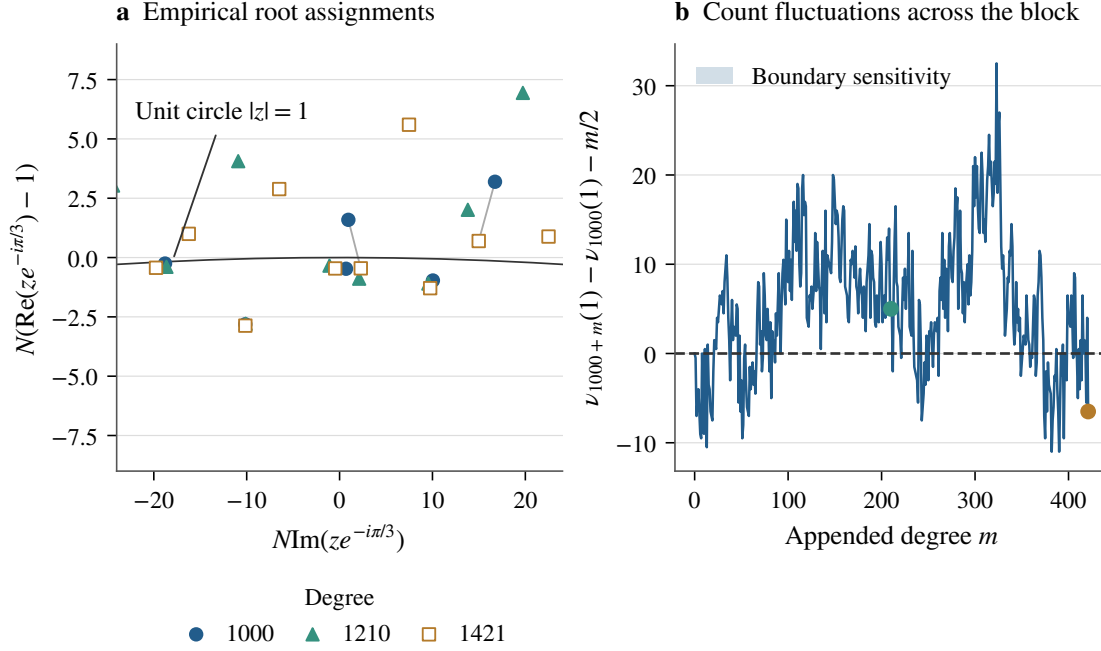


Figure 9: Zeros of one sequence at degrees 1000, 1210 and 1421 near $e^{i\pi/3}$, joined by an uncertified empirical assignment (a), and the change in the count across the block (b).

Hence the probability in the above requirement is at most $\mathbb{P}(E_{K,2,0,j}) \leq p_k(j)$ when $j \geq j_0(K)$, and at most $1 = p_k(j)$ when $j < j_0(K)$. Moreover, $a_k \rightarrow 0$ as $k \rightarrow \infty$, and, for fixed k , $b_k(j) \rightarrow 0$ as $j \rightarrow \infty$ and $\sum_j p_k(j) < \infty$, since only finitely many of its terms equal one and the others are summable by Step 6 of the proof of Theorem 1.1. By Lemma B.7, there are therefore deterministic $\varepsilon_j \rightarrow 0$ and events E_j with $\sum_j \mathbb{P}(E_j) < \infty$, off which the normalized block increment is at most ε_j . We may take E_j to be the event on which the normalized block increment exceeds ε_j , since it is contained in the event given by the lemma. (In the proof of Lemma B.7 the given events are this event for $j \geq J_1$ and Ω for $j < J_1$, where J_1 is the first of the integers J_k chosen there, and $\sum_{j \geq J_1} \mathbb{P}(E_j) \leq 1$.) All the events here are defined through the random signs on the original probability space. $\square$

With $n_j = j^8$ and $X_n = \nu_n(1)$, the numbers ε_j and the events E_j of Remark 6.1 are those required by Theorem B.6.

Figure 9 shows one block, from degree 1000 to degree 1421, of one sequence.

7 Other coefficient distributions

The radial law extends to Steinhaus, real and complex Gaussian, and nondegenerate bounded centrally symmetric coefficients. Besides the degrees of the prefixes, the proof of Section 6 uses the coefficients only through five inputs, which we state as hypotheses (H1)–(H5) on the coefficient law. We verify them for each law in Proposition 7.4 and prove Theorem 1.1 and Corollary 1.2 under them at the end of the section.

We take one sequence of iid coefficients with one of these laws, use it at every degree, and write

$$f_N(z) = \sum_{k=0}^N \xi_k z^k, \quad \sigma_N(r)^2 = \mathbb{E}|\xi_0|^2 \sum_{k=0}^N r^{2k},$$

and $X_r(\theta) = f_N(re^{i\theta})/\sigma_N(r)$, as in Section 3. In the three classes for which we prove a root law, the coefficients have mean zero and positive second moment. The logarithmic integral J_N and the closed-disk zero count are defined as before.

Hypotheses (H1)–(H5). Let the ξ_k be independent with a common law, normalized by $\mathbb{E}|\xi_0|^2 = 1$, let $L = N^{1/64}$, and let X_r , A_t , $B_{K,N}$ and the tails of Lemma 4.6 be defined as in Sections 3 and 4, with the ξ_k in place of the signs. For every $K \geq 2$ there are constants and numbers $\pi_K(N)$ with $\sum_j \pi_K(j^8) < \infty$ such that, for all sufficiently large N and uniformly for $r \in [1 - K/N, 1 + K/N]$, the following hold.

Hypothesis	Content	Where used below
(H1) Local counts	Except on an event of probability at most $N^{-3} + \pi_K(N)$, every disk $\{z : z - z_0 \leq 2K/N\}$ with $ z_0 - 1 \leq 1/N$ contains at most $C'_1(K) \log N$ zeros of f_N .	Steps 3 and 6
(H2) Derivative and tails	For an amplitude factor $B_0 \geq 1$ that does not depend on K , outside an event of probability at most $\pi_K(N)$, the conclusions of Lemmas 4.3 and 4.6 hold with their bounds multiplied by B_0 , each outside a further event of probability at most N^{-10} .	Steps 1–6
(H3) Occupation	For all $t \geq 0$, we have $ \mathbb{E}A_t - F_G(t) \leq H(K)N^{-1/2}$ and $\text{Var}(A_t) \leq H(K)N^{-1/2}$.	(52), Steps 3 and 4

Hypothesis	Content	Where used below
(H4) Moments and energy	There are constants $C(K)$ and $B_E(K)$ and an event E_r with $\mathbb{P}(E_r^c) \leq \pi_K(N)$ such that $\int_{E_r} \int \log X_r ^p d\mu d\mathbb{P} \leq (C(K)p)^{6p}$ for $p \geq 1$, and $\int X_r ^2 d\mu \leq B_E(K)$ on E_r .	(52), Steps 3, 4 and 6
(H5) Small derivatives	Outside the exceptional events of (H1), with $K+2$ in place of K , and of the second-derivative bound in (H2), and outside a further event of probability at most $C(K)N^{-5/16}(\log N)^2$, we have $B_{K,N} \leq C(K)NL^{-4} \log N$.	Step 3

For signs, all with $\pi_K(N) = 0$, (H1) is Lemma 4.2, (H2) holds with $B_0 = 1$ by Lemmas 4.3 and 4.6, (H3) is (16), (H4) holds with $E_r = \Omega$ by (17) and (18), and (H5) is the proof of Lemma 4.5, in the form (62). The hypotheses (H3) and (H4) are those of Lemma 3.2.

We first bound the local zero counts, which gives (H1). Steinhaus and bounded centrally symmetric coefficients are independent signs times bounded amplitudes, so Lemma 4.1 applies to them. Gaussian amplitudes are unbounded, and Lemma 7.3 treats them separately.

Lemma 7.1 (Local zero counts for Steinhaus coefficients). *Let the ξ_k be independent and uniform on the unit circle. Fix $K \geq 2$, and put*

$$A = \max(1, 1024\pi/c_{\text{HW}}), \quad t_N = \frac{A \log N}{N}, \quad C_1(K) = \frac{4A + 4K + 4}{\log 2}.$$

Let $N \geq e$ with $t_N \leq 1$. Except on an event of probability at most N^{-3} , every disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C_1(K) \log N$ zeros of f_N .

Proof. We write each coefficient as a sign times a point of a fixed semicircle and apply Lemma 4.1. The event in the lemma depends only on the law of $(\xi_0, \dots, \xi_N)$. Let the Z_k be uniform on a fixed semicircle, independent of each other and of the signs. Since $-e^{i\varphi} = e^{i(\varphi+\pi)}$, the products $\epsilon_k Z_k$ are independent and uniform on the unit circle, so we may take $\xi_k = \epsilon_k Z_k$. Since $|Z_k| = 1$ and $\sum_{k=0}^N |Z_k|^2 = N+1$, we may take $a_k = Z_k$, $B = 1$, $\eta = 1$ and $\mathcal{E} = \Omega$ in that lemma. Then $P_N = f_N$, and, since $\log 1 = 0$, its constants are

$$A(1, 1) = \max(1, 1024\pi/c_{\text{HW}}) = A, \quad t_N(1, 1) = t_N,$$

$$C_1(K, 1, 1) = \frac{4A + 4K + 4}{\log 2} = C_1(K).$$

This proves the lemma. $\square$

Lemma 7.2 (Local zero counts for bounded coefficients). *Let the ξ_k be independent with a fixed nondegenerate distribution on $\mathbb{R}$ or $\mathbb{C}$ that is centrally symmetric, that is, $\text{Law}(\xi_0) = \text{Law}(-\xi_0)$, and normalize*

$$\mathbb{E}|\xi_0|^2 = 1, \quad |\xi_0| \leq B, \quad B \geq 1.$$

Fix $K \geq 2$, and put

$$A_B = \max(1, 2048\pi B^2/c_{\text{HW}}), \quad t_{N,B} = \frac{A_B \log N}{N},$$

$$C_{1,B}(K) = \frac{4A_B + 4K + 4 + \log B}{\log 2}.$$

Let $N \geq e$ with $t_{N,B} \leq 1$. Except on an event of probability at most $\exp[-(N+1)/(2B^4)] + N^{-3}$, every disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C_{1,B}(K) \log N$ zeros of f_N .

Proof. We write each coefficient as a sign times an independent copy of itself, bound the probability that the amplitudes have small total energy, and apply Lemma 4.1 with $\eta = \frac{1}{2}$. The event in the lemma depends only on the law of $(\xi_0, \dots, \xi_N)$. Let $a_0, \dots, a_N$ be independent with the coefficient distribution, and independent of the signs. By symmetry, $\text{Law}(\epsilon_k a_k) = \frac{1}{2} \text{Law}(a_k) + \frac{1}{2} \text{Law}(-a_k) = \text{Law}(\xi_k)$, and the products are independent, so we may take $\xi_k = \epsilon_k a_k$. Let $\mathcal{E} = \{\sum_{k=0}^N |a_k|^2 \geq (N+1)/2\}$. Since $|a_k|^2 \in [0, B^2]$ and $\mathbb{E}|a_k|^2 = 1$, Hoeffding's inequality [10, Theorem 2] for the $N+1$ variables $|a_k|^2$ at the deviation $(N+1)/2$ gives

$$\begin{aligned} \mathbb{P}(\mathcal{E}^c) &= \mathbb{P}\left\{\sum_{k=0}^N (|a_k|^2 - 1) < -\frac{N+1}{2}\right\} \\ &\leq \exp\left[-\frac{2((N+1)/2)^2}{(N+1)B^4}\right] = \exp\left[-\frac{N+1}{2B^4}\right]. \end{aligned}$$

We apply Lemma 4.1 with these a_k , B and $\mathcal{E}$, and $\eta = \frac{1}{2}$. Then $P_N = f_N$, and the constants of that lemma are

$$A(B, \tfrac{1}{2}) = \max\left(1, \frac{1024\pi B^2}{c_{\text{HW}}/2}\right) = \max\left(1, \frac{2048\pi B^2}{c_{\text{HW}}}\right) = A_B,$$

$$t_N(B, \tfrac{1}{2}) = \frac{A_B \log N}{N} = t_{N,B},$$

$$C_1(K, B, \tfrac{1}{2}) = \frac{4A_B + 4K + 4 + \log B}{\log 2} = C_{1,B}(K).$$

Hence the probability that $\mathcal{E}$ occurs and some disk contains more than $C_{1,B}(K) \log N$ zeros is at most N^{-3} . Adding $\mathbb{P}(\mathcal{E}^c)$ proves the lemma. $\square$

Lemma 7.3 (Local zero counts for Gaussian coefficients). *Let $\xi_k = g_k$, where the g_k are independent standard real Gaussians, or let $\xi_k = (g_k + ih_k)/\sqrt{2}$, where all g_k, h_k are independent standard real Gaussians. Fix $K \geq 2$, let t_N and $C_1(K)$ be as in Lemma 7.1, and put*

$$C_{1,G}(K) = C_1(K) + \frac{2}{\log 2}.$$

Let $N \geq e$ with $t_N \leq 1$. Except on an event of probability at most $N^{-3} + e^{-3(N+1)/32}$, every disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C_{1,G}(K) \log N$ zeros of f_N .

Proof. Lemma 4.1 does not apply to unbounded amplitudes, but its proof works once $|f_N|$ is bounded outside an event of exponentially small probability. Let J , the arcs I and the real Gram matrix A_I be those of the proof of Lemma 4.1 with $a_k = 1$, $B = 1$ and $\eta = 1$, so that $J \leq 8N$, $\mu(I) = t_N/\pi$, the eigenvalues of A_I lie in $[0, 1]$, $\text{tr } A_I = (N+1)t_N/\pi$, $\|A_I\| \leq 1$ and $\|A_I\|_{\text{HS}}^2 \leq \text{tr } A_I$. Let $Q_I = \int_I |f_N(e^{i\theta})|^2 d\mu(\theta)$.

Real coefficients. With $g = (g_0, \dots, g_N)$, we have $Q_I = g^T A_I g$ and $\mathbb{E}Q_I = \text{tr } A_I$, since $\mathbb{E}g_j g_k = \mathbf{1}\{j = k\}$. Standard real Gaussian coordinates have ψ_2 norm at most two in the moment convention of Section 4.1. Indeed, $y^{p/2} e^{-y/4} \leq (2p/e)^{p/2}$ for $y \geq 0$, and $\mathbb{E}e^{g_k^2/4} = (1 - \frac{1}{2})^{-1/2} = \sqrt{2}$, so that, writing $|g_k|^p = (g_k^2)^{p/2} e^{-g_k^2/4} \cdot e^{g_k^2/4}$, for $p \geq 1$ we have

$$p^{-1/2}(\mathbb{E}|g_k|^p)^{1/p} \leq p^{-1/2}(\sqrt{2}(2p/e)^{p/2})^{1/p} = 2^{1/(2p)}(2/e)^{1/2} \leq 2/\sqrt{e} < 2.$$

Therefore (23) holds with g in place of ϵ , and the Hanson–Wright chain in the proof of Lemma 4.1 at $B = \eta = 1$ gives $\mathbb{P}\{Q_I < \frac{1}{2} \text{tr } A_I\} \leq 2N^{-16}$.

Complex coefficients. For an arc I , the complex Gram matrix is given by

$$M_{jk} = \int_I e^{i(k-j)\theta} d\mu(\theta), \quad 0 \leq j, k \leq N,$$

so that, expanding the square,

$$Q_I = \int_I \left| \sum_{k=0}^N \xi_k e^{ik\theta} \right|^2 d\mu(\theta) = \sum_{j,k} \bar{\xi}_j M_{jk} \xi_k.$$

For a complex vector x , the same expansion and Parseval's identity show that $\sum_{j,k} \overline{x_j} M_{jk} x_k = \int_I |\sum_k x_k e^{ik\theta}|^2 d\mu$ lies between 0 and $\int |\sum_k x_k e^{ik\theta}|^2 d\mu = \|x\|^2$, so that the eigenvalues of M lie in $[0, 1]$, and $\text{tr } M = (N+1)\mu(I)$. We write the arc energy as a quadratic form in the real vector $(g, h) = (g_0, \dots, g_N, h_0, \dots, h_N)$ with the matrix

$$\widetilde{M} = \frac{1}{2} \begin{pmatrix} \Re M & -\Im M \\ \Im M & \Re M \end{pmatrix}.$$

Indeed, since $\xi_k = (g_k + ih_k)/\sqrt{2}$, Q_I is real and $M = \Re M + i\Im M$, taking real parts in $Q_I = \frac{1}{2}(g - ih)^T M (g + ih)$ we get

$$Q_I = \frac{1}{2} \{g^T (\Re M) g + h^T (\Re M) h + h^T (\Im M) g - g^T (\Im M) h\} = (g, h)^T \widetilde{M} (g, h).$$

Let λ_ℓ be the eigenvalues of M . If $Mv = \lambda_\ell v$, then taking real and imaginary parts shows that $(\Re v, \Im v)$ and $(-\Im v, \Re v)$ are eigenvectors of $\widetilde{M}$ with eigenvalue $\lambda_\ell/2$, and for an orthonormal eigenbasis of M these $2N+2$ real vectors are orthonormal. Hence each $\lambda_\ell/2$ occurs twice as an eigenvalue of $\widetilde{M}$. Since $\lambda_\ell^2 \leq \lambda_\ell$ for $\lambda_\ell \in [0, 1]$, it follows that $\widetilde{M} \geq 0$, $\text{tr } \widetilde{M} = \text{tr } M$, $\|\widetilde{M}\| \leq \frac{1}{2}$ and $\|\widetilde{M}\|_{\text{HS}}^2 = \frac{1}{2} \sum_\ell \lambda_\ell^2 \leq \text{tr } M/2$. In particular $\mathbb{E}Q_I = \text{tr } \widetilde{M} = \text{tr } M$, since the $2N+2$ coordinates of (g, h) are independent standard Gaussians. As in the real case, the g_k and h_k have moment ψ_2 norm at most two, so the Hanson–Wright inequality at deviation $\text{tr } M/2$ and the bounds on $\widetilde{M}$ give, for each arc,

$$\begin{aligned} \mathbb{P}\{Q_I < \text{tr } M/2\} &\leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{(\text{tr } M/2)^2}{16\|\widetilde{M}\|_{\text{HS}}^2}, \frac{\text{tr } M/2}{4\|\widetilde{M}\|} \right\} \right] \\ &\leq 2 \exp[-c_{\text{HW}} \min\{\text{tr } M/32, \text{tr } M/4\}] = 2e^{-c_{\text{HW}} \text{tr } M/32} \\ &\leq 2e^{-c_{\text{HW}} \text{tr } M/64} = 2 \exp \left[-\frac{c_{\text{HW}}(N+1)t_N}{64\pi} \right] \leq 2N^{-16}, \end{aligned}$$

since $\text{tr } M = (N+1)\mu(I) = (N+1)t_N/\pi$, as in the real case.

Both cases. In both cases $\mathbb{E}Q_I = (N+1)\mu(I)$, and, by the union bound over at most $8N$ arcs, $\mathbb{P}\{Q_I < \frac{1}{2}\mathbb{E}Q_I \text{ for some arc } I\} \leq (8N)(2N^{-16}) = 16N^{-15} \leq N^{-3}$. On the complement of this event, averaging over each closed arc gives $\max_{\theta \in I} |f_N(e^{i\theta})|^2 \geq Q_I/\mu(I) \geq (N+1)/2$, and by continuity there exists $z_1 = e^{i\theta_1}$ on the arc such that $|f_N(z_1)|^2 \geq (N+1)/2$. Given z_0 , we take such a z_1 on the arc of the angular center nearest to $z_0/|z_0|$. As in the proof of Lemma 4.1, the target disk then lies strictly within $\{|z - z_1| < R\}$, where $R = (2K+1)/N + 2t_N$.

For the energy, let $S = \sum_{k=0}^N |\xi_k|^2$, which is $\sum_{k=0}^N g_k^2$ in the real case. For $0 \leq y \leq 1/8$ and real coefficients $\xi_k = g_k$, we have $\mathbb{E}e^{yg_k^2} = (1-2y)^{-1/2}$, and hence

$$\log \mathbb{E}e^{y(g_k^2-1)} = -y - \frac{1}{2} \log(1-2y) = \sum_{j \geq 2} \frac{2^{j-1}y^j}{j} \leq \frac{y^2}{1-2y} \leq 2y^2,$$

since $2^{j-1}/j \leq 2^{j-2}$ for $j \geq 2$, $\sum_{j \geq 2} 2^{j-2} y^j = y^2/(1-2y)$ and $1-2y \geq 3/4$. For complex coefficients, $|\xi_k|^2$ has the exponential distribution with mean one, so that $\mathbb{E}e^{y|\xi_k|^2} = (1-y)^{-1}$ and

$$\log \mathbb{E}e^{y(|\xi_k|^2-1)} = -y - \log(1-y) = \sum_{j \geq 2} \frac{y^j}{j} \leq \frac{y^2}{2(1-y)} \leq 2y^2,$$

since $1/j \leq 1/2$ for $j \geq 2$ and $1-y \geq 7/8$. In both cases, by independence, the exponential Markov inequality at $y = 1/8$ and the bound $2y^2 = 2/64$ for each of the $N+1$ factors, we have

$$\begin{aligned} \mathbb{P} \left\{ \sum_{k=0}^N |\xi_k|^2 > 2(N+1) \right\} &\leq e^{-2(N+1)/8} \mathbb{E}e^{S/8} = e^{-(N+1)/8} \mathbb{E}e^{(S-(N+1))/8} \\ &= e^{-(N+1)/8} \prod_{k=0}^N \mathbb{E}e^{(|\xi_k|^2-1)/8} \\ &\leq \exp \left[-\frac{N+1}{8} + \frac{2(N+1)}{64} \right] = e^{-3(N+1)/32}. \end{aligned}$$

On the complement of this event, the Cauchy-Schwarz inequality bounds the polynomial in the local Jensen disk by

$$\begin{aligned} |f_N(z)| &\leq \left(\sum_{k=0}^N |\xi_k|^2 \right)^{1/2} \left(\sum_{k=0}^N |z|^{2k} \right)^{1/2} \\ &\leq \sqrt{2(N+1)} \sqrt{N+1} e^{2NR} = \sqrt{2}(N+1)e^{2NR}, \end{aligned}$$

for $|z - z_1| \leq 2R$, where $|z| \leq 1 + 2R \leq e^{2R}$, so that $\sum_{k=0}^N |z|^{2k} \leq (N+1)e^{4NR}$. By the zero-count step [23, §8.21] and Jensen's formula [23, §3.61] at the nonzero center value, as in the proof of Lemma 4.1, and by the two bounds on $|f_N|$ above, we have

$$\begin{aligned} \#\{\alpha : |\alpha - z_1| < R\} &\leq \frac{\log(\sqrt{2}(N+1)e^{2NR}) - \frac{1}{2} \log((N+1)/2)}{\log 2} \\ &\leq C_1(K) \log N + \frac{1}{2} \leq C_{1,G}(K) \log N. \end{aligned}$$

The middle step is the last display of that proof at $B = \eta = 1$, whose maximum bound $(N+1)e^{2NR}$ lacks the factor $\sqrt{2}$; this factor adds $\log \sqrt{2}/\log 2 = \frac{1}{2}$. The last step holds since $\frac{1}{2} \leq (2/\log 2) \log N$ for $N \geq 2$. $\square$

In the radial estimates below, $K \geq 2$ is fixed, N is sufficiently large, and $r \in [1 - K/N, 1 + K/N]$.

Proposition 7.4 (Transfer to other coefficient distributions). *Let the coefficient distribution be Steinhaus, standard real Gaussian, standard complex Gaussian, or a fixed nondegenerate bounded centrally symmetric distribution on $\mathbb{R}$ or $\mathbb{C}$. For the bounded distribution, normalize*

$$\mathbb{E}|\xi_0|^2 = 1, \quad |\xi_0| \leq B, \quad B \geq 1.$$

The distribution satisfies (H1)–(H5), with the amplitude factor $B_0 = B$ for the bounded distribution and $B_0 = 1$ for the others. The occupation and jet estimates of Sections 3 and 4, and the local-count, derivative and maximal-tail estimates, hold with the model constants and exceptional probabilities in Section A.6. These probabilities are summable along $N = j^8$.

Put $V = \int |X_r|^2 d\mu$ and, for a bounded distribution, put

$$C_B = 2\{C_* + \max(\tfrac{1}{2} \log 2, \log B) + 1\}.$$

For $p \geq 1$, the table gives an upper bound for

$$\int_E \int |\log |X_r||^p d\mu d\mathbb{P}.$$

The bounds in the first two rows also hold without restriction to E .

Coefficient distribution	Logarithmic	
	moment	Energy on E
Steinhaus	$\leq (C_* p)^{6p}$	$E = \Omega, V = 1$
Standard real or complex Gaussian	$\leq (16p)^p$	$E = \{V \leq 2\}$
Bounded centrally symmetric	$\leq (C_B p)^{6p}$	$E = \{V \geq 1/2\}, V \leq B^2$

The corresponding energy exceptions satisfy

$$\mathbb{P}(E^c) \leq \begin{cases} 0, & \text{Steinhaus,} \\ \exp[-3N/(32e^{6K})], & \text{Gaussian,} \\ \exp[-N/(2B^4e^{6K})], & \text{bounded centrally symmetric.} \end{cases}$$

In particular, for all sufficiently large N ,

$$\begin{aligned} & \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > C(K)N^{-1/16}(\log N)^7\} \\ & \leq C(K)N^{-3/8} + N^{-12} + C(K)N^{-3/8}(\log N)^{-12} + \mathbb{P}(E^c). \end{aligned}$$

Here the constant $C(K)$ and the degree cutoff may depend on the coefficient distribution.

Proof. The idea is to check the inputs of Sections 3 and 4 for each distribution, to redo every computation in which an input changes, and to apply Lemma 3.2 for the logarithmic concentration. For each distribution, (H1) is the local-count lemma above, (H2) follows from the tail bounds at the samples in the proofs of Lemmas 4.3 and 4.6, (H3) is the occupation bound, (H4) is the moment and energy bound on E in the statement, and (H5) follows from the proof of Lemma 4.5 with these inputs and the jet bounds. Table 6 collects the inputs that change. The coefficients enter the proofs for signs only as follows.

- The one- and two-point bounds of Section 3.1 and the jet bounds in the proof of Lemma 4.5: through the independence and centering of the summands, their covariance matrices and their third moments.
- The proof of Theorem 3.1: through (16), (17) and (18).
- The proofs of Lemmas 4.3 and 4.6: through a tail bound at each sample and a deterministic bound on a derivative.

Since $\mathbb{E}|\xi_0|^2 = 1$, the normalization $\sigma_N(r)$ and the radial bounds (12) are those of the sign case.

Quantity	Steinhaus	Gaussian	Bounded	Source
Local count	$C_1(K+2)$	$C_{1,G}(K+2)$	$C_{1,B}(K+2)$	Lemmas 7.1–7.3
Exception beyond N^{-3}	none	$e^{-3(N+1)/32}$	$e^{-(N+1)/(2B^4)}$	Lemmas 7.1–7.3
f_N'' and tail bounds	signs	signs	B times signs	Lemmas 4.3, 4.6
Envelope exception	none	$2(2N+1)e^{-N^2/2}$	none	Lemmas 4.3, 4.6
Occupation, mean	signs	$2 + 320e^{8K}$	$2 + c_B$	(16)
Occupation, variance	signs	$14 + 960e^{8K}$	$14 + 3c_B$	(16)
Small derivatives	(29)	(48)	(49)	Lemma 4.5

Table 6: The inputs of the proof of Proposition 7.4 that depend on the law, with the sign-case result each row transfers: the first two rows give (H1), the next two (H2), the occupation rows (H3) and the last row (H5). “Signs” means the sign-case value, the Gaussian column covers the real and complex cases, the occupation constants multiply $N^{-1/2}$, and the bounds on f_N'' and on the tails fail with probability at most N^{-10} each, for Gaussian coefficients on the envelope event $\max_{k \leq 2N} |\xi_k| \leq N$.

Covariances. For $c \in \mathbb{C}$ let $M(c)$ be the real 2×2 matrix of the multiplication by c , so that, in the notation of (10), $M(e^{i\psi}) = R(\psi)$ and $F(\psi) = M(e^{i\psi})F(0)$. Let Q be the covariance matrix of $(\Re \xi_0, \Im \xi_0)$, and let $\zeta = \mathbb{E}\xi_0^2$. Since $\text{tr } Q = \mathbb{E}|\xi_0|^2 = 1$,

$\mathbb{E}(\Re \xi_0)^2 - \mathbb{E}(\Im \xi_0)^2 = \Re \zeta$ and $2\mathbb{E}[\Re \xi_0 \Im \xi_0] = \Im \zeta$, we have the first identity below. The real vector of $\xi_0 e^{ia}$ is $R(a)(\Re \xi_0, \Im \xi_0)^T$, and $M(\zeta)$ commutes with $R(a)$. Hence $R(a)R(b)^T = R(a-b)$ and $R(a)F(0)R(b)^T = F(a+b)$ give the cross covariance of the real vectors of $\xi_0 e^{ia}$ and $\xi_0 e^{ib}$:

$$(44) \quad Q = \frac{1}{2}I_2 + \frac{1}{2}M(\zeta)F(0), \quad R(a)QR(b)^T = \frac{1}{2}R(a-b) + \frac{1}{2}M(\zeta)F(a+b).$$

The laws have the following Q and ζ , where the real laws are the real Gaussian law and the real bounded laws, with signs as their Rademacher case.

Law	Q	ζ	Since
Steinhaus	$\frac{1}{2}I_2$	0	$\xi_0 = e^{i\omega}$, and 2ω is uniform modulo 2π
Real	$\text{diag}(1, 0)$	1	$\Im \xi_0 = 0$
Complex Gaussian	$\frac{1}{2}I_2$	0	$\Re \xi_0, \Im \xi_0$ are independent, of variance $\frac{1}{2}$
Bounded	$\frac{1}{2}I_2 + \frac{1}{2}M(\zeta)F(0)$	$\mathbb{E}\xi_0^2$	(44), with $ \zeta \leq 1$

For signs, $\zeta = 1$, and (44) summed over k gives the matrices V of Section 3.1.

Steinhaus coefficients. Let ξ_k be independent uniform points on the unit circle, and write $\xi_k = e^{i\omega_k}$, with independent ω_k uniform on $[0, 2\pi)$. Then $\mathbb{E}\xi_k = 0$, $|\xi_k|^2 = 1$ and $\mathbb{E}|\xi_k|^3 = 1$, and $Q = \frac{1}{2}I_2$ and $\zeta = 0$. Let $z = re^{i\theta}$ and $w = se^{i\phi}$, with $r, s \in [1 - K/N, 1 + K/N]$. Since the ξ_k are independent and centered, multiplying (44) at $a = k\theta$, $b = k\phi$ by $(rs)^k/(\sigma_N(r)\sigma_N(s))$ and summing over k shows, since $\sum_k r^{2k} = \sigma_N(r)^2$, that the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta))$ is $\frac{1}{2}I_2$, and that the cross covariance of $(\Re X_r(\theta), \Im X_r(\theta))$ and $(\Re X_s(\phi), \Im X_s(\phi))$ is

$$\frac{1}{2\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k R(k(\theta - \phi)).$$

Each entry of this matrix is, up to sign, $N/(2\sigma_N(r)\sigma_N(s))$ times the real or the imaginary part of the sum in (11) with $\beta = 0$ and $\psi = \theta - \phi$. For retained pairs, by (11) and the lower bound in (12) we have

$$\frac{N}{2\sigma_N(r)\sigma_N(s)} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k e^{ik(\theta - \phi)} \right| \leq \frac{e^{4K}}{2} \cdot \frac{10e^{4K}}{N \cdot N^{-1/2}} = 5e^{8K} N^{-1/2} \leq 10e^{8K} N^{-1/2}.$$

Hence the covariance matrices of Section 3.1 satisfy the entry bounds (13), with the one-point error equal to zero. The eigenvalue bound and the total variation bound (14) of that section use the covariance only through these entry bounds, so they hold with the same constants.

We apply Raič's theorem to independent centered real vectors obtained from each complex coefficient. The k th summand of the real vector with the coordinates

$\Re X_r(\theta)$, $\Im X_r(\theta)$, $\Re X_s(\phi)$ and $\Im X_s(\phi)$ is

$$\begin{pmatrix} r^k \cos(\omega_k + k\theta)/\sigma_N(r) \\ r^k \sin(\omega_k + k\theta)/\sigma_N(r) \\ s^k \cos(\omega_k + k\phi)/\sigma_N(s) \\ s^k \sin(\omega_k + k\phi)/\sigma_N(s) \end{pmatrix}.$$

These summands are independent and centered, their covariance matrices sum to that of the vector, and the squared norm of the k th summand is $r^{2k}/\sigma_N(r)^2 + s^{2k}/\sigma_N(s)^2$, as for signs. After the whitening of Section 3.1, the sums of their third moments are therefore at most $46e^{9K}N^{-1/2}$ and, for one point, at most $16e^{9K}N^{-1/2}$, as computed there. Hence the normal-approximation errors have the bounds of the sign case, and (15) and (16) hold for Steinhaus coefficients with the same constants.

For the jets in the proof of Lemma 4.5, we have $W_l(z) = N^{-1/2} \sum_{k=0}^N \xi_k (k/N)^l z^k$ for $l \in \{0, 1\}$. Multiplying (44) at $a = k\theta$, $b = k\phi$ by $(k/N)^{l+m} (rs)^k / N$ and summing over k shows that the cross covariance of $(\Re W_l(z), \Im W_l(z))$ and $(\Re W_m(w), \Im W_m(w))$ is

$$\frac{1}{2N} \sum_{k=0}^N (rs)^k \left(\frac{k}{N}\right)^{l+m} R(k(\theta - \phi)), \quad l, m \in \{0, 1\}.$$

For $w = z$, the covariance matrix of $W(z)$ is therefore the nonoscillatory matrix of that proof, with no oscillatory part. Its least eigenvalue is at least $e^{-4K}/32$, which is larger than the bound $e^{-4K}/80$ of (26) used there. For a retained pair, every entry of the cross covariance of $W(z)$ and $W(w)$ is, up to sign, half the real or the imaginary part of $N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta - \phi)}$, with $\beta = l + m \in \{0, 1, 2\}$, which by (11) is at most $5e^{4K}N^{-1/2} \leq 10e^{4K}N^{-1/2}$ in absolute value, the bound used there. The summands of $(W(z), W(w))$ are independent and centered, and the squared norm of the k th summand is $(r^{2k} + s^{2k})(1 + (k/N)^2)/N$, as for signs. Hence the Raič, small-ball and interpolation steps of that proof hold with the same constants.

For complex numbers c_k , the variables $\Re(\xi_k c_k)$ are independent and centered, with $|\Re(\xi_k c_k)| \leq |c_k|$. Hence, by Hoeffding's inequality [10, Theorem 2] for both tails, for $u > 0$,

$$\mathbb{P}\left\{\left|\Re \sum_k \xi_k c_k\right| \geq u\right\} \leq 2 \exp\left[-\frac{u^2}{2 \sum_k |c_k|^2}\right],$$

and the same bound holds for the imaginary part, since $|\Im(\xi_k c_k)| \leq |c_k|$. This is the bound used in the proofs of Lemmas 4.3 and 4.6, whose deterministic bounds use only $|\xi_k| \leq 1$. Hence these lemmas hold for Steinhaus coefficients with the same constants and exceptional probabilities, which is (H2) with $B_0 = 1$.

Write $\xi_k = \epsilon_k Z_k$ as in the proof of Lemma 7.1, and denote expectations over the signs and over the Z_k by $\mathbb{E}_\epsilon$ and $\mathbb{E}_Z$. Conditional on all Z_k , let $b_k = Z_k r^k / \sigma_N(r)$, so that $X_r(\theta) = \sum_k \epsilon_k b_k e^{ik\theta}$ and, since $|Z_k| = 1$, $\sum_{k=0}^N |b_k|^2 = \sigma_N(r)^{-2} \sum_{k=0}^N |Z_k|^2 r^{2k} = 1$. Hence (17) for the normalized coefficients b_k , whose constant C_* does not depend on them, gives $\mathbb{E}_\epsilon \int |\log |\sum_{k=0}^N \epsilon_k b_k e^{ik\theta}||^p d\mu \leq (C_* p)^{6p}$, and, by Fubini's theorem, $\mathbb{E} \int |\log |X_r||^p d\mu = \mathbb{E}_Z [\mathbb{E}_\epsilon \int |\log |X_r||^p d\mu] \leq (C_* p)^{6p}$.

For the local count (H1) we use Lemma 7.1 (Table 6). In the proof of Lemma 4.5, the coefficients enter only through this local count, Lemma 4.3, the covariances and the summand norms. These hold here with the same constants, so (29) holds for Steinhaus coefficients, which is (H5). By Parseval's identity and $|\xi_k| = 1$, $\int |X_r|^2 d\mu = \sigma_N(r)^{-2} \sum_k |\xi_k|^2 r^{2k} = 1$ for every realization, which is (18). With (16), (17) and (18), the proof of Theorem 3.1 applies word for word, and we obtain (9) with the same constants, which is the concentration bound of the proposition, with $E = \Omega$ and $\mathbb{P}(E^c) = 0$. In particular, (H3) and (H4) hold with the constants of the sign case and $E_r = \Omega$.

Real Gaussian coefficients. Let the coefficients be independent standard real Gaussians g_k . The real vector $(\Re X_r(\theta), \Im X_r(\theta))$ and the real vector with the coordinates $\Re X_r(\theta)$, $\Im X_r(\theta)$, $\Re X_s(\phi)$ and $\Im X_s(\phi)$ are the sums over k of g_k times the deterministic summands of Section 3.1. Since $\mathbb{E} g_k = 0$ and $\mathbb{E} g_j g_k = \mathbf{1}\{j = k\}$, as for signs, these vectors have the laws of G_Σ for the covariance matrices Σ of that section. Hence the normal approximation is exact, and only the total variation bound (14) of that section remains. For retained angles and pairs, $t, t' \geq 0$ and the product of disks D of Section 3.1, we get

$$(45) \quad \begin{aligned} \left| \mathbb{P}\{|X_r(\theta)| \leq t, |X_s(\phi)| \leq t'\} - F_G(t)F_G(t') \right| &= \left| \mathbb{P}(G_\Sigma \in D) - \mathbb{P}(G_{I_4/2} \in D) \right| \\ &\leq \text{TV}(G_\Sigma, G_{I_4/2}) \leq 320e^{8K}N^{-1/2}. \end{aligned}$$

Here the equality holds since the two coordinate pairs of $G_{I_4/2}$ are independent copies of G , and the last step is (14), which uses Σ only through the bound $40e^{8K}N^{-1/2}$ on the norm of Σ minus half the identity. With the disk of radius t and the one-point covariance, the same three steps show that $|\mathbb{P}\{|X_r(\theta)| \leq t\} - F_G(t)| \leq 320e^{8K}N^{-1/2}$. As in Section 3.1, the removed angles and pairs have measures at most $2N^{-1/2}$ and $10N^{-1/2}$, and there the integrands differ from $F_G(t)$ and $F_G(t)^2$ by at most one. Hence, by Tonelli's theorem, as in (16), uniformly for $t \geq 0$,

$$\begin{aligned} |\mathbb{E} A_t - F_G(t)| &\leq (2 + 320e^{8K})N^{-1/2}, \\ |\mathbb{E} A_t^2 - F_G(t)^2| &\leq (10 + 320e^{8K})N^{-1/2}, \\ \text{Var}(A_t) &= \mathbb{E} A_t^2 - F_G(t)^2 - (\mathbb{E} A_t - F_G(t))(\mathbb{E} A_t + F_G(t)) \end{aligned}$$

$$\begin{aligned}
&\leq |\mathbb{E}A_t^2 - F_G(t)^2| + |\mathbb{E}A_t - F_G(t)| |\mathbb{E}A_t + F_G(t)| \\
&\leq (10 + 320e^{8K})N^{-1/2} + 2(2 + 320e^{8K})N^{-1/2} \\
&= (14 + 960e^{8K})N^{-1/2},
\end{aligned}$$

where the last line uses the first two and $0 \leq \mathbb{E}A_t + F_G(t) \leq 2$. These are the occupation bounds (16) for real Gaussian coefficients, which give (H3).

The vector $(W(z), W(w))$ is the sum over k of g_k times the deterministic summands in the proof of Lemma 4.5. Hence it has the law of $G_{\Sigma_{z,w}}$, where $\Sigma_{z,w}$ is its covariance matrix (the matrix Σ of that proof), and $W(z)$ has the law of G_{Σ_z} , where Σ_z is the matrix A_z of that proof. Hence the covariance bounds of that proof hold, and its Raič steps are exact. Let $D' \subset \mathbb{R}^4$ be the product of the two disks of radii $C(K)N^{-1/32}/\sqrt{\log N}$ and $3N^{-1/64}$, the bounds (25) on the thresholds u and v_0 , where $N^{1/64}$ is the parameter L of that proof, so that D' contains the product of disks U of that proof. Bounding the density of G_{Σ_z} and the volume of D' as in the proof of Theorem B.4, with $(\det \Sigma_z)^{-1/2} \leq (e^{-4K}/80)^{-2} = 6400e^{8K}$ by (26), we get

$$\begin{aligned}
\mathbb{P}\{W(z) \in D'\} &= \mathbb{P}\{G_{\Sigma_z} \in D'\} \leq (2\pi)^{-2}(\det \Sigma_z)^{-1/2} \cdot \pi^2 \frac{C(K)^2 N^{-1/16}}{\log N} \cdot 9N^{-1/32} \\
&\leq 14400e^{8K}C(K)^2 N^{-3/32}/\log N \leq C(K)N^{-3/32}/\log N.
\end{aligned}$$

Since $N^{-3/32} = (N^{1/64})^{-6}$, this is (27) without its term $C(K)N^{-1/2}$. For separated pairs, with the block diagonal matrix Σ_0 of that proof, we have

$$\begin{aligned}
&\left| \text{Cov}(\mathbf{1}\{W(z) \in D'\}, \mathbf{1}\{W(w) \in D'\}) \right| \\
&= \left| \mathbb{P}\{G_{\Sigma_{z,w}} \in D' \times D'\} - \mathbb{P}\{G_{\Sigma_z} \in D'\} \mathbb{P}\{G_{\Sigma_w} \in D'\} \right| \\
&\leq \text{TV}(G_{\Sigma_{z,w}}, G_{\Sigma_0}) \leq 6400e^{8K}N^{-1/2},
\end{aligned}$$

where the first inequality holds since the product of the last two probabilities is $\mathbb{P}\{G_{\Sigma_0} \in D' \times D'\}$, by the independence of the two blocks of G_{Σ_0} , and the last is (28) with the eigenvalue bound $e^{-4K}/80$ of (26).

At every angle $\mathbb{E}|X_r(\theta)|^2 = \sigma_N(r)^{-2} \sum_k r^{2k} = 1$, so that the two real covariance eigenvalues are nonnegative and sum to one, and the larger one is at least $1/2$. Since $|X_r|$ is at least the absolute value of its projection onto an eigenvector with variance $v \geq 1/2$, and the Gaussian density is at most $1/\sqrt{2\pi v}$, we have $\mathbb{P}\{|X_r| \leq u\} \leq 2u/\sqrt{2\pi v} \leq 2u$ for $u \geq 0$. With $u = e^{-\tau}$, layer-cake integration yields the negative logarithmic moments

$$\mathbb{E}(\log^- |X_r|)^p = p \int_0^\infty \tau^{p-1} \mathbb{P}\{|X_r| < e^{-\tau}\} d\tau$$

$$\leq 2p \int_0^\infty \tau^{p-1} e^{-\tau} d\tau = 2\Gamma(p+1), \quad p \geq 1.$$

For the positive part, write $R = \sqrt{2}|G|$. Then $\sqrt{2}\Re G$ and $\sqrt{2}\Im G$ are independent standard Gaussians, and R is the norm of the vector they form. In a covariance eigenbasis, the coordinates of $(\Re X_r, \Im X_r)$ are independent centered Gaussians with variances at most one, so we can couple X_r so that $|X_r| \leq R$. Since $\log^+ x \leq x$, and $R^2/2$ has the exponential distribution with mean one, we have $\mathbb{E}(\log^+ |X_r|)^p \leq \mathbb{E}R^p = 2^{p/2}\Gamma(1+p/2)$. For $a > 0$, since $\max_{y \geq 0} y^a e^{-y/2} = (2a/e)^a$ and $\int_0^\infty e^{-y/2} dy = 2$, we have $\Gamma(a+1) = \int_0^\infty y^a e^{-y/2} dy \leq 2(2a/e)^a$. Taking $a = p$ and $a = p/2$, and using $4 \leq (4e)^p$ and $2 \leq (32ep)^{p/2}$ for $p \geq 1$, we obtain $2\Gamma(p+1) \leq 4(2p/e)^p \leq (8p)^p$ and $2^{p/2}\Gamma(1+p/2) \leq 2(2p/e)^{p/2} \leq (8p)^p$. Adding the two logarithmic parts and integrating over angle yields

$$(46) \quad \mathbb{E} \int |\log |X_r||^p d\mu \leq 2(8p)^p \leq (16p)^p, \quad p \geq 1.$$

For the angular energy we set $V = \sum_{k=0}^N w_k g_k^2$, with $w_k = r^{2k}/\sigma_N(r)^2$, which is $\int |X_r|^2 d\mu$ by Parseval's identity. For the weights, $\sum_k w_k = 1$ and $w_{\max} = \max_k w_k \leq e^{6K}/N$, by $r^{2k} \leq e^{2K}$ and the lower bound in (12). By the bound on $\log \mathbb{E}e^{y(g_k^2-1)}$ for $0 \leq y \leq 1/8$ in the proof of Lemma 7.3, with $y = \lambda w_k$, and by independence and $\mathbb{E}e^{\lambda w_k g_k^2} = (1 - 2\lambda w_k)^{-1/2}$, we have, for $0 \leq \lambda \leq (8w_{\max})^{-1}$,

$$\log \mathbb{E}e^{\lambda(V-1)} = \sum_k \{-\lambda w_k - \tfrac{1}{2} \log(1 - 2\lambda w_k)\} \leq 2\lambda^2 \sum_k w_k^2.$$

In the exponential Markov inequality we choose the largest admissible value $\lambda = (8w_{\max})^{-1}$. It is the best one: the exponent $-\lambda + 2\lambda^2 w_{\max}$ below has derivative $-1 + 4\lambda w_{\max} < 0$ for $0 \leq \lambda < (4w_{\max})^{-1}$, so it decreases on the admissible range. Then, since $\sum_k w_k^2 \leq w_{\max} \sum_k w_k = w_{\max}$,

$$(47) \quad \begin{aligned} \mathbb{P}(V > 2) &\leq e^{-\lambda} \mathbb{E}e^{\lambda(V-1)} \leq \exp\left(-\lambda + 2\lambda^2 \sum_k w_k^2\right) \leq \exp(-\lambda + 2\lambda^2 w_{\max}) \\ &= \exp[-3/(32w_{\max})] \leq \exp[-3N/(32e^{6K})]. \end{aligned}$$

Laurent and Massart [16] give sharper tail bounds for such sums.

By (46), $\mathbb{E} \int |\log |X_r||^p d\mu \leq (16p)^p \leq (16p)^{6p}$ for $p \geq 1$, and hence this bound also holds for the integral over any event. With this moment bound and the occupation bounds above, whose constants satisfy $2 + 320e^{8K} < 14 + 960e^{8K}$, Lemma 3.2 applies on the energy event $E = \{V \leq 2\}$, with

$$H = 14 + 960e^{8K}, \quad C = 16, \quad B_E = 2,$$

so that $4eC = 64e$ and $B_E/2 + 1 = 2$. Its error is at most $C(K)N^{-1/16}(\log N)^7$, and its probability bound, with $\mathbb{P}(E^c) = \mathbb{P}(V > 2)$ bounded in (47), is the concentration bound of the proposition. The moment bound on E , the energy bound $V \leq 2$ on E and (47) are (H4), with $E_r = \{V \leq 2\}$.

For the local count (H1) we use the real case of Lemma 7.3 (Table 6). For real numbers c_k , the sum $Y = \sum_k c_k g_k$ is a centered real Gaussian of variance $v = \sum_k c_k^2$. For $v > 0$ and a level $u > 0$, by the exponential Markov inequality at $\lambda = u/v$ we have

$$\mathbb{P}\{Y \geq u\} \leq e^{-\lambda u} \mathbb{E}e^{\lambda Y} = \exp(-\lambda u + \lambda^2 v/2) = e^{-u^2/(2v)},$$

and hence $\mathbb{P}\{|Y| \geq u\} \leq 2e^{-u^2/(2v)}$, since $-Y$ has the law of Y . This is the form of Hoeffding's inequality used in the proofs of Lemmas 4.3 and 4.6. The real and imaginary parts of $f_N''(z)$ and of a tail $\sum_{k=N+1}^n g_k z^k$ of Lemma 4.6, with $N < n \leq N + m_{\max}$, are such sums, with variances at most $\sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}$ and $\sum_{k=N+1}^n |z|^{2k}$. Applying the above bound to the real and the imaginary part at the level divided by $\sqrt{2}$, at each sample z we have

$$\begin{aligned} \mathbb{P}\{|f_N''(z)| \geq 50e^{K+4}N^{5/2}\sqrt{\log N}\} \\ \leq 4 \exp \left[-\frac{(50e^{K+4}N^{5/2}\sqrt{\log N})^2}{4 \sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}} \right] \leq 4N^{-100}, \end{aligned}$$

and, for $N < n \leq N + m_{\max}$ and $\rho = 1 + K/N$,

$$\begin{aligned} \mathbb{P}\left\{ \left| \sum_{k=N+1}^n g_k z^k \right| \geq 10e^{2K}\sqrt{m_{\max} \log N} \right\} \\ \leq 4 \exp \left[-\frac{(10e^{2K}\sqrt{m_{\max} \log N})^2}{4 \sum_{k=N+1}^n \rho^{2k}} \right] \leq 4N^{-25}, \end{aligned}$$

where the levels are half the threshold $100e^{K+4}N^{5/2}\sqrt{\log N}$ of Lemma 4.3 and two thirds of the amplitude $15e^{2K}\sqrt{m_{\max} \log N}$ of Lemma 4.6, and the last steps are those of the sign case. For interpolation, we consider the event $\max_{k \leq 2N} |g_k| \leq N$, whose complement has probability at most $2(2N+1)e^{-N^2/2}$, by the above bound with $v = 1$ and $u = N$ and a union bound over the $2N+1$ coefficients. On this event, the deterministic bounds in the proofs of Lemmas 4.3 and 4.6, which used $|\epsilon_k| = 1$, gain the factor N :

$$\begin{aligned} \sup_{|z| \leq 1+(K+1)/N} |f_N'''(z)| &\leq N \cdot e^{K+1}N^4 = e^{K+1}N^5, \\ \sup_{|z| \leq \rho} \left| \sum_{k=N+1}^n k g_k z^{k-1} \right| &\leq N \cdot 2N m_{\max} e^{2K} = 2N^2 m_{\max} e^{2K}. \end{aligned}$$

Each interpolation error is the arc-length spacing of the samples times such a supremum, so the spacing must shrink by the factor N to keep the interpolation errors of the sign case. We therefore use $\lceil N^4 \rceil \leq 2N^4$ equally spaced boundary samples in both proofs, in place of $\lceil N^3 \rceil$. Then every point of the circle $|z| = 1 + (K+1)/N$, of length at most 8π , lies within arc length $4\pi/N^4$ of a sample, and every point of the circle $|z| = \rho$, of length at most 4π , within $2\pi/N^4$. Therefore the interpolation errors are

$$\frac{4\pi}{N^4} \cdot e^{K+1} N^5 = 4\pi e^{K+1} N, \quad \frac{2\pi}{N^4} \cdot 2N^2 m_{\max} e^{2K} = \frac{4\pi m_{\max} e^{2K}}{N^2},$$

which are those of the sign case. There are at most $2N^4$ samples, and at most $2N^4 \cdot N$ pairs of a sample and one of the at most N nonempty tails, so that, by the union bound, the two unions have probabilities at most $(2N^4)(4N^{-100}) = 8N^{-96} \leq N^{-10}$ and $(2N^4 \cdot N)(4N^{-25}) = 8N^{-20} \leq N^{-10}$. We conclude that, on the event $\max_{k \leq 2N} |g_k| \leq N$, the bounds of Lemmas 4.3 and 4.6 hold for real Gaussian coefficients with the same constants, each outside an event of probability at most N^{-10} , which is (H2) with $B_0 = 1$ and the additional probability $2(2N+1)e^{-N^2/2}$ of the envelope event.

In the proof of Lemma 4.5, the cell assignment and the sector cover now use the local count of Lemma 7.3, with $K+2$ in place of K . From them we obtain

$$B_{K,N} \leq C_{1,G}(K+2)(\log N) \cdot \#\{\text{successful tests}\} + 40C_{1,G}(K+2)\sqrt{N} \log N,$$

where the successful tests are the retained mesh points z_i at which the event E_i of Theorem B.4 occurs, and we take $t = NL^{-4}$ there, as in that proof. Since the normal approximation is exact and the total variation bound above holds for every product of disks, the application of Theorem B.4 in that proof holds with $\delta = \delta_* = 0$ and its constants $C(K)$, which now also depend on $C_{1,G}(K+2)$. Adding the exceptional probabilities of the local count, of the derivative bound and of the event $\max_{k \leq 2N} |g_k| \leq N$, we get

$$(48) \quad \begin{aligned} \mathbb{P}\{B_{K,N} > C(K)NL^{-4} \log N\} \\ \leq N^{-3} + e^{-3(N+1)/32} + N^{-10} + 2(2N+1)e^{-N^2/2} + C(K)N^{-5/16}(\log N)^2, \end{aligned}$$

which is (H5). Since Gaussian coefficients are nonzero almost surely, every prefix has degree exactly N , and $f_N(0) = g_0 \neq 0$.

Standard complex Gaussian coefficients. The conservative bounds of the real case also hold for $\xi_k = (g_k + ih_k)/\sqrt{2}$, where all g_k, h_k are independent standard real Gaussians. We have $\mathbb{E}|\xi_k|^2 = 1$, and $|\xi_k|^2$ has the exponential distribution with

mean one. By the table of covariances, $Q = \frac{1}{2}I_2$ and $\zeta = 0$, as for Steinhaus coefficients. The covariance matrices of the real vectors of Section 3.1 and of the proof of Lemma 4.5 are computed from (44) and the independence of the coefficients alone. Hence they are those of the Steinhaus case, and they satisfy the entry and eigenvalue bounds shown there. These vectors are linear images of $(g_0, \dots, g_N, h_0, \dots, h_N)$, and hence centered Gaussian vectors, so the normal approximation is again exact. The chain (45) and the occupation bounds after it use Σ only through the norm bound $40e^{8K}N^{-1/2}$, so they hold word for word. In the real Gaussian case, the jet bounds use only the least eigenvalue $e^{-4K}/80$ of (26) and the total variation bound (28) in the proof of Lemma 4.5, so they also hold word for word.

For nonnegative weights with $\sum_k w_k = 1$, let $w_{\max} = \max_k w_k$ as before. For $0 \leq \lambda w_k \leq 1/8$, by independence, $\mathbb{E}e^{\lambda w_k |\xi_k|^2} = (1 - \lambda w_k)^{-1}$, $\sum_k w_k = 1$ and the bound on $\log \mathbb{E}e^{y(|\xi_k|^2 - 1)}$ in the proof of Lemma 7.3 with $y = \lambda w_k$, we have

$$\begin{aligned} \log \mathbb{E} \exp \left[\lambda \left(\sum_k w_k |\xi_k|^2 - 1 \right) \right] &= \sum_k \{ -\lambda w_k - \log(1 - \lambda w_k) \} \\ &\leq 2\lambda^2 \sum_k w_k^2 \leq 2\lambda^2 w_{\max}. \end{aligned}$$

With $\lambda = (8w_{\max})^{-1}$, the largest admissible value, as in the real case, Markov's inequality gives

$$\begin{aligned} \mathbb{P} \left\{ \sum_k w_k |\xi_k|^2 > 2 \right\} &\leq e^{-\lambda} \mathbb{E} \exp \left[\lambda \left(\sum_k w_k |\xi_k|^2 - 1 \right) \right] \\ &\leq \exp(-\lambda + 2\lambda^2 w_{\max}) = \exp[-3/(32w_{\max})]. \end{aligned}$$

For radial weights, $w_{\max} \leq e^{6K}/N$. Since $\sum_k w_k |\xi_k|^2 = V$ by Parseval's identity, this yields the bound on the probability of the energy exception in (47). For uniform weights, we get the local-count exception $e^{-3(N+1)/32}$ of Lemma 7.3.

For d coefficients, since $\mathbb{P}\{|\xi_k| > N\} = e^{-N^2}$, the common envelope satisfies $\mathbb{P}\{\max_{k < d} |\xi_k| > N\} \leq de^{-N^2} \leq 2de^{-N^2/2}$. A linear combination $Y = \sum_k c_k \xi_k$ is circular Gaussian with $v = \sum_k |c_k|^2$, so that $Y = 0$ almost surely if $v = 0$, and, if $v > 0$, $\mathbb{P}\{|Y| > u\} = e^{-u^2/v} \leq 4e^{-u^2/(4v)}$ for $u \geq 0$, since $|Y|^2/v$ has the exponential distribution with mean one. With $d = 2N + 1$, the envelope event is $\max_{k \leq 2N} |\xi_k| \leq N$, and its complement has the probability bound $2(2N + 1)e^{-N^2/2}$ of the real case. The sample events are $|Y| \geq u$ with $Y = f_N''(z)$, for which $v = \sum_{k=2}^N k^2(k-1)^2 |z|^{2k-4}$, and with Y a tail $\sum_{k=N+1}^n \xi_k z^k$, for which $v = \sum_{k=N+1}^n |z|^{2k}$. At the two levels of the real case, $4e^{-u^2/(4v)}$ is the bound in the first line of the corresponding displays there, so that the sample bounds are again $4N^{-100}$ and

$4N^{-25}$. The numbers of samples and of tails are also those of the real case, and hence the same sample thresholds and union estimates apply. On the envelope event, $|\xi_k| \leq N$ for $k \leq 2N$, which is all that the deterministic bounds on f_N''' and on the derivatives of the tails use, so that the N^4 grid yields the same interpolation bounds and the same derivative and tail amplitudes.

For the local count we use the complex case of Lemma 7.3, whose constant and exceptional probability are those of the real case (Table 6), and so are the derivative bound on the envelope event and the jet bounds. It follows that (48) also holds for these coefficients. For the logarithmic moments, X_r is a standard complex Gaussian, that is, circular with $\mathbb{E}|X_r|^2 = 1$, so that $|X_r|^2$ has the exponential distribution with mean one. Hence, for $u \geq 0$ and $p \geq 1$, we have $\mathbb{P}\{|X_r| \leq u\} = 1 - e^{-u^2} \leq \min(u^2, 1) \leq 2u$, and, with $u = e^{-\tau}$ as in the real case, $\log^+ x \leq x$ and $\mathbb{E}|X_r|^p = \mathbb{E}(|X_r|^2)^{p/2}$,

$$\begin{aligned} \mathbb{E}(\log^- |X_r|)^p &= p \int_0^\infty \tau^{p-1} \mathbb{P}\{|X_r| < e^{-\tau}\} d\tau \leq 2\Gamma(p+1) \leq (8p)^p, \\ \mathbb{E}(\log^+ |X_r|)^p &\leq \mathbb{E}|X_r|^p = \Gamma(1+p/2) \leq 2^{p/2}\Gamma(1+p/2) \leq (8p)^p, \end{aligned}$$

where the last inequalities are those of the real Gaussian case. Adding the two parts and integrating over angle, we obtain $\mathbb{E} \int |\log |X_r||^p d\mu \leq 2(8p)^p \leq (16p)^p$. Since the occupation bounds, the moment bound $(16p)^p$ and the bound (47) on $\mathbb{P}(V > 2)$ hold here with the constants of the real case, Lemma 3.2 applies with the parameters of the real case and gives the same exceptional probabilities. Hence (H1)–(H5) hold with the constants of the real case.

Centrally symmetric bounded coefficients. Recall that a real or complex coefficient distribution is centrally symmetric if $\text{Law}(\xi) = \text{Law}(-\xi)$, and nondegenerate if it is not a point mass. A centrally symmetric point mass sits at zero, so a centrally symmetric distribution is nondegenerate exactly when $\mathbb{P}(|\xi| > 0) > 0$, which for a bounded distribution is equivalent to $\mathbb{E}|\xi|^2 > 0$. We first bound the covariances and the third moments, then the derivatives and tails, and then the logarithmic estimates, which use the occupation bounds. For complex coefficients, symmetry implies that $\mathbb{E}\xi = 0$, and $\zeta = \mathbb{E}\xi^2$ satisfies $|\zeta| = |\mathbb{E}\xi^2| \leq \mathbb{E}|\xi|^2 = 1$. For $z = re^{i\theta}$ and $w = se^{i\phi}$, the ordinary and unconjugated covariance kernels are

$$\begin{aligned} \mathcal{K}(z, w) &= \mathbb{E}[X_r(\theta)\overline{X_s(\phi)}] = \frac{1}{\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k e^{ik(\theta-\phi)}, \\ \mathcal{P}(z, w) &= \mathbb{E}[X_r(\theta)X_s(\phi)] = \frac{\zeta}{\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k e^{ik(\theta+\phi)}. \end{aligned}$$

In real coordinates their cross covariance is

$$\frac{1}{2} \begin{pmatrix} \Re(\mathcal{K}(z, w) + \mathcal{P}(z, w)) & \Im(\mathcal{P}(z, w) - \mathcal{K}(z, w)) \\ \Im(\mathcal{P}(z, w) + \mathcal{K}(z, w)) & \Re(\mathcal{K}(z, w) - \mathcal{P}(z, w)) \end{pmatrix}.$$

Indeed, summing (44) over k as for Steinhaus coefficients gives $\frac{1}{2}M(\mathcal{K}(z, w)) + \frac{1}{2}M(\mathcal{P}(z, w))F(0)$, since $R(\psi) = M(e^{i\psi})$ and $F(\psi) = M(e^{i\psi})F(0)$, and this is the matrix above. For $w = z$ we have $\mathcal{K}(z, z) = 1$, since $\sum_k r^{2k} = \sigma_N(r)^2$, so that the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta))$ is

$$\frac{1}{2}I_2 + \frac{1}{2} \begin{pmatrix} \Re \mathcal{P}(z, z) & \Im \mathcal{P}(z, z) \\ \Im \mathcal{P}(z, z) & -\Re \mathcal{P}(z, z) \end{pmatrix}, \quad \mathcal{P}(z, z) = \frac{\zeta}{\sigma_N(r)^2} \sum_{k=0}^N r^{2k} e^{2ik\theta}.$$

Hence every entry of this matrix minus $\frac{1}{2}I_2$ is at most $\frac{1}{2}|\mathcal{P}(z, z)|$ in absolute value, and every entry of the cross covariance is at most $\frac{1}{2}(|\mathcal{K}(z, w)| + |\mathcal{P}(z, w)|)$. For retained angles and pairs, by $|\zeta| \leq 1$, (11) and the lower bound in (12) we have

$$\begin{aligned} \frac{1}{2}|\mathcal{P}(z, z)| &\leq \frac{N}{2\sigma_N(r)^2} \left| \frac{1}{N} \sum_{k=0}^N r^{2k} e^{2ik\theta} \right| \\ &\leq \frac{e^{4K}}{2} \cdot \frac{10e^{4K}}{N \cdot 2N^{-1/2}} \leq 10e^{8K} N^{-1/2}, \\ \frac{1}{2}(|\mathcal{K}(z, w)| + |\mathcal{P}(z, w)|) &\leq \frac{N}{2\sigma_N(r)\sigma_N(s)} \sum_{\pm} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k e^{ik(\theta \pm \phi)} \right| \\ &\leq \frac{e^{4K}}{2} \cdot 2 \cdot \frac{10e^{4K}}{N \cdot N^{-1/2}} = 10e^{8K} N^{-1/2}. \end{aligned}$$

These are the entry bounds (13). Since $|\zeta| \leq 1$, the Abel bounds for the oscillatory terms are unchanged, including the case $|\zeta| = 1$, when the coefficient distribution is supported on a line. For the derivative coordinates, write, as in the proof of Lemma 4.5, $W_0(z) = f_N(z)/\sqrt{N}$ and $W_1(z) = zf'_N(z)/N^{3/2}$. For $l, m \in \{0, 1\}$, by independence and centering we have

$$\begin{aligned} \mathbb{E}[W_l(z)\overline{W_m(w)}] &= \frac{1}{N} \sum_{k=0}^N (rs)^k (k/N)^{l+m} e^{ik(\theta-\phi)}, \\ \mathbb{E}[W_l(z)W_m(w)] &= \frac{\zeta}{N} \sum_{k=0}^N (rs)^k (k/N)^{l+m} e^{ik(\theta+\phi)}. \end{aligned}$$

Each real covariance block has the same form as the above 2×2 matrix, with these two sums in the places of $\mathcal{K}(z, w)$ and $\mathcal{P}(z, w)$. The nonoscillatory moment matrix still has the normalization $\mathbb{E}|\xi|^2 = 1$. It follows that, for $w = z$, the first sum is real, and that the covariance matrix Σ_z of $W(z)$ is the nonoscillatory matrix of the proof of Lemma 4.5 plus a matrix each entry of which is, up to sign, half the

real or the imaginary part of $\zeta N^{-1} \sum_{k=0}^N r^{2k} (k/N)^\beta e^{2ik\theta}$, with $\beta \in \{0, 1, 2\}$. Since $|\zeta| \leq 1$, these entries satisfy the bounds of that proof, so that this matrix has norm at most $20e^{4K} N^{-1/2}$ and the least eigenvalue of Σ_z is at least $e^{-4K}/80$, as in (26). For a retained pair, (11) and $|\zeta| \leq 1$ bound every entry of the cross covariance of $W(z)$ and $W(w)$ in absolute value by

$$\begin{aligned} & \frac{1}{2} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta-\phi)} \right| + \frac{|\zeta|}{2} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta+\phi)} \right| \\ & \leq \frac{1}{2} \cdot 10e^{4K} N^{-1/2} + \frac{1}{2} \cdot 10e^{4K} N^{-1/2} = 10e^{4K} N^{-1/2}, \end{aligned}$$

which is the bound of that proof. We conclude that the same angular exclusions and covariance lower bounds apply.

Raič's theorem yields occupation and mesh estimates with constants multiplied by the finite third moment. Indeed, the k th summand of the real vector with the coordinates $\Re X_r(\theta)$, $\Im X_r(\theta)$, $\Re X_s(\phi)$ and $\Im X_s(\phi)$ is the image of ξ_k under a real linear map. These summands are independent and centered, their covariance matrices sum to that of the vector, and the norm of the k th summand is $|\xi_k|$ times the value for signs, since $|\xi_k e^{ik\theta}| = |\xi_k|$. By the entry bounds, the eigenvalues of the covariance matrix Σ are at least $1/4$, as in Section 3.1, so $\|\Sigma^{-1/2}\| \leq 2$. Therefore the whitened k th summand has norm at most $2\sqrt{2}|\xi_k|e^{3K}N^{-1/2}$, and the third-moment sum in Raič's theorem is at most

$$\begin{aligned} \sum_{k=0}^N \mathbb{E} \left(2\sqrt{2}|\xi_k|e^{3K}N^{-1/2} \right)^3 &= (N+1) \mathbb{E}|\xi_0|^3 16\sqrt{2}e^{9K}N^{-3/2} \\ &\leq 32\sqrt{2} \mathbb{E}|\xi_0|^3 e^{9K}N^{-1/2} \leq 46B^3 e^{9K}N^{-1/2}. \end{aligned}$$

For one point, the whitened k th summand has norm at most $2|\xi_k|e^{3K}N^{-1/2}$, since the squared norm of the summand before whitening is then $|\xi_k|^2 r^{2k} / \sigma_N(r)^2 \leq |\xi_k|^2 e^{6K} / N$, and

$$\sum_{k=0}^N \mathbb{E} \left(2|\xi_k|e^{3K}N^{-1/2} \right)^3 = (N+1) \mathbb{E}|\xi_0|^3 8e^{9K}N^{-3/2} \leq 16B^3 e^{9K}N^{-1/2}.$$

For retained angles and pairs, the triangle inequality, Raič's theorem and the total variation bound therefore give, as in (15),

$$\begin{aligned} & \left| \mathbb{P}\{|X_r(\theta)| \leq t, |X_s(\phi)| \leq t'\} - F_G(t)F_G(t') \right| \\ & \leq (42 \cdot 4^{1/4} + 16) \cdot 46B^3 e^{9K}N^{-1/2} + 320e^{8K}N^{-1/2} \leq C(K, B)N^{-1/2}, \end{aligned}$$

and the same bound holds for $|\mathbb{P}\{|X_r(\theta)| \leq t\} - F_G(t)|$, since the one-point bound is $(42 \cdot 2^{1/4} + 16) \cdot 16B^3 e^{9K}N^{-1/2} + 320e^{8K}N^{-1/2}$, which is smaller. Here and below,

$C(K, B)$ denotes a constant depending only on K and B , which may increase from one estimate to the next. Let $c_B = (42 \cdot 4^{1/4} + 16) \cdot 46B^3e^{9K} + 320e^{8K}$, so that the above two-point bound, and the smaller one-point bound, are at most $c_B N^{-1/2}$. The derivation of (16) from (15), written out above for real Gaussian coefficients, gives, uniformly for $t \geq 0$,

$$\begin{aligned} |\mathbb{E}A_t - F_G(t)| &\leq (2 + c_B)N^{-1/2}, \\ |\mathbb{E}A_t^2 - F_G(t)^2| &\leq (10 + c_B)N^{-1/2}, \\ \text{Var}(A_t) &\leq |\mathbb{E}A_t^2 - F_G(t)^2| + 2|\mathbb{E}A_t - F_G(t)| \\ &\leq (10 + c_B)N^{-1/2} + 2(2 + c_B)N^{-1/2} = (14 + 3c_B)N^{-1/2}, \end{aligned}$$

which give (H3). In the proof of Lemma 4.5, the k th summand of $(W(z), W(w))$ has $|\xi_k|$ times the norm of the k th summand for signs, and $\|\Sigma_{z,w}^{-1/2}\| \leq (160e^{4K})^{1/2}$, since the least eigenvalue of the covariance matrix $\Sigma_{z,w}$ of $(W(z), W(w))$ is at least $e^{-4K}/160$, as there. It follows that its third-moment sum is at most

$$\mathbb{E}|\xi_0|^3 \cdot 16e^{3K}(160e^{4K})^{3/2}N^{-1/2} \leq 16B^3e^{3K}(160e^{4K})^{3/2}N^{-1/2},$$

and that its Raič bound $2000e^{3K}(160e^{4K})^{3/2}N^{-1/2}$ is multiplied by at most B^3 .

For the derivative and tail estimates, we apply Hoeffding's inequality, as in the Steinhaus case, to $\Re(\xi_k c_k)$ and $\Im(\xi_k c_k)$, now with $|\Re(\xi_k c_k)|, |\Im(\xi_k c_k)| \leq B|c_k|$. In the proof of Lemma 4.3, at each sample and at B times the level used there, we obtain

$$\begin{aligned} \mathbb{P}\{|f_N''(z)| \geq 50Be^{K+4}N^{5/2}\sqrt{\log N}\} \\ \leq 4 \exp \left[-\frac{(50Be^{K+4}N^{5/2}\sqrt{\log N})^2}{4B^2 \sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}} \right] \\ = 4 \exp \left[-\frac{(50e^{K+4}N^{5/2}\sqrt{\log N})^2}{4 \sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}} \right] \leq 4N^{-100}, \end{aligned}$$

where the last step is that of the sign case. As in the sign case, the union over the at most $2N^3$ samples has probability at most $8N^{-97} \leq N^{-10}$. Since $|\xi_k| \leq B$, the deterministic bound on f_N''' gains the factor B , so that the interpolation error is at most $4\pi Be^{K+1}N < 50Be^{K+4}N^{5/2}\sqrt{\log N}$. Hence $\sup_{|z| \leq 1+(K+1)/N} |f_N''(z)| \leq 100Be^{K+4}N^{5/2}\sqrt{\log N}$ outside an event of probability at most N^{-10} . In the proof of Lemma 4.6, at each sample, for $N < n \leq N + m_{\max}$ and at B times the level used there, we have

$$\mathbb{P}\left\{ \left| \sum_{k=N+1}^n \xi_k z^k \right| \geq 10Be^{2K}\sqrt{m_{\max} \log N} \right\}$$

$$\begin{aligned}
&\leq 4 \exp \left[-\frac{(10Be^{2K}\sqrt{m_{\max}\log N})^2}{4B^2\sum_{k=N+1}^n \rho^{2k}} \right] \\
&= 4 \exp \left[-\frac{(10e^{2K}\sqrt{m_{\max}\log N})^2}{4\sum_{k=N+1}^n \rho^{2k}} \right] \leq 4N^{-25}.
\end{aligned}$$

As in the sign case, the union over the at most $2N^4$ pairs of a sample and a nonempty tail has probability at most $8N^{-21} \leq N^{-10}$. Similarly, the deterministic bound on the derivatives of the tails gains the factor B , so that the interpolation error is at most $4\pi Bm_{\max}e^{2K}/N^2 < 5Be^{2K}\sqrt{m_{\max}\log N}$. Hence, outside an event of probability at most N^{-10} , $\max_{N < n \leq N+m_{\max}} \sup_{|z| \leq \rho} |\sum_{k=N+1}^n \xi_k z^k| < 15Be^{2K}\sqrt{m_{\max}\log N}$. This is (H2) with $B_0 = B$. The derivative and tail constants depend on B as in Section A.6.

With these bounds, the proof of Lemma 4.5 applies with constants depending on K and B . Its second-derivative bound is now $M_B N^{5/2} \sqrt{\log N}$, with $M_B = 100Be^{K+4}$. The bounds (25) require that the mesh constant of that proof, here $\eta \leq 1$, satisfy $M_B \eta \leq 1/2$, and the spacing is $h = \eta/(NL\sqrt{\log N})$ as there. For an annular root α with $|f'_N(\alpha)| < N^{3/2}/L$, outside the two real sectors, and its assigned mesh point z , Taylor's theorem at α , as in the proof of Theorem B.4, with $|z - \alpha| \leq h$, $|z| \leq 2$ and the bound $M_B N^{5/2} \sqrt{\log N}$ on $|f''_N|$, gives

$$\begin{aligned}
\frac{|f_N(z)|}{\sqrt{N}} &\leq \frac{Nh}{L} + \frac{M_B}{2} N^2 h^2 \sqrt{\log N} = \frac{\eta + M_B \eta^2/2}{L^2 \sqrt{\log N}} \leq \frac{M_B}{L^2 \sqrt{\log N}}, \\
\frac{|zf'_N(z)|}{N^{3/2}} &\leq \frac{2(1 + M_B \eta)}{L} \leq \frac{3}{L},
\end{aligned}$$

since $Nh = \eta/(L\sqrt{\log N})$, $\eta + M_B \eta^2/2 \leq 5\eta/4 \leq M_B$ and $M_B \eta \leq 1/2$. It follows that the thresholds u and v_0 of Theorem B.4, with $M_2 = M_B N^{5/2} \sqrt{\log N}$, satisfy (25) with $u_B = M_B/(L^2 \sqrt{\log N})$ in place of its bound on u and the same bound $3/L$ on v_0 . As for real Gaussian coefficients, the successful tests are the retained mesh points z_i at which the event E_i of Theorem B.4 occurs. By the mesh count of that proof, there are at most $53(K+1)\eta^{-2}NL^2 \log N = C(K, B)N^{33/32} \log N$ points, since $L^2 = N^{1/32}$. With the eigenvalue bounds of that proof and its two Raič bounds multiplied by at most B^3 , as shown above, the chain (27) becomes

$$\begin{aligned}
\mathbb{P}(E_i) &\leq \frac{u_B^2(3/L)^2}{4(e^{-4K}/80)^2} + 2000B^3e^{3K}(160e^{4K})^{3/2}N^{-1/2} \\
&= \frac{9 \cdot 80^2 e^{8K} M_B^2}{4} \cdot \frac{L^{-6}}{\log N} + 2000B^3e^{3K}(160e^{4K})^{3/2}N^{-1/2} \\
&\leq C(K, B)L^{-6}/\log N + C(K, B)N^{-1/2},
\end{aligned}$$

where the first step collects the four-dimensional Raič bound and the density and volume bounds of that chain, with $2/(e^{-4K}/80) = 160e^{4K}$. For a separated pair,

the bound $\delta_* + \chi + 2\delta$ on the covariance of the indicators of E_i and E_j in the proof of Theorem B.4 holds with its Raič terms δ_* and δ multiplied by at most B^3 and with the total variation bound χ of (28) unchanged, so that this covariance is at most $C(K, B)N^{-1/2}$. Let $m_B \leq C(K, B)NL^2 \log N$ be the number of retained mesh points and S the number of successful tests. Then, by the mean and variance bounds in the proof of Theorem B.4, we have

$$\mathbb{E}S \leq m_B \left\{ \frac{C(K, B)L^{-6}}{\log N} + C(K, B)N^{-1/2} \right\} \leq C(K, B)\{NL^{-4} + N^{1/2}L^2 \log N\},$$

$$\text{Var } S \leq 10m_B^2 N^{-1/2} + C(K, B)m_B^2 N^{-1/2} \leq C(K, B)N^{3/2}L^4(\log N)^2,$$

where the variance bound counts the covariances of the at most $10m_B^2 N^{-1/2}$ ordered pairs in the removed bands as one and the others as $C(K, B)N^{-1/2}$. For the local count (H1) we use Lemma 7.2, with $K+2$ in place of K , so that, on the local-count and derivative events,

$$B_{K,N} \leq C_{1,B}(K+2)(\log N)S + 40C_{1,B}(K+2)\sqrt{N} \log N.$$

As in the proof of Lemma 4.5, we take the deviation $t = NL^{-4}$. Since $L = N^{1/64}$, we have $\sqrt{N} \leq NL^{-4}$ and $N^{1/2}L^2 \log N \leq NL^{-4}$ for $N \geq 2$, so that $\mathbb{E}S + t \leq C(K, B)NL^{-4}$, and hence, on these events, $S \leq \mathbb{E}S + t$ implies

$$\begin{aligned} B_{K,N} &\leq C_{1,B}(K+2)(\log N)C(K, B)NL^{-4} + 40C_{1,B}(K+2)\sqrt{N} \log N \\ &\leq C(K, B)NL^{-4} \log N. \end{aligned}$$

By Chebyshev's inequality at deviation t , and since $L^{12} = N^{3/16}$,

$$\mathbb{P}\{S > \mathbb{E}S + t\} \leq \frac{\text{Var } S}{t^2} \leq C(K, B) \frac{N^{3/2}L^4(\log N)^2}{N^2L^{-8}} = C(K, B)N^{-5/16}(\log N)^2.$$

Adding the exceptional probabilities of Lemma 7.2 and of the above derivative bound, we obtain

$$\begin{aligned} (49) \quad \mathbb{P}\{B_{K,N} > C(K, B)NL^{-4} \log N\} \\ \leq N^{-3} + \exp[-(N+1)/(2B^4)] + N^{-10} + C(K, B)N^{-5/16}(\log N)^2, \end{aligned}$$

which is (H5).

Write $\xi_k = \epsilon_k a_k$ as in the proof of Lemma 7.2, with independent, possibly complex, amplitudes a_k with the coefficient distribution, independent of the signs. Conditional on the amplitudes, with the weights $w_k = r^{2k}/\sigma_N(r)^2$ of the Gaussian case, by Parseval's identity and $|\xi_k| = |a_k|$ we have $V = \sum_{k=0}^N w_k |a_k|^2$ and $\sum_{k=0}^N w_k =$

1. Since $0 \leq |a_k|^2 \leq B^2$, we have $V \leq B^2$ and $\mathbb{E}V = \sum_k w_k \mathbb{E}|a_k|^2 = 1$. By Hoeffding's inequality for the independent variables $w_k |a_k|^2$, with values in intervals of lengths $w_k B^2$, and since $\sum_k w_k^2 \leq w_{\max} \sum_k w_k = w_{\max} \leq e^{6K}/N$, we have

$$(50) \quad \begin{aligned} \mathbb{P}(V < 1/2) &\leq \mathbb{P}\{V - \mathbb{E}V \leq -1/2\} \leq \exp \left[-\frac{2(1/2)^2}{\sum_k (w_k B^2)^2} \right] \\ &= \exp \left[-\frac{1}{2B^4 \sum_k w_k^2} \right] \leq \exp[-1/(2B^4 w_{\max})] \leq \exp[-N/(2B^4 e^{6K})]. \end{aligned}$$

On the event $E = \{V \geq 1/2\}$ of the statement, let $b_k = r^k a_k / (\sigma_N(r) \sqrt{V})$. Then $\sum_{k=0}^N |b_k|^2 = 1$, $X_r(\theta) = \sqrt{V} \sum_{k=0}^N \epsilon_k b_k e^{ik\theta}$, and $|\log \sqrt{V}| \leq \max(\frac{1}{2} \log 2, \log B)$, since $1/2 \leq V \leq B^2$ on E . Denote expectations over signs and amplitudes by $\mathbb{E}_\epsilon$ and $\mathbb{E}_a$. By NNS, $\mathbb{E}_\epsilon \int |\log |X_r / \sqrt{V}||^p d\mu \leq (C_* p)^{6p}$ for $p \geq 1$. Hence, by Fubini's theorem, since E depends only on the amplitudes, and by $|\log |X_r|| \leq |\log |X_r / \sqrt{V}|| + |\log \sqrt{V}|$ and $(x + y)^p \leq 2^{p-1}(x^p + y^p)$ for $x, y \geq 0$, we have

$$\begin{aligned} \int_E \int |\log |X_r||^p d\mu d\mathbb{P} &= \mathbb{E}_a \left[1_E \mathbb{E}_\epsilon \int |\log |X_r||^p d\mu \right] \\ &\leq 2^{p-1} \{ (C_* p)^{6p} + \max(\tfrac{1}{2} \log 2, \log B)^p \} \\ &\leq 2^p \left[(C_* + \max(\tfrac{1}{2} \log 2, \log B) + 1)p \right]^{6p} \\ &\leq [2(C_* + \max(\tfrac{1}{2} \log 2, \log B) + 1)p]^{6p}, \end{aligned}$$

where the third step holds since the base $(C_* + \max(\frac{1}{2} \log 2, \log B) + 1)p$ is at least 1, at least $C_* p$ and at least $\max(\frac{1}{2} \log 2, \log B)$.

By the definition of C_B , the last bound is $(C_B p)^{6p}$, and $V \leq B^2$ on E . With the occupation bounds above, whose constants satisfy $2 + c_B < 14 + 3c_B$, Lemma 3.2 therefore applies on E , with

$$H = 14 + 3c_B, \quad C = C_B, \quad B_E = B^2.$$

Its error is at most $C(K, B)N^{-1/16}(\log N)^7$, and its probability bound, with $\mathbb{P}(E^c) \leq \exp[-N/(2B^4 e^{6K})]$ by (50), is the concentration bound of the proposition. The moment bound on E , the bound $V \leq B^2$ and (50) are (H4), with $E_r = \{V \geq 1/2\}$. The explicit bound is computed in Section A.6.

Summability. At $N = j^8$ the polynomial terms of the exceptional probabilities are summable, by the last column of Table 4. The remaining terms are

$$e^{-3(N+1)/32}, \quad e^{-3N/(32e^{6K})}, \quad 2(2N+1)e^{-N^2/2}, \quad e^{-(N+1)/(2B^4)}, \quad e^{-N/(2B^4 e^{6K})}.$$

All but the third are at most $e^{-\delta N}$ for a constant $\delta > 0$ depending only on K and B . Since $e^{-y} \leq 2y^{-2}$ for $y > 0$, we have, for every $\delta > 0$,

$$(51) \quad \begin{aligned} \sum_{j \geq 1} e^{-\delta j^8} &\leq \frac{2}{\delta^2} \sum_{j \geq 1} j^{-16} < \infty, \\ \sum_{j \geq 1} 2(2j^8 + 1)e^{-j^{16}/2} &\leq \sum_{j \geq 1} 6j^8 \cdot 8j^{-32} = 48 \sum_{j \geq 1} j^{-24} < \infty, \end{aligned}$$

where in the second chain we used $2(2j^8 + 1) \leq 6j^8$ and $e^{-j^{16}/2} \leq 2(j^{16}/2)^{-2} = 8j^{-32}$. Hence the exceptional probabilities are summable along $N = j^8$, which completes the proof. $\square$

Proof of Theorem 1.1 and Corollary 1.2 for other distributions. Multiplying every coefficient by a nonzero constant does not change the zeros of f_n . A nondegenerate centrally symmetric distribution is not concentrated at zero, so it has positive second moment, and we may assume $\mathbb{E}|\xi_0|^2 = 1$ for a bounded distribution, as in Proposition 7.4; the other three distributions satisfy it already. By that proposition, each of the four distributions satisfies (H1)–(H5). We show that the proof of Section 6 applies to any sequence of independent coefficients with a common law that satisfies $\mathbb{E}|\xi_0|^2 = 1$ and (H1)–(H5), writing out every step in which a constant, a degree or an exceptional probability changes. Constants $C(K)$ may now depend on the coefficient law. Since $\mathbb{E}|\xi_0|^2 = 1$, we have $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$ as for signs, and, as in Proposition 2.2, the variance profile is

$$F(x) = \frac{1}{2} \log \int_0^1 e^{2xt} dt, \quad F' = \Phi,$$

and Lemma 2.3 applies unchanged. The other deterministic results used there (Lemmas 2.4, B.3 and 5.4, Theorem 5.1 and Corollary 5.3) do not involve the coefficient law.

For $K \geq 2$, let $H(K)$ be the constant of (H3), and let

$$\Pi_K(N) = H(K)N^{-3/8} + N^{-12} + \frac{H(K)}{256}N^{-3/8}(\log N)^{-12}$$

be the probability bound of Lemma 3.2 without its term $\mathbb{P}(E^c)$. By (H3) and (H4), Lemma 3.2 applies at every radius $r \in [1 - K/N, 1 + K/N]$, with the event E_r of (H4), and gives

$$(52) \quad \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > C(K)N^{-1/16}(\log N)^7\} \leq \Pi_K(N) + \mathbb{P}(E_r^c),$$

where $C(K)$ is the bound on the error ε of that lemma. This is (9) with the additional term $\mathbb{P}(E_r^c)$. Outside its exceptional event, $J_N(r)$ is finite, so $f_N \neq 0$.

Step 1. Let $B_0 \geq 1$ be the amplitude factor of (H2). We replace the tail amplitude a and the second-derivative bound M_2 of Step 1 of Section 6 by $B_0 a$ and $B_0 M_2$, which (H2) provides, with $K + 2$ in place of K for the tails, and we keep ρ , d_0 , L and h . Since (31) does not involve the coefficients, the bounds of that step become those of the table: the sign-case bounds times B_0 , except for B_0^2 in the fourth row and the unchanged term $9|x|N^{-7/64}$.

Quantity	Bound for the law	For signs ($B_0 = 1$)
$B_0 a$	$45B_0 e^{2(K+2)} N^{7/16} \sqrt{\log N}$	(32)
$2B_0 a/d_0$	$90B_0 e^{2(K+2)} N^{-67/64} \sqrt{\log N}$	(33)
$4NB_0 a/d_0$	$180B_0 e^{2(K+2)} N^{-3/64} \sqrt{\log N}$	(34)
$4(B_0 M_2)(B_0 a)/d_0^2$	$180 \cdot 100B_0^2 e^{K+4} e^{2(K+2)} N^{-1/32} \log N$	(35)
$\frac{N}{h} \left(\frac{2B_0 a}{d_0} + \frac{ x m_j}{N^2} \right)$	$90B_0 e^{2(K+2)} N^{-1/32} \sqrt{\log N} + 9 x N^{-7/64}$	(40)

The bounds in the last three rows tend to zero, so the conditions of Table 2, with $B_0 a$ and $B_0 M_2$ in place of a and M_2 , hold for all N large enough, depending on K , x and B_0 . For Steinhaus and Gaussian coefficients $B_0 = 1$, and these parameters and bounds are unchanged.

Step 2. Outside the exceptional events of the second-derivative bound in (H2), we have $|f_N''| \leq B_0 M_2$ on $|z| \leq 1 + (K + 1)/N$. This disk contains the closed disk of radius $4B_0 a/d_0$ about every annular root, since $4B_0 a/d_0 \leq 1/N$ by Step 1. Lemma 5.4, applied to the good annular roots with $P = f_N$, the common amplitude $B_0 a$, $|f_N'(\alpha)| \geq d_0$ and $4(B_0 M_2)(B_0 a) \leq d_0^2$, shows that the selected disks, now of radius $2B_0 a/d_0$, are pairwise disjoint, that each of them contains exactly one zero, and that $|f_N| \geq 3B_0 a/2$ on their boundaries.

Step 3. The proof of Corollary 3.3 applies the deterministic Lemma 2.4, which requires a nonzero polynomial of degree at most N , outside the exceptional events of Theorem 3.1 at four radii. With (52) in place of that theorem, and since $f_N \not\equiv 0$ outside these events, (21) holds with the constant $C(K)$ of the distribution, outside four events of probabilities at most $\Pi_K(N) + \mathbb{P}(E_r^c)$, one at each radius r of that lemma. By (H5), outside the exceptional events of (H1), with $K + 2$ in place of K , and of the second-derivative bound in (H2), and outside a further event of probability at most $C(K)N^{-5/16}(\log N)^2$, we have $B_{K,N} \leq C(K)NL^{-4} \log N$. Hence (36) holds with the constant $C(K)$ of the distribution.

Step 4. At the five radii r_q of Step 4, of width R , (52) with R in place of K replaces Theorem 3.1, so that (37), (38) and (39) hold with the constant $C(R)$

of the distribution, outside five events of probabilities at most $\Pi_R(N) + \mathbb{P}(E_{r_q}^c)$, where $\Pi_R(N)$ is $\Pi_K(N)$ with R in place of K . The band condition now involves the radius $2B_0a/d_0$ of the selected disks, and it holds by the last bound of Step 1. Hence the number $c(r_N, 1 + x/n)$ of selected disks whose closures meet the closed annulus between the two target circles is again at most $2\varepsilon_N N$.

Step 5. By (H2), with $K + 2$ in place of K and $m_{\max} = m_j$, outside one maximal-tail event we have $|f_n - f_N| < B_0a$ on $|z| \leq \rho$ for all $N < n \leq N + m_j$, while $|f_N| \geq 3B_0a/2$ on the boundaries of the selected disks by Step 2. Theorem 5.1 needs nonzero polynomials. Outside the events of Step 3 we have $f_N \not\equiv 0$, and hence $f_n \not\equiv 0$, since the first $N + 1$ coefficients of f_n are those of f_N . If $\mathbb{P}(\xi_0 = 0) = 0$, every coefficient is nonzero almost surely, so $\deg f_n = n$ and $\deg f_N = N$, and (41) holds as before.

Suppose now that $\mathbb{P}(\xi_0 = 0) > 0$, which, of the four distributions, only a bounded one allows. Since the probabilities of the trailing-run events below are summable along $N = j^8$ once $8D|\log \mathbb{P}(\xi_0 = 0)| > 1$, any $D > 1/(8|\log \mathbb{P}(\xi_0 = 0)|)$ would do. We choose $D > 4/|\log \mathbb{P}(\xi_0 = 0)|$, which keeps a margin. Let the trailing-run event be the event that the trailing run of zero coefficients at $N = j^8$ exceeds $\lceil D \log N \rceil$. Since $0 < \mathbb{P}(\xi_0 = 0) < 1$, this event has probability at most $\mathbb{P}(\xi_0 = 0)^{\lceil D \log N \rceil} \leq \mathbb{P}(\xi_0 = 0)^{D \log N} = N^{-D|\log \mathbb{P}(\xi_0 = 0)|}$, and, since $8D|\log \mathbb{P}(\xi_0 = 0)| > 32$, we also have $\sum_{j=1}^{\infty} j^{-8D|\log \mathbb{P}(\xi_0 = 0)|} < \infty$. Outside the trailing-run event we have $\deg f_N \geq N - \lceil D \log N \rceil$ and $\deg f_n \leq n$, so that, for every $N \leq n \leq N + m_j$,

$$\begin{aligned} \deg f_n - \deg f_N &\leq n - N + \lceil D \log N \rceil, \\ 0 &\leq \frac{\deg f_n - \deg f_N}{N} \leq \frac{m_j}{N} + \frac{\lceil D \log N \rceil}{N} \longrightarrow 0, \end{aligned}$$

where $0 \leq$ holds since the prefix degrees are nondecreasing, and the limit follows from $n - N \leq m_j$ and (31). Thus the actual-degree term in Corollary 5.3, divided by N , tends to zero. Since each selected disk contains exactly one root of f_N and U_N counts the other roots of f_N , we still have $\deg f_N - s = U_N$, although $\deg f_N$ may now be smaller than N . Hence, by Corollary 5.3, Step 4 and the degree bound above, for $N \leq n \leq N + m_j$ we have

$$\begin{aligned} |\nu_n(1 + x/n) - \nu_N(r_N)| &\leq c(r_N, 1 + x/n) + (\deg f_N - s) + \max(0, \deg f_n - \deg f_N) \\ &\leq 2\varepsilon_N N + U_N + m_j + \lceil D \log N \rceil, \end{aligned}$$

and the three-term split of (41), with this bound in place of the matching bound there, gives

$$(53) \quad \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq 3\varepsilon_N + \frac{U_N}{N} + \frac{2m_j}{N} + \frac{[D \log N]}{N}.$$

The initial zero multiplicity, the index of the first nonzero coefficient, is finite almost surely, since the coefficients are independent and $\mathbb{P}(\xi_0 \neq 0) > 0$. Divided by N , it tends to zero, and the bound on U_N/N already counts these zeros at the origin as unselected roots. There are only finitely many zero polynomials, and their counts do not affect the limit.

Step 6. We now also include in the event $E_{K,R,x,j}$:

- the energy exceptions E_r^c of (H4) at the four radii of the width K and at the five radii of the width R ;
- the additional exceptional events of (H1), with $K + 2$ in place of K , and of (H2), which the derivative, tail and small-derivative bounds share;
- if $\mathbb{P}(\xi_0 = 0) > 0$, the trailing-run event.

Then $\mathbb{P}(E_{K,R,x,j})$ is at most the right-hand side of (42), with the constants of the distribution, plus these additional probabilities, which are summable along $N = j^8$ by (H1), (H2), (H4) and the sum above. For Steinhaus coefficients there are no additional events, and (42) holds as it stands. For Gaussian coefficients, by Proposition 7.4 and Table 6, we have, for all large j ,

$$\begin{aligned} \mathbb{P}(E_{K,R,x,j}) &\leq 4\Pi_K(N) + 5\Pi_R(N) + N^{-3} + 2N^{-10} + C(K)N^{-5/16}(\log N)^2 \\ &\quad + 4e^{-3N/(32e^{6K})} + 5e^{-3N/(32e^{6R})} + e^{-3(N+1)/32} + 2(2N+1)e^{-N^2/2}, \end{aligned}$$

and for bounded coefficients,

$$\begin{aligned} \mathbb{P}(E_{K,R,x,j}) &\leq 4\Pi_K(N) + 5\Pi_R(N) + N^{-3} + 2N^{-10} + C(K)N^{-5/16}(\log N)^2 \\ &\quad + 4e^{-N/(2B^4e^{6K})} + 5e^{-N/(2B^4e^{6R})} + e^{-(N+1)/(2B^4)} + N^{-D|\log \mathbb{P}(\xi_0=0)|}, \end{aligned}$$

where the last term is present only if $\mathbb{P}(\xi_0 = 0) > 0$. In both bounds, the first two exponential terms are the energy exceptions of Proposition 7.4 at the widths K and R , and the third is the additional exception of Lemma 7.3 or 7.2. At $N = j^8$ the terms of (42) are summable as in Step 6 of Section 6, the exponential terms by (51), and the trailing-run term by the sum above. Hence the Borel–Cantelli argument and the limit $K \rightarrow \infty$ of Step 6 apply, with the additional term $[D \log N]/N \rightarrow 0$ of (53) when $\mathbb{P}(\xi_0 = 0) > 0$, and Step 7 gives Theorem 1.1. Corollary 1.2 is its case $x = 0$, as in Section 6. $\square$

8 Quantitative refinements and weighted profiles

Appendix A makes every constant of Sections 2–6 explicit in the width K , so K may grow with the degree; this gives the logarithmic rate. The other subsections vary the block exponent, the circles and the coefficient weights. The coefficients are one nested Rademacher sequence unless stated otherwise, and the constants and thresholds are in Appendix A.

8.1 A logarithmic almost-sure rate

We obtain the constant in Theorem 1.3 from the explicit bounds below. The deterministic analytic thresholds depend on C_* , c_{HW} , while the last exceptional block is random.

Proof of Theorem 1.3. The idea is to run the proof of Corollary 1.2 in Section 6, that is, the proof of Theorem 1.1 at $x = 0$ with the probe width $R = 2$, at a width $K = K_N$ that grows with N , while C_* and c_{HW} stay fixed, and to bound the roots outside the annulus by Lemma 2.4 with $\eta = 1/K_N$, that is, with secant radii $1 \pm (K_N - 1)/N$ and $1 \pm K_N/N$. The leading term of that lemma is then $\log(K_N/(K_N - 1)) \leq 1/(K_N - 1)$, in place of the $2 \log 2/K$ of (36), so that an error of order $1/\log n$ requires K_N to grow at least like a multiple of $\log N$, while the other constants grow exponentially in K , which bounds $K_N/\log N$ from above. We derive these upper bounds, choose K_N , and then follow Steps 1–6 of that proof at the width K_N , with the parameters of Table 7.

Parameter	Value	Role
K_N	$\lfloor (\log N)/128 - \log \log N \rfloor$	annular width, with $(\log N)/(K_N - 1) \rightarrow 128$
η	$1/K_N$	Lemma 2.4 at the secant radii $1 \pm (K_N - 1)/N$ and $1 \pm K_N/N$, with leading term $\log(K_N/(K_N - 1)) \leq 1/(K_N - 1)$
$-\log \log N$	correction in K_N	gives $e^{bK_N} \leq N^{b/128}(\log N)^{-b}$, which makes the binding row $Z_K N^{-1/16} \log N$ of Table 8 $o(1/\log N)$
T	$(\log N)/32$	clipping level of Lemma 3.2

Table 7: The parameters of the proof of Theorem 1.3. The probes of Step 4 keep the fixed width $R = 2$.

The constants of the argument that depend on K are listed in Section A.7. Each bound in Table 8 is at most of the form $C_0(K+1)^p(\log N)^q e^{bK} N^{-\beta}$, with C_0 independent of K , real p, q , and $b, \beta > 0$. Since $e^{bK} N^{-\beta} = N^{bK/\log N - \beta}$, and since powers of $K+1 \leq \log N$ and of $\log N$ are dominated by any power of N , if $K \leq c \log N$ then

$$C_0(K+1)^p(\log N)^q e^{bK} N^{-\beta} \begin{cases} \text{tends to zero} & \text{if } c < \beta/b, \\ \text{is summable along } N = j^8 & \text{if } c < (\beta - 1/8)/b, \end{cases}$$

where in the second case the exponent of $j = N^{1/8}$ stays below -1 . At $c = \beta/b$ the bound still tends to zero if $K \leq (\beta/b) \log N - a \log \log N$ with $ab > p+q$, since then $e^{bK} \leq N^\beta (\log N)^{-ab}$ and $K+1 \leq \log N$. In the first part of Table 8, the two terms of

Bound	e^{bK}	$N^{-\beta}$	$K/\log N$ below
<i>Must tend to zero</i>			
Small-derivative count $Z_K N^{-1/16} \log N$	e^{8K}	$N^{-1/16}$	1/128
Taylor ratio $4M_2 a/d_0^2$	e^{3K}	$N^{-1/32}$	1/96
$N^{65/64} \cdot 2a/d_0$	e^{2K}	$N^{-1/32}$	1/64
Terms $H_K N^{-1/2} (\log N)^7$ of $\varepsilon_K^*(N)$	e^{20K}	$N^{-1/2}$	1/40
Second count term $Z'_K N^{-15/32} (\log N)^2$	e^{11K}	$N^{-15/32}$	15/352
$(6400e^{8K})^2/N$	e^{16K}	N^{-1}	1/16
D_K/N	e^{4K}	N^{-1}	1/4
<i>Must be summable along $N = j^8$</i>			
Occupation failure $H_K N^{-3/8}$	e^{20K}	$N^{-3/8}$	1/80
Clipped-integral failure $\frac{H_K}{256} N^{-3/8} (\log N)^{-12}$	e^{20K}	$N^{-3/8}$	1/80
Small-derivative failure $C_K N^{-5/16} (\log N)^2$	e^{13K}	$N^{-5/16}$	3/208

Table 8: The bounds of the proof in Section 8.1 that grow exponentially with the width K , and the upper bound on $K/\log N$ that each requires: β/b in the first part and $(\beta - 1/8)/b$ in the second.

the small-derivative count, the terms of $\varepsilon_K^*(N)$ and D_K/N enter the bound on U_N/N below through $Z_K N^{-1/16} \log N + Z'_K N^{-15/32} (\log N)^2 + 4(\varepsilon_K^*(N) + D_K/N)$, which must be $o(1/\log N)$. The other rows of the first part are ratios of Table 15, which must be at most one for the degree thresholds of Section A.7 to hold. The second part contains the exceptional probabilities in (67) that depend on K . The remaining ratio of Table 15, $2K/N$, has no exponential factor and tends to zero whenever $K \leq \log N$. The strongest condition is $K/\log N < 1/128$, which comes from the

small-derivative count. At $c = 1/128$ this row is at most $Z_0(K+1)^2 e^{8K} N^{-1/16} \log N$, by the bound on Z_K in Section A.7, and it must be $o(1/\log N)$. Hence $\log N$ times this row, with $p = 2$, $q = 2$ and $b = 8$ above, must tend to zero, and $a = 1$ suffices.

We choose

$$K_N = \left\lfloor \frac{\log N}{128} - \log \log N \right\rfloor, \quad \eta = \frac{1}{K_N},$$

where $N = j^8$ as in Section 6. For large N we have $K_N \geq 2$, and since $K_N \leq (\log N)/128 - \log \log N$ and $N \geq e$,

$$e^{bK_N} \leq N^{b/128} (\log N)^{-b} \leq N^{b/128}, \quad b \geq 0.$$

Then $K_N/\log N \leq 1/128$, which is below every other bound of Table 8, so that every condition there holds. At $K = K_N$ the error bounds and deterministic thresholds of Section A.7 become (68), and every threshold ratio tends to zero, so that the degree thresholds of that section hold for all large j . The proof of Corollary 1.2 applies at this width, with the four radial events taken at the radii $1 \pm (K_N - 1)/N$ and $1 \pm K_N/N$ of Lemma 2.4 at $\eta = 1/K_N$, which lie in $[1 - K_N/N, 1 + K_N/N]$. Its exceptional probability is bounded by (67) with $K = K_N$, $R = 2$ and $x = 0$, that is, by

$$4p_{K_N}^*(N) + 5p_2^*(N) + N^{-3} + 2N^{-10} + C_{K_N} N^{-5/16} (\log N)^2,$$

in the notation of Appendix A. Here the term N^{-3} is the local-count exception, with $K_N + 2$ in place of K . Since this parameter grows with N , we take it from Lemma 4.1 with $a_k = 1$, $B = 1$, $\mathcal{E} = \Omega$ and the parameter η of that lemma equal to 1. Then A and $t_N = A \log N/N$ are as in the proof of Lemma 4.2, the thresholds $N \geq e$ and $t_N \leq 1$ do not depend on K , and the constant is $C_1(K_N + 2) = (4A + 4K_N + 12)/\log 2$. The factors e^{20K} of H_K and e^{13K} of C_K give, with $N = j^8$,

$$\begin{aligned} N^{-3/8} e^{20K_N} &\leq N^{-3/8 + 20/128} = N^{-7/32} = j^{-7/4}, \\ N^{-5/16} e^{13K_N} &\leq N^{-5/16 + 13/128} = N^{-27/128} = j^{-27/16}. \end{aligned}$$

These are the two slowly decaying powers in (68) that come from the variable width. By the third and fourth lines of (68), with $K_N + 1 \leq \log N = 8 \log j$, the terms of (67) that depend on K_N satisfy

$$\begin{aligned} &4p_{K_N}^*(N) + C_{K_N} N^{-5/16} (\log N)^2 \\ &\leq 4H_0 N^{-7/32} + 4N^{-12} + \frac{H_0}{64} N^{-7/32} (\log N)^{-12} \\ &\quad + C_{\text{bad}} (K_N + 1)^2 N^{-27/128} (\log N)^2 \end{aligned}$$

$$\begin{aligned} &\leq 4H_0j^{-7/4} + 4j^{-96} + \frac{H_0}{64}j^{-7/4}(8\log j)^{-12} \\ &\quad + C_{\text{bad}}(8\log j)^4j^{-27/16}. \end{aligned}$$

Their exponents in j are less than -1 , and their remaining factors are polynomial in $\log j$. The other terms, $5p_2^*(N) + N^{-3} + 2N^{-10}$, do not depend on K_N , and by Table 4 at the width 2 they are summable along $N = j^8$. Thus every term in (67) is summable.

The bounds (31)–(35) of Step 1 in Section 6 are deterministic for $j \geq 60$. At the width K_N the constant of Lemma 4.3 is $100e^{K_N+4}$, and (33), (35) and $e^{bK_N} \leq N^{b/128}$ give

$$\begin{aligned} \frac{2a}{d_0} &\leq 90e^{2(K_N+2)}N^{-67/64}\sqrt{\log N} \leq 90e^4N^{-67/64+1/64}\sqrt{\log N}, \\ N^{65/64}\frac{2a}{d_0} &\leq 90e^4N^{-1/32+1/64}\sqrt{\log N} = 90e^4\sqrt{\log N}N^{-1/64}, \\ \frac{4M_2a}{d_0^2} &\leq 180 \cdot 100e^{K_N+4}e^{2(K_N+2)}N^{-1/32}\log N \\ &= 18000e^8e^{3K_N}N^{-1/32}\log N \leq 18000e^8(\log N)N^{-1/128}. \end{aligned}$$

Since the last two right-hand sides tend to zero, for all large j we have $2a/d_0 < N^{-65/64}$, so that $4a/d_0 < 2N^{-65/64} \leq 1/N$, and $4M_2a \leq d_0^2$. These are the last two rows of Table 15, in which $180S_0e^4 = 18000e^8$. In particular, the band condition $2a/d_0 < h/N = N^{-65/64}$ of Step 4 at $x = 0$ holds at the width K_N .

The probes of Step 4 at $x = 0$ lie at the radii $1 + qN^{-65/64}$, $q = -2, \dots, 2$, and their five events are taken at the fixed width 2. By (66), the explicit form of (38) at $x = 0$ and $R = 2$, we have

$$\max_{q=-1,0,1} \left| \frac{\nu_N(1 + qN^{-65/64})}{N} - \frac{1}{2} \right| \leq \delta_2^*(N) = O(N^{-1/64}),$$

since $\varepsilon_2^*(N)N^{1/64} = O(N^{-3/64}(\log N)^7)$ by (55). By the band condition above, the crossing count of Step 4 at the width K_N is at most $2\delta_2^*(N)N$.

By the first line of (36), the number U_N of unselected roots is bounded by the number of roots with $||\alpha| - 1| > K_N/N$ plus $B_{K_N,N}$. Outside the four radial events, Lemma 2.4 with $\eta = 1/K_N$ applies with $\varepsilon = \varepsilon_{K_N}^*(N)$, by (55), and with $D = D_{K_N}$, by Lemma 2.3 and Section A.1, and its bound is

$$\log \frac{K_N}{K_N - 1} + 4\left(\varepsilon_{K_N}^*(N) + \frac{D_{K_N}}{N}\right) \leq \frac{1}{K_N - 1} + 4\left(\varepsilon_{K_N}^*(N) + \frac{D_{K_N}}{N}\right),$$

since $1/(\eta K_N) = 1$ and $\log(1 + y) \leq y$. Outside the event of (62) we have $B_{K_N,N} \leq Z_{K_N}N^{15/16}\log N + Z'_{K_N}N^{17/32}(\log N)^2$. This is (69). Dividing it by N

and using (68), we get

$$\begin{aligned}
\frac{U_N}{N} &\leq \frac{1}{K_N - 1} + 4\left(\varepsilon_{K_N}^*(N) + \frac{D_{K_N}}{N}\right) + Z_{K_N} N^{-1/16} \log N + Z'_{K_N} N^{-15/32} (\log N)^2 \\
&\leq \frac{1}{K_N - 1} + 4E_0 N^{-1/16} (\log N)^7 + 40(K_N + 1)^2 N^{-31/32} \\
&\quad + Z_0(K_N + 1)^2 (\log N)^{-7} + Z'_0(K_N + 1)^2 N^{-49/128} (\log N)^2 \\
&\leq \frac{1}{K_N - 1} + o(1/\log N),
\end{aligned}$$

since $K_N + 1 \leq \log N$, so that $\log N$ times each term other than $1/(K_N - 1)$ tends to zero; for the fourth term, $(\log N) Z_0(K_N + 1)^2 (\log N)^{-7} \leq Z_0 (\log N)^{-4}$. By the normalized matching inequality (41) of Step 5 in Section 6, at $x = 0$ and with $\delta_2^*(N)$ in place of ε_N , we therefore have, uniformly for $N \leq n \leq N + m_j$,

$$\begin{aligned}
\left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq 3\delta_2^*(N) + \frac{U_N}{N} + \frac{2m_j}{N} \\
&\leq 3\delta_2^*(N) + \frac{1}{K_N - 1} + o(1/\log N) + 18N^{-1/8} \\
&= \frac{1}{K_N - 1} + o(1/\log N),
\end{aligned}$$

where $2m_j/N \leq 18N^{-1/8}$ by (31), and $(\log N)\delta_2^*(N) \rightarrow 0$. With every error term written out, this inequality is (70). Since the exceptional probabilities are summable, it follows from the Borel–Cantelli lemma that, almost surely, only finitely many exceptional events occur. Since $m_j \leq N$, we have $\log N \leq \log n \leq \log(2N)$, so that $\log n/\log N \rightarrow 1$ uniformly within the block. Since $(\log N)/128 - \log \log N - 1 < K_N \leq (\log N)/128$, we also have $K_N/\log N \rightarrow 1/128$ and $(\log N)/(K_N - 1) \rightarrow 128$. Almost surely, every large n lies in a block $N \leq n \leq N + m_j$ whose exceptional event does not occur, and therefore, by the bound above in the block of n ,

$$\begin{aligned}
\limsup_{n \rightarrow \infty} (\log n) \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq \limsup_{j \rightarrow \infty} \log(2N) \left(\frac{1}{K_N - 1} + o(1/\log N) \right) \\
&= \limsup_{j \rightarrow \infty} \frac{\log(2N)}{\log N} \left(\frac{\log N}{K_N - 1} + o(1) \right) = 128.
\end{aligned}$$

□

8.2 Choice of the sparse degree sequence

Proposition 8.1 is stated for the original parameter choice of this argument, with deviation $N^{-1/32}$ and exceptional root count $N^{31/32}$; with the powers of Sections 3–6 the same argument gives $16/5 < q < 16$, as shown at the end of its proof.

Proposition 8.1 (Sparse-degree exponents). *With logarithmic cutoff $(\log N)/32$, deviation $N^{-1/32}$, derivative parameter $L = N^{1/64}$, exceptional root count $N^{31/32}$, and thin-band width $N^{-1-1/64}$, one may replace j^8 by $\lfloor j^q \rfloor$ for every*

$$\frac{8}{3} < q < 16.$$

In particular j^3 works. After retuning these powers, every real $q > 2$ works for the fixed- K strong law and radial-profile proof. For this construction, the second-moment conditions require $q > 2$.

Proof. The idea is to rerun the proof of Theorem 1.1 in Section 6, which contains that of Corollary 1.2 as its case $x = 0$, along the degrees $\lfloor j^q \rfloor$, with the powers of the statement left as parameters, and to collect the inequalities that each step imposes on these powers and on q . We solve them for every $q > 2$, and then for the fixed powers of the statement.

In this proof we use $N = \lfloor j^q \rfloor$ and $m = \lfloor (j+1)^q \rfloor - \lfloor j^q \rfloor$ in place of j^8 and m_j . We claim that if $q > 1$, then eventually $m \leq (2 + q2^q)N^{1-1/q}$ and $m \leq N$. Indeed, for every $j \geq 1$,

$$\begin{aligned} m &\leq (j+1)^q - j^q + 1 \leq q(j+1)^{q-1} + 1 \leq q2^{q-1}j^{q-1} + 1 \\ &\leq q2^{q-1}(2N)^{1-1/q} + 1 \leq q2^q N^{1-1/q} + 1 \leq (2 + q2^q)N^{1-1/q}, \end{aligned}$$

where we use $\lfloor (j+1)^q \rfloor \leq (j+1)^q$ and $\lfloor j^q \rfloor > j^q - 1$, the mean value theorem for the q -th power on $[j, j+1]$, where its derivative is at most $q(j+1)^{q-1}$ since $q > 1$, and $j^{q-1} = (j^q)^{1-1/q}$ with $j^q < N+1 \leq 2N$. The second bound follows once $N^{1/q} \geq 2 + q2^q$. Also $m \geq 1$, since $(j+1)^q - j^q \geq qj^{q-1} \geq 1$. Since $\lfloor j^q \rfloor \geq j^q/2$, a bound on an exceptional probability that is a power of N times a power of $\log N$ is summable over j when q times the exponent of N is less than -1 .

We choose a tail disk of radius $1 + (K+2)/N$. The maximal-tail estimate applies with $a = 15e^{2(K+2)}\sqrt{m \log N}$, and its exceptional event still has probability at most N^{-10} . This is Lemma 4.6 with $K+2$ in place of K and $m_{\max} = m$, which applies since $1 \leq m \leq N$ and $N \geq \max\{4, K+2\}$ for large j . By the bound on m , we have

$$a \leq 15e^{2(K+2)}(2 + q2^q)^{1/2}N^{1/2-1/(2q)}\sqrt{\log N}.$$

We take the derivative parameter N^ℓ in place of $L = N^{1/64}$, so that $d_0 = N^{3/2-\ell}$. For the deterministic localization bounds we have

$$\frac{2Na}{d_0} = 2N^{\ell-1/2}a = O_K(N^{\ell-1/(2q)}\sqrt{\log N}),$$

$$\frac{4aN^{2\ell}}{N^3} \sup |f_N''| \leq \frac{4aN^{2\ell}}{N^3} \cdot 100e^{K+4}N^{5/2}\sqrt{\log N} = O_K(N^{2\ell-1/(2q)} \log N),$$

where the supremum is over $|z| \leq 1 + (K+1)/N$ and is bounded by Lemma 4.3 outside its exceptional event, the two O_K bounds use the bound on a , and the implied constants also depend on the fixed q . It follows that Taylor isolation holds when $\ell < 1/(4q)$. Indeed, then $\ell - 1/(2q) < 2\ell - 1/(2q) < 0$, so both bounds tend to zero, and for large N they imply that $4a/d_0 \leq 1/N$ and $4M_2a \leq d_0^2$, where $M_2 = 100e^{K+4}N^{5/2}\sqrt{\log N}$ and $4aN^{2\ell}/N^3 = 4a/d_0^2$. A band of width $N^{-1-\kappa}$ contains the localization disks when $\kappa < 1/(2q) - \ell$, since $N^{1+\kappa} \cdot 2a/d_0 = N^\kappa \cdot 2Na/d_0 = O_K(N^{\kappa+\ell-1/(2q)}\sqrt{\log N})$. The moving target in Theorem 1.1 has scaled drift $O_x(N^{-1/q})$, which the same band absorbs. Indeed, for $N \leq n \leq N+m$ the target radius moves by $|x/n - x/N| \leq |x|m/N^2$, as in Step 4 of Section 6, and by the bound on m we have

$$N \cdot \frac{|x|m}{N^2} \leq (2 + q2^q)|x|N^{-1/q}, \quad N^{1+\kappa} \cdot \frac{|x|m}{N^2} \leq (2 + q2^q)|x|N^{\kappa-1/q}.$$

Since $\kappa < 1/(2q) - \ell < 1/q$, the second right-hand side tends to zero.

For the logarithmic clipping we set $T = t \log N$ and $u = N^{-w}$ with $w = t > 0$, and we apply Remark B.2, with moment parameter $6 \log N$ and threshold N^{-2t} . By (15), the one- and two-point errors of Theorem B.1 are at most $C(K)N^{-1/2}$, and the removed angles and pairs there have measures at most $2N^{-1/2}$ and $10N^{-1/2}$. The quantities d_1 and v of the remark are the one-point error plus the removed measure of angles, and the two-point error plus the removed measure of pairs plus twice d_1 . Both are therefore at most $C(K)N^{-1/2}$, with a larger $C(K)$, and the remark applies with its exponent equal to $1/2$ and with $C(K)$ for both of its constants. Here the moment bound is (17), whose constant is C_* and whose exponent is 6, and the energy identity is (18). Therefore the logarithmic error satisfies

$$\begin{aligned} \eta_{\log} &\leq N^{-t} + 2tC(K)(\log N)N^{-1/2} \\ &\quad + e(12C_* \log N)^6 \sqrt{2N^{-2t} + C(K)N^{-1/2} + \frac{3}{2}N^{-2t}} = O_K(N^{-t}(\log N)^6) \end{aligned}$$

for $0 < t \leq 1/4$, since then $N^{-1/2} \leq N^{-2t} \leq N^{-t}$, and $1 \leq \log N \leq (\log N)^6$ for $N \geq e$. The first condition of (S) below implies that $t < 1/8$. The probabilities of the three exceptional events are bounded by constants times $N^{-1/2+4t}$, N^{-12} and $N^{-1/2+2t}(\log N)^2$, the three terms of the probability bound of Remark B.2, with moment parameter $6 \log N$.

The mesh geometry and the normal approximation apply to a general derivative parameter N^ℓ . Indeed, in Section A.4 the bound $D'_K N L^2 \log N$ on the number of mesh points, the one-point probability (60) and the covariance bound

$(3\mathcal{B}_K + \mathcal{T}_K)N^{-1/2}$ for a pair are written with L as a parameter, and the normal-approximation errors in them, $\mathcal{B}_K N^{-1/2}$ and $\mathcal{T}_K N^{-1/2}$, do not involve L . For a root-count threshold N^{1-d} , the mean and variance bounds in Corollary B.5, with both of its exponents equal to $1/2$ and its sector count $b_0 = O_K(\sqrt{N} \log N)$, lead to conditions (iii)–(vi) of the table below. We use $d < 1/2$ for the excluded-sector bound $O_K(\sqrt{N} \log N)$, which counts at most $C(K+2) \log N$ roots in each of $O(\sqrt{N})$ disks by Lemma 4.2, with $K+2$ in place of K . For logarithmic concentration and the thin band, we impose

$$(S) \quad q \left(\frac{1}{2} - 4t \right) > 1, \quad q \left(\frac{1}{2} - 2t \right) > 1, \quad 0 < \kappa < \min \left(t, \frac{1}{2q} - \ell \right).$$

At the moving radii of Theorem 1.1, the probes of Step 4 of Section 6 have spacing $h = N^{-\kappa}$. Fix an integer $R \geq |x| + 2$, as in Section 6, let $N \geq 2R$, and let y be one of the probes $x - h$, x and $x + h$. We apply Lemma B.3 at y , with R in place of K , which applies since $|y| + h \leq |x| + 2h < R < N$. We take the centering $\frac{1}{2} \log N$, the profile F of Proposition 2.2, $M_F = 1/2$ and $B_F = 1$, since $0 \leq \Phi' \leq 1/2$ and $0 \leq \Phi \leq 1$ (Step 4 of Section 6). In place of ε we take the variance-profile error $(R^2 + e^{4R}/2)/N$ of Lemma 2.3, and we use $\eta_{\log} = O_x(N^{-t}(\log N)^6)$, which is the bound on $\eta_{\log}$ above with R in place of K , since $|y| < R$, so that its constant depends on x through R . Combined with $|\Phi(y) - \Phi(x)| \leq |y - x|/2$, the lemma yields

$$\begin{aligned} \left| \frac{\nu_N(1 + y/N)}{N} - \Phi(x) \right| &\leq \frac{|y - x|}{2} + \frac{h}{4} + \frac{R}{N} + \frac{2(\eta_{\log} + (R^2 + e^{4R}/2)/N)(1 + R/N)}{h} \\ &\leq \frac{7}{4}N^{-\kappa} + 4N^{\kappa}(\eta_{\log} + (R^2 + e^{4R}/2)N^{-1}) \\ &= O_x(N^{-\kappa} + N^{\kappa-t}(\log N)^6), \end{aligned}$$

where we use $|y - x| \leq h = N^{-\kappa}$, $R/N \leq N^{-\kappa}$ and $1 + R/N \leq 2$ for large N . This tends to zero since $\kappa < t$. The strict power inequalities dominate the above logarithmic factors and all fixed- K constants. We therefore obtain the nine conditions below, of which rows (ii), (vii), (viii) and (ix) together form (S).

	Condition	Where it arises
(i)	$\ell < 1/(4q)$	Taylor isolation
(ii)	$\kappa < 1/(2q) - \ell$	band containing the localization disks
(iii)	$0 < d < 4\ell$	mean threshold of Corollary B.5
(iv)	$d < \frac{1}{2} - 2\ell$	mean threshold of Corollary B.5
(v)	$d < \frac{1}{2}$	excluded sectors
(vi)	$q(\frac{1}{2} - 4\ell - 2d) > 1$	root-count failure, summable
(vii)	$q(\frac{1}{2} - 4t) > 1$	occupation failure, summable
(viii)	$q(\frac{1}{2} - 2t) > 1$	clipped-integral failure, summable
(ix)	$0 < \kappa < t$	thin band, probes at the moving radii

For every $q > 2$ we choose

$$\Delta = \frac{1}{2} - \frac{1}{q}, \quad t = w = \frac{\Delta}{16},$$

$$\ell = \min\left(\frac{\Delta}{32}, \frac{1}{16q}\right), \quad d = \ell, \quad \kappa = \min\left(\frac{t}{2}, \frac{1}{8q}\right).$$

Then $0 < \Delta < 1/2$, all four powers are positive, and the nine conditions follow from

$$\begin{aligned} \text{(iii)} \quad d = \ell < 4\ell, & \quad \text{(v)} \quad d \leq \frac{\Delta}{32} < \frac{1}{2}, & \quad \text{(iv)} \quad d + 2\ell = 3\ell \leq \frac{3\Delta}{32} < \frac{1}{2}, \\ \text{(vii)} \quad 4t = \frac{\Delta}{4} < \Delta, & \quad \text{(viii)} \quad 2t = \frac{\Delta}{8} < \Delta, & \quad \text{(vi)} \quad 4\ell + 2d = 6\ell \leq \frac{3\Delta}{16} < \Delta, \\ \text{(i)} \quad 2\ell \leq \frac{1}{8q} < \frac{1}{2q}, & \quad \text{(ix)} \quad \kappa \leq \frac{t}{2} < t, & \quad \text{(ii)} \quad \ell + \kappa \leq \frac{1}{16q} + \frac{1}{8q} = \frac{3}{16q} < \frac{1}{2q}, \end{aligned}$$

where the middle row implies the summability conditions (vii), (viii) and (vi), since $q(\frac{1}{2} - \Delta) = 1$, and subtracting from $\frac{1}{2}$ a number smaller than Δ leaves more than $\frac{1}{2} - \Delta$. The remaining exceptional probabilities N^{-3}, N^{-10}, N^{-12} are summable as well. The conclusion then follows from the fixed- K argument and the countable intersection over K .

For the fixed powers of the statement, $t = w = 1/32$, $\ell = 1/64$, $d = 1/32$ and $\kappa = 1/64$, the conditions read

$$\begin{aligned} \text{(vii)} \quad q\left(\frac{1}{2} - 4t\right) &= \frac{3q}{8}, & \text{(vi)} \quad q\left(\frac{1}{2} - 4\ell - 2d\right) &= \frac{3q}{8}, & \text{(viii)} \quad q\left(\frac{1}{2} - 2t\right) &= \frac{7q}{16}, \\ \text{(i)} \quad \frac{1}{4q} - \ell &= \frac{16 - q}{64q}, & \text{(ii)} \quad \frac{1}{2q} - \ell - \kappa &= \frac{16 - q}{32q}, \end{aligned}$$

where the three quantities of the first row must exceed one and the two of the second row must be positive, while rows (ix), (iii) and (iv), $\kappa < t$, $d < 4\ell$ and $d < \frac{1}{2} - 2\ell$, hold since $1/64 < 1/32 < 1/16 < 15/32$, as does (v). For the fixed

powers above, the slow exponent of the exceptional probabilities is $3/8$, so that summability requires $q > 8/3$. The Taylor condition requires $q < 16$. At $q = 8/3$ the mesh bound is of order $j^{-1}(\log j)^4$, and at $q = 16$ the Taylor ratio grows logarithmically. For $q \leq 2$ and $t > 0$, the inequality $q(\frac{1}{2} - 4t) > 1$ cannot hold, since $q(\frac{1}{2} - 4t) < q/2 \leq 1$. These endpoint restrictions belong to the present second-moment estimates.

The sections above use the same cutoff and stronger powers, those of Lemmas 3.2 and 4.5: the deviation $N^{-1/16}(\log N)^7$ and the exceptional root count $C(K)N^{15/16} \log N$, with the same $L = N^{1/64}$ and thin-band width $N^{-1-1/64}$. With these powers the same argument applies for every $16/5 < q < 16$, which contains the exponent 8 of Section 6. Indeed, the Taylor and band conditions (i) and (ii) do not involve t and w , and still require $q < 16$. The occupation and clipped-integral failures in Theorem 3.1 are of order $N^{-3/8}$, which requires $3q/8 > 1$, and the slowest failure is now $C(K)N^{-5/16}(\log N)^2$, that of Lemma 4.5, which requires $5q/16 > 1$. The probes require $\kappa < 1/16$, which holds for $\kappa = 1/64$. $\square$

8.3 Sublevel components meeting nearby circles

Corollary 8.2 (Sublevel components meeting a nearby circle). *For a fixed tail bound a_0 , and $r_{\pm} = 1 \pm N^{-1-\kappa}$ with fixed $0 < \kappa \leq 1/12$, the number of roots in components of $\{|f_N| < a_0\}$ meeting those circles is $o(N)$ outside a single exceptional sequence summable at $N = j^8$.*

Proof. The idea is to bound, for each fixed K , the roots in question by the roots of f_N in a thin band about the unit circle, which (22) controls, and by the unselected roots U_N of the proof of Corollary 1.2 in Section 6, and then to let K tend to infinity by Lemma B.7. With $N = j^8$, let Y_j denote the number of roots of f_N in components of $\{|f_N| < a_0\}$ that meet at least one of the circles $|z| = r_+$ and $|z| = r_-$. We show that there exist deterministic numbers $\varepsilon_j \rightarrow 0$ and events F_j with $\sum_j \mathbb{P}(F_j) < \infty$ such that $Y_j \leq \varepsilon_j N$ off F_j .

Fix an integer $K \geq 2$, and let $a = 15e^{2(K+2)}\sqrt{m_j \log N}$ be the maximal-tail level of Step 1 of Section 6. Since $a \geq 15\sqrt{\log N}$, we have $a_0 \leq a$ for large N , and every component of $\{|f_N| < a_0\}$ lies in a component of $\{|f_N| < a\}$. So it suffices to count the roots in components of $\{|f_N| < a\}$ that meet the circles. A component containing a good annular root α lies in its selected disk, of radius $2a/d_0$, which contains no other root (Lemma 5.4, as in Step 2 of Section 6); every other root is counted by U_N . If this component meets either nearby circle at a point z , then $||\alpha| - 1| \leq |\alpha - z| + ||z| - 1| < N^{-1-\kappa} + 2a/d_0$. We enlarge the band

to half-width $N^{-1-\kappa_*}$, which must contain this range for large N and to which (22) must apply. This requires, first, $0 < \kappa_* < \kappa$, so that $N^{-1-\kappa}$ is small compared with $N^{-1-\kappa_*}$, second, $\kappa_* < 3/64$, so that $2a/d_0$ is small compared with $N^{-1-\kappa_*}$ by (33), since $1 - 67/64 = -3/64$, and third, $\kappa_* \leq 1/64$, the range of (22). We choose $\kappa_* = \min(\kappa/2, 1/128)$. Since $\kappa_* \leq \kappa/2$ and $\kappa_* \leq 1/128$, (33) gives

$$\begin{aligned} N^{1+\kappa_*} (N^{-1-\kappa} + 2a/d_0) &\leq N^{\kappa_*-\kappa} + 90e^{2(K+2)} N^{1+\kappa_*-67/64} \sqrt{\log N} \\ &\leq N^{-\kappa/2} + 90e^{2(K+2)} N^{-5/128} \sqrt{\log N}, \end{aligned}$$

and the right-hand side tends to zero, so that $N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*}$ for all N large enough. Since $0 < \kappa_* \leq 1/128 \leq 1/64$, (22) applies with 2 in place of K and κ_* in place of κ . Outside five events, one at each radius $1 + qN^{-1-\kappa_*}$ with $q = -2, \dots, 2$, it yields

$$\left| \frac{\nu_N(1 \pm N^{-1-\kappa_*})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa_*} + C(2)N^{\kappa_*-1/16}(\log N)^7.$$

This five-radius estimate bounds the number of roots in this band by $o(N)$:

$$\begin{aligned} \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| < N^{-1-\kappa_*}\} &\leq \nu_N(1 + N^{-1-\kappa_*}) - \nu_N(1 - N^{-1-\kappa_*}) \\ &= N \left(\frac{\nu_N(1 + N^{-1-\kappa_*})}{N} - \frac{1}{2} \right) - N \left(\frac{\nu_N(1 - N^{-1-\kappa_*})}{N} - \frac{1}{2} \right) \\ &\leq 2N(2N^{-\kappa_*} + C(2)N^{\kappa_*-1/16}(\log N)^7) = o(N), \end{aligned}$$

where the first inequality holds as in (39), and the last step since $\kappa_* > 0$ and $\kappa_* - 1/16 \leq 1/128 - 8/128 = -7/128$. Each of the five events has probability at most the right-hand side of (9) at $K = 2$, for every radius. Let $E_{K,j}(\kappa_*)$ be the event $E_{K,2,0,j}$ of Step 6 of the proof of Theorem 1.1, at $x = 0$ and $R = 2$, with its five probe events replaced by these five thin-band events. Its probability is bounded by the right-hand side of (42) with $R = 2$ (explicitly, (67)), which is summable for fixed K . Let $j_0(K) \geq 60$ be a deterministic index beyond which the thresholds of that proof at the width K , the bound $a_0 \leq a$, the strict inequality $N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*}$ and the five-radius estimate hold. Each root counted by Y_j lies in a component of $\{|f_N| < a\}$ meeting one of the circles. Hence it is either a good annular root, which lies in the band $||\alpha| - 1| < N^{-1-\kappa_*}$ by the strict inequality, or it is counted by U_N . Therefore, for $j \geq j_0(K)$ and outside $E_{K,j}(\kappa_*)$, by the band count and (36) we have

$$\begin{aligned} \frac{Y_j}{N} &\leq \frac{1}{N} \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| < N^{-1-\kappa_*}\} + \frac{U_N}{N} \\ &\leq 4N^{-\kappa_*} + 2C(2)N^{\kappa_*-1/16}(\log N)^7 \end{aligned}$$

$$+ \frac{2 \log 2}{K} + C(K)N^{-1/16}(\log N)^7 + C(K)N^{-1/16} \log N.$$

To apply Lemma B.7, which concerns normalized block increments, with the exponent 8, so that $n_j = j^8$, we take $X_n = 0$ when n is an eighth power and $X_n = Y_j$ when $j^8 < n < (j+1)^8$. Since $X_{n_j} = 0$ and each block contains more than one degree, we have $\max_{n_j \leq n < n_{j+1}} |X_n - X_{n_j}|/n_j = Y_j/N$. For each integer $K \geq 2$ we take the index of the lemma to be $K-1$, which runs over the positive integers, and put

$$\begin{aligned} b_{K-1}(j) &= 4N^{-\kappa_*} + 2C(2)N^{\kappa_*-1/16}(\log N)^7 \\ &\quad + C(K)N^{-1/16}(\log N)^7 + C(K)N^{-1/16} \log N, \end{aligned}$$

and let $p_{K-1}(j)$ be the bound (42) for $\mathbb{P}(E_{K,j}(\kappa_*))$ when $j \geq j_0(K)$, and $p_{K-1}(j) = 1$ when $j < j_0(K)$. By the last display, $\mathbb{P}\{Y_j/N > 2 \log 2/K + b_{K-1}(j)\} \leq p_{K-1}(j)$ for every j . Moreover $2 \log 2/K \rightarrow 0$ as $K \rightarrow \infty$, and, for fixed K , $b_{K-1}(j) \rightarrow 0$ as $j \rightarrow \infty$, since $\kappa_* > 0$ and $\kappa_* - 1/16 \leq -7/128$. Also $\sum_j p_{K-1}(j) < \infty$, since only finitely many terms equal one and the others are summable by Step 6 of the proof of Theorem 1.1. These are the hypotheses of Lemma B.7 at the index $K-1$, with $2 \log 2/K$ as the term that tends to zero as the index grows, and the lemma gives deterministic numbers $\varepsilon_j \rightarrow 0$ and events F_j , the events E_j of the lemma, with $\sum_j \mathbb{P}(F_j) < \infty$ such that $Y_j \leq \varepsilon_j N$ off F_j . This proves the conclusion for both signs simultaneously. $\square$

8.4 Polynomially varying deterministic coefficients

Let $(\xi_k)_{k \geq 0}$ be random variables on one probability space, let (c_k) be deterministic complex weights, and consider the prefixes

$$f_N(z) = \sum_{k=0}^N \xi_k c_k z^k.$$

Every degree uses the same coefficient sequence, and for independent centered coefficients with unit variance the normalization is the square root of $\sum_{k=0}^N |c_k|^2 r^{2k}$.

Proposition 8.3 (Polynomially weighted logarithmic variance). *Fix $\tau > -1/2$, put $c_k = k^\tau$ for $k \geq 1$, and choose a fixed finite $c_0 \neq 0$. For the normalization*

$$\sigma_{N,\tau}(r)^2 = |c_0|^2 + \sum_{k=1}^N k^{2\tau} r^{2k},$$

we have

$$\log \sigma_{N,\tau}(1 + x/N) - (\tau + 1/2) \log N \longrightarrow F_\tau(x) := \frac{1}{2} \log \int_0^1 t^{2\tau} e^{2xt} dt$$

uniformly on compact intervals of x . Moreover,

$$F'_\tau(x) = \frac{\int_0^1 t^{2\tau+1} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt}, \quad F'_\tau(0) = \frac{2\tau+1}{2\tau+2}.$$

The four-secant formula in Lemma 2.1 applies, where $F_{N,\tau}(x) = \log \sigma_{N,\tau}(1 + x/N) - (\tau + 1/2) \log N$. With the centering $\frac{1}{2} \log N$, we add $\tau \log N$ to $F_{N,\tau}$, a term that cancels in every secant. Both the convention at $k = 0$ and the restriction $\tau > -1/2$ are needed for the integral profile.

Proof. Since every compact interval of x lies in some $[-K, K]$, we fix $K > 0$ and work with $|x| \leq K$. The idea is to show that $\sigma_{N,\tau}(1 + x/N)^2 / N^{2\tau+1}$ converges to $\int_0^1 t^{2\tau} e^{2xt} dt$ uniformly for $|x| \leq K$, and then to take logarithms. For $|x| \leq K$ and $0 < t \leq 1$, let

$$g_x(t) = t^{2\tau} e^{2xt}, \quad I_\tau(x) = \int_0^1 g_x(t) dt = \int_0^1 t^{2\tau} e^{2xt} dt.$$

For the weighted sum, we have

$$S_{N,\tau}(x) := \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} e^{2xk/N} = \frac{1}{N} \sum_{k=1}^N (k/N)^{2\tau} e^{2xk/N} = \frac{1}{N} \sum_{k=1}^N g_x(k/N).$$

Step 1: the radial factor. Let $|x| \leq K$ and $N \geq 2K$. Multiplying the comparison (6) by $k^{2\tau} e^{2xk/N} / N^{2\tau+1} > 0$ and summing over $1 \leq k \leq N$, we get, by the definition of $S_{N,\tau}$,

$$e^{-2K^2/N} S_{N,\tau}(x) \leq \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} (1 + x/N)^{2k} \leq e^{2K^2/N} S_{N,\tau}(x).$$

Step 2: the Riemann sums for $\tau \geq 0$. When $\tau \geq 0$, the integrand is continuous, since the map $(x, t) \mapsto g_x(t)$ extends continuously to the compact set $[-K, K] \times [0, 1]$, and hence it is uniformly continuous there. Let $\eta(\delta)$ be the supremum of $|g_x(t) - g_x(t')|$ over $|x| \leq K$ and $t, t' \in [0, 1]$ with $|t - t'| \leq \delta$, so that $\eta(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. For $|x| \leq K$ we have

$$\begin{aligned} |S_{N,\tau}(x) - I_\tau(x)| &= \left| \sum_{k=1}^N \frac{1}{N} g_x(k/N) - \sum_{k=1}^N \int_{(k-1)/N}^{k/N} g_x(t) dt \right| \\ &= \left| \sum_{k=1}^N \int_{(k-1)/N}^{k/N} (g_x(k/N) - g_x(t)) dt \right| \\ &\leq \sum_{k=1}^N \int_{(k-1)/N}^{k/N} |g_x(k/N) - g_x(t)| dt \leq \eta(1/N), \end{aligned}$$

where the last step uses $|k/N - t| \leq 1/N$ on the k -th interval, whose lengths add up to one.

Step 3: the Riemann sums for $-1/2 < \tau < 0$. When $-1/2 < \tau < 0$, we fix $0 < \varepsilon < 1$ and split the sum at $k_\varepsilon = \lfloor \varepsilon N \rfloor$, where $N \geq 2/\varepsilon$, so that $2 \leq k_\varepsilon < N$. Then

$$\begin{aligned} S_{N,\tau}(x) - I_\tau(x) &= \left[\frac{1}{N} \sum_{k=1}^{k_\varepsilon} g_x(k/N) - \int_0^{k_\varepsilon/N} g_x(t) dt \right] \\ &\quad + \left[\frac{1}{N} \sum_{k=k_\varepsilon+1}^N g_x(k/N) - \int_{k_\varepsilon/N}^1 g_x(t) dt \right]. \end{aligned}$$

Since $t \mapsto t^{2\tau}$ is decreasing, we have $\frac{1}{N}(k/N)^{2\tau} \leq \int_{(k-1)/N}^{k/N} t^{2\tau} dt$ for $k \geq 2$, and the term $k=1$ is $\frac{1}{N}(1/N)^{2\tau} = N^{-(2\tau+1)}$. Also $e^{2xt} \leq e^{2K}$ for $|x| \leq K$ and $0 \leq t \leq 1$, and $k_\varepsilon/N \leq \varepsilon$. Hence the normalized initial part satisfies

$$\begin{aligned} 0 \leq \frac{1}{N} \sum_{k=1}^{k_\varepsilon} g_x(k/N) &\leq \frac{e^{2K}}{N} \sum_{k=1}^{k_\varepsilon} (k/N)^{2\tau} = e^{2K} \left(N^{-(2\tau+1)} + \sum_{k=2}^{k_\varepsilon} \frac{1}{N} (k/N)^{2\tau} \right) \\ &\leq e^{2K} \left(N^{-(2\tau+1)} + \int_{1/N}^{k_\varepsilon/N} t^{2\tau} dt \right) \leq e^{2K} \left(\frac{\varepsilon^{2\tau+1}}{2\tau+1} + N^{-(2\tau+1)} \right), \end{aligned}$$

where the intervals $[(k-1)/N, k/N]$ with $2 \leq k \leq k_\varepsilon$ make up $[1/N, k_\varepsilon/N]$. By the same two facts, the corresponding integral satisfies

$$0 \leq \int_0^{k_\varepsilon/N} g_x(t) dt \leq e^{2K} \int_0^\varepsilon t^{2\tau} dt = e^{2K} \frac{\varepsilon^{2\tau+1}}{2\tau+1}.$$

Since both initial parts are nonnegative, the first bracket is at most $e^{2K}(\varepsilon^{2\tau+1}/(2\tau+1) + N^{-(2\tau+1)})$ in absolute value. For the remaining Riemann sum, every interval $[(k-1)/N, k/N]$ with $k \geq k_\varepsilon + 1$ lies in $[\varepsilon/2, 1]$, since $k_\varepsilon/N \geq \varepsilon - 1/N \geq \varepsilon/2$. On the compact set $[-K, K] \times [\varepsilon/2, 1]$ the map $(x, t) \mapsto g_x(t)$ is uniformly continuous. Let $\eta_\varepsilon(\delta)$ be the supremum of $|g_x(t) - g_x(t')|$ over $|x| \leq K$ and $t, t' \in [\varepsilon/2, 1]$ with $|t - t'| \leq \delta$, so that $\eta_\varepsilon(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. Then the second bracket satisfies

$$\begin{aligned} \left| \frac{1}{N} \sum_{k=k_\varepsilon+1}^N g_x(k/N) - \int_{k_\varepsilon/N}^1 g_x(t) dt \right| &= \left| \sum_{k=k_\varepsilon+1}^N \int_{(k-1)/N}^{k/N} (g_x(k/N) - g_x(t)) dt \right| \\ &\leq \sum_{k=k_\varepsilon+1}^N \int_{(k-1)/N}^{k/N} |g_x(k/N) - g_x(t)| dt \leq \eta_\varepsilon(1/N), \end{aligned}$$

where the last step holds as in Step 2, since k/N and t lie in $[\varepsilon/2, 1]$. Therefore

$$\sup_{|x| \leq K} |S_{N,\tau}(x) - I_\tau(x)| \leq e^{2K} \left(\frac{\varepsilon^{2\tau+1}}{2\tau+1} + N^{-(2\tau+1)} \right) + \eta_\varepsilon(1/N).$$

Since $2\tau + 1 > 0$, the right-hand side tends to $e^{2K}\varepsilon^{2\tau+1}/(2\tau + 1)$ as $N \rightarrow \infty$, so that the limit superior of the left side as $N \rightarrow \infty$ is at most this value for every $0 < \varepsilon < 1$, and it is therefore zero.

We conclude that in both cases $S_{N,\tau}(x) \rightarrow I_\tau(x)$ uniformly for $|x| \leq K$.

Step 4: the variance. Let $|x| \leq K$ and $N \geq 2K$. By the definition of $\sigma_{N,\tau}$ and Step 1, we have

$$\begin{aligned} \left| \frac{\sigma_{N,\tau}(1+x/N)^2}{N^{2\tau+1}} - I_\tau(x) \right| &\leq \frac{|c_0|^2}{N^{2\tau+1}} + \left| \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} (1+x/N)^{2k} - S_{N,\tau}(x) \right| \\ &\quad + |S_{N,\tau}(x) - I_\tau(x)| \\ &\leq \frac{|c_0|^2}{N^{2\tau+1}} + (e^{2K^2/N} - 1)S_{N,\tau}(x) + |S_{N,\tau}(x) - I_\tau(x)| \\ &\leq \frac{|c_0|^2}{N^{2\tau+1}} + (e^{2K^2/N} - 1) \left(\frac{e^{2K}}{2\tau + 1} + |S_{N,\tau}(x) - I_\tau(x)| \right) \\ &\quad + |S_{N,\tau}(x) - I_\tau(x)|, \end{aligned}$$

where the second inequality also uses $1 - e^{-2K^2/N} \leq e^{2K^2/N} - 1$, which holds since $e^u + e^{-u} \geq 2$, and the last uses $I_\tau(x) \leq e^{2K} \int_0^1 t^{2\tau} dt = e^{2K}/(2\tau + 1)$. Since $2\tau + 1 > 0$, the term with the fixed c_0 also vanishes after normalization, and by Steps 2 and 3 the other terms tend to zero uniformly for $|x| \leq K$, since $e^{2K^2/N} - 1 \rightarrow 0$. Thus $\sigma_{N,\tau}(1+x/N)^2/N^{2\tau+1} \rightarrow \int_0^1 t^{2\tau} e^{2xt} dt$ uniformly for $|x| \leq K$.

Step 5: the logarithm. For $|x| \leq K$, the integral is at least $e^{-2K}/(2\tau + 1) > 0$, since $I_\tau(x) \geq e^{-2K} \int_0^1 t^{2\tau} dt$. By Step 1 we have

$$\begin{aligned} \frac{\sigma_{N,\tau}(1+x/N)^2}{N^{2\tau+1}} &\geq \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} (1+x/N)^{2k} \geq e^{-2K^2/N} S_{N,\tau}(x) \\ &\geq e^{-2K^2/N} (I_\tau(x) - |S_{N,\tau}(x) - I_\tau(x)|) \\ &\geq e^{-2K^2/N} \left(\frac{e^{-2K}}{2\tau + 1} - |S_{N,\tau}(x) - I_\tau(x)| \right), \end{aligned}$$

where the first inequality drops $|c_0|^2/N^{2\tau+1} \geq 0$. Also $I_\tau(x)$ is at least the same lower bound, since $e^{-2K^2/N} \leq 1$. Since $|\log a - \log b| \leq |a - b|/\min(a, b)$ for $a, b > 0$, we get, for all N so large that this lower bound is positive for $|x| \leq K$,

$$\begin{aligned} &|\log \sigma_{N,\tau}(1+x/N) - (\tau + 1/2) \log N - F_\tau(x)| \\ &= \frac{1}{2} \left| \log \frac{\sigma_{N,\tau}(1+x/N)^2}{N^{2\tau+1}} - \log I_\tau(x) \right| \\ &\leq \frac{| \sigma_{N,\tau}(1+x/N)^2/N^{2\tau+1} - I_\tau(x) |}{2e^{-2K^2/N} (e^{-2K}/(2\tau + 1) - |S_{N,\tau}(x) - I_\tau(x)|)}, \end{aligned}$$

since $F_\tau = \frac{1}{2} \log I_\tau$ and both arguments of the logarithms are at least the lower bound above. By Steps 2 to 4, the numerator tends to zero and the denominator tends to $2e^{-2K}/(2\tau + 1)$, both uniformly for $|x| \leq K$. Hence the left side tends to zero uniformly for $|x| \leq K$, which proves the profile limit of the proposition.

By differentiating the positive integral, we get

$$\frac{d}{dx} \left(\frac{1}{2} \log \int_0^1 t^{2\tau} e^{2xt} dt \right) = \frac{1}{2} \cdot \frac{\int_0^1 2t \cdot t^{2\tau} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt} = \frac{\int_0^1 t^{2\tau+1} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt},$$

where differentiation under the integral sign is allowed since, for $|x| \leq K$, the x -derivative $2t^{2\tau+1}e^{2xt}$ of the integrand is at most $2e^{2K}t^{2\tau}$, which is integrable on $[0, 1]$ since $2\tau > -1$. At zero this quotient is $\int_0^1 t^{2\tau+1} dt / \int_0^1 t^{2\tau} dt = (2\tau + 1)/(2\tau + 2)$. $\square$

Remark 8.4. For $\tau \leq -1/2$, the integral defining F_τ diverges at zero. The finite-degree radial secant estimate still applies with the exact normalization. To deduce an almost-sure root law from a variance profile, one must also establish the occupation and logarithmic hypotheses and the degree-block estimates.

The integral diverges since its integrand is at least $e^{-2|x|}t^{2\tau}$ and $2\tau \leq -1$. The finite-degree radial secant estimate of the remark is (2) of Lemma 2.1, which allows any normalization $\sigma_N(r) > 0$, and in particular $\sigma_{N,\tau}$.

Proposition 8.5 (Weighted polynomials with random signs). *Take $c_0 = 1$ and $c_k = k^\tau$ for $k \geq 1$, where $\tau > -1/2$. For independent signs put*

$$f_N^{(\tau)}(z) = \sum_{k=0}^N \epsilon_k c_k z^k, \quad X_r(\theta) = \frac{f_N^{(\tau)}(re^{i\theta})}{\sigma_{N,\tau}(r)}.$$

The uniform-on-compact variance limit in Proposition 8.3 holds for these polynomials. Its candidate radial profile is

$$\Phi_\tau(x) = F'_\tau(x) = \frac{\int_0^1 t^{2\tau+1} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt}, \quad \Phi_\tau(0) = \frac{2\tau + 1}{2\tau + 2}.$$

The normalized logarithmic moments and angular energy satisfy

$$\mathbb{E} \int |\log |X_r(\theta)||^p d\mu(\theta) \leq (C_* p)^{6p}, \quad p > 1, \quad \int |X_r(\theta)|^2 d\mu(\theta) = 1.$$

The energy identity holds for every sign vector.

Proof. The idea is that X_r is a Rademacher Fourier series whose deterministic coefficients $c_k r^k / \sigma_{N,\tau}(r)$ have squared moduli summing to one, which is all that the logarithmic bound and the energy identity of Section 3.2 use. The variance

convergence follows from the preceding proposition. By the definition of $\sigma_{N,\tau}$ with $c_0 = 1$, the deterministic Fourier coefficients satisfy

$$\sum_{k=0}^N \left| \frac{c_k r^k}{\sigma_{N,\tau}(r)} \right|^2 = \frac{1 + \sum_{k=1}^N k^{2\tau} r^{2k}}{\sigma_{N,\tau}(r)^2} = 1.$$

By Corollary 1.2 of Nazarov–Nishry–Sodin, the above logarithmic bound holds with the same absolute constant C_* . Since that corollary holds for every $p \geq 1$, the logarithmic bound also holds for $p = 1$, as in (17). By Parseval's identity, $\epsilon_k^2 = 1$ and the preceding identity, we obtain for every sign vector the energy identity

$$\int |X_r(\theta)|^2 d\mu(\theta) = \frac{1}{\sigma_{N,\tau}(r)^2} \sum_{k=0}^N \epsilon_k^2 c_k^2 r^{2k} = 1.$$

□

The conclusions of Proposition 8.5 hold for any fixed finite nonzero c_0 , with $|c_0|^2$ in place of 1 in the variance.

9 Open problems

The proof uses the coefficient distribution only through (H1)–(H5), bounds its errors by low moments, and sees the zeros only at the scale $1/n$. This suggests three problems: the weakest moment condition, the size of the fluctuations, and the local statistics of the zeros.

Problem 9.1 (The moment threshold). *Let $(\xi_k)_{k \geq 0}$ be independent and identically distributed complex random variables, not almost surely constant, and let f_n and ν_n be as in Theorem 1.1. Does the conclusion of Theorem 1.1 hold whenever $\mathbb{E}|\xi_0|^2 < \infty$?*

Theorem 1.1 establishes the law for five families of coefficient distributions, and our proof uses the distribution only through the hypotheses (H1)–(H5) of Section 7, which hold for signs by Sections 3 and 4 and which Proposition 7.4 verifies for the other four families. The second moment cannot be weakened. If $\xi_n \neq 0$, Jensen's formula for $w^n f_n(1/w)$ on the circle $|w| = e^{1/n}$ and the concavity of the logarithm give

$$n - \nu_n(1) \leq \frac{n}{2} \log \left(1 + e^2 \sum_{k < n} \frac{|\xi_k|^2}{|\xi_n|^2} \right).$$

When $\mathbb{E}|\xi_0|^2 = \infty$, the ratio $|\xi_n|^2 / \sum_{k < n} |\xi_k|^2$ is unbounded almost surely, by a theorem of Kesten [15], and hence $\limsup_n \nu_n(1)/n = 1$. At a fixed degree, with

mean zero and a finite second moment, the zeros near each point of the unit circle converge in distribution at the scale $1/n$ [12], and equidistribution of the zeros on the unit circle needs only a logarithmic moment [11].

Problem 9.2 (Fluctuations). *For Rademacher coefficients, is $(\nu_n(1) - n/2)/\sqrt{n}$ asymptotically normal with a positive variance? Is*

$$\limsup_{n \rightarrow \infty} \frac{\nu_n(1) - n/2}{\sqrt{n \log \log n}}$$

a positive and finite constant almost surely?

Theorem 1.3 gives $\nu_n(1) - n/2 = O(n/\log n)$ almost surely. Positive answers would identify the exact almost-sure order $\sqrt{n \log \log n}$, as in the law of the iterated logarithm for sums of independent random variables. Along the nested sequence, the increments $\nu_{n+m}(1) - \nu_n(1)$ measure the flow of zeros through the unit circle as coefficients are appended, and their size is part of the question.

Problem 9.3 (Almost-sure local universality). *Let the coefficients be as in Theorem 1.1, with $\mathbb{E}|\xi_0|^2 = 1$, and let $G(\zeta) = \int_0^1 e^{t\zeta} dW(t)$, where W is a complex Brownian motion with $\mathbb{E}|W(t)|^2 = t$. Almost surely, do the correlation functions of the zeros of $\zeta \mapsto f_n(e^{i\theta}(1 + \zeta/n))$, averaged over θ , converge to those of the zeros of G ?*

The zeros of G have intensity $\Phi'(\operatorname{Re} \zeta)/(2\pi)$, and Theorem 1.1 is the one-point case of the problem. At a fixed degree, local universality of the zeros near the unit circle was proved by Tao and Vu [21], and in distribution under a finite second moment by Kabluchko, Khoruzhenko and Marynych [12]. The problem asks for it almost surely, at every degree of one nested sequence. In the proof of Theorem 1.1, root matching (Theorem 5.1) keeps each selected zero of f_N near the unit circle within $N^{-65/64} = o(1/N)$ of a zero of f_n , at every degree n of its block, so that local configurations at the sparse degrees persist at every degree.

10 Numerical methods

The figures show the radial law and the steps of its proof at finite degrees, for nested sequences of random signs. This section records how the roots and statistics behind them were computed, and how sensitive they are to numerical error. The limits that hold with probability one follow from the proofs in the previous sections, and these proofs do not use the computations below.

We use 100 independent sequences of random signs at the degrees $N = 1000$, 3000, 10000, 30000 and 100000, and the block length is $\lfloor N^{7/8} \rfloor$. We use the 77 sequences whose roots were computed at all five base degrees, all 421 intermediate degrees for 78 sequences at $N = 1000$, and the endpoint samples listed in Table 9.

N	Base samples	Endpoint samples
1000	92	78
3000	92	58
10000	92	57
30000	91	53
100000	77	27

Table 9: Completed base samples and endpoint samples, at the degree $N + \lfloor N^{7/8} \rfloor$. The $N = 1000$ samples include every intermediate degree.

The roots were computed with MPSolve [3], or, at the degrees 1210 and 1421 of Figure 9, with a companion-matrix eigensolver, and then refined by Newton’s method. The residuals were checked by independent evaluation, and the normalized residual $|f_N(z)| / \sum_{k=0}^N |z|^k$ is below 3×10^{-14} at every degree.

We determine the roots at 1 and -1 , with their multiplicities, by exact integer division. For the remaining roots, the band $||z| - 1| \leq 10^{-9}$ yields a sensitivity interval, whose two endpoints count every root in that band as outside or inside the disk. We check the counting procedure by direct evaluation and on polynomials with known boundary roots. Over all base samples, the numbers of the remaining roots in this band are 0, 0, 2, 28, 284, in increasing degree order, and they belong to 0, 0, 1, 12, 61 polynomials, respectively. Thus the largest unnormalized sensitivity intervals have widths 0, 0, 2, 4, 14.

We compute the angular occupations and the logarithmic integral on a shifted FFT grid. Doubling the grid changes the logarithmic average by at most 4×10^{-4} and the occupation by at most 2×10^{-3} .

The profile and root-count comparisons use the 77 sequences above, and the angular comparisons use all 100. The residuals and grid comparisons quantify numerical sensitivity rather than interval certification. The bands and intervals in the figures are pointwise 95% percentile intervals from 2000 bootstrap draws of whole sequences. Figure 10 compares the empirical profile $\hat{\Phi}_N(x) = \nu_N(1 + x/N)/N$ with $\Phi(x)$. Log-log fits for each statistic, which describe only the measured range, are in `numerics/publication/STATISTICS.md` of the repository [14].

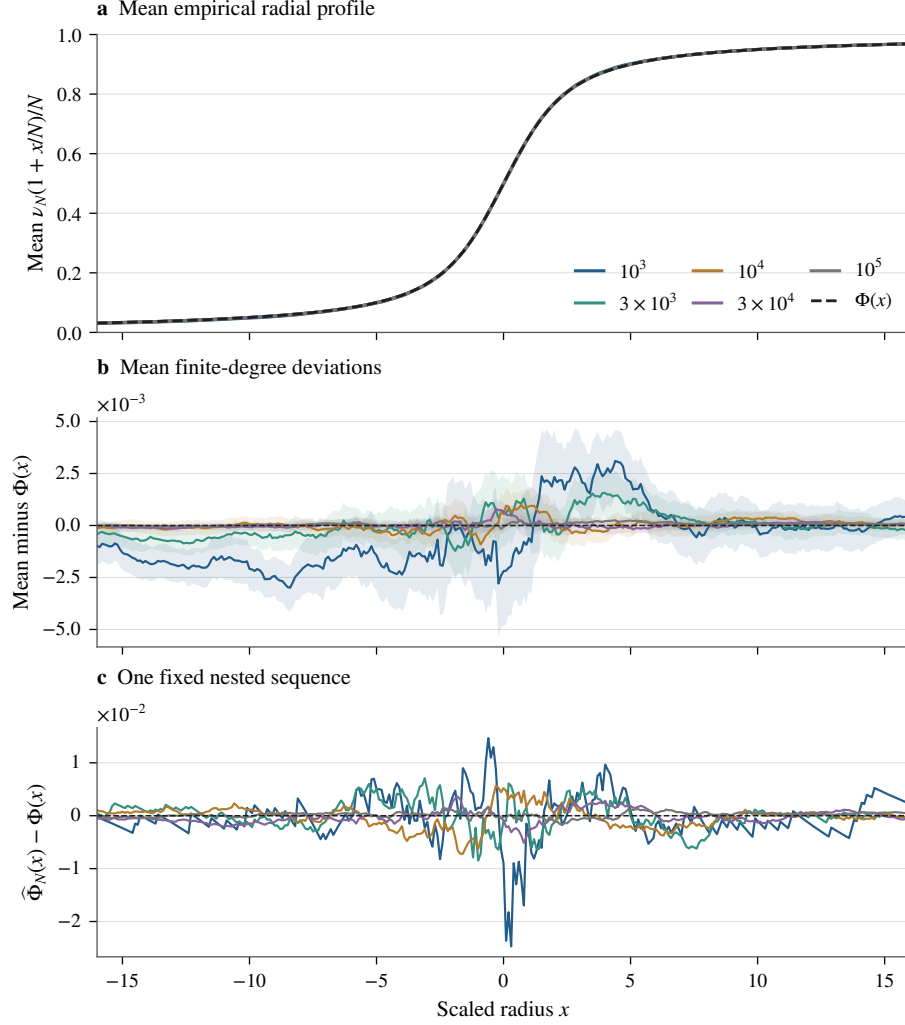


Figure 10: Mean empirical radial profile over 77 sequences at five degrees, which cannot be distinguished from Φ (a), and its deviations (b, c).

A Finite constants

The main text writes most constants as $C(K)$ or $O_K(\cdot)$. Here we make them explicit: each is an absolute number times a power of $K + 1$ and a factor e^{bK} , which gives explicit degree thresholds and lets Section 8.1 take the width K of order $\log N$. Let C_* be the positive universal constant in Nazarov–Nishry–Sodin, Corollary 1.2, and let $c_{\text{HW}} > 0$ be the constant in Rudelson–Vershynin, Theorem 1.1. We keep the dependence on both constants in all the bounds below. We choose, for fixed parameters, the least index beyond which every displayed inequality holds, so that all degree cutoffs are deterministic. Table 10 lists the degree thresholds and the

exceptional probability of each result of the main text, and Table 11 lists the named constants.

Result	Degree thresholds	Exceptional probability
A.1, Lemma 2.3	$N \geq 2K$	none
A.2, Theorem 3.1	$K \geq 2$ and $N \geq (160e^{8K})^2$, uniformly for $r \in [1 - K/N, 1 + K/N]$	$p_K^*(N)$
A.3, Corollary 3.3	those of A.2 and $N \geq 2K$	$4p_K^*(N)$ for (57), $5p_K^*(N)$ for (58), $5p_2^*(N)$ for its form at $K = 2$
Lemma 4.2	$t_N \leq 1$ and $N \geq e$	N^{-3}
Lemma 4.3	$N \geq \max\{2, (K + 1)/3\}$	N^{-10}
A.4, Lemma 4.5	$N \geq (6400e^{8K})^2$, and those of Lemma 4.2 and (63)	the right-hand side of (62)
Lemma 4.6	$j \geq 60$ and $N \geq \max\{4, K\}$	N^{-10}
Lemma 5.4	$4aL/N^{3/2} \leq 1/N$ and $4S_K aL^2 N^{-1/2} \sqrt{\log N} \leq 1$, that is, (63)	none
A.5, Section 6	$R \geq x + 2$, $K \geq R$, $N \geq 2R$ and $N \geq (160e^{8R})^2$, which implies $N > R$, those of the second to eighth rows at the width K , where Lemma 4.6 enters with $K + 2$ in place of K (so that $N \geq \max\{4, K + 2\}$), and $j \geq 60$	(67), with $R = 2$ and $x = 0$ for Corollary 1.2
A.7, Section 8.1	the conditions of Table 15, $j \geq 60$, $K \geq 2$, $A \log N/N \leq 1$ and $N \geq K + 2$	(67), with $K = K_N$, $R = 2$ and $x = 0$
A.8, Corollary 8.2	$0 < \kappa_* < \min(\kappa, 1/64)$ and $N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*}$	(67), with $R = 2$ and the thin-band events of A.3

Table 10: The degree thresholds and exceptional probabilities of the results of the main text, with the subsection of Appendix A, where there is one, that treats each. For other coefficient distributions, Section A.6 enlarges the constants and adds exceptional probabilities.

In Table 11, A , $C_1(K)$, S_K and λ_K are the constants of Lemmas 4.2 and 4.3 and of (26), where S_K is written $C(K)$ and λ_K is written λ . The law constants A_B , C_B , $C_{1,G}$ and $C_{1,B}$ are in Tables 12–14, and $A(B, \eta)$ and $C_1(K, B, \eta)$ in Lemma 4.1. The symbols A , C_0 and D_0 in Corollary B.5, B_0 in Section 7, and E_1, E_2 in Appendix B have other meanings.

Table 11: The named constants. Each is at most $C_0(K+1)^p e^{bK}$ with the listed (C_0, p, b) , and the absolute constants have $p = b = 0$.

Constant	Value	(C_0, p, b)	Defined	Used
D_K	$10(K+1)^2 e^{4K}$, and $D_2 = 90e^8$	$(10, 2, 4)$	A.1	A.3, A.5, A.7, 8.1, Table 8
H_K	$10^8 e^{20(K+1)}$, and $H_2 = 10^8 e^{60}$	$(H_0, 0, 20)$	A.2	A.3, A.6, A.7, Table 8
$C_1(K)$	$(4A + 4K + 4)/\log 2$, and $C_1(K+2) =$ $(4A + 4K + 12)/\log 2$	$(C_{10}, 1, 0)$ for $C_1(K+2)$	Lemma 4.2	4.2, 7, 8.1, A.4, A.6, A.7
S_K	$100e^{K+4}$	$(S_0, 0, 1)$	Lemma 4.3	A.4, A.5, A.6, A.7
λ_K	$e^{-4K}/80$	$(1/80, 0, -4)$	(26)	A.4, A.6, A.7
D'_K	$2^{18}(K+1)S_K^2$	$(D_0, 1, 2)$	A.4	A.6, A.7, Proposition 8.1
$\mathcal{B}_K$	$2000e^{3K}(2/\lambda_K)^{3/2}$	$(B_0, 0, 9)$	A.4	A.6, A.7, Proposition 8.1
$\mathcal{T}_K$	$80e^{4K}/\lambda_K = 6400e^{8K}$	$(6400, 0, 8)$	A.4	A.6, A.7, Proposition 8.1
V_K	$50 + 3\mathcal{B}_K + \mathcal{T}_K$	$(V_0, 0, 9)$	A.4	A.6, A.7
c_K	$9/(1024S_K^2\lambda_K^2)$	$(Q_0, 0, 6)$	A.4	A.6, A.7
C_K	$V_K(D'_K)^2$	$(C_{\text{bad}}, 2, 13)$	A.4	(62), (67), A.7, 8.1, Table 8
Z_K	$C_1(K+2)(1 + D'_K c_K)$	$(Z_0, 2, 8)$	A.4	(62), A.5, A.7, 8.1, Table 8
Z'_K	$C_1(K+2)(40 + D'_K \mathcal{B}_K)$	$(Z'_0, 2, 11)$	A.4	(62), A.5, A.7, 8.1, Table 8
C_*	constant of (17) , $C_* \geq 1$		(17)	A.2, E₁, E₂, A.6
c_{HW}	constant of (23)		(23)	A, A.6
A	$\max(1, 1024\pi/c_{\text{HW}})$		Lemma 4.2	$C_1(K)$, C_{10} , 8.1, A.7
E_1	$\frac{5}{2} + 2e^{1/2}(4eC_*)^6$		(56)	E_0
E_2	$\frac{1}{16} + e^{1/2}(4eC_*)^6$		(56)	E_0
H_0	$10^8 e^{20}$		A.7	E_0 , (68) , 8.1
S_0	$100e^4$		A.7	D_0, Q_0 , Table 15, 8.1
D_0	$2^{18}S_0^2$		A.7	$C_{\text{bad}}, Z_0, Z'_0$
B_0	$2000 \cdot 160^{3/2}$		A.7	V_0, Z'_0
V_0	$50 + 3B_0 + 6400$		A.7	C_{bad}
Q_0	$9 \cdot 80^2/(1024S_0^2)$		A.7	Z_0
C_{10}	$(4A + 12)/\log 2$		A.7	Z_0, Z'_0
C_{bad}	$V_0 D_0^2$		A.7	(68) , 8.1
Z_0	$C_{10}(1 + D_0 Q_0)$		A.7	(68) , 8.1
Z'_0	$C_{10}(40 + D_0 B_0)$		A.7	(68) , 8.1
E_0	$E_1 + E_2 H_0$		A.7	(68) , 8.1

A.1 The variance profile

For the constant of Lemma 2.3 we set, for an integer $K \geq 2$,

$$D_K = 10(K+1)^2 e^{4K}.$$

For $N \geq 2K$ and $|x| \leq K$, Lemma 2.3 and $1 \leq 9(K+1)^2$ give

$$\begin{aligned} \left| \log \sigma_N(1+x/N) - \frac{1}{2} \log N - F(x) \right| &\leq \frac{K^2 + e^{4K}/2}{N} \leq \frac{(K+1)^2 e^{4K} + e^{4K}}{N} \\ &\leq \frac{10(K+1)^2 e^{4K}}{N} = \frac{D_K}{N}. \end{aligned}$$

A.2 Concentration of the angular logarithm

The one- and two-point bounds give the occupation constant H_K of Theorem 3.1, and Lemma 3.2 gives its error $\varepsilon_K^*(N)$ and exceptional probability $p_K^*(N)$. Let $K \geq 2$ be an integer, set

$$H_K = 10^8 e^{20(K+1)},$$

and recall that $T = (\log N)/32$ is the clipping level of Lemma 3.2. The proof of Theorem 3.1 uses $N \geq 2K$, in (11) and (12), $N \geq (160e^{8K})^2$, so that $40e^{8K}N^{-1/2} \leq 1/4$ for the covariance matrices of Section 3.1, and $N \geq e$, in Lemma 3.2. All three hold for $N \geq (160e^{8K})^2$, since $(160e^{8K})^2 = 25600e^{16K}$ exceeds $2K$ and e .

Let $K \geq 2$, $N \geq (160e^{8K})^2$ and $r \in [1 - K/N, 1 + K/N]$. In (15), the line before the last is $N^{-1/2}$ times $(42d^{1/4} + 16) \cdot 46e^{9K} + 320e^{8K}$, where the dimension is $d \in \{2, 4\}$, so that $d^{1/4} \leq 4^{1/4}$. Since $42 \cdot 4^{1/4} + 16 = 42\sqrt{2} + 16 < 75.4$, $e^{8K} = e^{-K}e^{9K} \leq e^{-2}e^{9K}$, $75.4 \cdot 46 = 3468.4$ and $320e^{-2} < 44$, we have

$$\begin{aligned} (42 \cdot 4^{1/4} + 16) \cdot 46e^{9K} + 320e^{8K} &< 75.4 \cdot 46e^{9K} + 320e^{-2}e^{9K} \\ &< 3469e^{9K} + 44e^{9K} < 4000e^{9K}. \end{aligned}$$

Hence the one- and two-point differences in (15) are at most $4000e^{9K}N^{-1/2}$ on the retained angles and pairs. As in (16), where the removed sets have measures at most $2N^{-1/2}$ and $10N^{-1/2}$, we get, uniformly for $t \geq 0$,

$$|\mathbb{E}A_t - F_G(t)| \leq (2 + 4000e^{9K})N^{-1/2}, \quad |\mathbb{E}A_t^2 - F_G(t)^2| \leq (10 + 4000e^{9K})N^{-1/2}.$$

Since $12000e^{9K} + 14 \leq 12014e^{9K} < 10^8 e^{20(K+1)} = H_K$ and $F_G(t) = 1 - e^{-t^2}$, these bounds, the variance identity of (16) and $0 \leq \mathbb{E}A_t + F_G(t) \leq 2$ give

$$\begin{aligned} |\mathbb{E}A_t - (1 - e^{-t^2})| &\leq (2 + 4000e^{9K})N^{-1/2} \leq H_K N^{-1/2}, \\ \text{Var}(A_t) &\leq |\mathbb{E}A_t^2 - F_G(t)^2| + 2|\mathbb{E}A_t - F_G(t)| \end{aligned}$$

$$\begin{aligned}
&\leq (10 + 4000e^{9K})N^{-1/2} + 2(2 + 4000e^{9K})N^{-1/2} \\
(54) \quad &= (14 + 12000e^{9K})N^{-1/2} \leq H_K N^{-1/2}.
\end{aligned}$$

We take the deviation for the truncated logarithmic integral, the parameter u of Theorem B.1, to be $N^{-1/16}(\log N)^7$, and we set

$$\begin{aligned}
\varepsilon_K^*(N) &= N^{-1/16}(\log N)^7 + 2H_K T N^{-1/2} \\
&\quad + e^{1/2}(4eC_*)^6(2N^{-1/16} + H_K N^{-1/2})(\log N)^7 + \frac{3}{2}N^{-1/16}, \\
p_K^*(N) &= H_K N^{-3/8} + N^{-12} + 4H_K T^2 N^{-3/8}(\log N)^{-14}.
\end{aligned}$$

Then

$$\begin{aligned}
(55) \quad &\mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > \varepsilon_K^*(N)\} \leq p_K^*(N), \\
&\varepsilon_K^*(N) = O_K(N^{-1/16}(\log N)^7).
\end{aligned}$$

The first line of (55) is Lemma 3.2 with $H = H_K$, $C = C_*$, $E = \Omega$ and $B_E = 1$, which applies for $N \geq (160e^{8K})^2$ by (54), (17) and (18). Its error is $\varepsilon_K^*(N)$, since $B_E/2 + 1 = \frac{3}{2}$, and its probability bound is $p_K^*(N)$, since $4T^2 = (\log N)^2/256$. The second line follows from the bound on the error in Lemma 3.2, which at $C = C_*$ and $B_E = 1$ reads

$$\begin{aligned}
(56) \quad &\varepsilon_K^*(N) \leq E_1 N^{-1/16}(\log N)^7 + E_2 H_K N^{-1/2}(\log N)^7, \\
&E_1 = \frac{5}{2} + 2e^{1/2}(4eC_*)^6, \\
&E_2 = \frac{1}{16} + e^{1/2}(4eC_*)^6,
\end{aligned}$$

where E_1 and E_2 are absolute constants, since C_* is. Only the second term depends on K , and it carries the factor $N^{-1/2}$.

A.3 Annular mass and thin bands

Next we compute the constants of Corollary 3.3. As in its proof, the four secant radii $1 \pm K/N$ and $1 \pm K/(2N)$ lie in $[1 - K/N, 1 + K/N]$. Hence their variance errors are at most D_K/N for $N \geq 2K$, by Section A.1, and each logarithmic error is at most $\varepsilon_K^*(N)$, outside probability $p_K^*(N)$, for $N \geq (160e^{8K})^2$, by (55). Lemma 2.4 at $\eta = 1/2$, with $\varepsilon = \varepsilon_K^*(N)$ and $D = D_K$, then applies outside the four logarithmic exceptional events, whose union has probability at most $4p_K^*(N)$, and gives

$$(57) \quad \frac{\#\{\alpha : ||\alpha| - 1| > K/N\}}{N} \leq \frac{2 \log 2}{K} + \frac{8}{K}(\varepsilon_K^*(N) + D_K/N).$$

We use all five radii $1 + qN^{-1-\kappa}$, $q = -2, \dots, 2$, for $0 < \kappa \leq 1/64$, and $N \geq (160e^{8K})^2$ implies the condition $N \geq 4$ of Lemma 2.5. By that lemma with

$\varepsilon = \varepsilon_K^*(N)$, whose hypothesis holds outside the five events of (55) at these radii, for $q = -1, 0, 1$ the same sample satisfies

$$(58) \quad \left| \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa} + 4\varepsilon_K^*(N)N^\kappa$$

outside a union of probability at most $5p_K^*(N)$. The probabilities of the exceptional events are summable along $N = j^8$, since $4T^2(\log N)^{-14} = (\log N)^{-12}/256 \leq 1$ for $N \geq e$, so that the three terms of $p_K^*(j^8)$ are at most $H_K j^{-3}$, j^{-96} and $H_K j^{-3}$.

At $K = 2$ we have $H_2 = 10^8 e^{20(2+1)} = 10^8 e^{60}$, and $\varepsilon_2^*(N)$ and $p_2^*(N)$ are $\varepsilon_K^*(N)$ and $p_K^*(N)$ with $10^8 e^{60}$ in place of H_K , with the degree threshold $N \geq (160e^{16})^2$. Since the five radii lie in $[1 - 2/N, 1 + 2/N]$, (58) holds at $K = 2$, with $\varepsilon_2^*(N)$, outside probability $5p_2^*(N)$. This form is used in the proof of Corollary 8.2.

A.4 Few small derivatives

We now compute the constants of Lemma 4.5, following its proof at the annular width K , with the local-count and tail lemmas at the width $K + 2$.

Let $S_K = 100e^{K+4}$ be the constant of Lemma 4.3, so that (24) holds with $C(K) = S_K$, and let

$$C_1(K + 2) = \frac{4A + 4K + 12}{\log 2}$$

be the constant $C_1(K)$ of Lemma 4.2 at the width $K + 2$. We set

$$\begin{aligned} \lambda_K &= e^{-4K}/80, & D'_K &= 2^{18}(K + 1)S_K^2, \\ \mathcal{B}_K &= 2000e^{3K}(2/\lambda_K)^{3/2}, & \mathcal{T}_K &= 80e^{4K}/\lambda_K, \\ V_K &= 50 + 3\mathcal{B}_K + \mathcal{T}_K, & c_K &= \frac{9}{1024S_K^2\lambda_K^2}, \\ C_K &= V_K(D'_K)^2, & Z_K &= C_1(K + 2)(1 + D'_K c_K), \\ & & Z'_K &= C_1(K + 2)(40 + D'_K \mathcal{B}_K). \end{aligned}$$

Here $\mathcal{B}_K$ is the Raič bound in the proof of Lemma 4.5, which bounds the normal-approximation error and the one-point error, and $\mathcal{T}_K$ bounds the Gaussian comparison error.

In that proof, the mesh constant must be at most $1/(2S_K)$ for the bounds (25) and at most one for the mesh count. Both conditions hold for the value $1/(64S_K)$, since $S_K \cdot 1/(64S_K) = 1/64 \leq 1/2$ and $S_K \geq 1$. With $L = N^{1/64}$, let

$$h = \frac{1}{64S_K N L \sqrt{\log N}}.$$

We use exactly $\lceil 2K/(Nh) \rceil + 1$ equally spaced radial rows, including both endpoints, and $\lceil 8\pi/h \rceil$ angles on each row. This mesh has spacing h , and by the count of mesh points in the proof of Lemma 4.5, with $1/(64S_K)$ as its mesh constant, and $53 \cdot 64^2 = 217088 < 262144 = 2^{18}$, its total size satisfies

$$\begin{aligned} m &\leq 53(K+1)(64S_K)^2 NL^2 \log N = 53 \cdot 64^2 (K+1) S_K^2 NL^2 \log N \\ &\leq 2^{18} (K+1) S_K^2 NL^2 \log N = D'_K NL^2 \log N. \end{aligned}$$

Since $Nh = 1/(64S_K L \sqrt{\log N})$, $N^2 h^2 \sqrt{\log N} = 1/(4096 S_K^2 L^2 \sqrt{\log N})$ and $1/64 + 1/8192 < 1/16$, in (25), with $C(K) = S_K$ and the mesh constant $1/(64S_K)$, we have

$$\frac{Nh}{L} + \frac{S_K}{2} N^2 h^2 \sqrt{\log N} = \left(\frac{1}{64} + \frac{1}{8192} \right) \frac{1}{S_K L^2 \sqrt{\log N}} \leq \frac{1}{16 S_K L^2 \sqrt{\log N}},$$

and $2(1 + S_K/(64S_K))/L = 2(1 + 1/64)/L = 65/(32L) < 3/L$. It follows that the thresholds of Theorem B.4 at $\tau = h$, $R = 2$ and $M_2 = S_K N^{5/2} \sqrt{\log N}$ satisfy

$$(59) \quad u \leq (16S_K)^{-1} L^{-2} (\log N)^{-1/2}, \quad v_0 \leq 3/L.$$

The proof of Lemma 4.5 uses $N \geq 2K$ (in (11) and for the radial weights), $N \geq K$ (for $|z| \leq 2$), $N \geq 4$ (for the retained angles on each row), $N \geq e$ (for $h \leq 1/N$) and $N^{1/2} \geq 6400e^{8K}$ (for the covariance lower bounds λ_K in dimension four and $\lambda_K/2$ in dimension eight). Since $6400^2 e^{16K}$ exceeds $2K$, 4 and e , all of these hold if

$$N \geq (6400e^{8K})^2.$$

By the proof of Lemma 4.5, Theorem B.4 applies with these thresholds, with $p_2 = N^{-10}$, $q_0 = C_1(K+2) \log N$, $b_0 = 40C_1(K+2) \sqrt{N} \log N$ and $p_0 = N^{-3}$, and with $\lambda = \lambda_K$, $\delta = \delta_* = \mathcal{B}_K N^{-1/2}$, $\chi = \mathcal{T}_K N^{-1/2}$ and $b_* = 10N^{-1/2}$. Since $u^2 \leq (16S_K)^{-2} L^{-4} (\log N)^{-1}$ and $v_0^2 \leq 9L^{-2}$ by (59), we have

$$\frac{u^2 v_0^2}{4\lambda_K^2} \leq \frac{1}{4\lambda_K^2} \cdot \frac{9}{256S_K^2} \cdot \frac{L^{-6}}{\log N} = \frac{c_K L^{-6}}{\log N},$$

so that the line $u^2 v_0^2 / (4\lambda^2) + 2000e^{3K} (2/\lambda)^{3/2} N^{-1/2}$ of (27) gives at $\lambda = \lambda_K$ the one-point probability

$$(60) \quad c_K L^{-6} / \log N + \mathcal{B}_K N^{-1/2}.$$

For a retained separated pair, the covariance of the two indicators in the proof of Theorem B.4 is at most $\delta_* + \chi + 2\delta = (3\mathcal{B}_K + \mathcal{T}_K) N^{-1/2}$, and $V_* = (10 + 3\mathcal{B}_K + \mathcal{T}_K) N^{-1/2} \leq V_K N^{-1/2}$. Hence, by (60) and the bound on m ,

$$(61) \quad \mu_* \leq D'_K \{c_K NL^{-4} + \mathcal{B}_K N^{1/2} L^2 \log N\}, \quad m^2 V_* \leq V_K (D'_K)^2 N^{3/2} L^4 (\log N)^2.$$

We take $t = NL^{-4}$ in (78). By (61) and $\sqrt{N} \leq N^{1/2}L^2 \log N$ for $N \geq e$, we keep the terms with NL^{-4} apart from those with $N^{1/2}L^2$:

$$\begin{aligned} b_0 + q_0(\mu_* + t) &\leq C_1(K+2)(\log N) \left[40\sqrt{N} + NL^{-4} \right. \\ &\quad \left. + D'_K \{c_K NL^{-4} + \mathcal{B}_K N^{1/2}L^2 \log N\} \right] \\ &\leq Z_K NL^{-4} \log N + Z'_K N^{1/2}L^2 (\log N)^2. \end{aligned}$$

Since $t^2 = N^2L^{-8}$ and $L^{12} = N^{3/16}$, (61) gives

$$\frac{m^2 V_*}{t^2} \leq V_K (D'_K)^2 N^{-1/2} L^{12} (\log N)^2 = C_K N^{-5/16} (\log N)^2,$$

and hence, by (78), with $p_0 = N^{-3}$ the probability of the exceptional local-count event of Lemma 4.2 and $p_2 = N^{-10}$ that of the derivative event of Lemma 4.3,

$$\begin{aligned} (62) \quad &\mathbb{P}\{B_{K,N} > Z_K NL^{-4} \log N + Z'_K N^{1/2}L^2 (\log N)^2\} \\ &\leq N^{-3} + N^{-10} + C_K N^{-5/16} (\log N)^2. \end{aligned}$$

In this union, the local and derivative events are counted once. Since $NL^{-4} = N^{15/16}$, $N^{1/2}L^2 = N^{17/32}$ and $N^{17/32} \log N \leq N^{15/16}$, the count $B_{K,N}$ is at most $Z_K N^{15/16} \log N + Z'_K N^{17/32} (\log N)^2 \leq (Z_K + Z'_K) N^{15/16} \log N$ outside this union. The constant Z_K of the leading term grows like $(K+1)^2 e^{8K}$, and Z'_K like $(K+1)^2 e^{11K}$ (Table 11). No further threshold on N is needed here beyond $N \geq (6400e^{8K})^2$ above, the local-count thresholds $t_N \leq 1$ and $N \geq e$ of Lemma 4.2, and the localization thresholds (63).

A.5 The radial law

In this subsection we write out in finite form the bounds of the proof of Theorem 1.1 in Section 6, and hence of Corollary 1.2, which is its case $x = 0$ with $R = 2$. Recall that $N = j^8$, $m_j = (j+1)^8 - j^8$, $L = N^{1/64}$, $h = N^{-1/64}$ and $r_N = 1 + x/N$. Fix x and integers $R \geq |x| + 2$ and $K \geq R$, and let H_R , D_R , $\varepsilon_R^*(N)$ and $p_R^*(N)$ be H_K , D_K , $\varepsilon_K^*(N)$ and $p_K^*(N)$ at the width R , with the degree threshold $N \geq (160e^{8R})^2$. At $R = 2$ these are $\varepsilon_2^*(N)$ and $p_2^*(N)$ of Section A.3.

Let $a = 15e^{2(K+2)} \sqrt{m_j \log N}$ be the tail amplitude of Step 1 of Section 6, that is, that of Lemma 4.6 at the width $K+2$ with $m_{\max} = m_j$. With $M_2 = S_K N^{5/2} \sqrt{\log N}$ by (24) and $d_0 = N^{3/2}/L$, the localization conditions $4a/d_0 \leq 1/N$ and $4M_2 a \leq d_0^2$ of Table 2 read

$$(63) \quad 4aL/N^{3/2} \leq 1/N, \quad 4S_K a L^2 N^{-1/2} \sqrt{\log N} \leq 1,$$

since

$$\frac{4a}{d_0} = \frac{4aL}{N^{3/2}}, \quad \frac{4M_2a}{d_0^2} = \frac{4S_K N^{5/2} \sqrt{\log N} aL^2}{N^3} = 4S_K aL^2 N^{-1/2} \sqrt{\log N}.$$

The second condition implies the first: it gives $a \leq N^{15/32}/(4S_K \sqrt{\log N})$, and then $d_0 = N^{95/64}$ gives $4a/d_0 \leq N^{-65/64}/(S_K \sqrt{\log N}) < 1/N$.

By the first line of the bound on U_N in Step 3 of Section 6, by (57), multiplied by N , and by (62), on the radial and derivative events the number U_N of unselected roots satisfies

$$\begin{aligned} U_N &\leq \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\} + B_{K,N} \\ &\leq Z_K N^{15/16} \log N + Z'_K N^{17/32} (\log N)^2 \\ (64) \quad &+ N \left[\frac{2 \log 2}{K} + \frac{8}{K} \{\varepsilon_K^*(N) + D_K/N\} \right]. \end{aligned}$$

We apply the logarithmic estimates at $x + qh$, $q = -2, \dots, 2$, and Lemma B.3 at $x + qh$, $q = -1, 0, 1$, with R in place of K , logarithmic error $\eta_{\log} = \varepsilon_R^*(N)$, profile error $\varepsilon = D_R/N$, $F' = \Phi$, $M_F = 1/2$ and $B_F = 1$, since $0 \leq \Phi \leq 1$ and $0 \leq \Phi' = 2 \operatorname{Var}_x(t) \leq 1/2$ by Step 4 of Section 6, which also implies that $|\Phi(x + qh) - \Phi(x)| \leq |q|h/2$. For $N \geq 2R$ we get

$$\left| \frac{\nu_N(r_N + qh/N)}{N} - \Phi(x) \right| \leq \frac{|q|h}{2} + \frac{h}{4} + \frac{R}{N} + \frac{2(\varepsilon_R^*(N) + D_R/N)(1 + R/N)}{h}.$$

Since $|q|h/2 + h/4 \leq 2h$ for $|q| \leq 1$ and $2(1 + R/N) \leq 4$, we obtain

$$\begin{aligned} (65) \quad \left| \frac{\nu_N(r_N + qh/N)}{N} - \Phi(x) \right| &\leq \delta_R^*(N), \quad q = -1, 0, 1, \\ \delta_R^*(N) &:= 2h + \frac{R}{N} + \frac{4(\varepsilon_R^*(N) + D_R/N)}{h}. \end{aligned}$$

Here the exceptional probability is at most $5p_R^*(N)$, and $\delta_R^*(N) \rightarrow 0$. For Corollary 1.2, at $x = 0$ and $R = 2$, this reads

$$\begin{aligned} (66) \quad \left| \frac{\nu_N(1 + qN^{-65/64})}{N} - \frac{1}{2} \right| &\leq \delta_2^*(N), \quad q = -1, 0, 1, \\ \delta_2^*(N) &= 2N^{-1/64} + \frac{2}{N} + 4\left(\varepsilon_2^*(N) + \frac{D_2}{N}\right)N^{1/64}. \end{aligned}$$

By (65) with $q = \pm 1$, as in (39), the strict band $||\alpha| - r_N| < h/N$ contains at most $2\delta_R^*(N)N$ roots. This band contains the root of every selected disk whose closure meets the closed annulus between the circles of radii r_N and $1 + x/n$, for $N \leq n \leq N + m_j$, once $2a/d_0 + |x|m_j/N^2 < h/N$, as in Step 4 of Section 6. At the

width $K \geq R$ this follows from the thresholds. Indeed, since $d_0 = N^{3/2}/L = N^{95/64}$, and $a \leq N^{15/32}/(4S_K\sqrt{\log N})$ by the second condition of (63) with $L^2 = N^{1/32}$, we have

$$N^{65/64} \cdot \frac{2a}{d_0} = 2aN^{-15/32} \leq \frac{1}{2S_K\sqrt{\log N}} < \frac{1}{2},$$

since $S_K\sqrt{\log N} > 1$, and hence, by $N/h = N^{65/64}$ and (31),

$$\begin{aligned} \frac{N}{h} \left(\frac{2a}{d_0} + \frac{|x|m_j}{N^2} \right) &= N^{65/64} \cdot \frac{2a}{d_0} + |x|m_j N^{-63/64} \\ &< \frac{1}{2} + 9|x|N^{-7/64} \leq \frac{1}{2} + \frac{1}{2} = 1, \end{aligned}$$

since $18|x| < 18R < 3e^{7R/4} < 160^{7/32}e^{7R/4} \leq N^{7/64}$, by $|x| < R$, $18R < 3e^{7R/4}$ for $R \geq 2$, $160^{7/32} > 3$ and $N \geq (160e^{8R})^2$. Therefore the number $c(r_N, 1 + x/n)$ of these disks is at most $2\delta_R^*(N)N$, and the matching bound of Step 5 of Section 6 becomes $|\nu_n(1 + x/n) - \nu_N(r_N)| \leq 2\delta_R^*(N)N + U_N + m_j$. The chain (41), with $\delta_R^*(N)$ in place of ε_N , then gives

$$\left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq 3\delta_R^*(N) + \frac{U_N}{N} + \frac{2m_j}{N}, \quad N \leq n \leq N + m_j.$$

Collecting the exceptional probabilities, we find that the common exceptional event $E_{K,R,x,j}$ has probability at most

$$(67) \quad 4p_K^*(N) + 5p_R^*(N) + N^{-3} + 2N^{-10} + C_K N^{-5/16} (\log N)^2.$$

This union consists of the four radial logarithmic events, used for U_N , the five probe events at the width R , used in (65), and the local-count, derivative, mesh-deviation and maximal-tail events. The shared local-count and derivative events are counted once each, and a single maximal-tail event controls every degree in the block. For Corollary 1.2 we take $x = 0$ and $R = 2$. For every fixed K, R , the series of (67) over $N = j^8$ is finite, since its terms are, up to constants, the probabilities of Table 4, where $p_K^*(N)$ and $p_R^*(N)$ are explicit forms of $\Pi_K(N)$ and $\Pi_R(N)$, and every exponent of j in the last column of that table is less than -1 .

A.6 Other coefficient distributions

This subsection gives the constants with which the other distributions satisfy (H1)–(H5) in Proposition 7.4, as used in the proof of Theorem 1.1 and Corollary 1.2 for these distributions at the end of Section 7. For each distribution we recompute the constants of the sign case from the inputs that depend on it: its logarithmic-moment constant, its energy bound, its local-count constant and, for unbounded coefficients, an envelope on their size. Tables 12, 13 and 14 list these constants.

Input	Value
Logarithmic moments (C, β)	$(C_*, 6)$
Energy	$E = \Omega, B_E = 1, p_E = 0$
Occupation constant	H_K
Local count	$C_1(K + 2)$, exceptional probability N^{-3}
All other constants and thresholds	those of the sign case

Table 12: Constants for Steinhaus coefficients.

Steinhaus coefficients. These coefficients have the constants of the sign case (Table 12), since their amplitudes have modulus one, so that conditional signs yield the same C_* and $C_1(K + 2)$.

Input	Value
Logarithmic moments (C, β)	$(16, 6)$
Energy	$E = \{V \leq B_E\}, B_E = 2, p_E = \exp[-3N/(32e^{6K})]$
Occupation constant	H_K
Local count	$C_{1,G}(K + 2) = C_1(K + 2) + 2/\log 2 = (4A + 4(K + 2) + 6)/\log 2 = (4A + 4K + 14)/\log 2$, exceptional probability $N^{-3} + \exp[-3(N + 1)/32]$
Small derivatives	$Z_{K,G} = C_{1,G}(K + 2)(1 + D'_K c_K)$, $Z'_{K,G} = C_{1,G}(K + 2)(40 + D'_K \mathcal{B}_K)$; $D'_K, c_K, \mathcal{B}_K, V_K$ and $C_K = V_K(D'_K)^2$ unchanged
Derivative and tails	amplitudes of the sign case, $\lceil N^4 \rceil \leq 2N^4$ boundary samples, on the envelope event $\max_{k \leq 2N} g_k \leq N$ of probability at least $1 - 2(2N + 1)e^{-N^2/2}$
Thresholds	$N \geq (160e^{8K})^2, N \geq e^2, \rho \leq 2$, and those of the sign case

Table 13: Constants for standard real and complex Gaussian coefficients.

Real and complex Gaussian coefficients. By the proof of Proposition 7.4, with Lemma 7.3 for the local count, standard complex Gaussian coefficients have the constants of the real ones (Table 13).

The choice $(C, \beta) = (16, 6)$ is valid, since $(16p)^p \leq (16p)^{6p}$ for $p \geq 1$, so that, by (46), the logarithmic moments satisfy (17) with 16 in place of C_* . In the proof of Proposition 7.4, the Gaussian occupation constants $2 + 320e^{8K}$ and $14 + 960e^{8K}$ are at most H_K , since $2 + 320e^{8K} < 14 + 960e^{8K} \leq 974e^{9K} < 10^8 e^{20(K+1)} = H_K$,

so that (54) holds for these coefficients. Hence Lemma 3.2 applies with $H = H_K$, $C = 16$, $E = \{V \leq B_E\}$ and $B_E = 2$, where $V = \int |X_r|^2 d\mu$ as in Proposition 7.4, and $\mathbb{P}(E^c) \leq p_E$ by (47).

This is the computation of Section A.2 with C_* replaced by 16, $B_E = 1$ by 2 and the term p_E added: since $4e \cdot 16 = 64e$, $B_E/2 + 1 = 2$ and $4T^2 = (\log N)^2/256$, it gives, for $N \geq (160e^{8K})^2$ and $r \in [1 - K/N, 1 + K/N]$,

$$\begin{aligned} & \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > N^{-1/16}(\log N)^7 + 2H_K T N^{-1/2} \\ & \quad + e^{1/2}(64e)^6(2N^{-1/16} + H_K N^{-1/2})(\log N)^7 + 2N^{-1/16}\} \\ & \leq H_K N^{-3/8} + N^{-12} + \frac{H_K}{256} N^{-3/8} (\log N)^{-12} + p_E. \end{aligned}$$

For these coefficients, the error and the probability on the two sides of this display replace $\varepsilon_K^*(N)$ and $p_K^*(N)$ at each radius in (57), (58), (64), (66), (67) and (65), and in the radial thresholds of Section A.3.

The bounds of Lemmas 4.3 and 4.6 keep the amplitudes of the sign case on the envelope event, by the proof of Proposition 7.4: with $\lceil N^4 \rceil$ samples in place of $\lceil N^3 \rceil$, the derivative bounds gain the factor N and the sample spacing loses it. The thresholds $N \geq e^2$ and $\rho \leq 2$ imply the conditions $N \geq 4$ and $\rho \leq 2$ of the interpolation step in the proof of Lemma 4.6. In Z_K and Z'_K we replace $C_1(K+2)$ by $C_{1,G}(K+2)$, while the other constants of Section A.4 remain upper bounds, since the normal approximation is exact. On the common coefficient-envelope event, the bound (62) is then replaced by

$$\begin{aligned} & \mathbb{P}\{B_{K,N} > Z_{K,G} N L^{-4} \log N + Z'_{K,G} N^{1/2} L^2 (\log N)^2\} \\ & \leq N^{-3} + \exp[-3(N+1)/32] + N^{-10} + C_K N^{-5/16} (\log N)^2. \end{aligned}$$

All other exceptional probabilities are those of the sign case.

Centrally symmetric bounded coefficients. Assume that the coefficient distribution is nondegenerate and centrally symmetric, as defined in Section 7. We normalize $\mathbb{E}|\xi_0|^2 = 1$ and assume $|\xi_0| \leq B$, so that $B \geq 1$. We use V , $E = \{V \geq 1/2\}$, J_N and C_B as in Proposition 7.4 and its proof, and the constants of Table 14, where a is the tail amplitude of Step 1 of Section 6. The local-count constant is that of Lemma 7.2 with $K+2$ in place of K , and the conditional trace event adds $\exp[-(N+1)/(2B^4)]$ to its exceptional probability. Since $\mathcal{B}_{K,B} = B^3 \mathcal{B}_K$, $2/\lambda_K = 160e^{4K}$, $(160e^{4K})^{3/2} = 160^{3/2}e^{6K}$, $80e^{4K}/\lambda_K = 6400e^{8K}$, $D'_{K,B} = B^2 D'_K$, $(D'_K)^2 = 2^{36}(K+1)^2 S_K^4$ and $S_K^4 = 10^8 e^{4K+16}$, and since $D'_{K,B} c_{K,B} = D'_K c_K$ and

Input	Value
Logarithmic moments (C, β)	$(C_B, 6)$, with $C_B = 2\{C_* + \max(\frac{1}{2} \log 2, \log B) + 1\}$
Energy	$E = \{V \geq 1/2\}$, $B_E = B^2$, $p_H = \exp[-N/(2B^4 e^{6K})]$
Occupation constant	$H_{K,B} = (1 + B^3)H_K$
Derivative and mesh	$S_{K,B} = BS_K$, $D'_{K,B} = B^2 D'_K$, $\mathcal{B}_{K,B} = B^3 \mathcal{B}_K$, $c_{K,B} = c_K/B^2$; λ_K and $\mathcal{T}_K$ unchanged
Tail amplitude	Ba , with $a = 15e^{2(K+2)}\sqrt{m_j \log N}$
Local count	$A_B = \max(1, 2048\pi B^2/c_{\text{HW}})$, $C_{1,B}(K+2) = (4A_B + 4(K+2) + 4 + \log B)/\log 2$, exceptional probability $N^{-3} + \exp[-(N+1)/(2B^4)]$
Small derivatives	$V_{K,B}$, $C_{K,B}$, $Z_{K,B}$, $Z'_{K,B}$: the formulas for V_K , C_K , Z_K , Z'_K with these constants and $C_{1,B}(K+2)$
Thresholds	$A_B \log N/N \leq 1$, $N \geq e^2$, and the enlarged thresholds below

Table 14: Constants for a bounded centrally symmetric distribution with $\mathbb{E}|\xi_0|^2 = 1$ and $|\xi_0| \leq B$.

$$D'_{K,B} \mathcal{B}_{K,B} = B^5 D'_K \mathcal{B}_K,$$

$$\begin{aligned}
V_{K,B} &= 50 + 3\mathcal{B}_{K,B} + \mathcal{T}_K = 50 + 6000B^3 e^{3K} (2/\lambda_K)^{3/2} + \frac{80e^{4K}}{\lambda_K} \\
&= 50 + 6000 \cdot 160^{3/2} B^3 e^{9K} + 6400e^{8K}, \\
C_{K,B} &= V_{K,B} (D'_{K,B})^2 = B^4 V_{K,B} (D'_K)^2 \\
&= 2^{36} \cdot 10^8 B^4 (K+1)^2 e^{4K+16} V_{K,B}, \\
Z_{K,B} &= C_{1,B}(K+2)(1 + D'_{K,B} c_{K,B}) = C_{1,B}(K+2)(1 + D'_K c_K), \\
Z'_{K,B} &= C_{1,B}(K+2)(40 + D'_{K,B} \mathcal{B}_{K,B}) = C_{1,B}(K+2)(40 + B^5 D'_K \mathcal{B}_K).
\end{aligned}$$

For the occupation bounds, the constant c_B of the proof of Proposition 7.4 satisfies, as in Section A.2 with the factor B^3 on the first term only and with $B \geq 1$ in the last step,

$$c_B = (42 \cdot 4^{1/4} + 16) \cdot 46B^3 e^{9K} + 320e^{8K} < 3469B^3 e^{9K} + 44e^{9K} < 4000B^3 e^{9K}.$$

Hence, by $B^3 e^{9K} \geq 1$, $12014 < 10^8$ and $9K < 20(K+1)$, the occupation constants $2 + c_B < 14 + 3c_B$ of that proof satisfy $14 + 3c_B < 14 + 12000B^3 e^{9K} \leq 12014B^3 e^{9K} < B^3 H_K < H_{K,B}$, so that (54) holds with $H_{K,B}$ in place of H_K . Hence Lemma 3.2 applies with $H = H_{K,B}$, $C = C_B$, $E = \{V \geq 1/2\}$ and $B_E = B^2$, by the moment bound on E in Proposition 7.4 and since $V \leq B^2$. With $B_E/2 + 1 = B^2/2 + 1$ and

$4T^2 = (\log N)^2/256$, its error is

$$\begin{aligned}\varepsilon_{K,B}^*(N) &= N^{-1/16}(\log N)^7 + 2H_{K,B}TN^{-1/2} \\ &\quad + e^{1/2}(4eC_B)^6(2N^{-1/16} + H_{K,B}N^{-1/2})(\log N)^7 + (B^2/2 + 1)N^{-1/16},\end{aligned}$$

and it gives

$$\begin{aligned}\mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > \varepsilon_{K,B}^*(N)\} &\leq H_{K,B}N^{-3/8} + N^{-12} \\ &\quad + \frac{H_{K,B}}{256}N^{-3/8}(\log N)^{-12} + \exp[-N/(2B^4e^{6K})],\end{aligned}$$

where the last term is the probability of E^c from (50). As in the real Gaussian case, $\varepsilon_{K,B}^*(N)$ and the right-hand side of the last display replace $\varepsilon_K^*(N)$ and $p_K^*(N)$ at each radius. Similarly, by the proof of Proposition 7.4 with the deviation $t = NL^{-4}$, the bound (62) is replaced by

$$\begin{aligned}\mathbb{P}\{B_{K,N} > Z_{K,B}NL^{-4}\log N + Z'_{K,B}N^{1/2}L^2(\log N)^2\} \\ \leq N^{-3} + \exp[-(N+1)/(2B^4)] + N^{-10} + C_{K,B}N^{-5/16}(\log N)^2.\end{aligned}$$

We require $A_B \log N/N \leq 1$ and $N \geq e^2$, and all previous thresholds with the enlarged constants. Since $S_{K,B} \cdot Ba = B^2 S_K a$, these are

$$\begin{aligned}N &\geq (6400e^{8K})^2, & N &\geq 2K, \\ 4BaL/N^{3/2} &\leq 1/N, & 4B^2 S_K a L^2 N^{-1/2} \sqrt{\log N} &\leq 1,\end{aligned}$$

together with the thresholds $N \geq \max\{2, (K+1)/3\}$, $j \geq 60$ and $N \geq \max\{4, K+2\}$, which do not change. These inequalities hold eventually for each fixed distribution, with cutoffs that depend on B .

If $\mathbb{P}(\xi_0 = 0) > 0$, the trailing-zero allowance $\lceil D \log N \rceil$, with $D > 4/|\log \mathbb{P}(\xi_0 = 0)|$, and the bound $N^{-D|\log \mathbb{P}(\xi_0 = 0)|}$ on the probability of its exceptional event at each degree are those of the proof of Theorem 1.1 and Corollary 1.2 for other distributions in Section 7, which also treats the initial zero multiplicity and the actual degrees in Corollary 5.3.

A.7 Bounds for a growing annulus

In this subsection we compute the constants of the proof in Section 8.1, at the width $K = K_N$, where $K_N = \lfloor (\log N)/128 - \log \log N \rfloor$ is the width of that section, with $N = j^8$ and $L = N^{1/64}$ as before, and with the local-count and tail lemmas at

the width $K + 2$. The following bounds hold whenever the displayed deterministic conditions hold. Set

$$\begin{aligned} H_0 &= 10^8 e^{20}, & S_0 &= 100 e^4, & D_0 &= 2^{18} S_0^2, \\ B_0 &= 2000 \cdot 160^{3/2}, & V_0 &= 50 + 3B_0 + 6400, & Q_0 &= \frac{9 \cdot 80^2}{1024 S_0^2}, \\ A &= \max(1, 1024\pi/c_{\text{HW}}), & C_{10} &= (4A + 12)/\log 2, & C_{\text{bad}} &= V_0 D_0^2, \\ Z_0 &= C_{10}(1 + D_0 Q_0), & Z'_0 &= C_{10}(40 + D_0 B_0), & E_0 &= E_1 + E_2 H_0, \end{aligned}$$

where E_1 and E_2 are the absolute constants of (56). These constants are independent of K, N . By the definitions we have, for $K \geq 0$,

$$\begin{aligned} H_K &= H_0 e^{20K}, & S_K &= S_0 e^K, & D'_K &= D_0(K + 1)e^{2K}, \\ \mathcal{B}_K &= B_0 e^{9K}, & \mathcal{T}_K &= 6400 e^{8K}, & V_K &\leq V_0 e^{9K}, \\ c_K &= Q_0 e^{6K}, & C_1(K + 2) &\leq C_{10}(K + 1), & C_K &\leq C_{\text{bad}}(K + 1)^2 e^{13K}, \\ Z_K &\leq Z_0(K + 1)^2 e^{8K}, & Z'_K &\leq Z'_0(K + 1)^2 e^{11K}. \end{aligned}$$

Indeed, the first three hold since $H_K = 10^8 e^{20} e^{20K}$, $S_K = 100 e^4 e^K$ and $D'_K = 2^{18}(K + 1)S_K^2 = 2^{18}S_0^2(K + 1)e^{2K}$. For the others we use $1/\lambda_K = 80e^{4K}$, so that $2/\lambda_K = 160e^{4K}$ and $\mathcal{T}_K = 80e^{4K} \cdot 80e^{4K} = 6400e^{8K}$, and, for the bounds, $e^{8K} \leq e^{9K}$, $1 \leq (K + 1)e^{bK}$, $4K \leq (4A + 12)K$ and $C_1(K + 2) = (4A + 4(K + 2) + 4)/\log 2$:

$$\begin{aligned} \mathcal{B}_K &= 2000 e^{3K} (160 e^{4K})^{3/2} = 2000 \cdot 160^{3/2} e^{9K} = B_0 e^{9K}, \\ V_K &= 50 + 3B_0 e^{9K} + 6400 e^{8K} \leq (50 + 3B_0 + 6400) e^{9K} = V_0 e^{9K}, \\ c_K &= \frac{9 \cdot 80^2 e^{8K}}{1024 S_0^2 e^{2K}} = Q_0 e^{6K}, \\ C_1(K + 2) &= \frac{4A + 4K + 12}{\log 2} \leq \frac{(4A + 12)(K + 1)}{\log 2} = C_{10}(K + 1), \\ C_K &= V_K (D'_K)^2 \leq V_0 e^{9K} \cdot D_0^2 (K + 1)^2 e^{4K} = C_{\text{bad}}(K + 1)^2 e^{13K}, \\ Z_K &= C_1(K + 2)(1 + D'_K c_K) \leq C_{10}(K + 1)\{1 + D_0 Q_0 (K + 1)e^{8K}\} \\ &\leq Z_0(K + 1)^2 e^{8K}, \\ Z'_K &= C_1(K + 2)(40 + D'_K \mathcal{B}_K) \leq C_{10}(K + 1)\{40 + D_0 B_0 (K + 1)e^{11K}\} \\ &\leq Z'_0(K + 1)^2 e^{11K}. \end{aligned}$$

These are the forms $C_0(K + 1)^p e^{bK}$ of Table 11, and the following lemma evaluates such a form at $K = K_N$.

Lemma A.1 (Constants at the growing width). *Let $N \geq e$ and $K_N \geq 0$, and let $X_K \leq C_0(K + 1)^p e^{bK}$ with $C_0 \geq 0$, $p \geq 0$ and $b \geq 0$. Then*

$$X_{K_N} \leq C_0(K_N + 1)^p N^{b/128} (\log N)^{-b},$$

and if moreover $K_N + 1 \leq \log N$, then $X_{K_N} \leq C_0(\log N)^p N^{b/128}(\log N)^{-b}$.

Proof. By the floor, $K_N \leq (\log N)/128 - \log \log N$, and hence, since $b \geq 0$ and $\log N \geq 1$,

$$e^{bK_N} \leq N^{b/128}(\log N)^{-b} \leq N^{b/128}, \quad b \geq 0.$$

This gives the first bound, and the second follows since $0 < K_N + 1 \leq \log N$ and $p \geq 0$. $\square$

By Lemma A.1 with the forms above, keeping the factor $(K+1)^p$ and, only for Z_K , the factor $(\log N)^{-8}$, we get for $N \geq e$

$$\begin{aligned} \varepsilon_K^*(N) &\leq E_0 N^{-1/16} (\log N)^7, \\ D_K/N &\leq 10(K+1)^2 N^{-31/32}, \\ p_K^*(N) &\leq H_0 N^{-7/32} + N^{-12} + \frac{H_0}{256} N^{-7/32} (\log N)^{-12}, \\ C_K N^{-5/16} (\log N)^2 &\leq C_{\text{bad}} (K+1)^2 N^{-27/128} (\log N)^2, \\ Z_K N^{-1/16} \log N &\leq Z_0 (K+1)^2 (\log N)^{-7}, \\ (68) \quad Z'_K N^{-15/32} (\log N)^2 &\leq Z'_0 (K+1)^2 N^{-49/128} (\log N)^2. \end{aligned}$$

By Lemma A.1 with $b = 20$, we have $H_K = H_0 e^{20K} \leq H_0 N^{5/32}$, and $5/32 - 1/2 = -11/32$ and $5/32 - 3/8 = -7/32$. By (56) and $N^{-11/32} \leq N^{-1/16}$, the first line follows from

$$\varepsilon_K^*(N) \leq E_1 N^{-1/16} (\log N)^7 + E_2 H_0 N^{-11/32} (\log N)^7 \leq E_0 N^{-1/16} (\log N)^7.$$

The second line follows from $D_K/N = 10(K+1)^2 e^{4K} N^{-1} \leq 10(K+1)^2 N^{1/32-1} = 10(K+1)^2 N^{-31/32}$. Since $4T^2 = (\log N)^2/256$, the third line follows from

$$\begin{aligned} p_K^*(N) &= H_K N^{-3/8} + N^{-12} + \frac{H_K}{256} N^{-3/8} (\log N)^{-12} \\ &\leq H_0 N^{-7/32} + N^{-12} + \frac{H_0}{256} N^{-7/32} (\log N)^{-12}. \end{aligned}$$

By the bounds on C_K , Z_K and Z'_K above, and Lemma A.1 with $b = 13$, $b = 8$ and $b = 11$, the last three lines follow from $13/128 - 5/16 = -27/128$, $e^{8K} N^{-1/16} \leq (\log N)^{-8}$ and $11/128 - 15/32 = -49/128$.

The caps on $K/\log N$ in Table 8 are β/b in its first part and $(\beta - 1/8)/b$ in its second, where b is the exponent of the named constant of the row in Table 11, plus 2 for the factor $e^{2(K+2)} = e^4 e^{2K}$ of a in the two rows with a , and $b = 16$ for $(6400e^{8K})^2$.

Each ratio in Table 15 tends to zero, and the indicated condition holds once the ratio is at most one. Indeed, at $K = K_N$ every ratio other than $2K/N$ is a

constant times powers of $K_N + 1 \leq \log N$ and of $\log N$, times a negative power of N , and $2K_N/N \leq 2(\log N)/N$. These rates are compared in Section 8.1. We

Condition	Sufficient ratio bound
$(160e^{8K})^2 \leq N,$ $(6400e^{8K})^2 \leq N$	$6400^2 N^{-7/8}$
$2K \leq N$	$2K/N$
Taylor localization inequality	$180S_0e^4(\log N)N^{-1/128}$
$2a/d_0 < N^{-65/64}$	$90e^4\sqrt{\log N}N^{-1/64}$

Table 15: Sufficient ratio bounds for the degree conditions at the width $K = K_N$.

require the last ratio to be strictly below one. The condition $4a/d_0 \leq 1/N$ follows from the Taylor localization inequality, as shown after (63).

We obtain the ratios of the table by dividing each condition by its right-hand side and substituting the bounds above. For the first row, since $160 < 6400$ and by Lemma A.1 with $b = 16$,

$$\frac{(6400e^{8K})^2}{N} = 6400^2 e^{16K} N^{-1} \leq 6400^2 N^{1/8-1} = 6400^2 N^{-7/8}.$$

The last two rows, for the inequality $4S_K a L^2 N^{-1/2} \sqrt{\log N} \leq 1$ of Taylor localization in (63), which is $4M_2 a \leq d_0^2$, and for the strict band $N^{65/64} \cdot 2a/d_0 < 1$, are computed in the proof of Theorem 1.3 from (33) and (35), with $180S_0e^4 = 18000e^8$ and the exponents $7/16 + 1/32 - 1/2 = -1/32$, $3/128 - 1/32 = -1/128$, $7/16 - 15/32 = -1/32$ and $2/128 - 1/32 = -1/64$. They use the bound $m_j \leq 9N^{7/8}$ of (31) and the bound $a \leq 45e^4 e^{2K} N^{7/16} \sqrt{\log N}$ of (32), where $e^{2(K+2)} = e^4 e^{2K}$. The small-derivative lemma adds no ratio to the table, since its thresholds are $N \geq (6400e^{8K})^2$, those of Lemma 4.2 and (63).

We also impose

$$j \geq 60, \quad K \geq 2, \quad A \log N/N \leq 1, \quad N \geq K + 2.$$

These conditions are eventual. Beyond a fixed cutoff, the secant and probe radii belong to their required annuli. In this way we obtain all local-count, covariance, tail, localization and radial cutoffs used in the proof.

The proof in Section 8.1 bounds the number U_N of unselected roots at $K = K_N$, in place of (64), by

$$(69) \quad U_N \leq N \left[\frac{1}{K-1} + 4 \left(\varepsilon_K^*(N) + \frac{D_K}{N} \right) \right] + Z_K N^{15/16} \log N + Z'_K N^{17/32} (\log N)^2,$$

outside the four radial events at the radii $1 \pm (K_N - 1)/N$ and $1 \pm K_N/N$ and the event of (62). Its probes are those of (66), at the fixed width 2, so that the probe error at the unit circle is at most $\delta_2^*(N)$ and the crossing count at most $2\delta_2^*(N)N$. By (41) at $x = 0$, with $\delta_2^*(N)$ in place of ε_N as in Section A.5, and (69) divided by N , the bound that we use for the logarithmic rate is, for $N \leq n \leq N + m_j$,

$$\begin{aligned}
 \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq 3\delta_2^*(N) + \frac{U_N}{N} + \frac{2m_j}{N} \\
 &\leq 3 \left(2N^{-1/64} + \frac{2}{N} + 4 \left(\varepsilon_2^*(N) + \frac{D_2}{N} \right) N^{1/64} \right) + \frac{1}{K_N - 1} \\
 &\quad + 4 \left(\varepsilon_{K_N}^*(N) + \frac{D_{K_N}}{N} \right) + Z_{K_N} N^{-1/16} \log N \\
 (70) \quad &\quad + Z'_{K_N} N^{-15/32} (\log N)^2 + \frac{2m_j}{N}.
 \end{aligned}$$

Here $\varepsilon_{K_N}^*(N)$, $H_{K_N} = 10^8 e^{20(K_N+1)}$, $D_{K_N} = 10(K_N + 1)^2 e^{4K_N}$, Z_{K_N} and Z'_{K_N} are $\varepsilon_K^*(N)$, H_K , D_K , Z_K and Z'_K at $K = K_N$, and $D_2 = 90e^8$. For this bound, the exceptional probability is (67) with $K = K_N$, $R = 2$ and $x = 0$, where $C_{K_N} = V_{K_N}(D'_{K_N})^2$ and $p_{K_N}^*(N)$ is the function $p_K^*(N)$ at $K = K_N$. By the third and fourth lines of (68), with $4H_0/256 = H_0/64$, its terms $4p_{K_N}^*(N) + C_{K_N} N^{-5/16} (\log N)^2$ are bounded as in the proof of Theorem 1.3. By Lemma A.1, through (68), every term in (70) other than $1/(K_N - 1)$ is $o(1/\log N)$, as in Section 8.1, since $K_N + 1 \leq \log N$ and $m_j/N \leq 9N^{-1/8}$.

A.8 Nearby circles

Finally, we compute the constants of the proof of Corollary 8.2 in Section 8.3. For a target exponent $\kappa > 0$, we choose a wider band with

$$0 < \kappa_* < \min(\kappa, 1/64),$$

as does the choice $\kappa_* = \min(\kappa/2, 1/128)$ in that proof. Then the band $||z| - 1| < N^{-1-\kappa_*}$ contains the circles $|z| = 1 \pm N^{-1-\kappa}$, and κ_* lies in the range $0 < \kappa \leq 1/64$ of the thin-band estimate of Section A.3. We impose

$$N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*},$$

where $2a/d_0$ is the radius of the selected disks of Step 2 of Section 6. As in the proof of Corollary 8.2, this condition places every good annular root whose disk meets one of the circles in the band. By the form of (58) at $K = 2$ in Section A.3, with κ_* in place of κ ,

$$\left| \frac{\nu_N(1 + qN^{-1-\kappa_*})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa_*} + 4\varepsilon_2^*(N)N^{\kappa_*}, \quad q = -1, 0, 1,$$

outside probability $5p_2^*(N)$. Since $p_2^*(N)$ does not depend on the thin-band exponent, the exceptional probability of the block is (67) with $R = 2$, where the term $5p_2^*(N)$ now bounds these five thin-band events in place of the five probe events.

B General concentration and counting estimates

This appendix proves the general estimates that the main text applies with its own constants: concentration of the angular logarithm for general random coefficients, a deterministic local secant bound for radial counts, a second-moment count of roots with a small derivative, and the passage from a sparse subsequence of degrees to all degrees. The tools are clipping with Chebyshev's inequality, Jensen's formula, Taylor's theorem on a mesh, and the Borel–Cantelli lemma.

Let $(a_k)_{k \geq 0}$ be a sequence of random complex coefficients on one probability space, and put

$$f_N(z) = \sum_{k=0}^N a_k z^k.$$

The same coefficient sequence is used for all degrees, and in particular $\deg f_N \leq N$ when f_N is nonzero. Let $\sigma_N(r) > 0$ be deterministic, and write

$$J_N(r) = \int_{\mathbb{T}} \log |f_N(re^{i\theta})| d\mu(\theta),$$

where μ is the normalized angular measure. Deterministic counting statements concern nonzero polynomials, and in probabilistic applications we include the case of an identically zero polynomial in an explicit exceptional event.

B.1 Angular occupation and logarithmic integrals

One- and two-point occupation bounds. For fixed N, r , we set

$$X(\theta) = \frac{f_N(re^{i\theta})}{\sigma_N(r)}.$$

Let G have density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$. We set

$$F_G(t) = \mathbb{P}(|G| \leq t) = 1 - e^{-t^2}.$$

There are measurable exceptional sets $E_1 \subset \mathbb{T}$, $E_2 \subset \mathbb{T}^2$ with $\mu(E_1) \leq \eta_1$, $(\mu \otimes \mu)(E_2) \leq \eta_2$, such that, for all $s, t \geq 0$,

$$|\mathbb{P}(|X(\theta)| \leq t) - F_G(t)| \leq \delta_1 \quad (\theta \notin E_1),$$

$$|\mathbb{P}(|X(\theta)| \leq s, |X(\phi)| \leq t) - F_G(s)F_G(t)| \leq \delta_2 \quad ((\theta, \phi) \notin E_2).$$

In the two-point bound we compare with a pair of independent standard complex Gaussians. If the normal approximation yields a correlated Gaussian pair, then δ_2 includes the error of the comparison with this independent pair.

Logarithmic moments and energy. For fixed constants $C > 0$, $\beta > 0$, and for all real $p > 1$,

$$(71) \quad \mathbb{E} \int_{\mathbb{T}} |\log |X(\theta)||^p d\mu(\theta) \leq (Cp)^{\beta p}.$$

We will also use the following eventwise bound on an event E_H with $\mathbb{P}(E_H^c) \leq p_H$:

$$(72) \quad \mathbb{E} \left[\mathbf{1}_{E_H} \int_{\mathbb{T}} |\log |X(\theta)||^p d\mu(\theta) \right] \leq (Cp)^{\beta p} \quad (p > 1).$$

Here the expectation is the integral over E_H . We evaluate the integrand only on that event. This also defines the eventwise bound when the angular integral is infinite on E_H^c . The exact angular energy identity is

$$(73) \quad \int_{\mathbb{T}} |X(\theta)|^2 d\mu(\theta) = 1 \quad \text{almost surely.}$$

The upper-energy probability bound is

$$(74) \quad \mathbb{P} \left(\int_{\mathbb{T}} |X(\theta)|^2 d\mu(\theta) > B_E \right) \leq p_E \quad (B_E > 0).$$

The full bound (71) is the case $E_H = \Omega$ and $p_H = 0$ of (72). We use the tail form (74) when the coefficient moduli are random, and under (73) we can take $B_E = 1$, $p_E = 0$. For the coefficient distributions of Sections 3 and 7, these logarithmic bounds hold for every $p \geq 1$ (see (17), (46) and Proposition 7.4), but in the proof of Theorem B.1 we use them only for $p = 2q \geq 2$.

Theorem B.1 (Logarithmic integral concentration). *Assume the one- and two-point occupation bounds, the eventwise logarithmic bound (72), and the energy bound (74). We set*

$$d_1 = \delta_1 + \eta_1, \quad d_2 = \delta_2 + \eta_2, \quad v = d_2 + 2d_1.$$

For all $T > 0$, $u > 0$, $b > 0$, $q \geq 1$, and $\lambda > 1$, we set

$$(75) \quad \eta_{\log} = u + 2Td_1 + \lambda(2Cq)^{\beta} \left(e^{-2T} + d_1 + b \right)^{1-1/(2q)} + (B_E/2 + 1)e^{-2T},$$

$$(76) \quad p_{\text{fail}} = v/b^2 + \lambda^{-2q} + 4T^2 v/u^2 + p_E + p_H.$$

Then

$$\mathbb{P}(|J_N(r) - \log \sigma_N(r) - \mathbb{E} \log |G|| > \eta_{\log}) \leq p_{\text{fail}}, \quad \mathbb{E} \log |G| = -\gamma/2.$$

The same conclusion holds if, in place of the occupation bounds, $|\mathbb{E} A_t - F_G(t)| \leq d_1$ and $\text{Var}(A_t) \leq v$ for all $t \geq 0$ and some $d_1, v \geq 0$, where $A_t = \mu\{\theta : |X(\theta)| \leq t\}$. If moreover $\int_{\mathbb{T}} |X|^2 d\mu \leq B_E$ on E_H , then $p_E + p_H$ in (76) may be replaced by $\mathbb{P}(E_H^c)$.

Proof. The idea is to clip the logarithm at $\pm T$ and to bound the mean and the variance of the clipped integral by the occupation bounds. The logarithmic moments and the occupation bound at e^{-T} control the clipped lower tail, through Hölder's inequality, and the energy bound the upper tail. Table 16 lists the five exceptional events, one for each term of (76).

We set $A_t = \int_{\mathbb{T}} \mathbf{1}_{\{|X(\theta)| \leq t\}} d\mu(\theta)$. By Tonelli's theorem, we have

$$\begin{aligned} \mathbb{E}A_t &= \int_{\mathbb{T}} \mathbb{P}(|X(\theta)| \leq t) d\mu(\theta), \\ \mathbb{E}A_t^2 &= \mathbb{E} \int_{\mathbb{T}^2} \mathbf{1}_{\{|X(\theta)| \leq t\}} \mathbf{1}_{\{|X(\phi)| \leq t\}} d\mu(\theta) d\mu(\phi) \\ &= \int_{\mathbb{T}^2} \mathbb{P}(|X(\theta)| \leq t, |X(\phi)| \leq t) d\mu(\theta) d\mu(\phi). \end{aligned}$$

By these identities and the [one- and two-point occupation bounds](#), with $s = t$ in the second, and since the probabilities and $F_G(t)$ lie in $[0, 1]$, we have, for every $t \geq 0$,

$$\begin{aligned} |\mathbb{E}A_t - F_G(t)| &= \left| \int_{\mathbb{T}} (\mathbb{P}(|X(\theta)| \leq t) - F_G(t)) d\mu(\theta) \right| \\ &\leq \int_{\mathbb{T} \setminus E_1} \delta_1 d\mu + \int_{E_1} 1 d\mu \leq \delta_1 + \mu(E_1) \leq d_1, \\ |\mathbb{E}A_t^2 - F_G(t)^2| &= \left| \int_{\mathbb{T}^2} (\mathbb{P}(|X(\theta)| \leq t, |X(\phi)| \leq t) - F_G(t)^2) d(\mu \otimes \mu) \right| \\ &\leq \int_{\mathbb{T}^2 \setminus E_2} \delta_2 d(\mu \otimes \mu) + \int_{E_2} 1 d(\mu \otimes \mu) \leq \delta_2 + (\mu \otimes \mu)(E_2) \leq d_2. \end{aligned}$$

Since both $\mathbb{E}A_t$ and $F_G(t)$ belong to $[0, 1]$, we have

$$\begin{aligned} |(\mathbb{E}A_t)^2 - F_G(t)^2| &= |\mathbb{E}A_t - F_G(t)| |\mathbb{E}A_t + F_G(t)| \leq 2d_1, \\ \text{Var}(A_t) &= \mathbb{E}A_t^2 - (\mathbb{E}A_t)^2 \leq |\mathbb{E}A_t^2 - F_G(t)^2| + |(\mathbb{E}A_t)^2 - F_G(t)^2| \leq d_2 + 2d_1 = v. \end{aligned}$$

From here on, the occupation bounds enter only through the bounds $|\mathbb{E}A_t - F_G(t)| \leq d_1$ and $\text{Var}(A_t) \leq v$, which proves the first variant in the statement.

Since $F_G(e^{-T}) = 1 - \exp(-e^{-2T}) \leq e^{-2T}$, taking $t = e^{-T}$ in these bounds and applying Chebyshev's inequality, we obtain

$$\begin{aligned} \mathbb{E}A_{e^{-T}} &\leq F_G(e^{-T}) + d_1 \leq e^{-2T} + d_1, \\ \mathbb{P}(A_{e^{-T}} > e^{-2T} + d_1 + b) &\leq \mathbb{P}(A_{e^{-T}} - \mathbb{E}A_{e^{-T}} > b) \leq \frac{\text{Var}(A_{e^{-T}})}{b^2} \leq \frac{v}{b^2}. \end{aligned}$$

We call the event $\{A_{e^{-T}} > e^{-2T} + d_1 + b\}$ the Chebyshev event for $A_{e^{-T}}$.

On E_H , we set $W = \int_{\mathbb{T}} |\log |X(\theta)||^{2q} d\mu(\theta)$. By (72) with $p = 2q$, we have $\mathbb{E}[\mathbf{1}_{E_H} W] \leq (2Cq)^{2\beta q}$, and applying Markov's inequality to W on E_H , we get

$$\mathbb{P}\{\omega \in E_H : W > \lambda^{2q} (2Cq)^{2\beta q}\} \leq \frac{\mathbb{E}[\mathbf{1}_{E_H} W]}{\lambda^{2q} (2Cq)^{2\beta q}} \leq \frac{(2Cq)^{2\beta q}}{\lambda^{2q} (2Cq)^{2\beta q}} = \lambda^{-2q}.$$

We call the event on the left the Markov event for W .

We now truncate the logarithm and set $\ell_T(z) = \max(-T, \min(T, \log |z|))$ and $L_T = \int_{\mathbb{T}} \ell_T(X) d\mu$. For every $z \in \mathbb{C}$, including $z = 0$, we have the layer-cake identity

$$\ell_T(z) = T - \int_{-T}^T \mathbf{1}_{\{|z| \leq e^\tau\}} d\tau,$$

since the set of $\tau \in [-T, T]$ with $|z| \leq e^\tau$ is $[\max(-T, \log |z|), T]$ if $\log |z| \leq T$ (with $\log 0 = -\infty$), and is empty otherwise. By the layer-cake identity at $z = X(\theta)$ and at $z = G$, Tonelli's theorem and $\mathbb{P}(|G| \leq e^\tau) = F_G(e^\tau)$, we have

$$\begin{aligned} L_T &= \int_{\mathbb{T}} \left(T - \int_{-T}^T \mathbf{1}_{\{|X(\theta)| \leq e^\tau\}} d\tau \right) d\mu(\theta) = T - \int_{-T}^T A_{e^\tau} d\tau, \\ \mathbb{E}L_T - \mathbb{E}\ell_T(G) &= \int_{-T}^T (F_G(e^\tau) - \mathbb{E}A_{e^\tau}) d\tau, \\ L_T - \mathbb{E}L_T &= - \int_{-T}^T (A_{e^\tau} - \mathbb{E}A_{e^\tau}) d\tau. \end{aligned}$$

By these identities, the bounds $|\mathbb{E}A_t - F_G(t)| \leq d_1$ and $\text{Var}(A_t) \leq v$, and the Cauchy–Schwarz inequality, we have

$$\begin{aligned} |\mathbb{E}L_T - \mathbb{E}\ell_T(G)| &\leq \int_{-T}^T |\mathbb{E}A_{e^\tau} - F_G(e^\tau)| d\tau \leq 2Td_1, \\ |\text{Cov}(A_{e^\tau}, A_{e^{\tau'}})| &\leq \sqrt{\text{Var}(A_{e^\tau}) \text{Var}(A_{e^{\tau'}})} \leq v, \end{aligned}$$

and hence, by Fubini's theorem, which applies since $|A_{e^\tau} - \mathbb{E}A_{e^\tau}| \leq 1$,

$$\begin{aligned} \text{Var}(L_T) &= \mathbb{E} \left(\int_{-T}^T (A_{e^\tau} - \mathbb{E}A_{e^\tau}) d\tau \right)^2 \\ &= \int_{-T}^T \int_{-T}^T \mathbb{E}[(A_{e^\tau} - \mathbb{E}A_{e^\tau})(A_{e^{\tau'}} - \mathbb{E}A_{e^{\tau'}})] d\tau d\tau' \\ &= \int_{-T}^T \int_{-T}^T \text{Cov}(A_{e^\tau}, A_{e^{\tau'}}) d\tau d\tau' \leq \int_{-T}^T \int_{-T}^T |\text{Cov}(A_{e^\tau}, A_{e^{\tau'}})| d\tau d\tau' \leq 4T^2v. \end{aligned}$$

Applying Chebyshev's inequality again, we get

$$\mathbb{P}(|L_T - \mathbb{E}L_T| > u) \leq \frac{\text{Var}(L_T)}{u^2} \leq \frac{4T^2v}{u^2}.$$

We call the event on the left the Chebyshev event for L_T .

On E_H , outside the Chebyshev event for $A_{e^{-T}}$ and the Markov event for W , by Hölder's inequality with the exponents $2q$ and $2q/(2q-1)$ we bound the negative truncation error by

$$\int_{\{|X| < e^{-T}\}} (-\log |X| - T) d\mu \leq \int_{\{|X| < e^{-T}\}} |\log |X|| d\mu$$

$$\begin{aligned}
&\leq \left(\int_{\mathbb{T}} |\log |X||^{2q} d\mu \right)^{1/(2q)} \left(\int_{\mathbb{T}} \mathbf{1}_{\{|X| < e^{-T}\}} d\mu \right)^{1-1/(2q)} \\
&\leq W^{1/(2q)} A_{e^{-T}}^{1-1/(2q)} \leq \lambda(2Cq)^\beta (e^{-2T} + d_1 + b)^{1-1/(2q)}.
\end{aligned}$$

On the energy event $\{\int_{\mathbb{T}} |X|^2 d\mu \leq B_E\}$, by Tonelli's theorem applied to $(y - T)_+ = \int_T^\infty \mathbf{1}_{\{y > \tau\}} d\tau$ at $y = \log |X|$ and by Markov's inequality for μ , we bound the positive truncation error by

$$\begin{aligned}
\int_{\{|X| > e^T\}} (\log |X| - T) d\mu &= \int_T^\infty \mu\{|X| > e^\tau\} d\tau \leq \int_T^\infty e^{-2\tau} \left(\int_{\mathbb{T}} |X|^2 d\mu \right) d\tau \\
&\leq \int_T^\infty B_E e^{-2\tau} d\tau = \frac{B_E}{2} e^{-2T}.
\end{aligned}$$

For the Gaussian tails, $\int_T^\infty \mathbb{P}(|G| < e^{-\tau}) d\tau$ and $\int_T^\infty \mathbb{P}(|G| > e^\tau) d\tau$ are at most $\frac{1}{2}e^{-2T}$ by (19). On E_H , outside the Markov event for W , the integral $\int_{\mathbb{T}} \log |X| d\mu$ is finite, since $\int_{\mathbb{T}} |\log |X||^{2q} d\mu = W \leq \lambda^{2q}(2Cq)^{2\beta q}$ and $2q \geq 1$. Since

$$|\ell_T(z) - \log |z|| = (-\log |z| - T)_+ + (\log |z| - T)_+,$$

on E_H and on the energy event, outside the Chebyshev event for $A_{e^{-T}}$ and the Markov event for W , it follows from the layer-cake identity, the two truncation bounds and the two Gaussian tail bounds that

$$\begin{aligned}
\left| \int_{\mathbb{T}} \log |X| d\mu - L_T \right| &\leq \int_{\mathbb{T}} |\log |X| - \ell_T(X)| d\mu \\
&= \int_{\{|X| < e^{-T}\}} (-\log |X| - T) d\mu + \int_{\{|X| > e^T\}} (\log |X| - T) d\mu \\
&\leq \lambda(2Cq)^\beta (e^{-2T} + d_1 + b)^{1-1/(2q)} + \frac{B_E}{2} e^{-2T},
\end{aligned}$$

$$\begin{aligned}
|\mathbb{E}\ell_T(G) - \mathbb{E}\log |G|| &\leq \mathbb{E}|\ell_T(G) - \log |G|| = \mathbb{E}(-\log |G| - T)_+ + \mathbb{E}(\log |G| - T)_+ \\
&= \int_T^\infty \mathbb{P}(|G| < e^{-\tau}) d\tau + \int_T^\infty \mathbb{P}(|G| > e^\tau) d\tau \leq e^{-2T}.
\end{aligned}$$

Finally, $J_N(r) - \log \sigma_N(r) = \int \log |X| d\mu$, and outside the five exceptional events of Table 16, by the previous display and the bounds on $|L_T - \mathbb{E}L_T|$ and $|\mathbb{E}L_T - \mathbb{E}\ell_T(G)|$, we have

$$\begin{aligned}
\left| \int \log |X| d\mu - \mathbb{E}\log |G| \right| &\leq \left| \int \log |X| d\mu - L_T \right| + |L_T - \mathbb{E}L_T| \\
&\quad + |\mathbb{E}L_T - \mathbb{E}\ell_T(G)| + |\mathbb{E}\ell_T(G) - \mathbb{E}\log |G|| \\
&\leq \lambda(2Cq)^\beta (e^{-2T} + d_1 + b)^{1-1/(2q)} + \frac{B_E}{2} e^{-2T} + u + 2Td_1 + e^{-2T} = \eta_{\log}.
\end{aligned}$$

Under (73) and (71) the last two events are absent. If $\int_{\mathbb{T}} |X|^2 d\mu \leq B_E$ on E_H , then the complement of the energy event is contained in E_H^c . The last two events

Exceptional event	Defined by	Probability at most
Chebyshev event for $A_{e^{-T}}$	$A_{e^{-T}} > e^{-2T} + d_1 + b$	v/b^2
Markov event for W	$\omega \in E_H,$ $W > \lambda^{2q}(2Cq)^{2\beta q}$	λ^{-2q}
Chebyshev event for L_T	$ L_T - \mathbb{E}L_T > u$	$4T^2v/u^2$
Complement of the energy event	$\int_{\mathbb{T}} X ^2 d\mu > B_E$	p_E
Complement of E_H		p_H

Table 16: The exceptional events in the proof of Theorem B.1. The sum of the bounds on their probabilities is (76).

then have probability at most $\mathbb{P}(E_H^c)$ together, which proves the second variant in the statement. By (72), the logarithmic integral is finite almost surely on E_H , even for polynomials with roots on the integration circle. Since $|G|^2$ has the exponential distribution with mean one, (20) gives $\mathbb{E} \log |G| = -\gamma/2$. $\square$

Remark B.2 (Polynomial choices of parameters). As in Proposition 8.1, let $d_1 \leq DN^{-1/2}$, $v \leq VN^{-1/2}$, $\beta = 6$, $T = t \log N$, $b = N^{-2t}$, $u = N^{-t}$, $q = 6 \log N \geq 1$ and $\lambda = e$, so $\lambda(2Cq)^6 = e(12C \log N)^6$. Under (71) and (73), take $p_H = 0$, $B_E = 1$ and $p_E = 0$. Then the terms of (75) and (76), other than u and $p_E = p_H = 0$, satisfy

$$\begin{aligned}
2Td_1 &\leq 2t(\log N)DN^{-1/2}, & (B_E/2 + 1)e^{-2T} &= \frac{3}{2}N^{-2t}, \\
e^{-2T} + d_1 + b &\leq N^{-2t} + DN^{-1/2} + N^{-2t} = 2N^{-2t} + DN^{-1/2}, \\
v/b^2 &\leq VN^{-1/2}/N^{-4t} = VN^{4t-1/2}, & \lambda^{-2q} &= e^{-12 \log N} = N^{-12}, \\
4T^2v/u^2 &\leq 4t^2(\log N)^2VN^{-1/2}/N^{-2t} = 4t^2V(\log N)^2N^{2t-1/2}.
\end{aligned}$$

Since $2N^{-2t} + DN^{-1/2} \leq 1$ for large N and $1 - 1/(2q) \geq 1/2$, the power in (75) is then at most the square root of its base. By substituting these bounds, we get

$$\eta_{\log} \leq N^{-t} + 2tD(\log N)N^{-1/2} + e(12C \log N)^6 \sqrt{2N^{-2t} + DN^{-1/2}} + \frac{3}{2}N^{-2t},$$

$$p_{\text{fail}} \leq VN^{4t-1/2} + N^{-12} + 4t^2V(\log N)^2N^{2t-1/2}.$$

The failure terms tend to zero when $4t < 1/2$. For $N = j^s$, the three failure terms are

$$Vj^{-s(1/2-4t)}, \quad j^{-12s}, \quad 4t^2Vs^2(\log j)^2j^{-s(1/2-2t)},$$

and they are summable if

$$s(\frac{1}{2} - 4t) > 1, \quad 12s > 1, \quad s(\frac{1}{2} - 2t) > 1.$$

In particular, take $t = 1/32$ and $s = 8$. Then $s(\frac{1}{2} - 4t) = 8(\frac{1}{2} - \frac{1}{8}) = 3$, $12s = 96$ and $s(\frac{1}{2} - 2t) = 8(\frac{1}{2} - \frac{1}{16}) = \frac{7}{2}$ all exceed one, and for $N \geq e$ with $2N^{-1/16} + DN^{-1/2} \leq 1$,

$$\begin{aligned} \eta_{\log} &\leq N^{-1/32} + \frac{D}{16}(\log N)N^{-1/2} + e(12C \log N)^6 \sqrt{2N^{-1/16} + DN^{-1/2}} + \frac{3}{2}N^{-1/16} \\ &\leq \left(\frac{5}{2} + \frac{D}{16} + e(12C)^6 \sqrt{2 + D} \right) N^{-1/32} (\log N)^6, \end{aligned}$$

by the bound on $\eta_{\log}$ above with $2tD = D/16$, $2t = 1/16$ and $u = N^{-1/32}$, and by $\log N \geq 1$ and $N^{-1/2} \leq N^{-1/16} \leq N^{-1/32}$. The failure is summable on j^8 , and the error is at most $N^{-1/32}(\log N)^7$ once $\log N$ is at least the above constant, a deterministic threshold.

Lemma 3.2 instead keeps the power $1 - 1/(2q)$, with $q = \lceil \log N \rceil$ and $\lambda = (2e \log N/q)^6 \log N \geq e^6 \log N$. Then $\lambda(2Cq)^6 = (4eC)^6 (\log N)^7$, $\lambda^{-2q} \leq N^{-12}$ and $x^{1-1/(2q)} \leq e^{1/2}x$ for $x \geq N^{-1}$. The lower tail is then of order $N^{-2t}(\log N)^7$ rather than $N^{-t}(\log N)^6$, which allows the deviation $u = N^{-2t}(\log N)^7$. That lemma takes $t = 1/32$, the clipping level $(\log N)/32$, so that the error is of order $N^{-1/16}(\log N)^7$ and the three failure terms are of order $N^{-3/8}$, N^{-12} and $N^{-3/8}(\log N)^{-12}$. The choice $q = 6 \log N$, $\lambda = e$ above gives the same failure bound N^{-12} with $\lambda(2Cq)^6 = e(12C \log N)^6$. The extra factor $\log N$ in λ is the difference between the exponents 7 and 6.

B.2 A local radial secant estimate

Lemma B.3 (Local approximation of radial counts). *Let f_N be a nonzero polynomial of degree at most N , and let $\sigma_N(r) > 0$ be deterministic. Suppose that the function $F_N(x)$, equal to $\log \sigma_N(1 + x/N)$ minus a centering independent of x , satisfies $|F_N(x) - F(x)| \leq \varepsilon$ on $[-K, K]$. Suppose also $|F'| \leq B_F$ and $|F'(x) - F'(y)| \leq M_F|x - y|$ there. Suppose the three logarithmic deviations*

$$|J_N(1 + y/N) - \log \sigma_N(1 + y/N) - \mathbb{E} \log |G||, \quad y = x - h, x, x + h,$$

are at most $\eta_{\log}$ outside their respective exceptional events. If $h > 0$ and $|x| + h \leq K < N$, then outside the union of those three events,

$$(77) \quad \left| \frac{\nu_N(1 + x/N)}{N} - F'(x) \right| \leq \frac{M_F h}{2} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h}.$$

In particular an error $\eta_{\log} + \varepsilon \leq N^{-1/32}(\log N)^7$, with $h = N^{-\kappa}$, $0 < \kappa \leq 1/64$, gives convergence to $F'(0)$ uniformly for deterministic $|x| \leq h$.

Proof. By Jensen's formula, $\nu_N(r)/N$ lies between two secant slopes of J_N in the variable $N \log \rho$, one on each side of r . The idea is to replace J_N by F at the three points, up to a constant which cancels, and then to compare the secant slopes of F with $F'(x)$. The three errors come from the curvature of F , the change of the denominators, and the deviations of J_N from F at the three points.

Let $r = 1 + x/N$ and set

$$d_- = N \log \frac{r}{r - h/N}, \quad d_+ = N \log \frac{r + h/N}{r}.$$

Since $|x| + h \leq K < N$, we have $r - h/N = 1 + (x - h)/N \geq 1 - K/N > 0$. By the integral form (4) of Jensen's formula, in which ν_N counts the roots in the closed disk, and since ν_N is nondecreasing, we have

$$\begin{aligned} J_N(r) - J_N(r - h/N) &= \int_{r-h/N}^r \frac{\nu_N(\rho)}{\rho} d\rho \leq \nu_N(r) \log \frac{r}{r - h/N} = \frac{\nu_N(r)d_-}{N}, \\ J_N(r + h/N) - J_N(r) &= \int_r^{r+h/N} \frac{\nu_N(\rho)}{\rho} d\rho \geq \nu_N(r) \log \frac{r + h/N}{r} = \frac{\nu_N(r)d_+}{N}. \end{aligned}$$

Dividing by $d_{\pm} > 0$, we obtain the Jensen bounds

$$\frac{J_N(r) - J_N(r - h/N)}{d_-} \leq \frac{\nu_N(r)}{N} \leq \frac{J_N(r + h/N) - J_N(r)}{d_+}.$$

Since $N \log(1 + u/N)$ has derivative $1/(1 + u/N)$ in u , we have

$$\begin{aligned} d_- &= N \log(1 + x/N) - N \log(1 + (x - h)/N) = \int_{x-h}^x \frac{du}{1 + u/N}, \\ d_+ &= N \log(1 + (x + h)/N) - N \log(1 + x/N) = \int_x^{x+h} \frac{du}{1 + u/N}. \end{aligned}$$

Since $|x| + h \leq K < N$, every $u \in [x - h, x + h]$ satisfies $0 < 1 - K/N \leq 1 + u/N \leq 1 + K/N$, and taking reciprocals and integrating over an interval of length h , we obtain

$$\begin{aligned} \frac{1}{1 + K/N} &\leq \frac{1}{1 + u/N} \leq \frac{1}{1 - K/N}, & \frac{h}{1 + K/N} &\leq d_{\pm} \leq \frac{h}{1 - K/N}, \\ 1 - \frac{K}{N} &\leq \frac{h}{d_{\pm}} \leq 1 + \frac{K}{N}, & \left| \frac{h}{d_{\pm}} - 1 \right| &\leq \frac{K}{N}. \end{aligned}$$

Since F' is Lipschitz on $[-K, K]$, which contains $x \pm t$ for $0 \leq t \leq h$, for the ordinary secants we have

$$\left| \frac{F(x + h) - F(x)}{h} - F'(x) \right| = \left| \frac{1}{h} \int_0^h (F'(x + t) - F'(x)) dt \right|$$

$$\begin{aligned}
&\leq \frac{1}{h} \int_0^h M_F t \, dt = \frac{M_F h}{2}, \\
\left| \frac{F(x) - F(x-h)}{h} - F'(x) \right| &= \left| \frac{1}{h} \int_0^h (F'(x-t) - F'(x)) \, dt \right| \\
&\leq \frac{1}{h} \int_0^h M_F t \, dt = \frac{M_F h}{2}.
\end{aligned}$$

Each ordinary secant has absolute value at most B_F , since it is the mean of F' over an interval in $[-K, K]$. Replacing its denominator h by $d_{\pm}$ multiplies it by $h/d_{\pm}$, and

$$\begin{aligned}
\left| \frac{F(x+h) - F(x)}{d_+} - \frac{F(x+h) - F(x)}{h} \right| &= \left| \frac{F(x+h) - F(x)}{h} \right| \left| \frac{h}{d_+} - 1 \right| \\
&\leq B_F \left| \frac{h}{d_+} - 1 \right| \leq \frac{B_F K}{N}, \\
\left| \frac{F(x) - F(x-h)}{d_-} - \frac{F(x) - F(x-h)}{h} \right| &= \left| \frac{F(x) - F(x-h)}{h} \right| \left| \frac{h}{d_-} - 1 \right| \\
&\leq B_F \left| \frac{h}{d_-} - 1 \right| \leq \frac{B_F K}{N}.
\end{aligned}$$

Let c be the centering, so that $F_N(x) = \log \sigma_N(1+x/N) - c$. For $y = x-h, x, x+h$, which lie in $[-K, K]$, we have

$$\begin{aligned}
&J_N(1+y/N) - \mathbb{E} \log |G| - c - F(y) \\
&= \left\{ J_N(1+y/N) - \log \sigma_N(1+y/N) - \mathbb{E} \log |G| \right\} + \left\{ F_N(y) - F(y) \right\}.
\end{aligned}$$

The constant $\mathbb{E} \log |G| + c$ does not depend on y , so it cancels from each secant. Outside the exceptional event at y , the first brace is at most $\eta_{\log}$ in absolute value and the second at most ε . Since $r \pm h/N = 1 + (x \pm h)/N$, subtracting the identity at two neighboring points gives

$$\begin{aligned}
\left| J_N(r+h/N) - J_N(r) - (F(x+h) - F(x)) \right| &\leq 2(\eta_{\log} + \varepsilon), \\
\left| J_N(r) - J_N(r-h/N) - (F(x) - F(x-h)) \right| &\leq 2(\eta_{\log} + \varepsilon).
\end{aligned}$$

By the lower bound $d_{\pm} \geq h/(1+K/N)$, the secant error is therefore at most $2(\eta_{\log} + \varepsilon)/d_{\pm} \leq 2(\eta_{\log} + \varepsilon)(1+K/N)/h$. Adding the three errors to the Jensen bounds, we obtain

$$\begin{aligned}
\frac{\nu_N(r)}{N} &\leq \frac{J_N(r+h/N) - J_N(r)}{d_+} \leq \frac{F(x+h) - F(x)}{d_+} + \frac{2(\eta_{\log} + \varepsilon)(1+K/N)}{h} \\
&\leq \frac{F(x+h) - F(x)}{h} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1+K/N)}{h}
\end{aligned}$$

$$\begin{aligned}
&\leq F'(x) + \frac{M_F h}{2} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h}, \\
\frac{\nu_N(r)}{N} &\geq \frac{J_N(r) - J_N(r - h/N)}{d_-} \geq \frac{F(x) - F(x - h)}{d_-} - \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h} \\
&\geq \frac{F(x) - F(x - h)}{h} - \frac{B_F K}{N} - \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h} \\
&\geq F'(x) - \frac{M_F h}{2} - \frac{B_F K}{N} - \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h},
\end{aligned}$$

which together give (77). The Jensen bounds include roots on the counting circle. Since the three exceptional events may be arbitrarily dependent, we add their three probabilities. To prove the last assertion, let $|x| \leq h = N^{-\kappa}$ and let N be so large that $2h \leq K$, so that $|x| + h \leq K$. Since $|F'(x) - F'(0)| \leq M_F |x| \leq M_F h$, it follows from (77) and $\eta_{\log} + \varepsilon \leq N^{-1/32}(\log N)^7$ that

$$\begin{aligned}
\left| \frac{\nu_N(1 + x/N)}{N} - F'(0) \right| &\leq \left| \frac{\nu_N(1 + x/N)}{N} - F'(x) \right| + |F'(x) - F'(0)| \\
&\leq \frac{M_F h}{2} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h} + M_F h \\
&= \frac{3M_F}{2} N^{-\kappa} + \frac{B_F K}{N} + 2 \left(1 + \frac{K}{N} \right) (\eta_{\log} + \varepsilon) N^\kappa \\
&\leq \frac{3M_F}{2} N^{-\kappa} + \frac{B_F K}{N} + 2 \left(1 + \frac{K}{N} \right) N^{\kappa-1/32} (\log N)^7,
\end{aligned}$$

which tends to zero since $\kappa \leq 1/64 < 1/32$. $\square$

B.3 A second-moment bound for small-derivative roots

Mesh and local multiplicity. Let $\mathcal{A}$ be an annulus contained in $\{|z| \leq R\}$ and let $z_1, \dots, z_m$ be retained mesh points. Apart from at most b_0 roots in explicitly excluded sectors, every root of f_N in $\mathcal{A}$ is within distance τ of a retained point. The segment between them lies in a specified domain D , and $|z_i| \leq R$. Except on events of probabilities at most p_2 and p_0 , respectively,

$$\sup_D |f_N''| \leq M_2, \quad \text{each mesh cell contains at most } q_0 \text{ roots.}$$

The probability p_0 also includes failure of the sector bound.

For the cell bound we can use a disk count, provided that each cell lies in one of the disks of a local count such as Lemma 4.1.

Jet approximation. We write the pair

$$W(z_i) = (W_0(z_i), W_1(z_i)) = (f_N(z_i)/\sqrt{N}, z_i f'_N(z_i)/N^{3/2})$$

as a real four-vector. For every retained i there is a mean-zero Gaussian four-vector $H_i = ((H_i)_0, (H_i)_1)$, written as a pair like $W(z_i)$, with covariance least eigenvalue at least $\lambda > 0$. For the product-of-disks event $E_i = \{|W_0(z_i)| \leq u, |W_1(z_i)| \leq v_0\}$, put $g_i = \mathbb{P}(|(H_i)_0| \leq u, |(H_i)_1| \leq v_0)$ and assume

$$|\mathbb{P}(E_i) - g_i| \leq \delta.$$

All but at most $b_* m^2$ ordered pairs (i, j) additionally obey

$$|\mathbb{P}(E_i \cap E_j) - g_i g_j| \leq \delta_* + \chi.$$

For example, the pair estimate can be obtained by an eight-dimensional Gaussian approximation with error δ_* , followed by a comparison with the independent marginal Gaussians with total variation error χ . We include in the exceptional count the ordinary and conjugate angular diagonals, the radial duplicates, and $i = j$, unless they satisfy the above pair estimate.

Theorem B.4 (Annular small-derivative count). *Assume the [mesh and local multiplicity](#) bounds and the [jet approximation](#), and denote by B_N the number of roots in $\mathcal{A}$ satisfying $|f'_N(\alpha)| < N^{3/2}/L$. In the jet hypothesis use*

$$u = \frac{N^{3/2}\tau/L + M_2\tau^2/2}{\sqrt{N}}, \quad v_0 = \frac{R(N^{3/2}/L + M_2\tau)}{N^{3/2}}.$$

We set

$$\mu_* = m \left(\frac{u^2 v_0^2}{4\lambda^2} + \delta \right), \quad V_* = b_* + \delta_* + \chi + 2\delta.$$

For every $t > 0$,

$$(78) \quad \mathbb{P}\{B_N > b_0 + q_0(\mu_* + t)\} \leq p_0 + p_2 + \frac{m^2 V_*}{t^2}.$$

Proof. The idea is to detect each counted root outside the excluded sectors at its assigned mesh point, where both components of W are small, and to bound the number of mesh points with small W by its mean and its variance, which the jet approximation controls. We then bound B_N by Chebyshev's inequality and the cell bound.

Let α be a root in B_N outside the excluded sectors, and let z_i be its assigned mesh point. Then $f_N(\alpha) = 0$, $|f'_N(\alpha)| < N^{3/2}/L$ and $|z_i - \alpha| \leq \tau$, and the segment $[\alpha, z_i]$ lies in D by the [mesh and local multiplicity](#) bounds. On the derivative event $\{\sup_D |f''_N| \leq M_2\}$, Taylor's theorem with integral remainder along the segment gives

$$|f_N(z_i)| = \left| f'_N(\alpha)(z_i - \alpha) + \int_{[\alpha, z_i]} (z_i - \zeta) f''_N(\zeta) d\zeta \right|$$

$$\begin{aligned} &\leq |f'_N(\alpha)| |z_i - \alpha| + \frac{M_2}{2} |z_i - \alpha|^2 \leq \frac{N^{3/2}\tau}{L} + \frac{M_2\tau^2}{2}, \\ |f'_N(z_i)| &= \left| f'_N(\alpha) + \int_{[\alpha, z_i]} f''_N(\zeta) d\zeta \right| \leq |f'_N(\alpha)| + M_2 |z_i - \alpha| \leq \frac{N^{3/2}}{L} + M_2\tau. \end{aligned}$$

Since $|z_i| \leq R$, dividing by the normalizations in $W(z_i)$ we obtain

$$\frac{|f_N(z_i)|}{\sqrt{N}} \leq \frac{N^{3/2}\tau/L + M_2\tau^2/2}{\sqrt{N}} = u, \quad \frac{|z_i f'_N(z_i)|}{N^{3/2}} \leq \frac{R(N^{3/2}/L + M_2\tau)}{N^{3/2}} = v_0,$$

so that the event E_i occurs at every assigned point.

We set $Z = \sum_{i=1}^m \mathbf{1}_{E_i}$. Let Σ_i be the covariance matrix of H_i , whose eigenvalues are at least λ , so that it has determinant at least λ^4 . Hence the density of H_i is bounded by $(2\pi)^{-2}\lambda^{-2}$, since at every $y \in \mathbb{R}^4$ it satisfies

$$(2\pi)^{-2}(\det \Sigma_i)^{-1/2} \exp\left(-\frac{1}{2}\langle \Sigma_i^{-1}y, y \rangle\right) \leq (2\pi)^{-2}(\det \Sigma_i)^{-1/2} \leq (2\pi)^{-2}\lambda^{-2}.$$

Since the product of the two complex disks has four-dimensional volume $\pi^2 u^2 v_0^2$, by the [jet approximation](#) we have

$$\begin{aligned} g_i &\leq (2\pi)^{-2}\lambda^{-2}\pi^2 u^2 v_0^2 = \frac{u^2 v_0^2}{4\lambda^2}, \\ \mathbb{E}Z &= \sum_{i=1}^m \mathbb{P}(E_i) \leq \sum_{i=1}^m (g_i + \delta) \leq m \left(\frac{u^2 v_0^2}{4\lambda^2} + \delta \right) = \mu_*. \end{aligned}$$

For an admissible ordered pair (i, j) , that is, one that obeys the pair estimate of the jet approximation, since $\mathbb{P}(E_i), g_j \leq 1$ we have

$$\begin{aligned} |\mathbb{P}(E_i)\mathbb{P}(E_j) - g_i g_j| &= |\mathbb{P}(E_i)(\mathbb{P}(E_j) - g_j) + g_j(\mathbb{P}(E_i) - g_i)| \\ &\leq \mathbb{P}(E_i)|\mathbb{P}(E_j) - g_j| + g_j|\mathbb{P}(E_i) - g_i| \leq 2\delta, \\ |\text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j})| &= |\mathbb{P}(E_i \cap E_j) - \mathbb{P}(E_i)\mathbb{P}(E_j)| \\ &\leq |\mathbb{P}(E_i \cap E_j) - g_i g_j| + |g_i g_j - \mathbb{P}(E_i)\mathbb{P}(E_j)| \leq \delta_* + \chi + 2\delta. \end{aligned}$$

On each of the at most $b_* m^2$ exceptional pairs we use the bound one, and summing over all m^2 ordered pairs, including the diagonal, we obtain

$$\begin{aligned} \text{Var } Z &= \sum_{i,j=1}^m \text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j}) \leq \sum_{i,j=1}^m |\text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j})| \\ &\leq b_* m^2 + m^2(\delta_* + \chi + 2\delta) = m^2 V_*. \end{aligned}$$

Since $\mathbb{E}Z \leq \mu_*$, Chebyshev's inequality yields

$$\mathbb{P}(Z > \mu_* + t) \leq \mathbb{P}(Z - \mathbb{E}Z > t) \leq \frac{\text{Var } Z}{t^2} \leq \frac{m^2 V_*}{t^2}.$$

On the two geometric good events, each cell receives at most q_0 roots, and the excluded sectors contain at most b_0 roots. On these events and on the derivative event, by the first part of the proof every counted root outside the excluded sectors is assigned to a retained point z_i at which E_i occurs. The cell of z_i contains the roots assigned to z_i , so the cell bound of the [mesh hypothesis](#) allows at most q_0 of them. Hence $B_N \leq b_0 + q_0 Z$, so that $B_N > b_0 + q_0(\mu_* + t)$ implies $Z > \mu_* + t$ outside the geometric and derivative exceptional events, and

$$\mathbb{P}\{B_N > b_0 + q_0(\mu_* + t)\} \leq p_0 + p_2 + \mathbb{P}(Z > \mu_* + t) \leq p_0 + p_2 + \frac{m^2 V_*}{t^2}.$$

□

Corollary B.5 (Polynomial parameters for the mesh bound). *Suppose the following bounds hold with fixed positive constants:*

$$\begin{aligned} R &\leq 2, & A &\geq S, & \tau &= (ANL\sqrt{\log N})^{-1}, \\ M_2 &= SN^{5/2}\sqrt{\log N}, & m &\leq DNL^2 \log N, & q_0 &\leq C_0 \log N. \\ \delta &\leq D_0 N^{-\beta_1}, & V_* &\leq VN^{-\beta}, & L &= N^\ell. \end{aligned}$$

Then

$$(79) \quad \mu_* \leq \frac{9D}{A^2\lambda^2} NL^{-4} + DD_0 N^{1-\beta_1} L^2 \log N.$$

For a chosen $d > 0$, impose the explicit deterministic thresholds

$$b_0 \leq \tfrac{1}{2}N^{1-d}, \quad C_0 \log N \left[\frac{9D}{A^2\lambda^2} NL^{-4} + DD_0 N^{1-\beta_1} L^2 \log N \right] \leq \tfrac{1}{4}N^{1-d}.$$

Then

$$(80) \quad \mathbb{P}(B_N > N^{1-d}) \leq p_0 + p_2 + 16C_0^2 D^2 V N^{-\beta+4\ell+2d} (\log N)^4.$$

The mean thresholds are eventual if $d < 4\ell$, $d < \beta_1 - 2\ell$, and $b_0 = o(N^{1-d})$. The last failure term is summable on $N = j^s$ whenever

$$s(\beta - 4\ell - 2d) > 1.$$

In particular $\beta = \beta_1 = 1/2$, $\ell = 1/64$, $d = 1/32$ give $B_N \leq N^{31/32}$. The probability of the exceptional event, in addition to $p_0 + p_2$, is at most

$$16C_0^2 D^2 V N^{-3/8} (\log N)^4.$$

Proof. We first substitute the parameters into the definitions of u , v_0 and μ_* in Theorem B.4, and then apply this theorem with a deviation t such that $b_0 + q_0(\mu_* + t) \leq N^{1-d}$.

With the chosen parameters, and since $S \leq A$ and $R \leq 2$, we have

$$\begin{aligned} \frac{N^{3/2}\tau}{L} &= \frac{N^{1/2}}{AL^2\sqrt{\log N}}, & \frac{M_2\tau^2}{2} &= \frac{SN^{1/2}}{2A^2L^2\sqrt{\log N}}, \\ u &= \left(\frac{1}{A} + \frac{S}{2A^2}\right) L^{-2}(\log N)^{-1/2} \leq \frac{3}{2A} L^{-2}(\log N)^{-1/2}, \\ M_2\tau &= \frac{S}{A} \cdot \frac{N^{3/2}}{L}, & v_0 &= \frac{R}{L} \left(1 + \frac{S}{A}\right) \leq \frac{4}{L}, \\ \frac{u^2v_0^2}{4\lambda^2} &\leq \frac{1}{4\lambda^2} \frac{9}{4A^2L^4\log N} \frac{16}{L^2} = \frac{9}{A^2\lambda^2\log N} L^{-6}. \end{aligned}$$

Using also $m \leq DNL^2\log N$ and $\delta \leq D_0N^{-\beta_1}$, we have

$$\mu_* \leq DNL^2\log N \left(\frac{9}{A^2\lambda^2\log N} L^{-6} + D_0N^{-\beta_1} \right) = \frac{9D}{A^2\lambda^2} NL^{-4} + DD_0N^{1-\beta_1} L^2\log N.$$

This proves (79).

We now choose the deviation t in Theorem B.4. The event $\{B_N > N^{1-d}\}$ is contained in the event of (78) when $b_0 + q_0(\mu_* + t) \leq N^{1-d}$. Since the probability bound m^2V_*/t^2 decreases in t , we take the largest t with $C_0t\log N \leq \frac{1}{4}N^{1-d}$,

$$t = \frac{N^{1-d}}{4C_0\log N}.$$

By the two deterministic thresholds, the mean bound (79) with $q_0 \leq C_0\log N$, and this choice of t , we have

$$\begin{aligned} b_0 + q_0(\mu_* + t) &= b_0 + q_0\mu_* + q_0t \\ &\leq \frac{1}{2}N^{1-d} + C_0\log N \left[\frac{9D}{A^2\lambda^2} NL^{-4} + DD_0N^{1-\beta_1} L^2\log N \right] \\ &\quad + C_0t\log N \leq \frac{1}{2}N^{1-d} + \frac{1}{4}N^{1-d} + \frac{1}{4}N^{1-d} = N^{1-d}. \end{aligned}$$

Hence, by (78), $m \leq DNL^2\log N$, $V_* \leq VN^{-\beta}$ and $L = N^\ell$, we have

$$\begin{aligned} \mathbb{P}(B_N > N^{1-d}) &\leq \mathbb{P}\{B_N > b_0 + q_0(\mu_* + t)\} \leq p_0 + p_2 + \frac{m^2V_*}{t^2}, \\ \frac{m^2V_*}{t^2} &\leq D^2N^2L^4(\log N)^2 VN^{-\beta} \frac{16C_0^2(\log N)^2}{N^{2-2d}} \\ &= 16C_0^2D^2VN^{-\beta}L^4N^{2d}(\log N)^4 = 16C_0^2D^2VN^{-\beta+4\ell+2d}(\log N)^4. \end{aligned}$$

We have thus proved (80). With $L = N^\ell$, the left-hand side of the second threshold equals

$$\begin{aligned} & \frac{9C_0D}{A^2\lambda^2} N^{1-4\ell} \log N + C_0DD_0N^{1-\beta_1+2\ell}(\log N)^2 \\ &= \frac{1}{4} N^{1-d} \left(\frac{36C_0D}{A^2\lambda^2} N^{d-4\ell} \log N + 4C_0DD_0N^{d-\beta_1+2\ell}(\log N)^2 \right). \end{aligned}$$

When $d < 4\ell$ and $d < \beta_1 - 2\ell$, the factor in parentheses tends to zero, and so the second threshold holds for all large N . The first threshold holds for all large N when $b_0 = o(N^{1-d})$. With $N = j^s$, the last failure term equals $16C_0^2D^2Vj^{-s(\beta-4\ell-2d)}(s \log j)^4$, which is summable when $s(\beta - 4\ell - 2d) > 1$. For the particular powers of the corollary, $4\ell = 2d = 1/16$, and the three exponents, the conditions $d < 4\ell$ and $d < \beta_1 - 2\ell$, and the summability at $N = j^8$ read

$$\begin{aligned} 1 - 4\ell &= \frac{15}{16}, & 1 - \beta_1 + 2\ell &= \frac{17}{32}, & -\beta + 4\ell + 2d &= -\frac{3}{8}, \\ 4\ell = \frac{1}{16} &> \frac{1}{32} = d, & \beta_1 - 2\ell &= \frac{15}{32} > d, & 8(\beta - 4\ell - 2d) &= 3 > 1. \end{aligned}$$

Since $1 - d = 31/32$, once the two thresholds hold, we obtain $B_N \leq N^{31/32}$ outside an event of probability at most $p_0 + p_2 + 16C_0^2D^2VN^{-3/8}(\log N)^4$. $\square$

In the proof of Lemma 4.5 we instead take $t = NL^{-4}$ in (78), of the order of the leading term of (79). This needs no mean threshold, and it gives the smaller count $C(K)NL^{-4} \log N$ with the failure term $C(K)N^{-5/16}(\log N)^2$.

B.4 From a sparse subsequence of degrees to the full sequence

Theorem B.6 (Sparse-to-dense convergence). *Let $s \geq 1$ be fixed, $n_j = \lfloor j^s \rfloor$, and let (X_n) be real random variables on a common probability space. Suppose $X_{n_j}/n_j \rightarrow c$ almost surely. If there are deterministic $\varepsilon_j \rightarrow 0$ and events E_j such that*

$$\sum_j \mathbb{P}(E_j) < \infty, \quad \max_{n_j \leq n < n_{j+1}} |X_n - X_{n_j}| \leq \varepsilon_j n_j \quad \text{on } E_j^c,$$

then $X_n/n \rightarrow c$ almost surely.

The numbers ε_j need not be monotone. In particular, the case $c = 1/2$ yields the strong law needed for the unweighted Rademacher model.

Proof. We compare each degree n of the block $n_j \leq n < n_{j+1}$ with n_j , and use the Borel–Cantelli lemma to discard the events E_j . For every J , by the union bound

we have

$$\mathbb{P}\left(\bigcup_{j \geq J} E_j\right) \leq \sum_{j \geq J} \mathbb{P}(E_j) \longrightarrow 0$$

as $J \rightarrow \infty$, since the series converges. It follows that almost surely only finitely many E_j occur, which is the Borel–Cantelli lemma. For $n_j \leq n < n_{j+1}$ we write

$$\frac{X_n}{n} - c = \frac{n_j}{n} \left(\frac{X_{n_j}}{n_j} - c \right) + \frac{X_n - X_{n_j}}{n} + c \left(\frac{n_j}{n} - 1 \right).$$

Outside E_j , by the block bound $|X_n - X_{n_j}| \leq \varepsilon_j n_j$, it follows that

$$\begin{aligned} \left| \frac{X_n}{n} - c \right| &\leq \frac{n_j}{n} \left| \frac{X_{n_j}}{n_j} - c \right| + \frac{|X_n - X_{n_j}|}{n} + |c| \left| \frac{n_j}{n} - 1 \right| \\ &\leq \frac{n_j}{n} \left| \frac{X_{n_j}}{n_j} - c \right| + \frac{\varepsilon_j n_j}{n} + |c| \frac{n - n_j}{n} \\ &\leq \frac{n_j}{n} \left| \frac{X_{n_j}}{n_j} - c \right| + \varepsilon_j + |c| \frac{n_{j+1} - n_j}{n_j}, \end{aligned}$$

where the last step uses $\varepsilon_j \geq 0$, which holds on E_j^c since there $0 \leq |X_n - X_{n_j}| \leq \varepsilon_j n_j$. Since $j^s - 1 < n_j \leq j^s$ for every j , we have, for $j \geq 2$,

$$\frac{n_{j+1}}{n_j} < \frac{(j+1)^s}{j^s - 1} = \frac{(1 + 1/j)^s}{1 - j^{-s}} \longrightarrow 1.$$

Since also $n_{j+1} \geq n_j$, it follows that $n_{j+1}/n_j \rightarrow 1$, and hence the last ratio, which equals $n_{j+1}/n_j - 1$, tends to zero. As $n \rightarrow \infty$, the index j with $n_j \leq n < n_{j+1}$ tends to infinity. Almost surely, $X_{n_j}/n_j \rightarrow c$ and E_j occurs for only finitely many j , and then the three terms of the last bound tend to zero, the first since $n_j/n \leq 1$ and the second since $\varepsilon_j \rightarrow 0$. This proves the assertion. $\square$

B.5 Deterministic choice of a vanishing block envelope

Lemma B.7 (Deterministic diagonalization). *Let $s \geq 1$, $n_j = \lfloor j^s \rfloor$, and let (X_n) be real random variables on a common probability space. Suppose that for each positive integer k ,*

$$\mathbb{P}\left\{\max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} > a_k + b_k(j)\right\} \leq p_k(j),$$

where $a_k \rightarrow 0$, $b_k(j) \rightarrow 0$ for fixed k , and $\sum_j p_k(j) < \infty$ for fixed k . Then there are deterministic numbers $\varepsilon_j \rightarrow 0$ and events E_j with $\sum_j \mathbb{P}(E_j) < \infty$ such that

$$\max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} \leq \varepsilon_j \quad \text{on } E_j^c.$$

In particular, if $X_{n_j}/n_j \rightarrow c$ almost surely, then $X_n/n \rightarrow c$ almost surely.

Proof. The idea is to let the index k grow slowly with j , along a deterministic diagonal $k(j) \rightarrow \infty$ chosen so that the probabilities of the exceptional events along the diagonal are still summable, and then to apply Theorem B.6. Choose increasing deterministic integers J_k , with $J_k \geq k$ and $J_k > J_{k-1}$, such that for all $j \geq J_k$ we have $b_k(j) \leq 1/k$ and $\sum_{j \geq J_k} p_k(j) \leq 2^{-k}$, which is possible since $b_k(j) \rightarrow 0$ and $\sum_j p_k(j) < \infty$. Choosing the least admissible integer makes each J_k deterministic, and this definition does not require a numerical bound for J_k . For $j \geq J_1$, we set

$$k(j) = \max\{k : J_k \leq j\}, \quad \varepsilon_j = a_{k(j)} + \frac{1}{k(j)}.$$

Since $J_k \geq k$ and the J_k increase strictly, $k(j)$ is finite and tends to infinity, and therefore $\varepsilon_j \rightarrow 0$. Let E_j be the event that the normalized block increment exceeds ε_j , and for $j < J_1$ take $E_j = \Omega$ and choose ε_j arbitrarily. For $j \geq J_1$ we have $J_{k(j)} \leq j$, hence $b_{k(j)}(j) \leq 1/k(j)$ and $\varepsilon_j \geq a_{k(j)} + b_{k(j)}(j)$. By the hypothesis with $k = k(j)$ we therefore have $\mathbb{P}(E_j) \leq p_{k(j)}(j)$, and these bounds satisfy

$$\sum_{j \geq J_1} p_{k(j)}(j) \leq \sum_{k \geq 1} \sum_{j \geq J_k} p_k(j) \leq \sum_{k \geq 1} 2^{-k} = 1,$$

where the first step holds since each $j \geq J_1$ contributes the single nonnegative term $p_{k(j)}(j)$, with $j \geq J_{k(j)}$, which appears in the inner sum for $k = k(j)$. Since only finitely many $j < J_1$ remain, it follows that $\sum_j \mathbb{P}(E_j) < \infty$. Finitely many initial degrees do not affect the limit, and the final assertion follows from Theorem B.6. The conclusion holds for arbitrary dependence between degrees, blocks, and exceptional events. $\square$

B.6 The mean for coefficients in $\{0, 1\}$

Remark B.8 (The mean for coefficients in $\{0, 1\}$). Let $K \geq 2$, $N \geq 2K$, $|r - 1| \leq K/N$ and $\text{dist}(\theta, 2\pi\mathbb{Z}) > 0$, let $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$, as in Section 2, and let $D_N(z) = \sum_{k=0}^N z^k$. Then

$$\frac{|D_N(re^{i\theta})|}{\sigma_N(r)} \leq \frac{10e^{4K}}{\sqrt{N} \text{dist}(\theta, 2\pi\mathbb{Z})}.$$

Indeed, $D_N(re^{i\theta}) = \{1 - (re^{i\theta})^{N+1}\}/(1 - re^{i\theta})$. For the numerator, $r \leq 1 + K/N$ gives $|1 - (re^{i\theta})^{N+1}| \leq 1 + (1 + K/N)^{N+1} \leq 1 + e^{2K} \leq 2e^{2K}$. For the denominator, $|1 - re^{i\theta}|^2 = (1 - r)^2 + 4r \sin^2(\theta/2) \geq 4r \sin^2(\theta/2)$, and $r \geq 1 - K/N \geq 1/2$ together with $|\sin(\theta/2)| \geq \text{dist}(\theta, 2\pi\mathbb{Z})/\pi$, by the concavity of the sine on $[0, \pi/2]$, gives $|1 - re^{i\theta}| \geq (\sqrt{2}/\pi) \text{dist}(\theta, 2\pi\mathbb{Z})$. For the normalization, every term of $\sigma_N(r)^2$ is at least $(1 - K/N)^{2N} \geq e^{-4K}$, by $\log(1 - u) \geq -2u$ for $0 \leq u \leq 1/2$, so that $\sigma_N(r) \geq \sqrt{N}e^{-2K}$. Combining the three bounds and $\sqrt{2}\pi < 10$ gives the claim.

C The Lean formalization

Theorem 1.1 and Corollary 1.2 for all coefficient distributions considered, Theorem 1.3, and the results of Section 8 are formalized in Lean 4 [5] with Mathlib [22] (Lean v4.34.0, Mathlib commit 5ed2965), in release 1.1.0 of [14] (Lean sources `swb:1:dir:c99b4d21420e01e2905fd093f3333dc2377ee298`). The formalization also uses the Hanson–Wright inequality from SLT [27] and Bentkus’s normal approximation [2] from ProbabilityApproximation [26]. The declarations are checked by the Lean kernel and depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The formal proofs follow the architecture of this paper, with different intermediate estimates and constants in places.

The principal declarations, in the namespace `Erdos522`, are

```
Corollary 1.2 (Erdős #522)  erdos_522
Theorem 1.1 (Rademacher)   radial_profile_compact_uniform_prefixes
Theorem 1.3                 ae_rademacher_logarithmic_root_rate_eventually
```

Theorem 1.3 is formalized in the equivalent form “for every $c > 128$, eventually $(\log n) |\nu_n(1)/n - 1/2| \leq c$ ”, and the radial laws for all degrees, obtained through the degrees $\lfloor j^q \rfloor$ of Proposition 8.1, are formalized at each fixed x . The other formal statements coincide with, or are stronger than, those of the paper. The file `lean/CORRESPONDENCE.md` in [14] lists the declaration of each result, including those for the other coefficient distributions and Section 8, and compares the formal and informal proofs.

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